

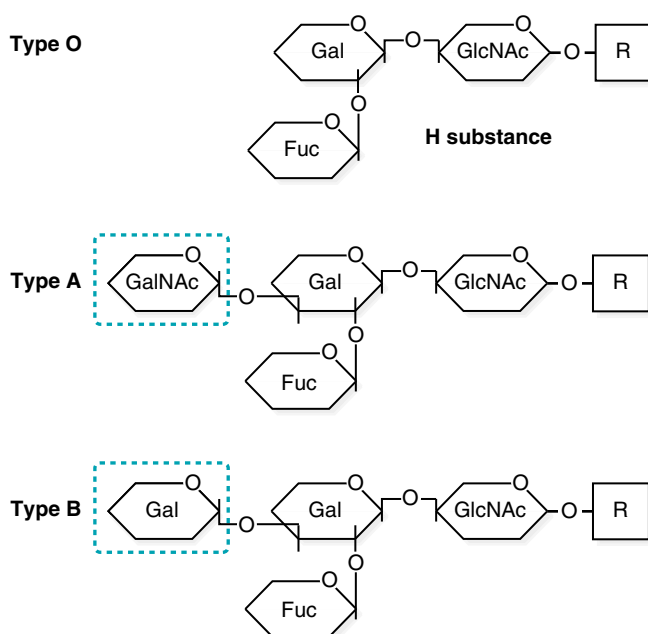
Blood Type

Fig. 30.16. Structures of the blood group substances. Note that these structures are the same except that type A has *N*-acetylgalactosamine (GalNAc) at the nonreducing end, type B has galactose (Gal), and type O has neither. R is either a protein or the lipid ceramide. Each antigenic determinant is boxed. Fuc = fucose; GlcNAc = *N*-acetylglucosamine; Gal = galactose.



The blood group substances are oligosaccharide components of glycolipids and glycoproteins found in most cell membranes. Those located on red blood cells have been studied extensively. A single genetic locus with two alleles determines an individual's blood type. These genes encode glycosyltransferases involved in the synthesis of the oligosaccharides of the blood group substances.

Most individuals can synthesize the H substance, an oligosaccharide that contains a fucose linked to a galactose at the nonreducing end of the blood group substance (see Fig. 30.16). Type A individuals produce an *N*-acetylgalactosamine transferase (encoded by the A gene) that attaches *N*-acetylgalactosamine to the galactose residue of the H substance. Type B individuals produce a galactosyltransferase (encoded by the B gene) that links galactose to the galactose residue of the H substance. Type AB individuals have both alleles and produce both transferases. Thus, some of the oligosaccharides of their blood group substances contain *N*-acetylgalactosamine and some contain galactose. Type O individuals produce a defective transferase, and, therefore, they do not attach either *N*-acetylgalactosamine or galactose to the H substance. Thus, individuals of blood type O have only the H substance.

B. Synthesis

Cerebrosides are synthesized from ceramide and UDP-glucose or UDP-galactose. They contain a single sugar (a monosaccharide). Gangliosides contain oligosaccharides produced from UDP-sugars and CMP-NANA, which is the precursor for the *N*-acetylneuraminic acid residues that branch from the linear chain. The synthesis of the sphingolipids is described in more detail in Chapter 33.

Sphingolipids are produced in the Golgi complex. Their lipid component becomes part of the membrane of the secretory vesicle that buds from the *trans* face of the Golgi. After the vesicle membrane fuses with the cell membrane, the lipid component of the glycolipid remains in the outer layer of the cell membrane, and the carbohydrate component extends into the extracellular space.



Cholera toxin binds to the carbohydrate portion of the GM1 ganglioside to allow its catalytic subunit to enter the cell.



Q: Erna Nemdy determined that a patient's blood type was AB. The new surgical resident was eager to give this patient a blood transfusion and, because AB blood is rare and an adequate amount was not available in the blood bank, he requested type A blood. Should Erna give him type A blood for his patient?



Jay Sakz has Tay-Sachs disease, which belongs to a group of gangliosidoses that include Fabry's and Gaucher's diseases. They mainly affect the brain, the skin, and the reticuloendothelial system (e.g., liver and spleen). In these diseases, complex lipids accumulate. Each of these lipids contains a ceramide as part of its structure (Table 30.3). The rate at which the lipid is synthesized is normal. However, the lysosomal enzyme required to degrade it is not very active, either because it is made in deficient quantities because of a mutation in a gene that specifically codes for the enzyme or because a critical protein required to activate the enzyme is deficient. Because the lipid cannot be degraded, it accumulates and causes degeneration of the affected tissues with progressive malfunction, such as the psychomotor deficits that occur as a result of the central nervous system involvement seen in most of these storage diseases.

Table 30.3. Defective Enzymes in the Gangliosidoses

Disease	Enzyme Deficiency	Accumulated Lipid
Fucosidosis	α -Fucosidase	Cer-Glc-Gal-GalNAc-Gal:Fuc H-isoantigen
Generalized gangliosidosis	G _{M1} - β -galactosidase	Cer-Glc-Gal(NeuAc)-GalNAc:Gal G _{M1} ganglioside
Tay-Sachs disease	Hexosaminidase A	Cer-Glc-Gal(NeuAc):GalNAc G _{M2} ganglioside
Tay-Sachs variant or Sandhoff disease	Hexosaminidase A and B	Cer-Glc-Gal-Gal:GalNAc Globoside plus G _{M2} ganglioside
Fabry's disease	α -Galactosidase	Cer-Glc-Gal:Gal Globotriaosylceramide
Ceramide lactoside lipidoses	Ceramide lactosidase (β -galactosidase)	Cer-Glc:Gal Ceramide lactoside
Metachromatic leukodystrophy	Arylsulfatase A	Cer-Gal:OSO ₃ 3-Sulfogalactosylceramide
Krabbe's disease	β -Galactosidase	Cer:Gal Galactosylceramide
Gaucher's disease	β -Glucosidase	Cer:Glc Glucosylceramide
Niemann-Pick disease	Sphingomyelinase	Cer:P-choline Sphingomyelin
Farber's disease	Ceramidase	Acyl: sphingosine Ceramide

NeuAc, *N*-acetylneuraminic acid; Cer, ceramide; Glc, glucose; Gal, galactose; Fuc, fucose : site of deficient enzyme reaction.

CLINICAL COMMENTS



During her stint in the hospital blood bank, **Erna Nemdy** learned that the importance of the ABO blood group system in transfusion therapy is based on two principles (Table 30.4). (a) Antibodies to A and to B antigens occur naturally in the blood serum of persons whose red blood cell surfaces lack the corresponding antigen (i.e., individuals with A antigens on their red blood cells have B antibodies in their serum and vice versa). These antibodies may arise as a result of previous exposure to cross-reacting antigens in bacteria and foods or to blood transfusions. (b) Antibodies to A and B are usually present in high titers and are capable of activating the entire complement system. As a result, these antibodies may cause intravascular destruction of a large number of incompatible red blood cells given during a blood transfusion. Individuals with type AB blood have both A and B antigens and do not produce antibodies to either. Hence, they are “universal” recipients. They can safely receive red blood cells from individuals of A, B, AB, or O blood type. (However, they cannot safely receive serum from these individuals because it contains antibodies to A or B antigens.) Those with type O blood do not have either antigen. They are “universal” donors; i.e., their red cells can safely be infused into type A, B, O, or AB individuals. (However, their serum contains antibodies to both A and B antigens and cannot safely be used.)



The patient could safely receive type A blood **cells** from another person because he has both A and B antigens on his own cells and does not have antibodies in his serum to either type A or B cells. However, he should not be given type A serum (or type A whole blood) because type A serum contains antibodies to type B antigens, which are present on his cells.

Table 30.4. Characteristics of the ABO Blood Groups

	O	A	B	AB
Red cell type	O	A	B	AB
Possible genotypes	OO	AA or AO	BB or BO	AB
Antibodies in serum	Anti-A and B	Anti-B	Anti-A	None
Frequency (in Caucasians)	45%	40%	10%	5%
Can accept blood types	O	A, O	B, O	A, B, AB, O

The second important red blood cell group is the Rh group. It is important because one of its antigenic determinants, the D antigen, is a very potent immunogen, stimulating the production of a large number of antibodies.

The unique carbohydrate composition of the glycoproteins that constitute the antigenic determinants on red blood cells in part contributes to the relative immunogenicity of the A, B, and Rh (D) red blood cell groups in human blood.



Tay-Sachs disease, the problem afflicting **Jay Sakz**, is an autosomal recessive disorder that is rare in the general population (1 in 300,000 births), but its prevalence in Jews of Eastern European extraction (who make up 90% of the Jewish population in the United States) is much higher (1 in 3,600 births). One in 28 Ashkenazi Jews carries this defective gene. Its presence can be discovered by measuring the tissue level of the protein produced by the gene (hexosaminidase A) or by recombinant DNA techniques. Skin fibroblasts of concerned couples planning a family are frequently used for these tests.

Carriers of the affected gene have a reduced but functional level of this enzyme that normally hydrolyzes a specific bond between an *N*-acetyl-D-galactosamine and a D-galactose residue in the polar head of the ganglioside.

No effective therapy is available. Enzyme replacement has met with little success because of the difficulties in getting the enzyme across the blood-brain barrier.

BIOCHEMICAL COMMENTS



Hexosaminidase A, the enzyme defective in Tay-Sachs disease, is actually composed of two subunits, an α and a β chain. The exact stoichiometry of the active enzyme is unknown, but it may be $\alpha_2\beta_2$. The α subunit is coded for by the HexA gene, whereas the β subunit is coded for by the HexB gene. In Tay-Sachs disease, the α subunit is defective, and hexosaminidase A activity is lost. However, the β subunit can form active tetramers in the absence of the α subunit, and this activity, named hexosaminidase B, which cleaves the glycolipid globoside, retains activity in children with Tay-Sachs disease. Thus, children with Tay-Sachs disease accumulate the ganglioside GM2, but not globoside (Fig. 30.17).

Mutation of the HexB gene, and production of a defective β subunit, leads to inactivation of both hexosaminidase A and B activity. Such a mutation leads to Sandhoff disease. Both activities are lost because both activities require a functional β subunit. The clinical course of this disease is similar to Tay-Sachs but with an accelerated timetable because of the initial accumulation of both GM2 and globoside in the lysosomes.

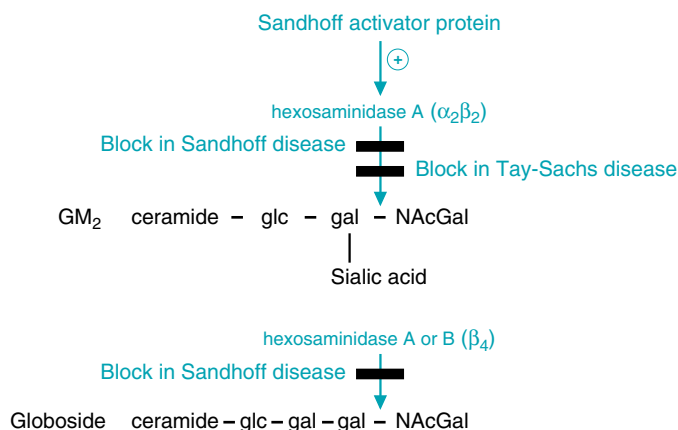


Fig. 30.17. Substrate specificities of hexosaminidase A, B, and the function of the activator protein. Glc = glucose; gal = galactose; NAcGal = N-acetylgalactosamine.

A third type of mutation also can lead to disease symptoms similar to those of Tay-Sachs disease. Children were identified with Tay-Sachs symptoms, but when both hexosaminidase A and B activities were measured in a test tube, they were normal. This disease, ultimately named Sandhoff activator disease, is caused by a mutation in a protein that is needed to activate hexosaminidase A activity. In the absence of the activator, hexosaminidase A activity is minimal, and GM2 initially accumulates in lysosomes. This mutation has no effect on hexosaminidase B activity.

When a glycolipid cannot be degraded because of an enzymatic mutation, it accumulates in residual bodies (vacuoles that contain material that lysosomal enzymes cannot digest). Normal cells contain a small number of residual bodies, but in diseases of lysosomal enzymes, large numbers of residual bodies accumulate within the cell, eventually interfering with normal cell function.

In 70% of the cases of Tay-Sachs disease in persons of Ashkenazai Jewish background, exon 11 of the gene for the α chain of hexosaminidase A contains a mutation. The normal gene sequence encodes a protein with the amino acids arg-ile-ser-tyr-gly-pro-asp in this region, as shown below:

$$\begin{array}{ccccccc} & & & & 10 & & 20 \\ 5' - & \text{CGTATATCCTATGGCCCTGAC} \\ & \text{arg - ile - ser - tyr - gly - pro - asp} \end{array}$$

The mutant DNA sequence for this area is shown below:

$$\begin{array}{ccccccc} & & & & 10 & & 20 \\ 5' - & \text{CGTATATCTATCCTATGGCCCTGAC} \\ & \text{arg - ile - ser - ile - leu - leu - trp - pro -} \end{array}$$

A four-base insertion (underlined) occurs in the mutated gene, which alters the reading frame of the protein, and also introduces a premature stop codon further down the protein, such that no functional α subunit can be produced.

Suggested References

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- Lloyd KO. The chemistry and immunochemistry of blood group A, B, H and Lewis antigens: past, present and future. *Glycoconjugate J* 2000:531–541.



REVIEW QUESTIONS—CHAPTER 30

- Which of the following best describes a mother with galactosemia caused by a deficiency of galactose 1-phosphate uridylyl transferase?
 - She can convert galactose to UDP-galactose for lactose synthesis during lactation.
 - She can form galactose 1-phosphate from galactose.
 - She can use galactose as a precursor to glucose production.
 - She can use galactose to produce glycogen.
 - She will have lower than normal levels of serum galactose after drinking milk.

2. The immediate carbohydrate precursors for glycolipid and glycoprotein synthesis are which of the following?
 - (A) Sugar phosphates
 - (B) Sugar acids
 - (C) Sugar alcohols
 - (D) Nucleotide sugars
 - (E) Acyl-sugars

3. A newborn is diagnosed with neonatal jaundice. In this patient, the bilirubin produced lacks which of the following carbohydrates?
 - (A) Glucose
 - (B) Gluconate
 - (C) Glucuronate
 - (D) Galactose
 - (E) Galactitol

4. The nitrogen donor for the formation of amino sugars is which of the following?
 - (A) Ammonia
 - (B) Asparagine
 - (C) Glutamine
 - (D) Adenine
 - (E) Dolichol

5. Which of the following glycolipids would accumulate in a patient with Sandhoff's disease?
 - (A) GM1
 - (B) Lactosyl-ceramide
 - (C) Globoside
 - (D) Glucocerebroside
 - (E) GM3

31 Gluconeogenesis and Maintenance of Blood Glucose Levels

*During fasting, many of the reactions of glycolysis are reversed as the liver produces glucose to maintain blood glucose levels. This process of **glucose production** is called **gluconeogenesis**.*

*Gluconeogenesis, which occurs primarily in the liver, is the pathway for the synthesis of glucose from compounds other than carbohydrates. In humans, the major precursors of glucose are **lactate**, **glycerol**, and **amino acids**, particularly **alanine**. Except for three key sequences, the reactions of gluconeogenesis are reversals of the steps of glycolysis (Fig. 31.1). The sequences of gluconeogenesis that do not use enzymes of glycolysis involve the irreversible, regulated steps of glycolysis. These three steps are the conversion of (a) pyruvate to phosphoenolpyruvate, (b) fructose 1,6-bisphosphate to fructose 6-phosphate, and (c) glucose 6-phosphate to glucose.*

Some tissues of the body, such as the brain and red blood cells, cannot synthesize glucose on their own, yet depend on glucose for energy. On a long-term basis, most tissues also require glucose for other functions such as the synthesis of the ribose moiety of nucleotides or the carbohydrate portion of glycoproteins and glycolipids. Therefore, to survive, humans must have mechanisms for maintaining blood glucose levels.

*After a meal containing carbohydrates, blood glucose levels rise (Fig. 31.2). Some of the glucose from the diet is stored in the liver as **glycogen**. After 2 or 3 hours of fasting, this glycogen begins to be degraded by the process of **glycogenolysis**, and glucose is released into the blood. As glycogen stores decrease, **adipose triacylglycerols** are also degraded, providing **fatty acids** as an alternative fuel and glycerol for the synthesis of glucose by gluconeogenesis. Amino acids are also released from the muscle to serve as gluconeogenic precursors.*

During an overnight fast, blood glucose levels are maintained by both glycogenolysis and gluconeogenesis. However, after approximately 30 hours of fasting, liver glycogen stores are mostly depleted. Subsequently, gluconeogenesis is the only source of blood glucose.

*Changes in the metabolism of glucose that occur during the switch from the fed to the fasting state are regulated by the hormones **insulin** and **glucagon**. Insulin is elevated in the fed state, and glucagon is elevated during fasting. Insulin stimulates the **transport** of glucose into certain cells such as those in muscle and adipose tissue. Insulin also alters the activity of key enzymes that regulate metabolism, stimulating the storage of fuels. Glucagon counters the effects of insulin, stimulating the release of stored fuels and the conversion of lactate, amino acids, and glycerol to glucose.*

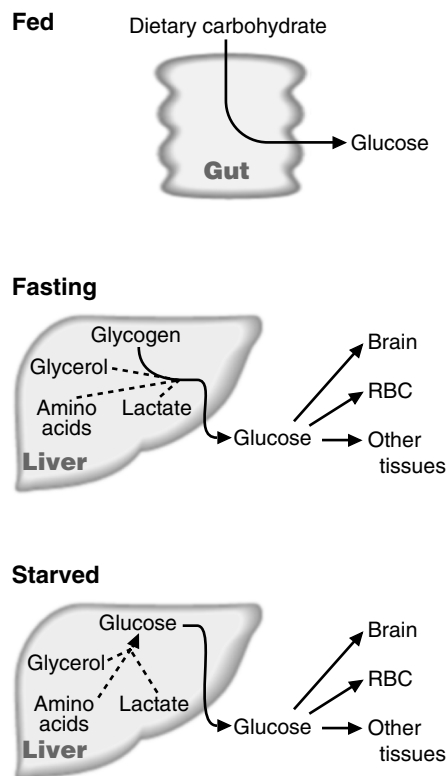


Fig. 31.2. Sources of blood glucose in the fed, fasting, and starved states. RBC = red blood cells.

Q: What clinical signs and symptoms help to distinguish a coma caused by an excess of blood glucose and ketone bodies due to a deficiency of insulin (diabetic ketoacidosis [DKA]) from a coma caused by a sudden lowering of blood glucose (hypoglycemic coma) induced by the inadvertent injection of excessive insulin—the current problem experienced by **Di Abietes**?

Blood glucose levels are maintained not only during fasting, but also during exercise, when muscle cells take up glucose from the blood and oxidize it for energy. During exercise, the liver supplies glucose to the blood by the processes of glycogenolysis and gluconeogenesis.



THE WAITING ROOM



Al Martini, a known alcoholic, was brought to the emergency room by his landlady, who stated that he had been drinking heavily for the past week. During this time his appetite had gradually diminished, and he had not eaten any food for the past 3 days. He was confused, combative, tremulous, and sweating profusely. His speech was slurred. His heart rate was rapid (110 beats per minute). As his blood pressure was being determined, he had a grand mal seizure. His blood glucose, drawn just before the onset of the seizure, was 28 mg/dL or 1.6 mM (reference range for overnight fasting blood glucose = 80–100 mg/dL or 4.4–5.6 mM). His blood ethanol level drawn at the same time was 295 mg/dL (intoxication level, i.e., “confused” stage = 150–300 mg/dL).



Emma Wheezer presented to the emergency room 3 days after discharge from the hospital following a 10-day admission for severe refractory bronchial asthma. She required high-dose intravenous dexamethasone (an anti-inflammatory synthetic glucocorticoid) for the first 8 days of her stay. After 2 additional days receiving oral dexamethasone, she was discharged on substantial pharmacologic doses of this steroid and instructed to return to her physician’s office in 5 days. She presented now with marked polyuria (increased urination), polydipsia (increased thirst), and muscle weakness. Her blood glucose was 275 mg/dL or 15 mM (reference range = 80–100 mg/dL or 4.4–5.6 mM).



Di Abietes could not remember whether she had taken her 6:00 PM insulin dose, when, in fact, she had done so. Unfortunately, she decided to give herself the evening dose (for the second time). When she did not respond to her alarm clock at 6:00 AM the following morning, her roommate tried unsuccessfully to awaken her. The roommate called an ambulance, and Di was rushed to the hospital emergency room in a coma. Her pulse and blood pressure at admission were normal. Her skin was flushed and slightly moist. Her respirations were slightly slow.



Ann O’Rexia continues to resist efforts on the part of her psychiatrist and family physician to convince her to increase her caloric intake. Her body weight varies between 97 and 99 lb, far below the desirable weight for a woman who is 5 feet 7 inches tall. In spite of her severe diet, her fasting blood glucose levels range from 55 to 70 mg/dL. She denies having any hypoglycemic symptoms.



Otto Shape has complied with his calorie-restricted diet and aerobic exercise program. He has lost another 7 lb and is closing in on his goal of weighing 154 lb. He notes increasing energy during the day, and remains alert during lectures and assimilates the lecture material noticeably better than he did before starting his weight loss and exercise program. He jogs for 45 minutes each morning before breakfast.

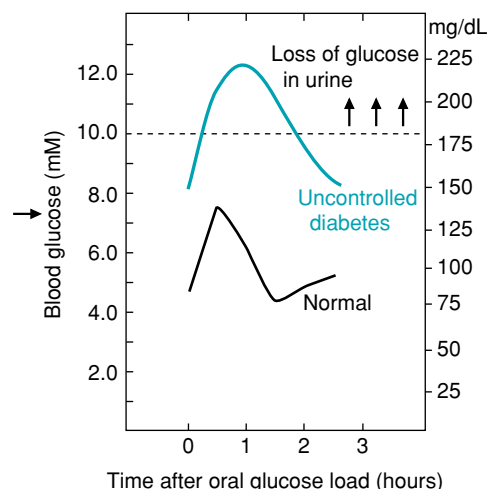


Diabetes mellitus (DM) should be suspected if a venous plasma glucose level drawn irrespective of when food was last eaten (a “random” sample of blood glucose) is “unequivocally elevated” (i.e., ≥ 200 mg/dL), particularly in a patient who manifests the classic signs and symptoms of chronic hyperglycemia (polydipsia, polyuria, blurred vision, headaches, rapid weight loss, sometimes accompanied by nausea and vomiting). To confirm the diagnosis, the patient should fast overnight (10–16 hours), and the blood glucose measurement should be repeated. Values of less than 110 mg/dL are considered normal. Values greater than 140 mg/dL are indicative of DM. Glycosylated hemoglobin should be measured to determine the extent of hyperglycemia over the past 4 to 8 weeks. Values of fasting blood glucose between 111 and 140 mg/dL are designated impaired fasting glucose tolerance (IGT), and further testing should be performed to determine whether these individuals will eventually develop overt diabetes mellitus.

Although the oral glucose tolerance test (OGTT) is contraindicated for patients who clearly have diabetes mellitus, it is used for patients with fasting blood glucose in the IGT range (between 115 and 140 mg/dL). In the OGTT, a nonpregnant patient who has fasted overnight drinks 75 g glucose in an aqueous solution. Blood samples are drawn before the oral glucose load and at 30, 60, 90, and 120 minutes thereafter. If any one of the 30-, 60-, and 90-minute samples and the 120-minute sample are greater than 200 mg/dL, overt DM is indicated.

The diagnosis of IGT and the more severe form of glucose intolerance (DM) is based on blood glucose levels because no more specific characteristic for the disorder exists. The distinction between IGT and DM is clouded by the fact that a patient’s blood glucose level may vary significantly with serial testing over time under the same conditions of diet and activity.

The renal tubular transport maximum in the average healthy subject is such that glucose will not appear in the urine until the blood glucose level exceeds 180 mg/dL. As a result, reagent tapes (Tes-Tape or Dextrostix) designed to detect the presence of glucose in the urine are not sensitive enough to establish a diagnosis of early DM.



A: Comatose patients in diabetic ketoacidosis have the smell of acetone (a derivative of the ketone body acetoacetate) on their breath. In addition, DKA patients have deep, relatively rapid respirations typical of acidotic patients (Kussmaul respirations). These respirations result from an acidosis-induced stimulation of the respiratory center in the brain. More CO_2 is exhaled in an attempt to reduce the amount of acid in the body: $\text{H}^+ + \text{HCO}_3^- \rightarrow \text{H}_2\text{CO}_3 \rightarrow \text{H}_2\text{O} + \text{CO}_2$ (exhaled).

The severe hyperglycemia of DKA also causes an osmotic diuresis (i.e., glucose entering the urine carries water with it), which, in turn, causes a contraction of blood volume. Volume depletion may be aggravated by vomiting, which is common in patients with DKA. DKA may cause dehydration (dry skin), a low blood pressure, and a rapid heartbeat. These respiratory and hemodynamic alterations are not seen in patients with hypoglycemic coma. The flushed, wet skin of hypoglycemic coma is in contrast to the dry skin observed in DKA.



Glucocorticoids are naturally occurring steroid hormones. In humans, the major glucocorticoid is cortisol. Glucocorticoids are produced in the adrenal cortex in response to various types of stress (see Chapter 43). One of their actions is to stimulate the degradation of muscle protein. Thus, increased amounts of amino acids become available as substrates for gluconeogenesis. **Emma Wheezer** noted muscle weakness, a result of the muscle-degrading action of the synthetic glucocorticoid dexamethasone, which she was taking for its anti-inflammatory effects.

I. GLUCOSE METABOLISM IN THE LIVER

Glucose serves as a fuel for most tissues of the body. It is the major fuel for certain tissues such as the brain and red blood cells. After a meal, food is the source of blood glucose. The liver oxidizes glucose and stores the excess as glycogen. The liver also uses the pathway of glycolysis to convert glucose to pyruvate, which provides carbon for the synthesis of fatty acids. Glycerol 3-phosphate, produced from glycolytic intermediates, combines with fatty acids to form triacylglycerols, which are secreted into the blood in very-low-density lipoproteins (VLDL; further explained in Chapter 32). During fasting, the liver releases glucose into the blood, so that glucose-dependent tissues do not suffer from a lack of energy. Two mechanisms are involved in this process: glycogenolysis and gluconeogenesis. Hormones, particularly insulin and glucagon, dictate whether glucose flows through glycolysis or whether the reactions are reversed and glucose is produced via gluconeogenesis.

II. GLUCONEOGENESIS

Gluconeogenesis, the process by which glucose is synthesized from noncarbohydrate precursors, occurs mainly in the liver under fasting conditions. Under the more extreme conditions of starvation, the kidney cortex also may produce glucose. For the most part, the glucose produced by the kidney cortex is used by the kidney medulla, but some may enter the bloodstream.

Starting with pyruvate, most of the steps of gluconeogenesis are simply reversals of those of glycolysis (Fig. 31.3). In fact, these pathways differ at only three points. Enzymes involved in catalyzing these steps are regulated so that either glycolysis or gluconeogenesis predominates, depending on physiologic conditions.

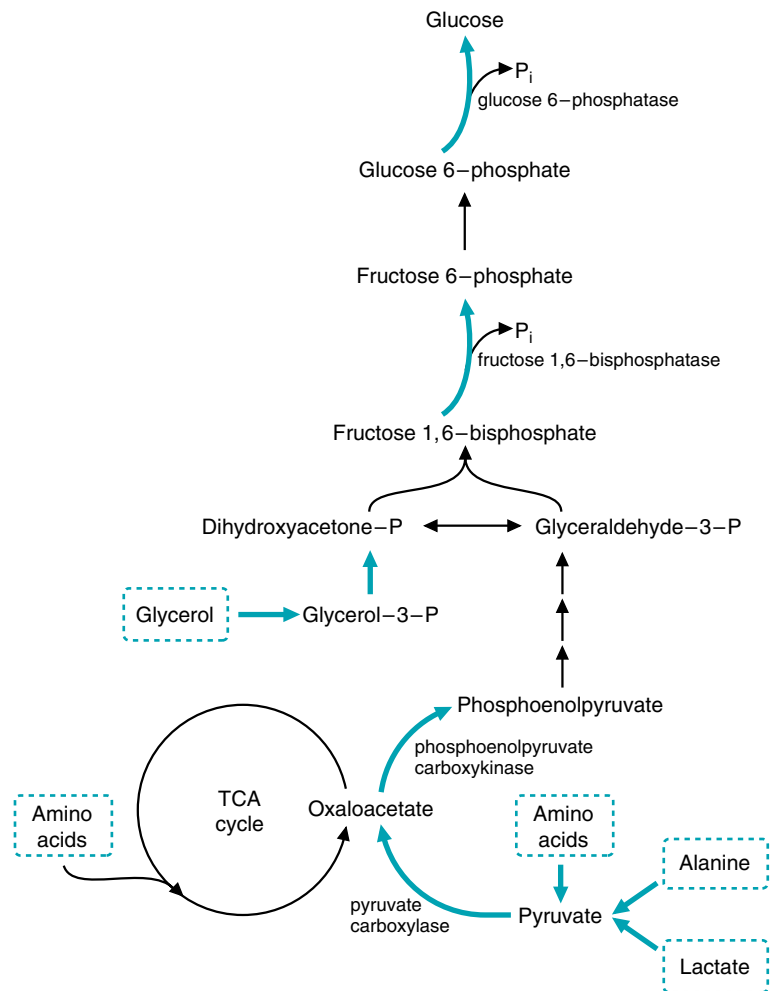


Fig. 31.3. Key reactions of gluconeogenesis. The precursors are amino acids (particularly alanine), lactate, and glycerol. Heavy arrows indicate steps that differ from those of glycolysis.



Because ethanol metabolism only gives rise to acetyl-CoA, the carbons of ethanol cannot be used for gluconeogenesis.

Most of the steps of gluconeogenesis use the same enzymes that catalyze the process of glycolysis. The flow of carbon, of course, is in the reverse direction. Three reaction sequences of gluconeogenesis differ from the corresponding steps of glycolysis. They involve the conversion of pyruvate to phosphoenolpyruvate (PEP) and the reactions that remove phosphate from fructose 1,6-bisphosphate to form fructose 6-phosphate and from glucose 6-phosphate to form glucose (see Fig. 31.3). The conversion of pyruvate to PEP is catalyzed during gluconeogenesis by a series of enzymes instead of the single enzyme used for glycolysis. The reactions that remove phosphate from fructose 1,6-bisphosphate and from glucose 6-phosphate each use single enzymes that differ from the corresponding enzymes of glycolysis. Although phosphate is added during glycolysis by kinases, which use adenosine triphosphate (ATP), it is removed during gluconeogenesis by phosphatases that release P_i via hydrolysis reactions.

A. Precursors for Gluconeogenesis

The three major carbon sources for gluconeogenesis in humans are lactate, glycerol, and amino acids, particularly alanine. Lactate is produced by anaerobic glycolysis in tissues such as exercising muscle or red blood cells, as well as by adipocytes during the fed state. Glycerol is released from adipose stores of

triacylglycerol, and amino acids come mainly from amino acid pools in muscle, where they may be obtained by degradation of muscle protein. Alanine, the major gluconeogenic amino acid, is produced in the muscle from other amino acids and from glucose (see Chapter 38).

B. Formation of Gluconeogenic Intermediates from Carbon Sources

The carbon sources for gluconeogenesis form pyruvate, intermediates of the tricarboxylic acid (TCA) cycle, or intermediates common both to glycolysis and gluconeogenesis.

1. LACTATE, AMINO ACIDS, AND GLYCEROL

Pyruvate is produced in the liver from the gluconeogenic precursors lactate and alanine. Lactate dehydrogenase oxidizes lactate to pyruvate, generating NADH (Fig. 31.4A), and alanine aminotransferase converts alanine to pyruvate (see Fig. 31.4B).

Although alanine is the major gluconeogenic amino acid, other amino acids, such as serine, serve as carbon sources for the synthesis of glucose because they also form pyruvate, the substrate for the initial step in the process. Some amino acids form intermediates of the TCA cycle (see Chapter 20), which can enter the gluconeogenic pathway.

The carbons of glycerol are gluconeogenic because they form dihydroxyacetone phosphate (DHAP), a glycolytic intermediate (see Fig. 31.4C)

2. PROPIONATE

Fatty acids with an odd number of carbon atoms, which are obtained mainly from vegetables in the diet, produce propionyl CoA from the three carbons at the ω -end of the chain (see Chapter 23). These carbons are relatively minor precursors of glucose in humans. Propionyl CoA is converted to methylmalonyl CoA, which is rearranged to form succinyl CoA, a 4-carbon intermediate of the TCA cycle that can be used for gluconeogenesis. The remaining carbons of an odd-chain fatty acid form acetyl CoA, from which no net synthesis of glucose occurs.

β -Oxidation of fatty acids produces acetyl CoA. Because the pyruvate dehydrogenase reaction is thermodynamically and kinetically irreversible, acetyl CoA does not form pyruvate for gluconeogenesis. Therefore, if acetyl CoA is to produce glucose, it must enter the TCA cycle and be converted to malate. For every two carbons of acetyl CoA that are converted to malate, two carbons are released as CO_2 : one in the reaction catalyzed by isocitrate dehydrogenase and the other in the reaction catalyzed by α -ketoglutarate dehydrogenase. Therefore, there is no *net* synthesis of glucose from acetyl CoA.



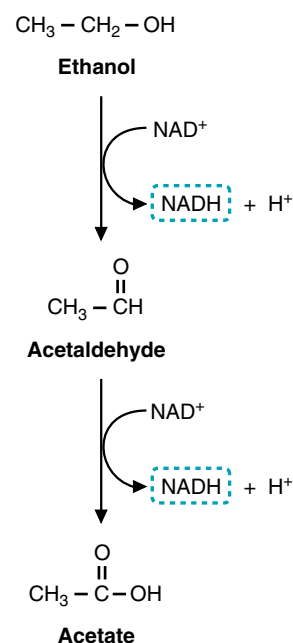
In some species, propionate is a major source of carbon for gluconeogenesis. Ruminants can produce massive amounts of glucose from propionate. In cows, the cellulose in grass is converted to propionate by bacteria in the rumen. This substrate is then used to generate more than 5 lb glucose each day by the process of gluconeogenesis.



In a fatty acid with 19 carbons, how many carbons (and which ones) form glucose?



Excessive ethanol metabolism will block the production of gluconeogenic precursors. Cells have limited amounts of NAD, which exist either as NAD^+ or as NADH. As the levels of NADH rise, those of NAD^+ fall, and the ratio of the concentrations of NADH and NAD^+ ($[\text{NADH}]/[\text{NAD}^+]$) increases. In the presence of ethanol, which is very rapidly oxidized in the liver, the $[\text{NADH}]/[\text{NAD}^+]$ ratio is much higher than it is in the normal fasting liver. High levels of NADH drive the lactate dehydrogenase reaction toward lactate. Therefore, lactate cannot enter the gluconeogenic pathway, and pyruvate that is generated from alanine is converted to lactate. Because glycerol is oxidized by NAD^+ during its conversion to DHAP, the conversion of glycerol to glucose is also inhibited when NADH levels are elevated. Consequently, the major precursors lactate, alanine, and glycerol are not used for gluconeogenesis under conditions in which alcohol metabolism is high.



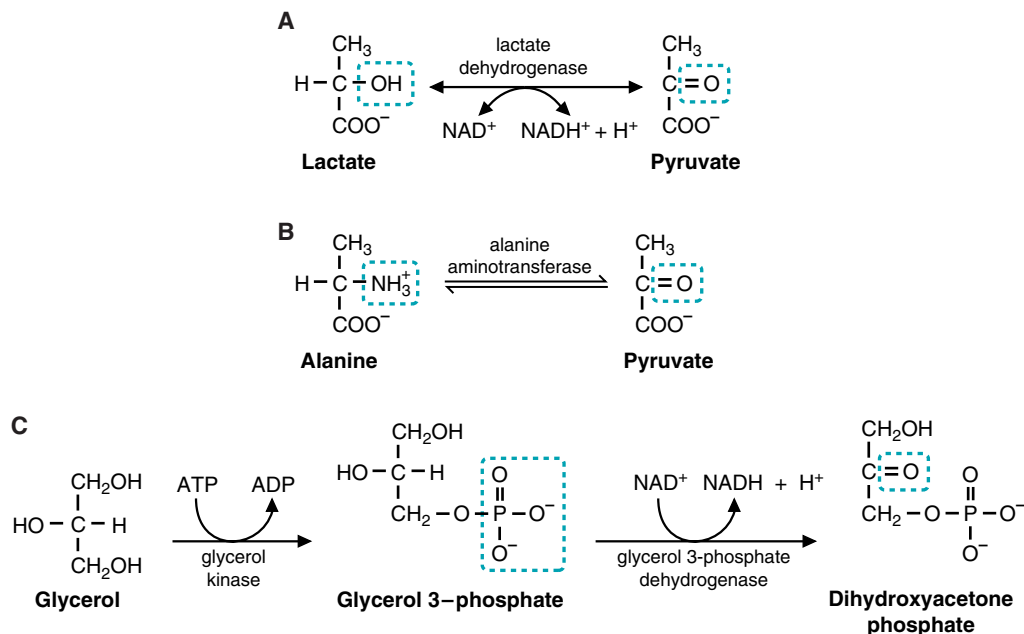


Fig. 31.4. Metabolism of gluconeogenic precursors. A. Conversion of lactate to pyruvate. B. Conversion of alanine to pyruvate. In this reaction, alanine aminotransferase transfers the amino group of alanine to α -ketoglutarate to form glutamate. The coenzyme for this reaction, pyridoxal phosphate, accepts and donates the amino group. C. Conversion of glycerol to dihydroxyacetone phosphate.



Amino acids that form intermediates of the TCA cycle are converted to malate, which enters the cytosol and is converted to oxaloacetate, which proceeds through gluconeogenesis to form glucose. When excessive amounts of ethanol are ingested, elevated NADH levels inhibit the conversion of malate to oxaloacetate in the cytosol. Therefore, carbons from amino acids that form intermediates of the TCA cycle cannot be converted to glucose as readily.

C. Pathway of Gluconeogenesis

Gluconeogenesis occurs by a pathway that reverses many, but not all, of the steps of glycolysis.

1. CONVERSION OF PYRUVATE TO PHOSPHOENOLPYRUVATE

In glycolysis, PEP is converted to pyruvate by pyruvate kinase. In gluconeogenesis, a series of steps are required to accomplish the reversal of this reaction (Fig. 31.5). Pyruvate is carboxylated by pyruvate carboxylase to form oxaloacetate. This enzyme, which requires biotin, is the catalyst of an anaplerotic (refilling) reaction of the TCA cycle (see Chapter 20). In gluconeogenesis, this reaction replenishes the oxaloacetate that is used for the synthesis of glucose (Fig. 31.6).

The CO_2 that was added to pyruvate to form oxaloacetate is released in the reaction catalyzed by phosphoenolpyruvate carboxykinase (PEPCK), which generates PEP (Fig. 31.7A). For this reaction, GTP provides a source of energy as well as the phosphate group of PEP. Pyruvate carboxylase is found in mitochondria. In various species, PEPCK is located either in the cytosol or in mitochondria, or it is distributed between these two compartments. In humans, the enzyme is distributed about equally in each compartment.

Oxaloacetate, generated from pyruvate by pyruvate carboxylase or from amino acids that form intermediates of the TCA cycle, does not readily cross the mitochondrial membrane. It is either decarboxylated to form PEP by the mitochondrial PEPCK or it is converted to malate or aspartate (see Figs. 31.7B and 31.7C). The conversion of oxaloacetate to malate requires NADH. PEP, malate, and aspartate can be transported into the cytosol.

After malate or aspartate traverse the mitochondrial membrane (acting as carriers of oxaloacetate) and enter the cytosol, they are reconverted to oxaloacetate by reversal of the reactions given above (see Figs. 31.7B and C). The conversion of malate to oxaloacetate generates NADH. Whether oxaloacetate is transported



Only the three carbons at the ω -end of an odd chain fatty acid that form propionyl CoA are converted to glucose. The remaining 16 carbons of a fatty acid with 19 carbons form acetyl CoA, which does not form any net glucose.

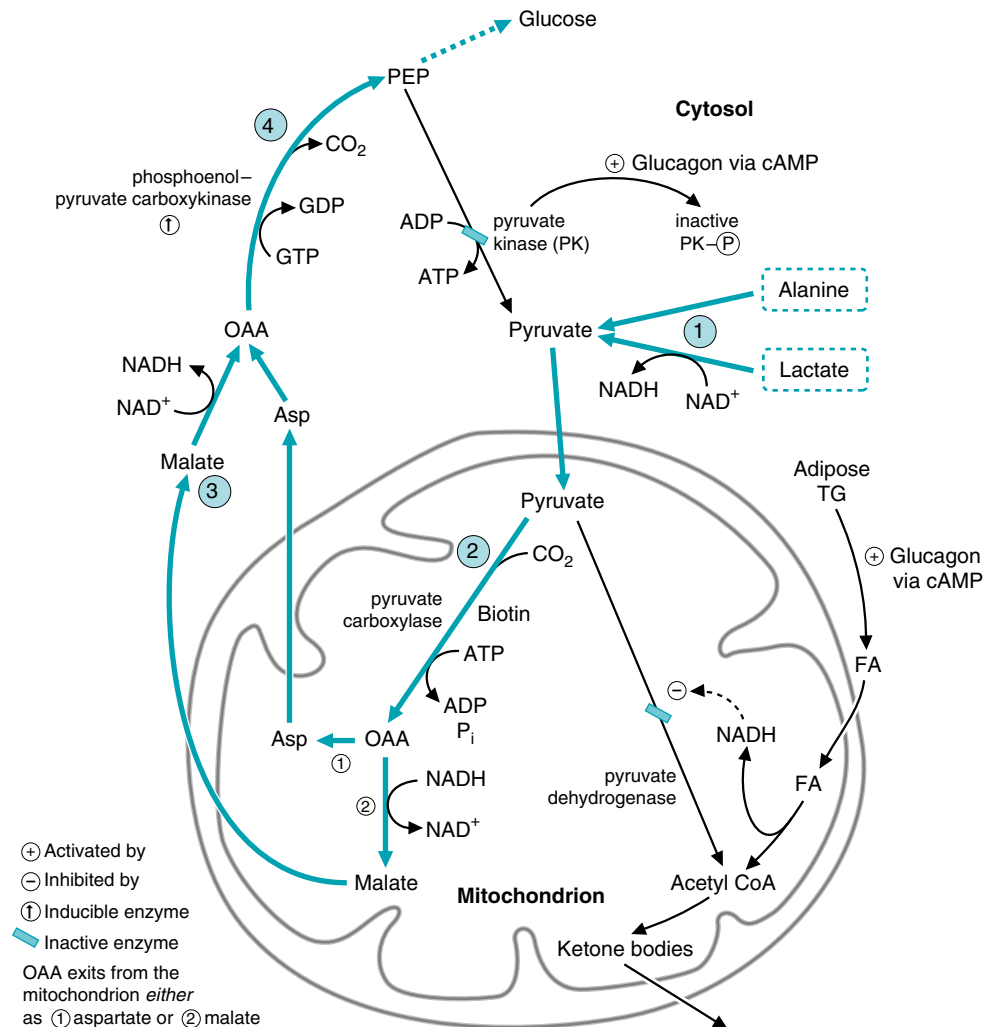


Fig. 31.5. Conversion of pyruvate to phosphoenolpyruvate (PEP). Follow the shaded circled numbers on the diagram, starting with the precursors alanine and lactate. The first step is the conversion of alanine and lactate to pyruvate. Pyruvate then enters the mitochondria and is converted to OAA (circle 2) by pyruvate carboxylase. Pyruvate dehydrogenase has been inactivated by both the NADH and acetyl-CoA generated from fatty acid oxidation, which allows oxaloacetate production for gluconeogenesis. The oxaloacetate formed in the mitochondria is converted to either malate or aspartate to enter the cytoplasm via the malate/aspartate shuttle. Once in the cytoplasm the malate or aspartate is converted back into oxaloacetate (circle 3), and phosphoenolpyruvate carboxykinase will convert it to PEP (circle 4). The white circled numbers are alternate routes for exit of carbon from the mitochondrion using the malate/aspartate shuttle. OAA = oxaloacetate; FA = fatty acid; TG = triacylglycerol.

across the mitochondrial membrane as malate or aspartate depends on the need for reducing equivalents in the cytosol. NADH is required to reduce 1,3-bisphosphoglycerate to glyceraldehyde 3-phosphate during gluconeogenesis.

Oxaloacetate, produced from malate or aspartate in the cytosol, is converted to PEP by the cytosolic PEPCK (see Fig. 31.7A).

2. CONVERSION OF PHOSPHOENOLPYRUVATE TO FRUCTOSE 1,6-BISPHOSPHATE

The remaining steps of gluconeogenesis occur in the cytosol (Fig. 31.8). Starting with PEP as a substrate, the steps of glycolysis are reversed to form glyceraldehyde 3-phosphate. For every two molecules of glyceraldehyde 3-phosphate that are formed, one is converted to dihydroxyacetone phosphate (DHAP). These two triose

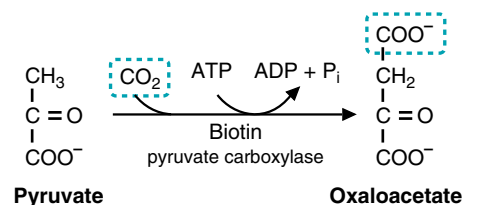


Fig. 31.6. Conversion of pyruvate to oxaloacetate.

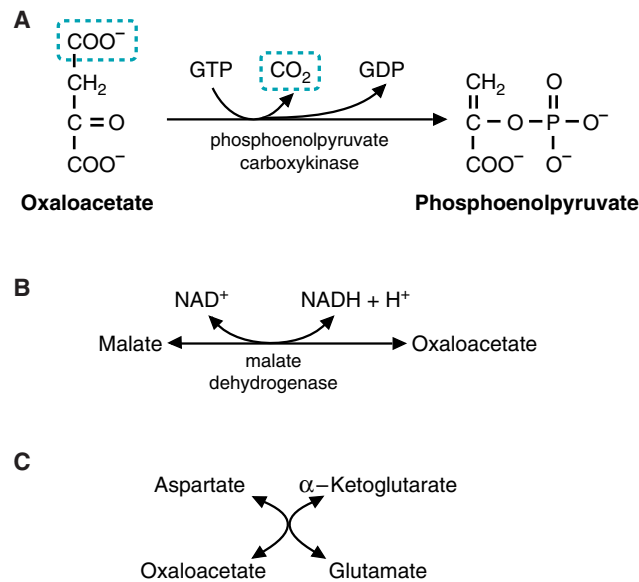


Fig. 31.7. The generation of PEP from gluconeogenic precursors. A. Conversion of oxaloacetate to phosphoenolpyruvate, using PEP carboxykinase. B. Interconversion of oxaloacetate and malate. C. Transamination of aspartate to form oxaloacetate. Note that the cytosolic reaction is the reverse of the mitochondrial reaction as shown in Figure 31.5.

phosphates, DHAP and glyceraldehyde 3-phosphate, condense to form fructose 1,6-bisphosphate by a reversal of the aldolase reaction.

Because glycerol forms DHAP, it enters the gluconeogenic pathway at this level.

3. CONVERSION OF FRUCTOSE 1,6-BISPHOSPHATE TO FRUCTOSE 6-PHOSPHATE

The enzyme fructose 1,6-bisphosphatase releases inorganic phosphate from fructose 1,6-bisphosphate to form fructose 6-phosphate. This is not a reversal of the PFK-1 reaction; ATP is not produced when the phosphate is removed from the 1 position of fructose 1,6-bisphosphate, because that is a low-energy phosphate bond. Rather, inorganic phosphate is released in this hydrolysis reaction. In the next reaction of gluconeogenesis, fructose 6-phosphate is converted to glucose 6-phosphate by the same isomerase used in glycolysis (phosphoglucosomerase).

4. CONVERSION OF GLUCOSE 6-PHOSPHATE TO GLUCOSE

Glucose 6-phosphatase hydrolyzes P_i from glucose 6-phosphate, and free glucose is released into the blood. As with fructose 1,6-bisphosphatase, this is not a reversal of the glucokinase reaction, because the phosphate bond in glucose 6-phosphate is a low-energy bond, and ATP is not generated at this step.

Glucose 6-phosphatase is located in the membrane of the endoplasmic reticulum. It is used not only in gluconeogenesis, but also to produce blood glucose from the breakdown of liver glycogen.



Al Martini had not eaten for 3 days, so he had no dietary source of glucose, and his liver glycogen stores were essentially depleted. He was solely dependent on gluconeogenesis to maintain his blood glucose levels. One of the consequences of ethanol ingestion and the subsequent rise in NADH levels is that the major carbon sources for gluconeogenesis cannot readily be converted to glucose. After his alcoholic binges, Mr. Martini became hypoglycemic. His blood glucose was 28 mg/dL.

D. Regulation of Gluconeogenesis

Although gluconeogenesis occurs during fasting, it is also stimulated during prolonged exercise, by a high-protein diet, and under conditions of stress. The factors that promote the overall flow of carbon from pyruvate to glucose include the availability of substrate and changes in the activity or amount of certain key enzymes of glycolysis and gluconeogenesis.

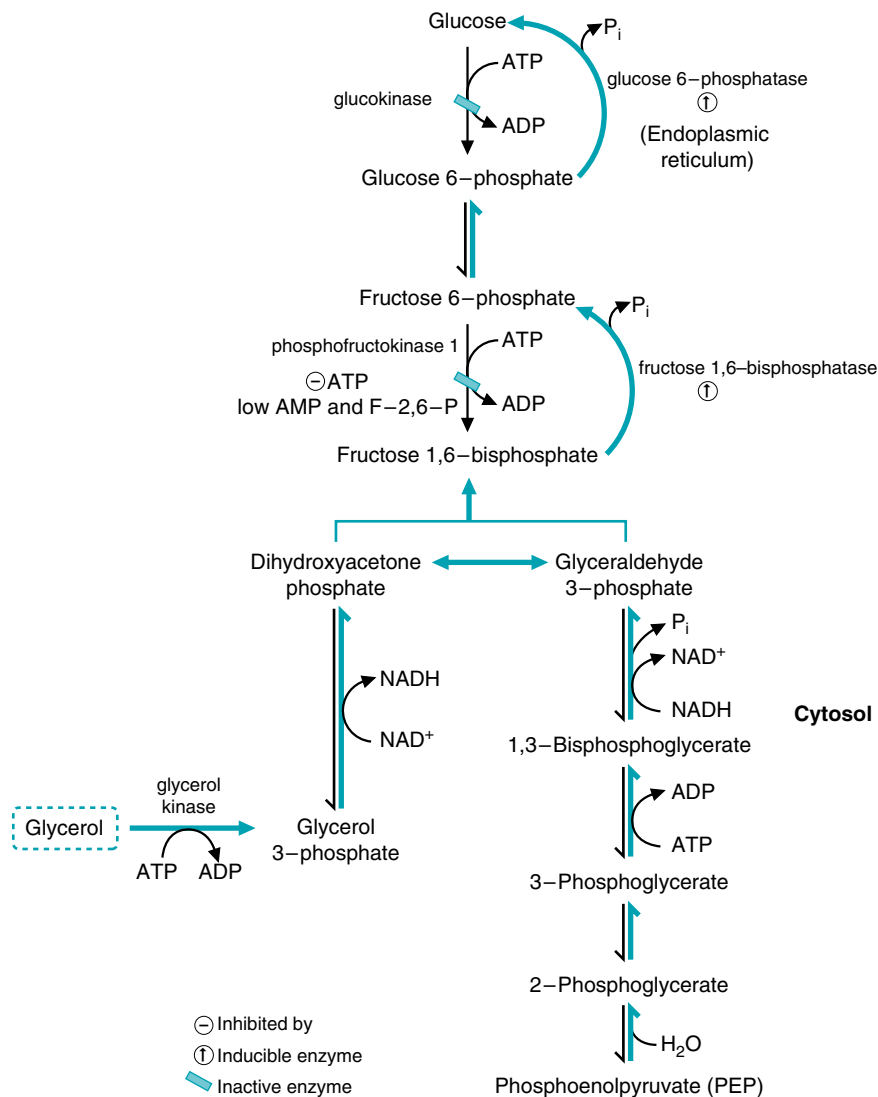


Fig. 31.8. Conversion of phosphoenolpyruvate and glycerol to glucose.

1. AVAILABILITY OF SUBSTRATE

Gluconeogenesis is stimulated by the flow of its major substrates from peripheral tissues to the liver. Glycerol is released from adipose tissue whenever the levels of insulin are low and the levels of glucagon or the “stress” hormones, epinephrine and cortisol (a glucocorticoid), are elevated in the blood (see Chapter 26). Lactate is produced by muscle during exercise and by red blood cells. Amino acids are released from muscle whenever insulin is low or when cortisol is elevated. Amino acids are also available for gluconeogenesis when the dietary intake of protein is high and intake of carbohydrate is low.

2. ACTIVITY OR AMOUNT OF KEY ENZYMES

Three sequences in the pathway of gluconeogenesis are regulated:

1. pyruvate → phosphoenolpyruvate
2. fructose 1,6-bisphosphate → fructose 6-phosphate
3. glucose 6-phosphate → glucose.

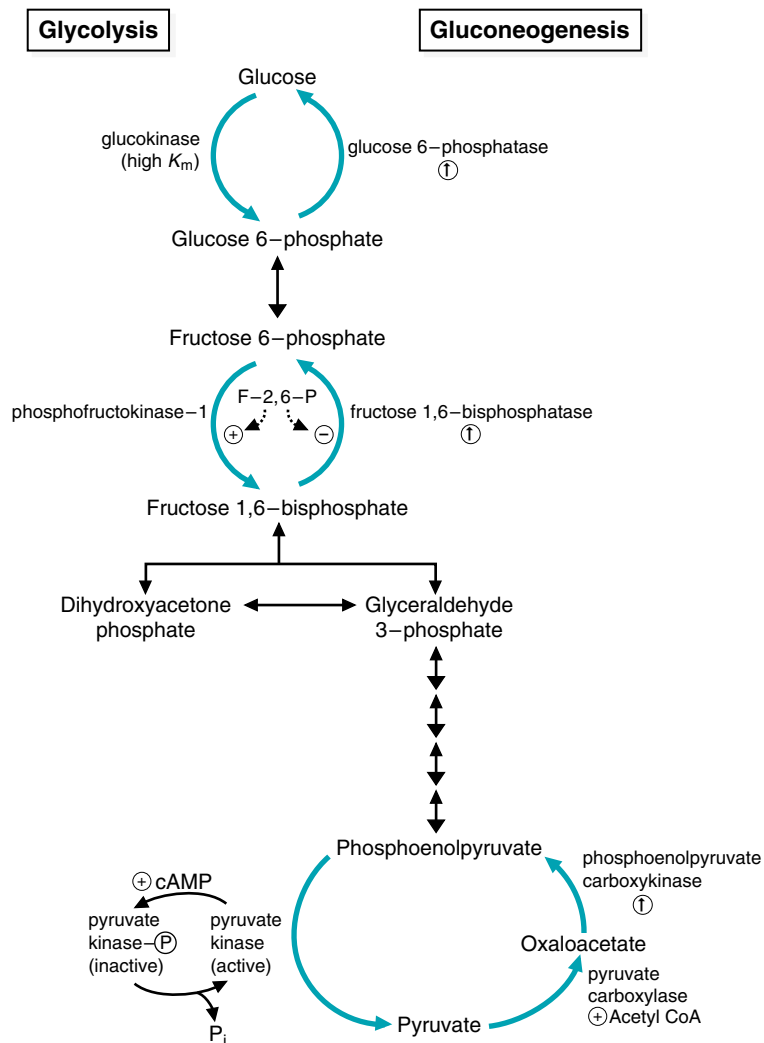


Fig. 31.9. Enzymes involved in regulating the substrate cycles of glycolysis and gluconeogenesis. Heavy arrows indicate the three substrate cycles. F-2,6-P, fructose 2,6-bisphosphate; +, activated by; -, inhibited by; circled ↑, inducible enzyme.



PEPCK is an example of an inducible enzyme. The gene for the cytosolic form of the enzyme contains regulatory sequences in its 5'-flanking region. cAMP, which is produced as a result of the binding of hormones such as glucagon or epinephrine to receptors on the cell surface, causes one of these sequences to be activated. Glucocorticoids affect a different regulatory sequence. cAMP or glucocorticoids independently cause increased transcription of the PEPCK gene. The mRNA produced during transcription travels to the cytosol combines with ribosomes and is translated, producing increased amounts of the enzyme PEPCK.

These steps correspond to those in glycolysis that are catalyzed by regulatory enzymes. The enzymes involved in these steps of gluconeogenesis differ from those that catalyze the reverse reactions in glycolysis. The net flow of carbon, whether from glucose to pyruvate (glycolysis) or from pyruvate to glucose (gluconeogenesis), depends on the relative activity or amount of these glycolytic or gluconeogenic enzymes (Fig. 31.9 and Table 31.1).

3. CONVERSION OF PYRUVATE TO PHOSPHOENOLPYRUVATE

Pyruvate, a key substrate for gluconeogenesis, is derived from lactate and amino acids, particularly alanine. Pyruvate is not converted to acetyl CoA under conditions favoring gluconeogenesis because pyruvate dehydrogenase is relatively inactive. Instead, pyruvate is converted to oxaloacetate by pyruvate carboxylase. Subsequently, oxaloacetate is converted to PEP by PEPCK. Because of the activity state of the enzymes discussed in subsequent sections, PEP reverses the steps of glycolysis, ultimately forming glucose.

Pyruvate dehydrogenase is inactive. Under conditions of fasting, insulin levels are low, and glucagon levels are elevated. Consequently, fatty acids and glycerol are

Table 31.1. Regulation of Enzymes of Glycolysis and Gluconeogenesis in Liver

A. Glycolytic Enzymes	Mechanism
Pyruvate kinase	Activated by F-1, 6-P Inhibited by ATP, alanine Inhibited by phosphorylation (glucagon and epinephrine lead to an increase in cAMP levels, which activates protein kinase A)
Phosphofructokinase-1	Activated by F-2,6-P, AMP Inhibited by ATP, citrate
Glucokinase	High K_m for glucose Induced by insulin
B. Gluconeogenic Enzymes	Mechanism
Pyruvate carboxylase	Activated by acetyl CoA
Phosphoenolpyruvate carboxykinase	Induced by glucagon, epinephrine, glucocorticoids Repressed by insulin
Fructose 1,6-bisphosphatase	Inhibited by F-2,6-P, AMP
Glucose 6-phosphatase	Induced during fasting

released from the triacylglycerol stores of adipose tissue. Fatty acids travel to the liver, where they undergo β -oxidation, producing acetyl CoA, NADH, and ATP. As a consequence, the concentration of ADP decreases. These changes result in the phosphorylation of pyruvate dehydrogenase to the inactive form. Therefore, pyruvate is not converted to acetyl CoA.

Pyruvate carboxylase is active. Acetyl CoA, which is produced by oxidation of fatty acids, activates pyruvate carboxylase. Therefore, pyruvate, derived from lactate or alanine, is converted to oxaloacetate.

Phosphoenolpyruvate carboxykinase is induced. Oxaloacetate produces PEP in a reaction catalyzed by PEPCK. Cytosolic PEPCK is an inducible enzyme, which means that the quantity of the enzyme in the cell increases because of increased transcription of its gene and increased translation of its mRNA. The major inducer is cyclic adenosine monophosphate (cAMP), which is increased by hormones that activate adenylate cyclase. Adenylate cyclase produces cAMP from ATP. Glucagon is the hormone that causes cAMP to rise during fasting, whereas epinephrine acts during exercise or stress. cAMP activates protein kinase A, which phosphorylates a set of specific transcription factors (CREB) that stimulate transcription of the PEPCK gene (see Chapter 16 and Fig. 16.18). Increased synthesis of mRNA for PEPCK results in increased synthesis of the enzyme. Cortisol, the major human glucocorticoid, also induces PEPCK.

Pyruvate kinase is inactive. When glucagon is elevated, pyruvate kinase is phosphorylated and inactivated by a mechanism involving cAMP and protein kinase A. Therefore, PEP is not reconverted to pyruvate. Rather, it continues along the pathway of gluconeogenesis. If PEP were reconverted to pyruvate, these substrates would simply cycle, causing a net loss of energy with no net generation of useful products. The inactivation of pyruvate kinase prevents such futile cycling and promotes the net synthesis of glucose.

4. CONVERSION OF FRUCTOSE 1,6-BISPHOSPHATE TO FRUCTOSE 6-PHOSPHATE

The carbons of PEP reverse the steps of glycolysis, forming fructose 1,6-bisphosphate. Fructose 1,6-bisphosphatase acts on this bisphosphate to release inorganic phosphate and produce fructose 6-phosphate. A futile substrate cycle is prevented at



The mechanism of action of steroid hormones on glucose homeostasis differs from that of glucagon or epinephrine (see Chapters 16 and 26). Glucocorticoids are steroid hormones that stimulate gluconeogenesis, in part because they induce the synthesis of PEPCK. **Emma Wheezer** had elevated levels of blood glucose because she was being treated with large pharmacologic doses of dexamethasone, a potent synthetic glucocorticoid.

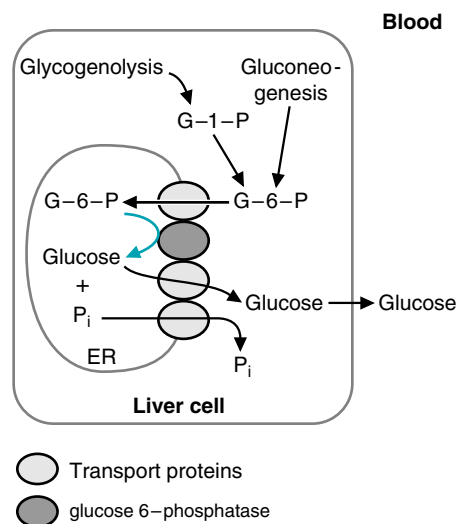


Fig. 31.10. Location and function of glucose 6-phosphatase. Glucose 6-phosphate travels on a transporter (shaded oval) into the endoplasmic reticulum (ER), where it is hydrolyzed by glucose 6-phosphatase (black oval) to glucose and P_i . These products travel back to the cytosol on transporters (shaded ovals).



Glucose 6-phosphatase is used in both glycogenolysis and gluconeogenesis.

this step because, under conditions that favor gluconeogenesis, the concentrations of the compounds that activate the glycolytic enzyme PFK-1 are low. These same compounds, fructose 2,6-bisphosphate (whose levels are regulated by insulin and glucagon) and AMP, are allosteric inhibitors of fructose 1,6-bisphosphatase. When the concentrations of these allosteric effectors are low, PFK-1 is less active, fructose 1,6-bisphosphatase is more active, and the net flow of carbon is toward fructose 6-phosphate and, thus, toward glucose. The synthesis of fructose 1,6-bisphosphatase is also induced during fasting.

5. CONVERSION OF GLUCOSE 6-PHOSPHATE TO GLUCOSE

Glucose 6-phosphatase catalyzes the conversion of glucose 6-phosphate to glucose, which is released from the liver cell (Fig. 31.10). The glycolytic enzyme glucokinase, which catalyzes the reverse reaction, is relatively inactive during gluconeogenesis. Glucokinase, which has a high $S_{0.5}$ (K_m) for glucose (see Fig. 9.3), is not very active during fasting because the blood glucose level is lower (approximately 5 mM) than the $S_{0.5}$ of the enzyme.

Glucokinase is also an inducible enzyme. The concentration of the enzyme increases in the fed state, when blood glucose and insulin levels are elevated, and decreases in the fasting state, when glucose and insulin are low.

E. Energy Is Required for the Synthesis of Glucose

During the gluconeogenic reactions, 6 moles of high-energy phosphate bonds are cleaved. Two moles of pyruvate are required for the synthesis of 1 mole of glucose. As 2 moles of pyruvate are carboxylated by pyruvate carboxylase, 2 moles of ATP are hydrolyzed. PEPCK requires 2 moles of GTP (the equivalent of 2 moles of ATP) to convert 2 moles of oxaloacetate to 2 moles of PEP. An additional 2 moles of ATP are used when 2 moles of 3-phosphoglycerate are phosphorylated, forming 2 moles of 1,3-bisphosphoglycerate. Energy in the form of reducing equivalents (NADH) is also required for the conversion of 1,3-bisphosphoglycerate to glyceraldehyde 3-phosphate. Under fasting conditions, the energy required for gluconeogenesis is obtained from β -oxidation of fatty acids.

III. CHANGES IN BLOOD GLUCOSE LEVELS AFTER A MEAL

The metabolic transitions that occur as a person eats a meal and progresses through the various stages of fasting have been described in detail in the previous chapters. This chapter summarizes the concepts presented in these previous chapters. Because a thorough understanding of these concepts is so critical to medicine, a summary is not only warranted but essential.

After a high-carbohydrate meal, blood glucose rises from a fasting level of approximately 80 to 100 mg/dL (~5 mM) to a level of approximately 120 to 140 mg/dL (8 mM) within a period of 30 minutes to 1 hour (Fig. 31.11). The concentration of glucose in the blood then begins to decrease, returning to the fasting range by approximately 2 hours after the meal (see also Chapter 26).

Blood glucose levels increase as dietary glucose is digested and absorbed. The values go no higher than approximately 140 mg/dL in a normal, healthy person because tissues take up glucose from the blood, storing it for subsequent use and oxidizing it for energy. After the meal is digested and absorbed, blood glucose levels decline because cells continue to metabolize glucose.

If blood glucose levels continued to rise after a meal, the high concentration of glucose would cause the release of water from tissues as a result of the osmotic effect of glucose. Tissues would become dehydrated, and their function would be affected. A hyperosmolar coma could result from dehydration of the brain.

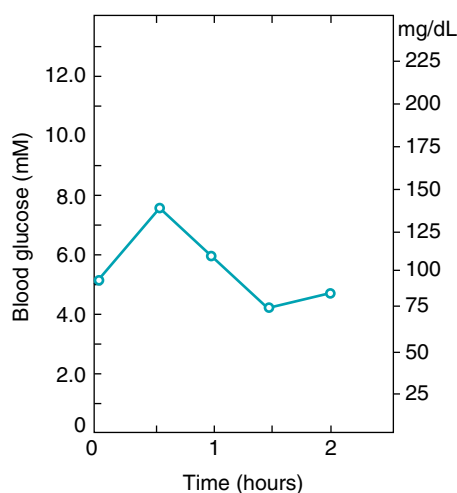


Fig. 31.11. Blood glucose concentrations at various times after a meal.

Conversely, if blood glucose levels continued to drop after a meal, tissues that depend on glucose would suffer from a lack of energy. If blood glucose levels dropped abruptly, the brain would not be able to produce an adequate amount of ATP. Light-headedness and dizziness would result, followed by drowsiness and, eventually, coma. Red blood cells would not be able to produce enough ATP to maintain the integrity of their membranes. Hemolysis of these cells would decrease the transport of oxygen to the tissues of the body. Eventually, all tissues that rely on oxygen for energy production would fail to perform their normal functions. If the problem were severe enough, death could result.

Devastating consequences of glucose excess or insufficiency are normally avoided because the body is able to regulate its blood glucose levels. As the concentration of blood glucose approaches the normal fasting range of 80 to 100 mg/dL roughly 2 hours after a meal, the process of glycogenolysis is activated in the liver. Liver glycogen is the primary source of blood glucose during the first few hours of fasting. Subsequently, gluconeogenesis begins to play a role as an additional source of blood glucose. The carbon for gluconeogenesis, a process that occurs in the liver, is supplied by other tissues. Exercising muscle and red blood cells provide lactate through glycolysis; muscle also provides amino acids by degradation of protein; and glycerol is released from adipose tissue as triacylglycerol stores are mobilized.

Even during a prolonged fast, blood glucose levels do not decrease dramatically. After 5 to 6 weeks of starvation, blood glucose levels decrease to only approximately 65 mg/dL (Table 31.2).

Table 31.2. Blood Glucose Levels at Various Stages of Fasting

Glucose (mg/dL)	
Glucose, 700 g/day IV	100
Fasting, 12 hr	80
Starvation, 3 days	70
Starvation, 5–6 weeks	65

Source of data: Ruderman NB, Aoki TT, Cahill GF Jr. Gluconeogenesis and its disorders in man. In: Hanson RW, Mehlerman MA, eds. *Gluconeogenesis: Its Regulation in Mammalian Species*. New York, John Wiley. 1976:517.

A. Blood Glucose Levels in the Fed State

The major factors involved in regulating blood glucose levels are the blood glucose concentration itself and hormones, particularly insulin and glucagon.

As blood glucose levels rise after a meal, the increased glucose concentration stimulates the β cells of the pancreas to release insulin (Fig. 31.12). Certain amino acids, particularly arginine and leucine, also stimulate insulin release from the pancreas.

Blood levels of glucagon, which is secreted by the α cells of the pancreas, may increase or decrease, depending on the content of the meal. Glucagon levels decrease in response to a high-carbohydrate meal, but they increase in response to a high-protein meal. After a typical mixed meal containing carbohydrate, protein, and fat, glucagon levels remain relatively constant, whereas insulin levels increase (Fig. 31.13).



When **Di Abietes** inadvertently injected an excessive amount of insulin, she caused an acute reduction in her blood glucose levels 4 to 5 hours later while she was asleep. Had she been awake, she would have first experienced symptoms caused by a hypoglycemia-induced hyperactivity of her sympathetic nervous system (e.g., sweating, tremulousness, palpitations). Eventually, as her hypoglycemia became more profound, she would have experienced symptoms of “neuroglycopenia” (inadequate glucose supply to the brain), such as confusion, speech disturbances, emotional instability, possible seizure activity, and, finally, coma. While sleeping, she had reached this neuroglycopenic stage of hypoglycemia and could not be aroused at 6:00 AM.

Ann O’ Rexia, whose intake of glucose and of glucose precursors has been severely restricted, has not developed any of these manifestations. Her lack of hypoglycemic symptoms can be explained by the very gradual reduction of her blood glucose levels as a consequence of near starvation and her ability to maintain blood glucose levels within an acceptable fasting range through hepatic gluconeogenesis. In addition, lipolysis of adipose triacylglycerols produces fatty acids, which are used as fuel and converted to ketone bodies by the liver. The oxidation of fatty acids and ketone bodies by the brain and muscle reduces the need for blood glucose.

In **Di Abiete’s** case, the excessive dose of insulin inhibited lipolysis and ketone body synthesis, so these alternative fuels were not available to spare blood glucose. The rapidity with which hypoglycemia was induced could not be compensated for quickly enough by hepatic gluconeogenesis, which was inhibited by the insulin, and hypoglycemia ensued.

A stat finger stick revealed that Di’s capillary blood glucose level was less than 20 mg/dL. An intravenous infusion of a 50% solution of glucose was started, and her blood glucose level was determined frequently. When Di regained consciousness, the intravenous solution was eventually changed to 10% glucose. After 6 hours, her blood glucose levels stayed in the upper normal range, and she was able to tolerate oral feedings. She was transferred to the metabolic unit for overnight monitoring. By the next morning, her previous diabetes treatment regimen was reestablished. The reasons that she had developed hypoglycemic coma were explained to Di, and she was discharged to the care of her family doctor.

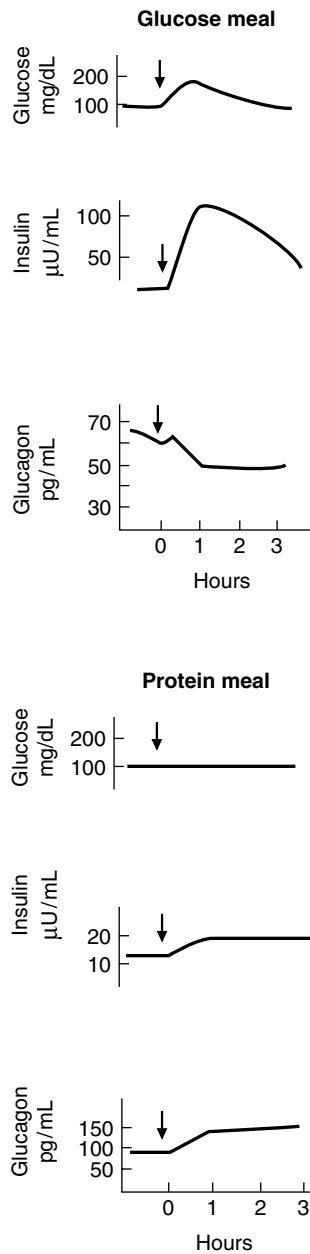


Fig. 31.12. Blood glucose, insulin, and glucagon levels after a high-carbohydrate and a high-protein meal.

1. FATE OF DIETARY GLUCOSE IN THE LIVER

After a meal, the liver oxidizes glucose to meet its immediate energy needs. Any excess glucose is converted to stored fuels. Glycogen is synthesized and stored in the liver, and glucose is converted to fatty acids and to the glycerol moiety that reacts with the fatty acids to produce triacylglycerols. These triacylglycerols are packaged in very-low-density lipoproteins (VLDL) and transported to adipose tissue, where the fatty acids are stored in adipose triacylglycerols.

Regulatory mechanisms control the conversion of glucose to stored fuels. As the concentration of glucose increases in the hepatic portal vein, the concentration of glucose in the liver may increase from the fasting level of 80 to 100 mg/dL (~ 5 mM) to a concentration of 180 to 360 mg/dL (10–20 mM). Consequently, the velocity of the glucokinase reaction increases because this enzyme has a high $S_{0.5}$ (K_m) for glucose (Fig. 31.14). Glucokinase is also induced by a high-carbohydrate diet; the quantity of the enzyme increases in response to elevated insulin levels.

Insulin promotes the storage of glucose as glycogen by countering the effects of glucagon-stimulated phosphorylation. The response to insulin activates the phosphatases that dephosphorylate glycogen synthase (which leads to glycogen synthase activation) and glycogen phosphorylase (which leads to inhibition of the enzyme) (Fig. 31.15A). Insulin also promotes the synthesis of the triacylglycerols that are released from the liver into the blood as VLDL. The regulatory mechanisms for this process are described in Chapter 33.

2. FATE OF DIETARY GLUCOSE IN PERIPHERAL TISSUES

Almost every cell in the body oxidizes glucose for energy. Certain critical tissues, particularly the brain, other nervous tissue, and red blood cells, especially depend on glucose for their energy supply. The brain requires approximately 150 g glucose per day. In addition, approximately 40 g/day glucose is required by other glucose-dependent tissues. Furthermore, all tissues require glucose for the pentose phosphate pathway, and many tissues use glucose for synthesis of glycoproteins and other carbohydrate-containing compounds.

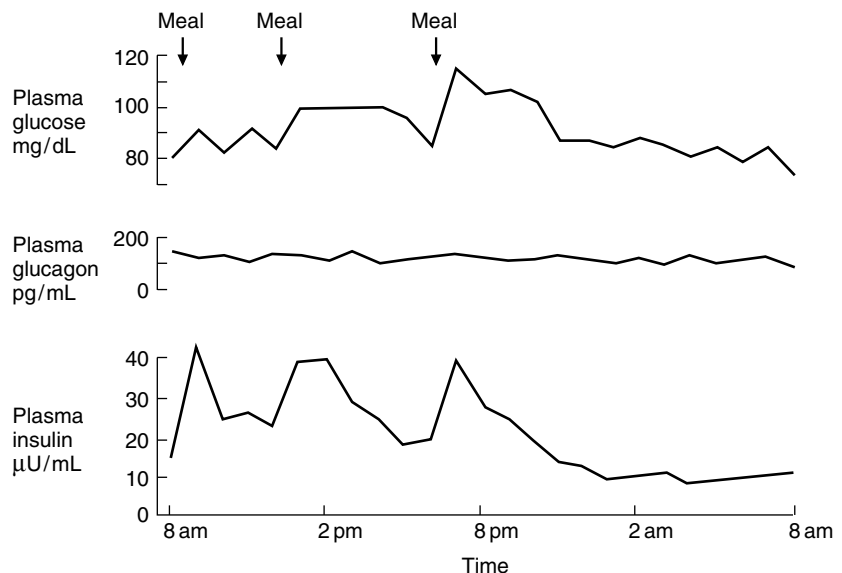


Fig. 31.13. Blood glucose, insulin, and glucagon levels over a 24-hour period in a normal person eating mixed meals. Reprinted with permission from Tasaka Y, Sekine M, Wakatsuki M, et al. *Horm Metab Res* 1975;7:206. Copyright © Thieme Medical Publishers, Inc.

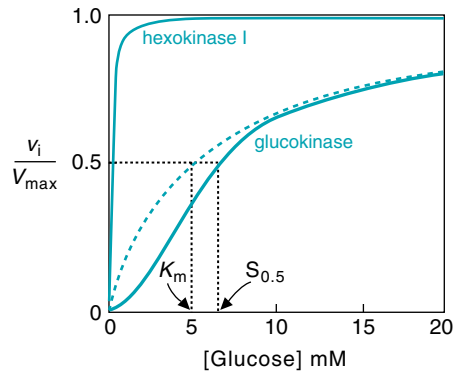


Fig. 31.14. Velocity of the glucokinase reaction.

Insulin stimulates the transport of glucose into adipose and muscle cells by promoting the recruitment of glucose transporters to the cell membrane (see Fig. 31.15C). Other tissues, such as the liver, brain, and red blood cells, have a different type of glucose transporter that is not as significantly affected by insulin.

In muscle, glycogen is synthesized after a meal by a mechanism similar to that in the liver (see Fig. 31.15B). Metabolic differences exist between these tissues (see Chapter 28), but, in essence, insulin stimulates glycogen synthesis in resting muscle as it does in the liver. A key difference between muscle and liver is that insulin greatly stimulates the transport of glucose into muscle cells but only slightly stimulates its transport into liver cells.

3. RETURN OF BLOOD GLUCOSE TO FASTING LEVELS

After a meal has been digested and absorbed, blood glucose levels reach a peak and then begin to decline. The uptake of dietary glucose by cells, particularly those in the liver, muscle, and adipose tissue, lowers blood glucose levels. By 2 hours after a meal, blood glucose levels return to the normal fasting level of less than 140 mg/dL.

B. Blood Glucose Levels in the Fasting State

1. CHANGES IN INSULIN AND GLUCAGON LEVELS

During fasting, as blood glucose levels decrease, insulin levels decrease, and glucagon levels rise. These hormonal changes cause the liver to degrade glycogen by the process of glycogenolysis and to produce glucose by the process of gluconeogenesis so that blood glucose levels are maintained.

2. STIMULATION OF GLYCOGENOLYSIS

Within a few hours after a high-carbohydrate meal, glucagon levels begin to rise. Glucagon binds to cell surface receptors and activates adenylate cyclase, causing cAMP levels in liver cells to rise (Fig. 31.16). cAMP activates protein kinase A, which phosphorylates and inactivates glycogen synthase. Therefore, glycogen synthesis decreases.

At the same time, protein kinase A stimulates glycogen degradation by a two-step mechanism. Protein kinase A phosphorylates and activates phosphorylase kinase. This enzyme, in turn, phosphorylates and activates glycogen phosphorylase.

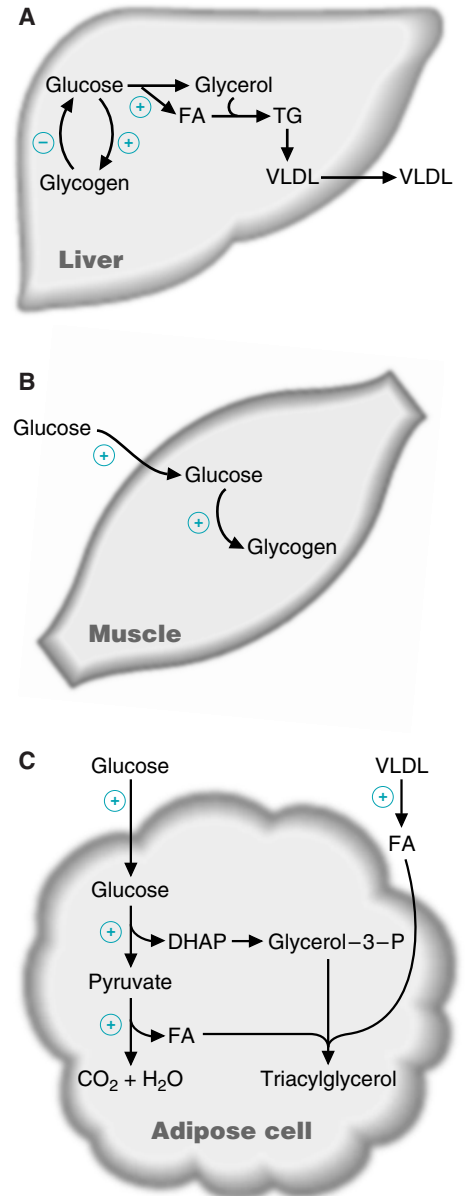


Fig. 31.15. Glucose metabolism in various tissues. A. Effect of insulin on glycogen synthesis and degradation and on VLDL synthesis in the liver. B. Glucose metabolism in resting muscle in the fed state. The transport of glucose into cells and the synthesis of glycogen are stimulated by insulin. C. Glucose metabolism in adipose tissue in the fed state. FA = fatty acids; DHAP = dihydroxyacetone phosphate. FA = fatty acids; TG = triacylglycerols; + = stimulated by insulin; - = inhibited by insulin.

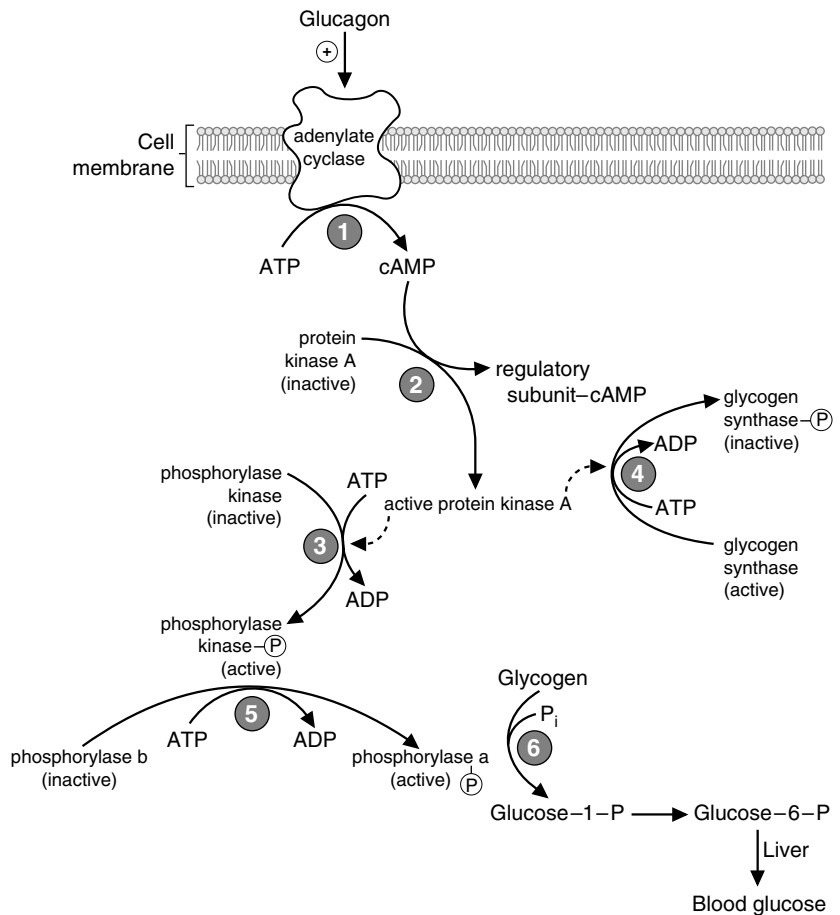


Fig. 31.16. Regulation of glycogenolysis in the liver by glucagon. 1. Glucagon binding to its receptor leads to the activation of adenylate cyclase, which leads to an increase in cAMP levels. 2. cAMP binds to the regulatory subunits of protein kinase A, thereby activating the catalytic subunit. 3. Active protein kinase A phosphorylates and activates phosphorylase kinase, while simultaneously inactivating glycogen synthase (4). 5. Active phosphorylase kinase converts glycogen phosphorylase b to glycogen phosphorylase a. 6. Phosphorylase a degrades glycogen, producing glucose 1-phosphate, which is converted to glucose 6-phosphate, then glucose, for export into the blood.



The pathophysiology leading to an elevation of blood glucose after a meal differs between patients with type 1 diabetes mellitus and those with type 2 diabetes mellitus. **Di Abietes**, who has type 1 disease, cannot secrete insulin adequately in response to a meal because of a defect in the β cells of her pancreas.

Ann Sulin, however, has type 2 disease. In this form of the disorder, the cause of glucose intolerance is more complex, involving at least a delay in the release of relatively appropriate amounts of insulin after a meal combined with a degree of resistance to the actions of insulin in skeletal muscle and adipocytes. Excessive hepatic gluconeogenesis occurs even though blood glucose levels are elevated.

Glycogen phosphorylase catalyzes the phosphorolysis of glycogen, producing glucose 1-phosphate, which is converted to glucose 6-phosphate. Dephosphorylation of glucose 6-phosphate by glucose 6-phosphatase produces free glucose, which then enters the blood.

3. STIMULATION OF GLUCONEOGENESIS

By 4 hours after a meal, the liver is supplying glucose to the blood not only by the process of glycogenolysis but also by the process of gluconeogenesis. Hormonal changes cause peripheral tissues to release precursors that provide carbon for gluconeogenesis, specifically lactate, amino acids, and glycerol.

Regulatory mechanisms promote the conversion of gluconeogenic precursors to glucose (Fig. 31.17). These mechanisms prevent the occurrence of potential futile cycles, which would continuously convert substrates to products while consuming energy but producing no useful result.

These regulatory mechanisms inactivate the glycolytic enzymes pyruvate kinase, phosphofructokinase-1 (PFK-1), and glucokinase during fasting and promote the flow of carbon to glucose via gluconeogenesis. These mechanisms operate at the three steps where glycolysis and gluconeogenesis differ:

1. Pyruvate (derived from lactate and alanine) is converted by the gluconeogenic pathway to phosphoenolpyruvate (PEP). PEP is not reconverted to pyruvate (a potential futile cycle) because glucagon-stimulated phosphorylation inactivates pyruvate kinase. Therefore, PEP reverses the steps of glycolysis and forms fructose 1,6-bisphosphate.
2. Fructose 1,6-bisphosphate is converted to fructose 6-phosphate by a bisphosphatase. Because the glycolytic enzyme PFK-1 is relatively inactive mainly as a result of low fructose 2,6-bisphosphate levels, fructose 6-phosphate is not converted back to fructose 1,6-bisphosphate, and a second potential futile cycle is avoided. The low fructose 2,6-bisphosphate levels are attributable in part to the phosphorylation of phosphofructokinase-2 by protein kinase A, which has been activated in response to glucagon. Fructose 6-phosphate is converted to glucose 6-phosphate.
3. Glucose 6-phosphate is dephosphorylated by glucose 6-phosphatase, forming free glucose. Because glucokinase has a high $S_{0.5}$ (K_m) for glucose, and glucose concentrations are relatively low in liver cells during fasting, glucose is released into the blood. Therefore, the third potential futile cycle does not occur.

Enzymes that participate in gluconeogenesis, but not in glycolysis, are active under fasting conditions. Pyruvate carboxylase is activated by acetyl CoA, derived from oxidation of fatty acids. Phosphoenolpyruvate carboxykinase, fructose 1,6-bisphosphatase, and glucose 6-phosphatase are induced; that is, the quantity of the enzymes increases. Fructose 1,6-bisphosphatase is also active because levels of fructose 2,6-bisphosphate, an inhibitor of the enzyme, are low.

4. STIMULATION OF LIPOLYSIS

The hormonal changes that occur during fasting stimulate the breakdown of adipose triacylglycerols (see Chapters 3, 33, and 43). Consequently, fatty acids and glycerol are released into the blood (Fig. 31.18). Glycerol serves as a source of carbon for gluconeogenesis. Fatty acids become the major fuel of the body and are oxidized to CO_2 and H_2O by various tissues, which enables these tissues to decrease their utilization of glucose. Fatty acids are also oxidized to acetyl CoA in the liver to provide energy for gluconeogenesis. In a prolonged fast, acetyl CoA is converted to ketone bodies, which enter the blood and serve as an additional fuel source for the muscle and the brain.

C. Blood Glucose Levels during Prolonged Fasting (Starvation)

During prolonged fasting, a number of changes in fuel utilization occur. These changes cause tissues to use less glucose than they use during a brief fast and to use predominantly fuels derived from adipose triacylglycerols (i.e., fatty acids and their derivatives, the ketone bodies). Therefore, blood glucose levels do not decrease drastically. In fact, even after 5 to 6 weeks of starvation, blood glucose levels are still in the range of 65 mg/dL (Fig. 31.19, see Table 31.2).

The major change that occurs in starvation is a dramatic elevation of blood ketone body levels after 3 to 5 days of fasting (see Fig. 31.19). At these levels, the brain and other nervous tissues begin to use ketone bodies and, consequently, they oxidize less glucose, requiring roughly one third as much glucose (approximately 40 g/day) as under normal dietary conditions. As a result of reduced glucose utilization, the rate of gluconeogenesis in the liver decreases, as does the production of urea (see Fig. 31.19). Because in this stage of starvation amino acids, obtained

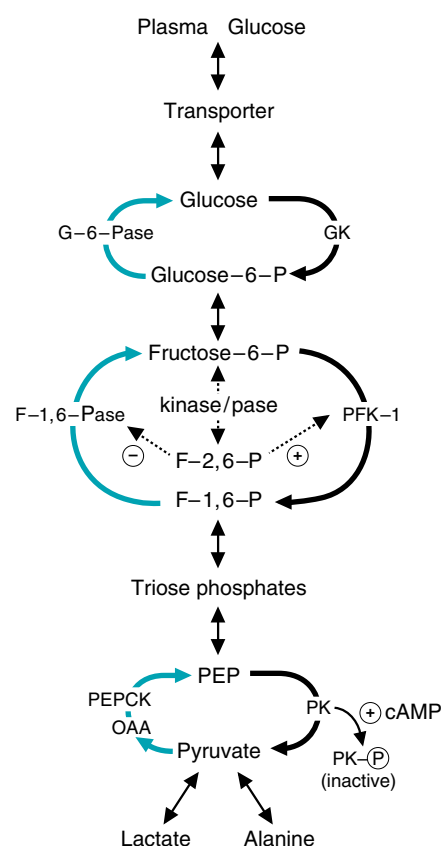


Fig. 31.17. Regulation of gluconeogenesis in the liver. GK = glucokinase; G-6-Pase = glucose 6-phosphatase; PK = pyruvate kinase; OAA = oxaloacetate; PEPCK = phosphoenolpyruvate carboxykinase; F-1,6-Pase = fructose 1,6-bisphosphatase; F-2,6-P = fructose 2,6-bisphosphate; PFK-1 = phosphofructokinase-1.

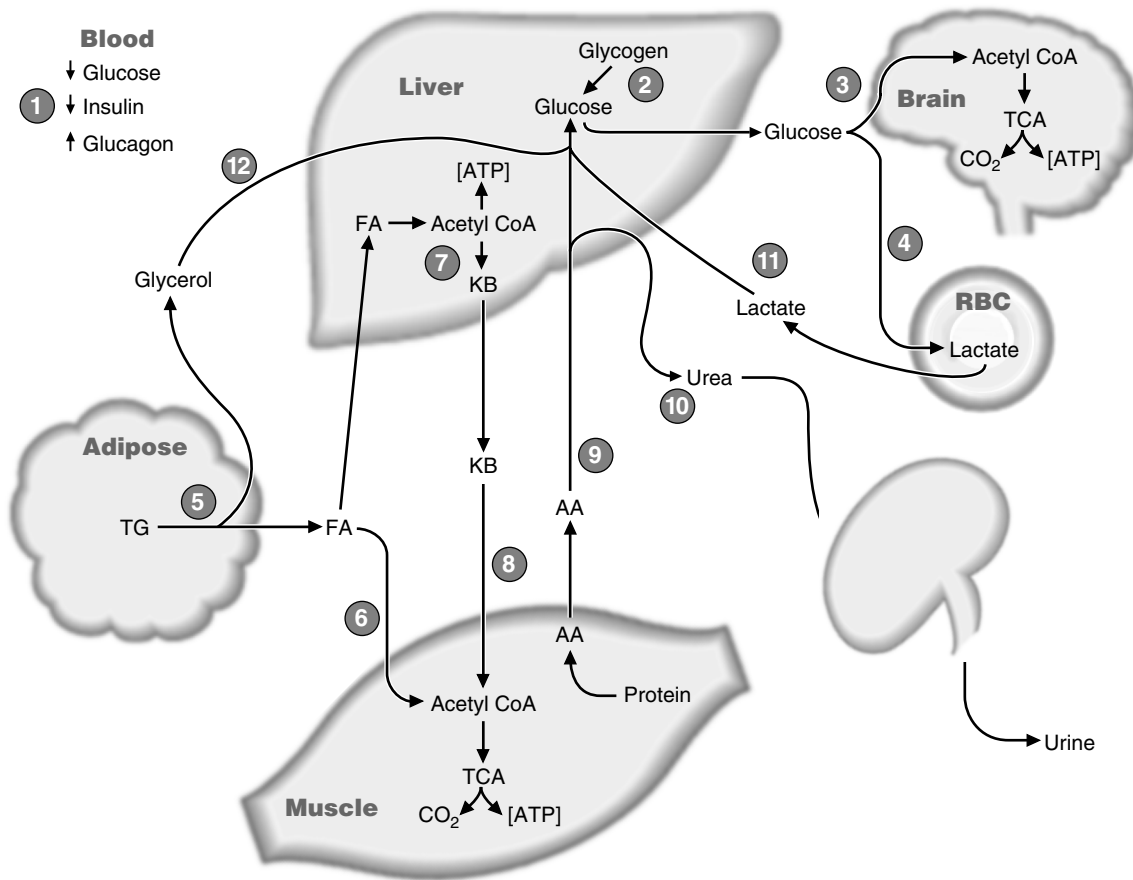


Fig. 31.18. Tissue interrelationships during fasting. 1. Blood glucose levels drop, decreasing insulin and raising blood glucagon levels. 2. Glycogenolysis is induced in the liver to raise blood glucose levels. 3. The brain uses the glucose released by the liver, as do the red blood cells (4). 5. Adipose tissues are induced to release free fatty acids and glycerol from stored triglycerides. 6. The muscle and liver use fatty acids for energy. 7. The liver converts fatty acid derived acetyl-CoA to ketone bodies for export, which the muscles (8) and brain can use for energy. 9. Protein turnover is induced in muscle, and amino acids leave the muscle and travel to the liver for use as gluconeogenic precursors. 10. The high rate of amino acid metabolism in the liver generates urea, which travels to the kidney for excretion. 11. Red blood cells produce lactate, which returns to the liver as a substrate for gluconeogenesis. 12. The glycerol released from adipose tissue is used by the liver set gluconeogenesis. KB = ketone bodies; FA = fatty acids; AA = amino acids; TG = triacylglycerols.

from the degradation of existing proteins, are the major gluconeogenic precursor, reducing glucose requirements in tissues reduces the rate of protein degradation, and, hence, the rate of urea formation. Protein from muscle and other tissues is therefore spared, because there is less need for amino acids for gluconeogenesis.

Body protein, particularly muscle protein, is not primarily a storage form of fuel in the same sense as glycogen or triacylglycerol; proteins have many functions beside fuel storage. For example, proteins function as enzymes, as structural proteins, and in muscle contraction. If tissue protein is degraded to too great an extent, body function can be severely compromised. If starvation continues and no other problems, such as infections, occur, a starving individual usually dies because of severe protein loss that causes malfunction of major organs, such as the heart. Therefore, the increase in ketone body levels that results in the sparing of body protein allows individuals to survive for extended periods without ingesting food.

D. Summary of Sources of Blood Glucose

Immediately after a meal, dietary carbohydrates serve as the major source of blood glucose (Fig. 31.20). As blood glucose levels return to the fasting range

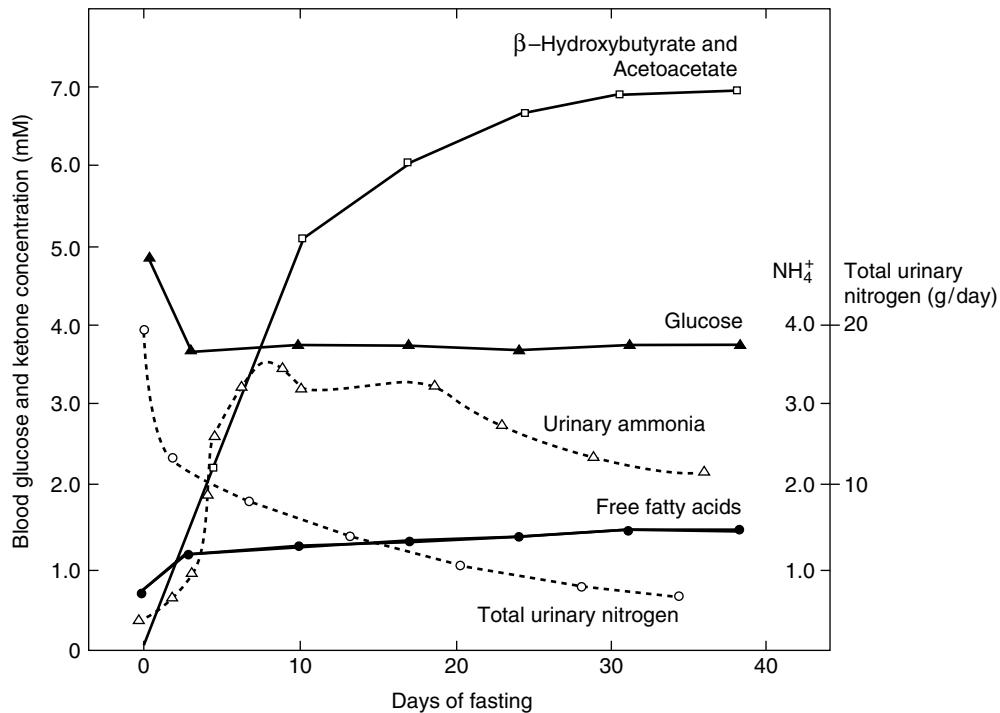


Fig. 31.19. Changes in blood fuels during fasting. The units for fatty acids, glucose, and ketone bodies are millimolar (on left) and for urinary nitrogen and ammonia are grams/day (on right). Modified from Linder MC. *Nutritional Biochemistry and Metabolism*, 2nd Ed. Stamford, CT: Appleton & Lange, 1991:103. © 1991 Appleton & Lange.

within 2 hours after a meal, glycogenolysis is stimulated and begins to supply glucose to the blood. Subsequently, glucose is also produced by gluconeogenesis.

During a 12-hour fast, glycogenolysis is the major source of blood glucose. Thus, it is the major pathway by which glucose is produced in the basal state (after a 12-hour fast). However, by approximately 16 hours of fasting, glycogenolysis and gluconeogenesis contribute equally to the maintenance of blood glucose.

By 30 hours after a meal, liver glycogen stores are substantially depleted. Subsequently, gluconeogenesis is the primary source of blood glucose.

The mechanisms that cause fats to be used as the major fuel and that allow blood glucose levels to be maintained during periods of food deprivation result in the conservation of body protein and, consequently, permit survival during prolonged fasting for periods often exceeding 1 or more months.

E. Blood Glucose Levels during Exercise

During exercise, mechanisms very similar to those that are used during fasting operate to maintain blood glucose levels. The liver maintains blood glucose levels through both glucagon- and epinephrine-induced glycogenolysis and gluconeogenesis. The use of fuels by muscle during exercise, including the uptake and use of blood glucose, is discussed in Chapter 47.



Otto Shape is able to jog for 45 minutes before eating breakfast without developing symptoms of hypoglycemia in spite of enhanced glucose utilization by skeletal muscle during exercise. He maintains his blood glucose level in an adequate range through hepatic glycogenolysis and gluconeogenesis.

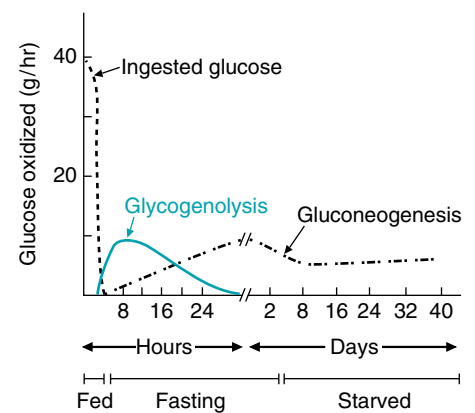


Fig. 31.20. Sources of blood glucose in fed, fasting, and starved states. Note that the scale changes from hours to days. From Ruderman NB, et al. In: Hanson RW, Mehlerman MA, eds. *Gluconeogenesis: Its Regulation in Mammalian Species*. 1976:518. © 1976 John Wiley & Sons.



Remember that muscle glycogen is not used to maintain blood glucose levels; muscle cells lack glucose 6-phosphatase, so glucose cannot be produced from glucose 6-phosphate for export.



Epidemiologic studies have correlated the results of the OGTT with the current and subsequent development of the microvascular complications of DM in the eye (diabetic retinopathy) and in the kidney (diabetic nephropathy). Their development is usually limited to those patients whose fasting blood glucose level exceeds 140 mg/dL (7.8 mM) or whose 2-hour postprandial blood glucose level exceeds 200 mg/dL (11.1 mM).

As a consequence, the success of therapy with diet alone or with diet plus oral hypoglycemic agents is often determined by measuring blood glucose levels during fasting, before breakfast, and 2 hours after a meal. For patients taking an intermediate-acting insulin, blood glucose also may be measured at other times, such as when insulin action is expected to peak after its injection (e.g., around 5:00 PM, when intermediate-acting insulin has been given at 7:00 AM the same day).

CLINICAL COMMENTS



The chronic excessive ingestion of ethanol concurrent with a recent reduction in nutrient intake caused **Al Martini's** blood glucose level to decrease to 28 mg/dL. This degree of hypoglycemia caused the release of a number of “counterregulatory” hormones into the blood, including glucagon, growth hormone, cortisol, and epinephrine (adrenaline).

Some of the patient's signs and symptoms are primarily the result of an increase in adrenergic nervous system activity after a rapid decrease in blood glucose. The subsequent increase in epinephrine levels in the blood leads to tremulousness, excessive sweating, and rapid heart rate. Other manifestations arise when the brain has insufficient glucose, hence the term “neuroglycopenic symptoms.” Mr. Martini was confused, combative, had slurred speech, and eventually had a grand mal seizure. If not treated quickly by intravenous glucose administration, Mr. Martini may have lapsed into a coma. Permanent neurologic deficits and even death may result if severe hypoglycemia is not corrected in 6 to 10 hours.



The elevation in blood glucose that occurred in **Emma Wheezer's** case was primarily a consequence of the large pharmacologic doses of a glucocorticoid that she received in an effort to reduce the intrabronchial inflammatory reaction characteristic of asthmatic bronchospasm. Although the development of hyperglycemia in this case could be classified as a “secondary” form of diabetes mellitus, most patients treated with glucocorticoids do not develop glucose intolerance. Ms. Wheezer, therefore, may have a predisposition to the eventual development of “primary” diabetes mellitus.

In hyperglycemia, increased amounts of glucose enter the urine, causing large amounts of water to be excreted. This “osmotic diuresis” is responsible for the increased volume of urine (polyuria) noted by the patient. Because of increased urinary water loss, the effective circulating blood volume is reduced. Therefore, less blood reaches volume-sensitive receptors in the central nervous system, which then trigger the sensation of thirst, causing increased drinking activity (polydipsia).

A diabetic diet and the tapering of her steroid dose over a period of several weeks gradually returned Ms. Wheezer's blood glucose level into the normal range.



Chronically elevated levels of glucose in the blood may contribute to the development of the microvascular complications of diabetes mellitus, such as diabetic retinal damage, kidney damage, and nerve damage, as well as macrovascular complications such as cerebrovascular, peripheral vascular, and coronary vascular insufficiency. The precise mechanism by which long-term hyperglycemia induces these vascular changes is not fully established.

One postulated mechanism proposes that nonenzymatic glycation (glycosylation) of proteins in vascular tissue alters the structure and functions of these proteins. A protein exposed to chronically increased levels of glucose will covalently bind glucose, a process called glycation or glycosylation. This process is not regulated by enzymes (see Chapter 9). These nonenzymatically glycosylated proteins slowly form cross-linked protein adducts (often called advanced glycosylation products) within the microvasculature and macrovasculature.

By cross-linking vascular matrix proteins and plasma proteins, chronic hyperglycemia may cause narrowing of the luminal diameter of the microvessels in the retina (causing diabetic retinopathy), the renal glomeruli (causing diabetic nephropathy), and the microvessels supplying peripheral and autonomic nerve fibers (causing diabetic neuropathy). The same process has been postulated to accelerate atherosclerotic change in the macrovasculature, particularly in the brain

(causing strokes), the coronary arteries (causing heart attacks), and the peripheral arteries (causing peripheral arterial insufficiency and gangrene). The abnormal lipid metabolism associated with poorly controlled diabetes mellitus also may contribute to the accelerated atherosclerosis associated with this metabolic disorder (see Chapters 33 and 34).

Until recently, it was argued that meticulous control of blood glucose levels in a diabetic patient would not necessarily prevent or even slow these complications of chronic hyperglycemia. The publication of the Diabetes Control and Complications Trial, however, suggests that maintaining long-term euglycemia (normal blood glucose levels) in diabetic patients slows the progress of unregulated glycation of proteins as well as corrects their dyslipidemia. In this way, careful control may favorably affect the course of the microvascular and macrovascular complications of diabetes mellitus in patients such as **Di Abietes** and **Ann Sulin**.

BIOCHEMICAL COMMENTS



Plants are the ultimate source of the earth's supply of glucose. Plants produce glucose from atmospheric CO_2 by the process of photosynthesis (Fig. 31.21A). In contrast to plants, humans cannot synthesize glucose by the fixation of CO_2 . Although we have a process called gluconeogenesis, the term may really be a misnomer. Glucose is not generated anew by gluconeogenesis; compounds produced from glucose are simply recycled to glucose. We obtain glucose from the plants, including bacteria, that we eat and, to some extent, from animals in our food supply. We use this glucose both as a fuel and as a source of carbon for the synthesis of fatty acids, amino acids, and other sugars (see Fig. 31.21B). We store glucose as glycogen, which, along with gluconeogenesis, provides glucose when needed for energy (see Fig. 31.21C).

Lactate, one of the carbon sources for gluconeogenesis, is actually produced from glucose by tissues that obtain energy by oxidizing glucose to pyruvate through glycolysis. The pyruvate is then reduced to lactate, released into the bloodstream, and reconverted to glucose by the process of gluconeogenesis in the liver. This process is known as the Cori cycle (Fig. 31.21D).

Carbons of alanine, another carbon source for gluconeogenesis, may be produced from glucose. In muscle, glucose is converted via glycolysis to pyruvate and transaminated to alanine. Alanine from muscle is recycled to glucose in the liver. This process is known as the glucose–alanine cycle (Fig. 31.21E). Glucose also may be used to produce nonessential amino acids other than alanine, which are subsequently reconverted to glucose in the liver by gluconeogenesis. Even the essential amino acids that we obtain from dietary proteins are synthesized in plants and bacteria using glucose as the major source of carbon. Therefore, all amino acids that are converted to glucose in humans, including the essential amino acids, were originally synthesized from glucose.

The production of glucose from glycerol, the third major source of carbon for gluconeogenesis, is also a recycling process. Glycerol is derived from glucose via the dihydroxyacetone phosphate intermediate of glycolysis. Fatty acids are then esterified to the glycerol and stored as triacylglycerol. When these fatty acids are released from the triacylglycerol, the glycerol moiety can travel to the liver and be reconverted to glucose (see Fig. 31.21F).

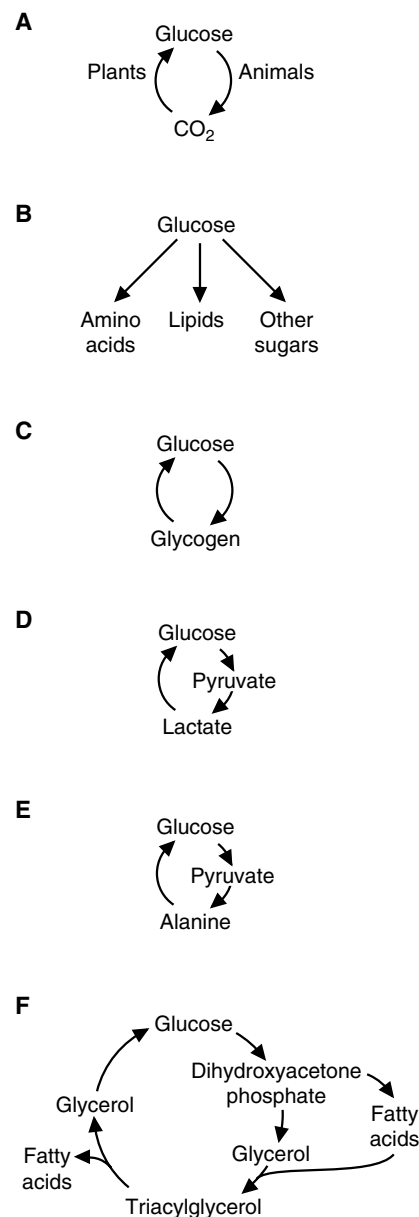


Fig. 31.21. Recycling of glucose.

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**REVIEW QUESTIONS—CHAPTER 31**

1. A common intermediate in the conversion of glycerol and lactate to glucose is which of the following?
 - (A) Pyruvate
 - (B) Oxaloacetate
 - (C) Malate
 - (D) Glucose 6-phosphate
 - (E) Phosphoenolpyruvate
2. A patient presented with a bacterial infection that produced an endotoxin that inhibits phosphoenolpyruvate carboxykinase. In this patient, then, under these conditions, glucose production from which of the following precursors would be inhibited?
 - (A) Alanine
 - (B) Glycerol
 - (C) Even-chain-number fatty acids
 - (D) Phosphoenolpyruvate
 - (E) Galactose
3. Which of the following statements best describes glucagon?
 - (A) It acts as an anabolic hormone.
 - (B) It acts on skeletal muscle, liver, and adipose tissue.
 - (C) It acts primarily on the liver and adipose tissue.
 - (D) Its concentration in the blood increases after a high-carbohydrate meal.
 - (E) Its concentration increases in the blood when insulin levels increase.
4. Which of the following is most likely to occur in a normal individual after ingesting a high-carbohydrate meal?
 - (A) Only insulin levels decrease.
 - (B) Only insulin levels increase.
 - (C) Only glucagon levels increase.
 - (D) Both insulin and glucagon levels decrease.
 - (E) Both insulin and glucagon levels increase.
5. A patient arrives at the hospital in an ambulance. She is currently in a coma. Before lapsing into the coma, her symptoms included vomiting, dehydration, low blood pressure, and a rapid heartbeat. She also had relatively rapid respirations, resulting in more carbon dioxide exhaled. These symptoms are consistent with which of the following conditions?
 - (A) The patient lacks a pancreas.
 - (B) Ketoalkolosis
 - (C) Hypoglycemic coma
 - (D) Diabetic ketoacidosis
 - (E) Insulin shock in a diabetic patient

Lipid Metabolism

Most of the lipids found in the body fall into the categories of fatty acids and triacylglycerols; glycerophospholipids and sphingolipids; eicosanoids; cholesterol, bile salts, and steroid hormones; and fat-soluble vitamins. These lipids have very diverse chemical structures and functions. However, they are related by a common property: their relative insolubility in water.

Fatty acids, which are stored as triacylglycerols, serve as fuels, providing the body with its major source of energy (Fig. VI.1). Glycerophospholipids and sphingolipids, which contain esterified fatty acids, are found in membranes and in blood lipoproteins at the interfaces between the lipid components of these structures and the surrounding water. These membrane lipids form hydrophobic barriers between subcellular compartments and between cellular constituents and the extracellular milieu. Polyunsaturated fatty acids containing 20 carbons form the eicosanoids, which regulate many cellular processes (Fig. VI.2).

Cholesterol adds stability to the phospholipid bilayer of membranes. It serves as the precursor of the bile salts, detergent-like compounds that function in the process of lipid digestion and absorption (Fig. VI.3). Cholesterol also serves as the precursor of the steroid hormones, which have many actions, including the regulation of metabolism, growth, and reproduction.

The fat-soluble vitamins are lipids that are involved in such varied functions as vision, growth, and differentiation (vitamin A), blood clotting (vitamin K), prevention of oxidative damage to cells (vitamin E), and calcium metabolism (vitamin D).

Triacylglycerols, the major dietary lipids, are digested in the lumen of the intestine (Fig. VI.4). The initial digestive products, free fatty acids and 2-monoacylglycerol, are reconverted to triacylglycerols in intestinal epithelial cells, packaged in lipoproteins known as chylomicrons (so they can safely enter the circulation), and secreted into the lymph. Ultimately, chylomicrons enter the blood, serving as one of the major blood lipoproteins.

Very low density lipoprotein (VLDL) is produced in the liver, mainly from dietary carbohydrate. Lipogenesis is an insulin-stimulated process through which glucose is converted to fatty acids, which are subsequently esterified to glycerol to form the triacylglycerols that are packaged in VLDL and secreted from the liver. Thus, chylomicrons primarily transport dietary lipids, and VLDL transports endogenously synthesized lipids.

The triacylglycerols of chylomicrons and VLDL are digested by lipoprotein lipase (LPL), an enzyme found attached to capillary endothelial cells (see Fig. VI.4). The fatty acids that are released are taken up by muscle and many other tissues and oxidized to CO_2 and water to produce energy (see Chapter 23). After a meal, these fatty acids are taken up by adipose tissue and stored as triacylglycerols.

LPL converts chylomicrons to chylomicron remnants and VLDL to intermediate density lipoprotein (IDL). These products, which have a relatively low triacylglycerol content, are taken up by the liver by the process of endocytosis and degraded by lysosomal action. IDL may also be converted to low density lipoprotein (LDL) by further digestion of triacylglycerol. Endocytosis of LDL occurs in peripheral tissues as well as the liver (Table VI.1), and is the major means of cholesterol transport and delivery to peripheral tissues.

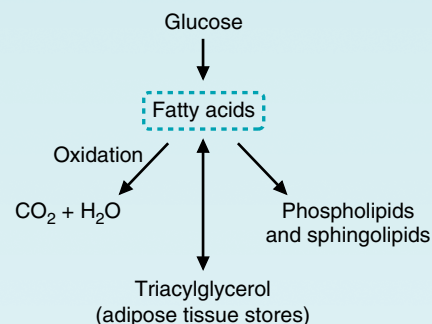


Fig. VI.1. Summary of fatty acid metabolism.

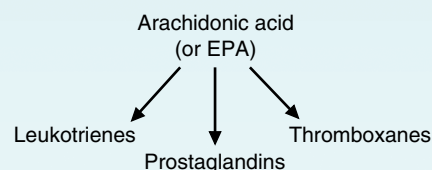


Fig. VI.2. Summary of eicosanoid synthesis.
EPA = eicosapentenoic acid.

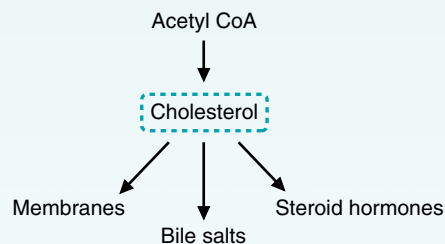


Fig. VI.3. Summary of cholesterol metabolism.

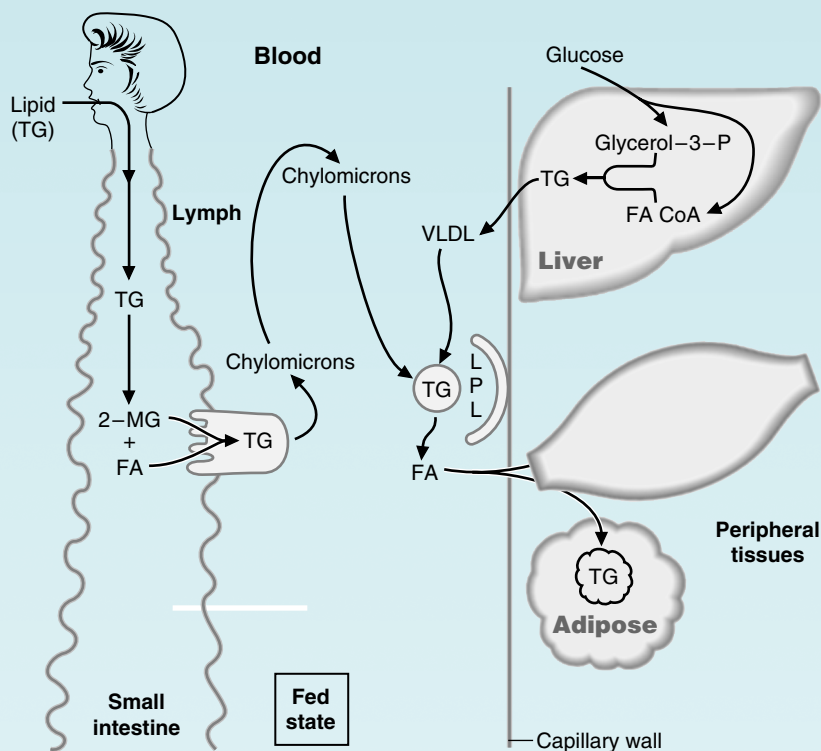


Fig. VI.4. Overview of triacylglycerol metabolism in the fed state. TG = triacylglycerol; 2-MG = 2-monoacylglycerol; FA = fatty acid; circled TG = triacylglycerols of VLDL and chylomicrons; LPL = lipoprotein lipase.

The principal function of high density lipoprotein (HDL) is to transport excess cholesterol obtained from peripheral tissues to the liver and to exchange proteins and lipids with chylomicrons and VLDL. The protein exchange converts "nascent" particles to "mature" particles.

During fasting, fatty acids and glycerol are released from adipose triacylglycerol stores (Fig. VI.5). The glycerol travels to the liver and is used for gluconeogenesis. Only the liver contains glycerol kinase, which is required for glycerol metabolism.

Table VI.1. Blood Lipoproteins

Chylomicrons
• produced in intestinal epithelial cells from dietary fat
• carries triacylglycerol in blood
VLDL (very low density lipoprotein)
• produced in liver mainly from dietary carbohydrate
• carries triacylglycerol in blood
IDL (intermediate density lipoprotein)
• produced in blood (remnant of VLDL after triacylglycerol digestion)
• endocytosed by liver or converted to LDL
LDL (low density lipoprotein)
• produced in blood (remnant of IDL after triacylglycerol digestion; endproduct of VLDL)
• contains high concentration of cholesterol and cholesterol esters
• endocytosed by liver and peripheral tissues
HDL (high density lipoprotein)
• produced in liver and intestine
• exchanges proteins and lipids with other lipoproteins
• functions in the return of cholesterol from peripheral tissues to the liver

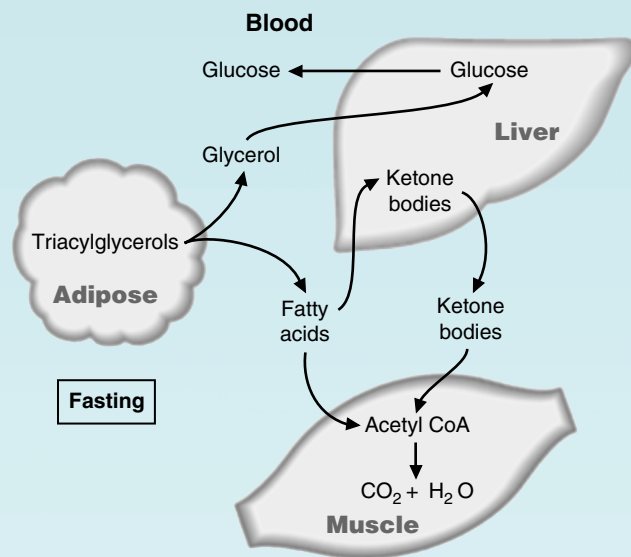


Fig. VI.5. Overview of triacylglycerol metabolism during fasting.

The fatty acids form complexes with albumin in the blood and are taken up by muscle, kidney, and other tissues, where ATP is generated by their oxidation to CO_2 and water. Liver also converts some of the carbon to ketone bodies, which are released into the blood. Ketone bodies are oxidized for energy in muscle, kidney, and other tissues during fasting, and in the brain during prolonged starvation (see Chapter 23).

32 Digestion and Transport of Dietary Lipids

Triacylglycerols are the major fat in the human diet, consisting of three **fatty acids** esterified to a **glycerol** backbone. Limited digestion of these lipids occurs in the mouth (**lingual lipase**) and stomach (**gastric lipase**) because of the low solubility of the substrate. In the intestine, however, the fats are emulsified by **bile salts** that are released from the gallbladder. This increases the available surface area of the lipids for **pancreatic lipase and colipase** to bind and to digest the triglycerides. Degradation products are **free fatty acids** and **2-monoacylglycerol**. When partially digested food enters the intestine, the hormone **cholecystokinin** is secreted by the intestine, which signals the gallbladder to contract and release bile acids, and the pancreas to release digestive enzymes.

In addition to triacylglycerols, phospholipids, cholesterol, and cholesterol esters (cholesterol esterified to fatty acids) are present in the foods we eat. Phospholipids are hydrolyzed in the intestinal lumen by **phospholipase A2**, and cholesterol esters are hydrolyzed by **cholesterol esterase**. Both of these enzymes are secreted from the pancreas.

The products of enzymatic digestion (free fatty acids, glycerol, **lysophospholipids**, cholesterol) form **micelles** with bile acids in the intestinal lumen. The micelles interact with the enterocyte membrane and allow diffusion of the lipid soluble components across the enterocyte membrane into the cell. The bile acids, however, do not enter the enterocyte at this time. They remain in the intestine, travel further down, and are then reabsorbed and sent back to the liver by the **enterohepatic circulation**. This allows the bile salts to be used multiple times in fat digestion.

The intestinal epithelial cells will resynthesize triacylglycerol from free fatty acids and 2-monacylglycerol and will package them with a protein, **apolipoprotein B-48**, phospholipids, and cholesterol esters into a soluble lipoprotein particle known as a **chylomicron**. The chylomicrons are secreted into the **lymph** and eventually end up in the circulation, where they can distribute dietary lipids to all tissues of the body.

Once in circulation, the newly released (“nascent”) chylomicrons interact with another lipoprotein particle, **HDL** (high-density lipoprotein) and acquire two **apoproteins** from HDL, apoprotein **CII** and **E**. This converts the nascent chylomicron to a “mature” chylomicron. The apoCII on the mature chylomicron activates the enzyme **lipoprotein lipase (LPL)**, which is located on the inner surface of the capillary endothelial cells of muscle and adipose tissue. The LPL digests the triglyceride in the chylomicron, producing free fatty acids and glycerol. The fatty acids enter the adjacent organs either for energy production (muscle) or fat storage (adipocyte). The glycerol that is released is metabolized in the liver.

As the chylomicron loses triglyceride, its density increases and it becomes a **chylomicron remnant**, which is taken up by the liver by receptors that recognize apolipoprotein E. In the liver, the chylomicron remnant is degraded into its component parts for further disposition by the liver.



The lymph system is a network of vessels that surround interstitial cavities in the body. Cells secrete various compounds into the lymph, and the lymph vessels transport these fluids away from the interstitial spaces in the body tissues and into the bloodstream. In the case of the intestinal lymph system, the lymph enters the bloodstream through the thoracic duct. These vessels are designed such that under normal conditions the contents of the blood cannot enter the lymphatic system. The lymph fluid is similar in composition to that of the blood but lacks the cells found in blood.



THE WAITING ROOM



Will Sichel had several episodes of mild back and lower extremity pain over the last year, probably caused by minor sickle cell crises. He then developed severe right upper abdominal pain radiating to his lower right chest and his right flank 36 hours before admission to the emergency room. He states that the pain is not like his usual crisis pain. Intractable vomiting began 12 hours after the onset of these new symptoms. He reports that his urine is the color of iced tea and his stool now has a light clay color.

On physical examination, his body temperature is slightly elevated, and his heart rate is rapid. The whites of his eyes (the sclerae) are obviously jaundiced (a yellow discoloration caused by the accumulation of bilirubin pigment). He is exquisitely tender to pressure over his right upper abdomen.

The emergency room physician suspects that Michael is not in sickle cell crisis but instead has either acute cholecystitis (gallbladder inflammation) or a gallstone lodged in his common bile duct, causing cholestasis (the inability of the bile from the liver to reach his small intestine). His hemoglobin level was low at 7.6 mg/dL (reference range = 12–16) but unchanged from his baseline 3 months earlier. His serum total bilirubin level was 3.2 mg/dL (reference range = 0.2–1.0), and his direct (conjugated) bilirubin level was 0.9 mg/dL (reference range = 0–0.2).

Intravenous fluids were started, he was not allowed to take anything by mouth, a nasogastric tube was passed and placed on constant suction, and symptomatic therapy was started for pain and nausea. When his condition had stabilized, Michael was sent for an ultrasonographic (ultrasound) study of his upper abdomen.



Al Martini has continued to abuse alcohol and to eat poorly. After a particularly heavy intake of vodka, a steady severe pain began in his upper mid-abdomen. This pain spread to the left upper quadrant and eventually radiated to his mid-back. He began vomiting nonbloody material and was brought to the hospital emergency room with fever, a rapid heart beat, and a mild reduction in blood pressure. On physical examination, he was dehydrated and tender to pressure over the upper abdomen. His vomitus and stool were both negative for occult blood.

Blood samples were sent to the laboratory for a variety of hematologic and chemical tests, including a measurement of serum amylase and lipase, digestive enzymes normally secreted from the exocrine pancreas through the pancreatic ducts into the lumen of the small intestine.



Currently, 38% of the calories (kcal) in the typical American diet come from fat. The content of fat in the diet increased from the early 1900s until the 1960s, and then decreased as we became aware of the unhealthy effects of a high-fat diet. According to current recommendations, fat should provide no more than 30% of the total calories of a healthy diet.

I. DIGESTION OF TRIACYLGLYCEROLS

Triacylglycerols are the major fat in the human diet because they are the major storage lipid in the plants and animals that constitute our food supply. Triacylglycerols contain a glycerol backbone to which three fatty acids are esterified (Fig. 32.1). The main route for digestion of triacylglycerols involves hydrolysis to fatty acids and 2-monoacylglycerols in the lumen of the intestine. However, the route depends to some extent on the chain length of the fatty acids. Lingual and gastric lipases are produced by cells at the back of the tongue and in the stomach, respectively. These lipases preferentially hydrolyze short- and medium-chain fatty acids (containing 12 or fewer carbon atoms) from dietary triacylglycerols. Therefore, they are most active in

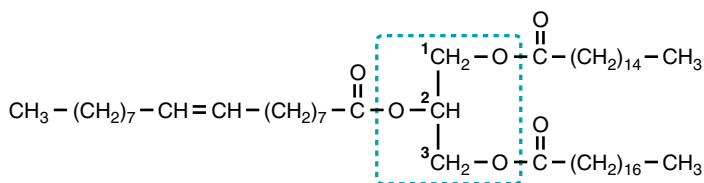


Fig. 32.1. Structure of a triacylglycerol. The glycerol moiety is highlighted, and its carbons are numbered.

infants and young children, who drink relatively large quantities of cow's milk, which contains triacylglycerols with a high percentage of short- and medium-chain fatty acids.

A. Action of Bile Salts

Dietary fat leaves the stomach and enters the small intestine, where it is emulsified (suspended in small particles in the aqueous environment) by bile salts (Fig. 32.2). The bile salts are amphipathic compounds (containing both hydrophobic and hydrophilic components), synthesized in the liver (see Chapter 34 for the pathway) and secreted via the gallbladder into the intestinal lumen. The contraction of the gallbladder and secretion of pancreatic enzymes are stimulated by the gut hormone cholecystokinin, which is secreted by the intestinal cells when stomach contents enter the intestine. Bile salts act as detergents, binding to the globules of dietary fat as they are broken up by the peristaltic action of the intestinal muscle. This emulsified fat, which has an increased surface area as compared with unemulsified fat, is attacked by digestive enzymes from the pancreas (Fig. 32.3).

B. Action of Pancreatic Lipase

The major enzyme that digests dietary triacylglycerols is a lipase produced in the pancreas. Pancreatic lipase is secreted along with another protein, colipase, along with bicarbonate, which neutralizes the acid that enters the intestine with partially digested food from the stomach. Bicarbonate raises the pH of the contents of the intestinal lumen into a range (pH ~ 6) that is optimal for the action of all of the digestive enzymes of the intestine. Bicarbonate secretion from the pancreas is stimulated by the hormone secretin, which is released from the intestine when acid enters the duodenum.



In patients such as **Will Sichel** who have severe and recurrent episodes of increased red blood cell destruction (hemolytic anemia), greater than normal amounts of the red cell pigment heme must be processed by the liver and spleen. In these organs, heme (derived from hemoglobin) is degraded to bilirubin, which is excreted by the liver in the bile.

If large quantities of bilirubin are presented to the liver as a consequence of acute hemolysis, the capacity of the liver to conjugate it, that is, convert it to the water-soluble bilirubin diglucuronide, can be overwhelmed. As a result, a greater percentage of the bilirubin entering the hepatic biliary ducts in patients with hemolysis is in the less water-soluble forms. In the gallbladder, these relatively insoluble particles tend to precipitate as gallstones rich in calcium bilirubinate. In some patients, one or more stones may leave the gallbladder through the cystic duct and enter the common bile duct. Most pass harmlessly into the small intestine and are later excreted in the stool. Larger stones, however, may become entrapped in the lumen of the common bile duct, where they cause varying degrees of obstruction to bile flow (cholestasis) with associated ductal spasm, producing pain. If adequate amounts of bile salts do not enter the intestinal lumen, dietary fats cannot readily be emulsified and digested.



The mammary gland produces milk, which is the major source of nutrients for the breastfed human infant. The fatty acid composition of human milk varies, depending on the diet of the mother. However, long-chain fatty acids predominate, particularly palmitic, oleic, and linoleic acids. Although the amount of fat contained in human milk and cow's milk is similar, cow's milk contains more short- and medium-chain fatty acids and does not contain the long-chain, polyunsaturated fatty acids found in human milk that are important in brain development.

Although the concentrations of pancreatic lipase and bile salts are low in the intestinal lumen of the newborn infant, the fat of human milk is still readily absorbed. This is true because lingual and gastric lipases produced by the infant partially compensate for the lower levels of pancreatic lipase. The human mammary gland also produces lipases that enter the milk. One of these lipases, which requires lower levels of bile salts than pancreatic lipase, is not inactivated by stomach acid and functions in the intestine for a number of hours.

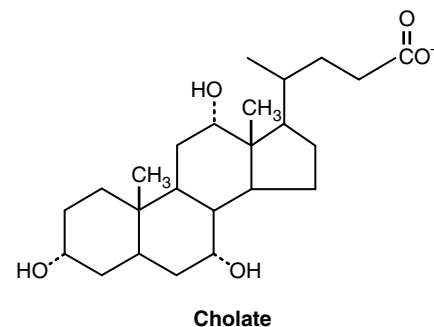


Fig. 32.2. Structure of a bile salt. The bile salts are derived from cholesterol and retain the cholesterol ring structure. They differ from cholesterol in that the rings in bile salts contain more hydroxyl groups and a polar side chain and lack a 5-6 double bond.



Al Martini's serum levels of pancreatic amylase (which digests dietary starch) and pancreatic lipase were elevated, a finding consistent with a diagnosis of acute and possibly chronic pancreatitis. The elevated levels of these enzymes in the blood are the result of their escape from the inflamed exocrine cells of the pancreas into the surrounding pancreatic veins. The cause of this inflammatory pancreatic process in this case was related to the toxic effect of acute and chronic excessive alcohol ingestion.

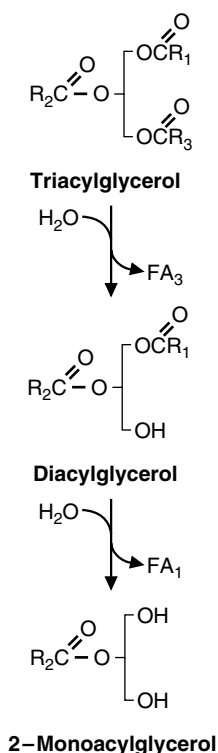


Fig. 32.4. Action of pancreatic lipase. Fatty acids (FA) are cleaved from positions 1 and 3 of the triacylglycerol, and a monoacylglycerol with a fatty acid at position 2 is produced.

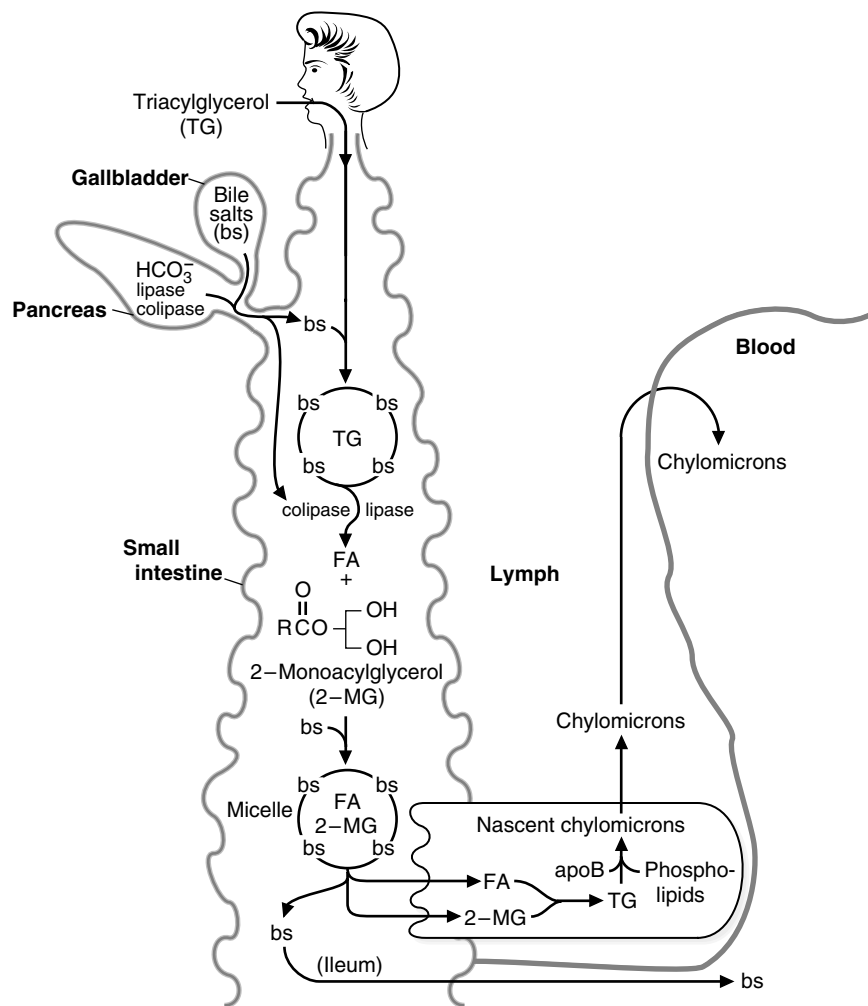


Fig. 32.3. Digestion of triacylglycerols in the intestinal lumen. TG = triacylglycerol; bs = bile salts; FA = fatty acid; 2-MG = 2-monoacylglycerol.



Bile salts inhibit pancreatic lipase activity by coating the substrate and not allowing the enzyme access to the substrate. The colipase relieves the bile salt inhibition, and allows the triglyceride to enter the active site of the lipase.



The exocrine pancreas secretes phospholipase A2 in an inactive zymogen form, prophospholipase A2. The enzyme is activated in the intestinal lumen by proteolytic cleavage by trypsin. Pancreatic lipase, however, is secreted in its active form, and only needs to bind colipase and substrate to be active.



When he was finally able to tolerate a full diet, **Al Martini's** stools became bulky, glistening, yellow-brown, and foul smelling. They floated on the surface of the toilet water. What caused this problem?

The colipase binds to the dietary fat and to the lipase, thereby increasing lipase activity. Pancreatic lipase hydrolyzes fatty acids of all chain lengths from positions 1 and 3 of the glycerol moiety of the triacylglycerol, producing free fatty acids and 2-monoacylglycerol, i.e., glycerol with a fatty acid esterified at position 2 (Fig. 32.4). The pancreas also produces esterases that remove fatty acids from compounds (such as cholesterol esters) and phospholipase A2 that digests phospholipids to a free fatty acid and a lysophospholipid (Fig. 32.5).

II. ABSORPTION OF DIETARY LIPIDS

The fatty acids and 2-monoacylglycerols produced by digestion are packaged into micelles, tiny microdroplets emulsified by bile salts (see Fig. 32.3). Other dietary lipids, such as cholesterol, lysophospholipids, and fat-soluble vitamins, are also packaged in these micelles. The micelles travel through a layer of water (the unstirred water layer) to the microvilli on the surface of the intestinal epithelial cells, where the fatty acids, 2-monoacylglycerols, and other dietary lipids are absorbed, but the bile salts are left behind in the lumen of the gut.

The bile salts are extensively resorbed when they reach the ileum. Greater than 95% of the bile salts are recirculated, traveling through the enterohepatic circulation

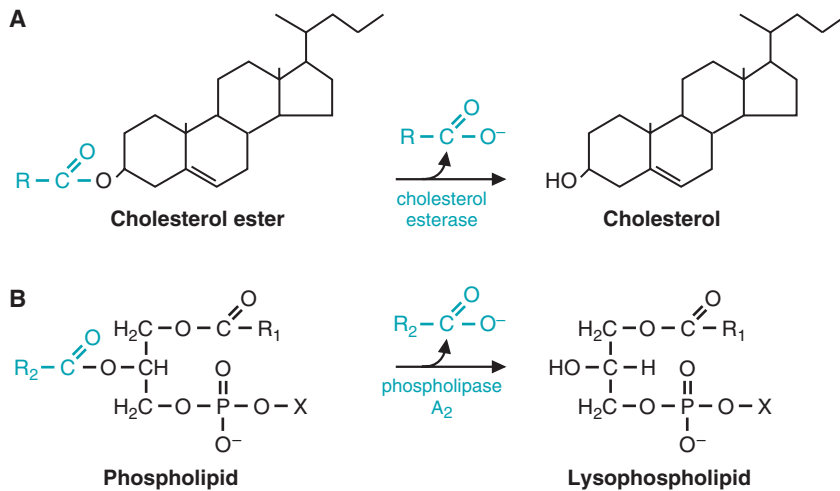


Fig. 32.5. Action of pancreatic esterases (A) and phospholipase A₂ (B).

to the liver, which secretes them into the bile for storage in the gallbladder and ejection into the intestinal lumen during another digestive cycle (Fig. 32.6).

Short- and medium-chain fatty acids (C₄ to C₁₂) do not require bile salts for their absorption. They are absorbed directly into intestinal epithelial cells. Because they do not need to be packaged to increase their solubility, they enter the portal blood (rather than the lymph) and are transported to the liver bound to serum albumin.

III. SYNTHESIS OF CHYLOMICRONS

Within the intestinal epithelial cells, the fatty acids and 2-monoacylglycerols are condensed by enzymatic reactions in the smooth endoplasmic reticulum to form triacylglycerols. The fatty acids are activated to fatty acyl CoA by the same process



Al Martini's stool changes are characteristic of steatorrhea (fat-laden stools caused by malabsorption of dietary fats), in this case caused by a lack of pancreatic secretions, particularly pancreatic lipase, which normally digests dietary fat.

Steatorrhea also may be caused by insufficient production or secretion of bile salts. Therefore, **Michael Sichel** might also develop this condition.



For bile salt micelles to form, the concentration of bile salts in the contents of the intestinal lumen must reach 5–15 $\mu\text{mol/mL}$. This critical concentration of bile salts is therefore required for optimal lipid absorption.

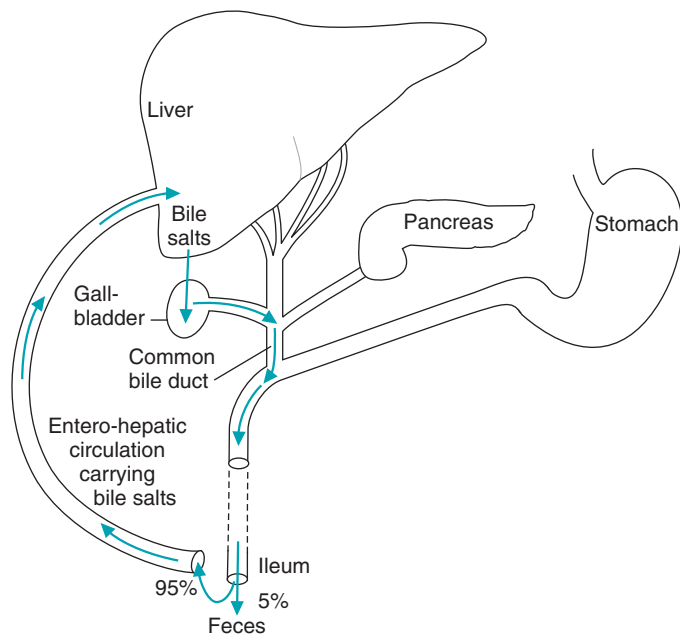


Fig. 32.6. Recycling of bile salts. Bile salts are synthesized in the liver, stored in the gallbladder, secreted into the small intestine, resorbed in the ileum, and returned to the liver via the enterohepatic circulation. Five percent or less of luminal bile acids are excreted in the stool under normal circumstances.

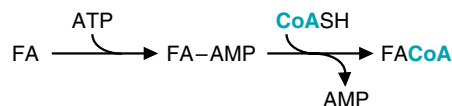
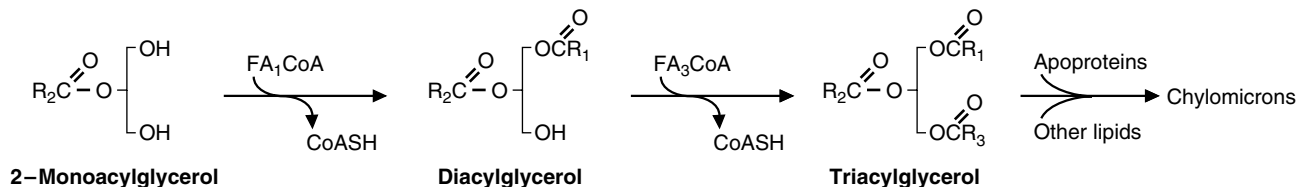
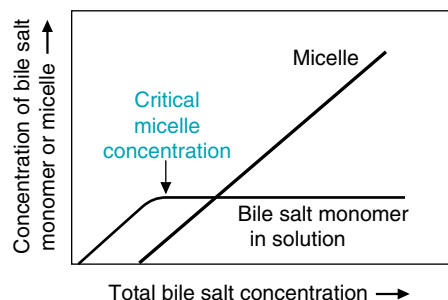
Activation of fatty acids**Triacylglycerol synthesis**

Fig. 32.7. Resynthesis of triacylglycerols in intestinal epithelial cells. Fatty acids (FA), produced by digestion, are activated in intestinal epithelial cells and then esterified to the 2-monoacylglycerol produced by digestion. The triacylglycerols are packaged in chylomicrons and secreted into the lymph.



Because the fat-soluble vitamins (A, D, E, and K) are absorbed from micelles along with the long-chain fatty acids and 2-monoacylglycerols, prolonged obstruction of the duct that carries exocrine secretions from the pancreas and the gallbladder into the intestine (via the common duct) could lead to a deficiency of these metabolically important substances. If the obstruction of **Michael Sichel's** common duct continues, he will eventually suffer from a fat-soluble vitamin deficiency. (Graph from Devlin T. Textbook of Biochemistry, 3rd Ed. 1992:1084. Copyright © John Wiley & Sons, Inc.)

used for activation of fatty acids before β -oxidation (see Chapter 23). A fatty acyl CoA then reacts with a 2-monoacylglycerol to form a diacylglycerol, which reacts with another fatty acyl CoA to form a triacylglycerol (Fig. 32.7). The reactions for triacylglycerol synthesis in intestinal cells differ from those in liver and adipose cells in that 2-monoacylglycerol is an intermediate in triacylglycerol in intestinal cells, whereas phosphatidic acid is the necessary intermediate in other tissues.

Triacylglycerols are transported in lipoprotein particles because they are insoluble in water. If triacylglycerols directly entered the blood, they would coalesce, impeding blood flow. Intestinal cells package triacylglycerols together with proteins and phospholipids in chylomicrons, which are lipoprotein particles that do not readily coalesce in aqueous solutions (Figs. 32.8 and 32.9). Chylomicrons also contain cholesterol and fat-soluble vitamins. The protein constituents of the lipoproteins are known as apoproteins.

The major apoprotein associated with chylomicrons as they leave the intestinal cells is B-48 (Fig. 32.10). The B-48 apoprotein is structurally and genetically

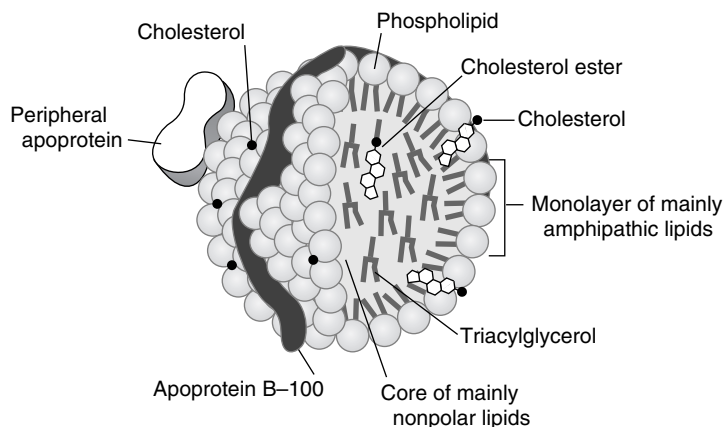


Fig. 32.8. Example of the structure of a blood lipoprotein. VLDL is depicted. Lipoproteins contain phospholipids and proteins on the surface, with their hydrophilic regions interacting with water. Hydrophobic molecules are in the interior of the lipoprotein. The hydroxyl group of cholesterol is near the surface. In cholesterol esters, the hydroxyl group is esterified to a fatty acid. Cholesterol esters are found in the interior of lipoproteins and are synthesized by reaction of cholesterol with an activated fatty acid (see Chapter 33).

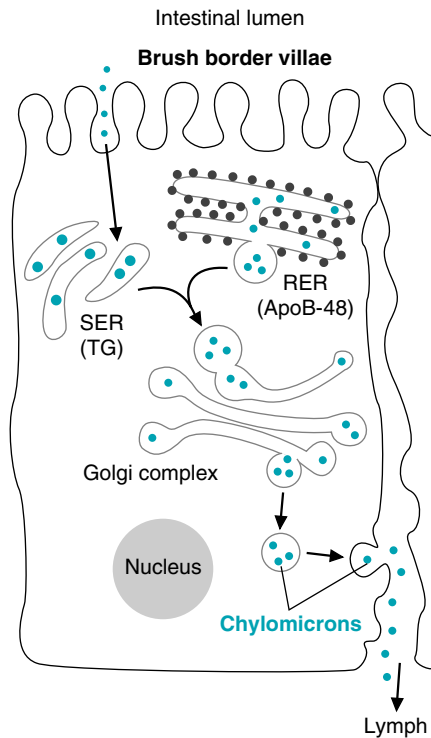


Fig. 32.10. Formation and secretion of chylomicrons. The triacylglycerol is produced in the smooth endoplasmic reticulum (SER) of intestinal epithelial cells from the digestive products, fatty acids, and 2-monoacylglycerols. The protein is synthesized in the rough endoplasmic reticulum (RER). The major apoprotein in chylomicrons is B-48. Assembly of the lipoproteins occurs in both the ER and the Golgi complex.

related to the B-100 apoprotein synthesized in the liver that serves as a major protein of another lipid carrier, very-low-density lipoprotein (VLDL). These two apoproteins are encoded by the same gene. In the intestine, the primary transcript of this gene undergoes RNA editing (Fig. 32.11 and see Chapter 15). A stop codon is generated that causes a protein to be produced in the intestine that is 48% of the size of the protein produced in the liver; hence the designations B-48 and B-100.

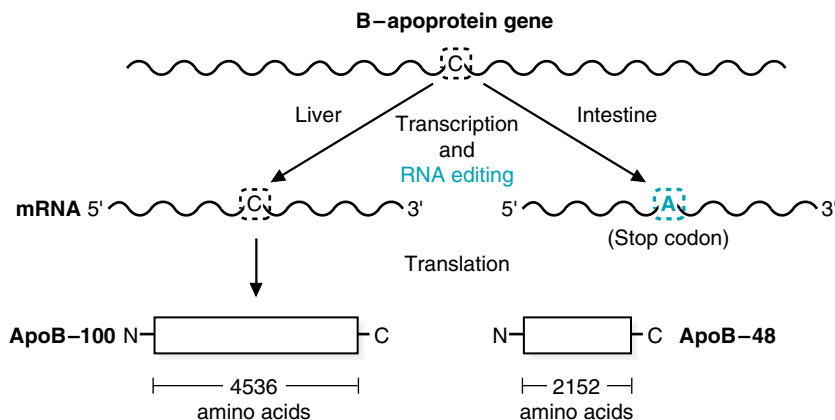


Fig. 32.11. B-apoprotein gene. The gene, located on chromosome 2, is transcribed and translated in liver to produce apoB-100, which is 4,536 amino acids in length (one of the longest single-polypeptide chains). In intestinal cells, RNA editing converts a cytosine (C) to an adenine (A), producing a stop codon. Consequently, the B-apoprotein of intestinal cells (apoB-48) contains only 2,152 amino acids. ApoB-48 is 48% of the size of apoB-100.

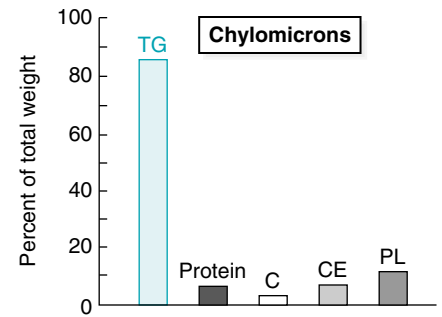
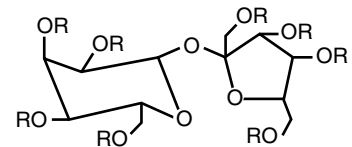


Fig. 32.9. Composition of a typical chylomicron. Although the composition varies to some extent, the major component is triacylglycerol (TG). C = cholesterol; CE = cholesterol ester; PL = phospholipid.



Olestra is an artificial fat substitute designed to allow individuals to obtain the taste and food consistency of fat, without the calories from fat. The structure of Olestra is shown below and consists of a sucrose molecule to which fatty acids are esterified to the hydroxyl groups.



The fatty acids attached to sucrose are resistant to hydrolysis by pancreatic lipase, so Olestra passes through the intestine intact and is eliminated in the feces. As a result, no useful calories can be obtained through the metabolism of Olestra.



Because of their high triacylglycerol content, chylomicrons are the least dense of the blood lipoproteins. When blood is collected from patients with certain types of hyperlipoproteinemias (high concentrations of lipoproteins in the blood) in which chylomicron levels are elevated, and the blood is allowed to stand in the refrigerator overnight, the chylomicrons float to the top of the liquid and coalesce, forming a creamy layer.



One manner in which individuals can lose weight is to inhibit the activity of pancreatic lipase. This would result in reduced fat digestion and absorption and a reduced caloric yield from the diet. The drug Orlistat is a chemically synthesized derivative of lipstatin, a natural lipase inhibitor found in certain bacteria. The drug works in the intestinal lumen and forms a covalent bond with the active site serine residue of both gastric and pancreatic lipase, thereby inhibiting their activities. Nondigested triglycerides are not absorbed by the intestine and are eliminated in the feces. Under normal use of the drug, approximately 30% of dietary fat absorption is inhibited. Because excessive nondigested fat in the intestines can lead to gastrointestinal distress related to excessive intestinal gas formation, individuals taking this drug need to follow a reduced daily intake of fat in their diet, which should be evenly distributed amongst the meals of the day.

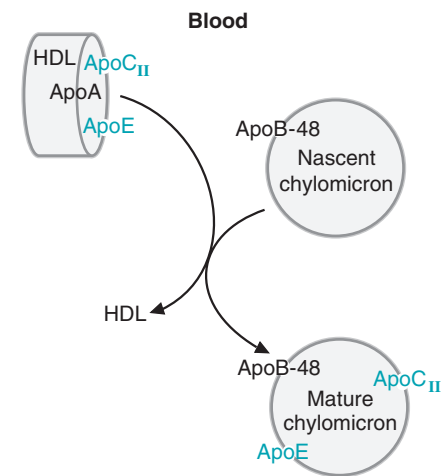


Fig. 32.12. Transfer of proteins from HDL to chylomicrons. Newly synthesized chylomicrons (nascent chylomicrons) mature as they receive apoproteins CII and E from HDL. HDL functions in the transfer of these apoproteins and also in transfer of cholesterol from peripheral tissues to the liver (see Table 1 in the introduction to this section).

The protein component of the lipoproteins is synthesized on the rough endoplasmic reticulum. Lipids, which are synthesized in the smooth endoplasmic reticulum, are complexed with the proteins to form the chylomicrons (see Fig. 32.10).

IV. TRANSPORT OF DIETARY LIPIDS IN THE BLOOD

By the process of exocytosis, chylomicrons are secreted by the intestinal epithelial cells into the chyle of the lymphatic system and enter the blood through the thoracic duct. Chylomicrons begin to enter the blood within 1 to 2 hours after the start of a meal; as the meal is digested and absorbed, they continue to enter the blood for many hours. Initially, the particles are called nascent (newborn) chylomicrons. As they accept proteins from HDL within the lymph and the blood, they become “mature” chylomicrons.

HDL transfers proteins to the nascent chylomicrons, particularly apoprotein E (apoE) and apoprotein CII (apoCII) (Fig. 32.12). ApoE is recognized by membrane receptors, particularly those on the surface of liver cells, allowing ApoE-bearing lipoproteins to enter these cells by endocytosis for subsequent digestion by lysosomes. ApoCII acts as an activator of LPL, the enzyme on capillary endothelial cells, primarily within muscle and adipose tissue, that digests the triacylglycerols of the chylomicrons and VLDL in the blood.

V. FATE OF CHYLOMICRONS

The triacylglycerols of the chylomicrons are digested by LPL attached to the proteoglycans in the basement membranes of endothelial cells that line the capillary walls (Fig. 32.13). LPL is produced by adipose cells, muscle cells (particularly cardiac muscle), and cells of the lactating mammary gland. The isozyme synthesized in adipose cells has a higher K_m than the isozyme synthesized in muscle cells. Therefore, adipose LPL is more active after a meal, when chylomicrons levels are elevated in the blood. Insulin stimulates the synthesis and secretion of adipose LPL, such that after a meal, when triglyceride levels increase in circulation, LPL has been upregulated (through insulin release) to facilitate the hydrolysis of fatty acids from the triglyceride.

The fatty acids released from triacylglycerols by LPL are not very soluble in water. They become soluble in blood by forming complexes with the protein albumin. The major fate of the fatty acids is storage as triacylglycerol in adipose tissue. However, these fatty acids also may be oxidized for energy in muscle and other tissues (see Fig. 32.13). The LPL in the capillaries of muscle cells has a lower K_m than adipose LPL. Thus, muscle cells can obtain fatty acids from blood lipoproteins whenever they are needed for energy, even if the concentration of the lipoproteins is low.

The glycerol released from chylomicron triacylglycerols by LPL may be used for triacylglycerol synthesis in the liver in the fed state.

The portion of a chylomicron that remains in the blood after LPL action is known as a chylomicron remnant. This remnant binds to receptors on hepatocytes (the major cells of the liver), which recognize apoprotein E, and is taken up by the process of endocytosis. Lysosomes fuse with the endocytic vesicles, and the



Heparin is a complex polysaccharide that is a component of proteoglycans (see Chapter 49). Isolated heparin is frequently used as an anticoagulant, because it binds to antithrombin III (ATIII), and the activated ATIII then binds factors necessary for clotting and inhibits them from working. As LPL is bound to the capillary endothelium through binding to proteoglycans, heparin also can bind to LPL and dislodge it from the capillary wall. This leads to loss of LPL activity and an increase of triglyceride content in the blood.

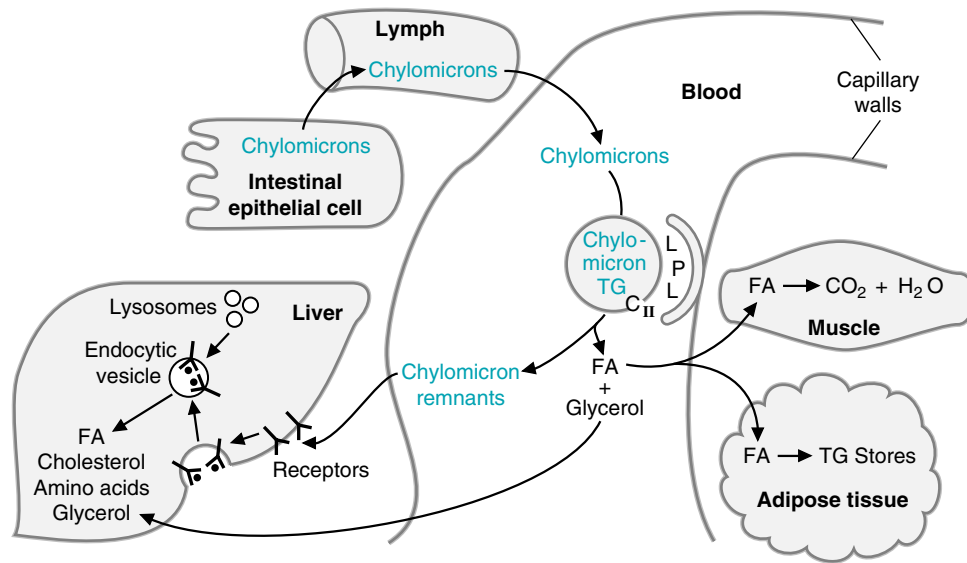


Fig. 32.13. Fate of chylomicrons. Chylomicrons are synthesized in intestinal epithelial cells, secreted into the lymph, pass into the blood, and become mature chylomicrons (see Fig. 32.11). On capillary walls in adipose tissue and muscle, lipoprotein lipase (LPL) activated by ApoC_{II} digests the triacylglycerols (TG) of chylomicrons to fatty acids and glycerol. Fatty acids (FA) are oxidized in muscle or stored in adipose cells as triacylglycerols. The remnants of the chylomicrons are taken up by the liver by receptor-mediated endocytosis. Lysosomal enzymes within the hepatocyte digest the remnants, releasing the products into the cytosol.

chylomicron remnants are degraded by lysosomal enzymes. The products of lysosomal digestion (e.g., fatty acids, amino acids, glycerol, cholesterol, phosphate) can be reused by the cell.

CLINICAL COMMENTS



The upper abdominal ultrasound study showed a large gallstone lodged in **Will Sichel's** common duct with dilation of this duct proximal to the stone. Michael was scheduled for endoscopic retrograde cholangiopancreatography (ERCP). (An ERCP involves cannulation of the common bile duct—and, if necessary, the pancreatic duct—through a tube placed through the mouth and stomach and into the upper small intestine.) With this technique, a stone can be snared in the common duct and removed to relieve an obstruction.

If common duct obstruction is severe enough, bilirubin flows back into the venous blood draining from the liver. As a consequence, serum bilirubin levels, particularly the indirect (unconjugated) fraction, increase. Tissues such as the sclerae of the eye take up this pigment, which causes them to become yellow (jaundiced). **Will Sichel's** condition was severe enough to cause jaundice by this mechanism.



Alcohol excess may produce proteinaceous plugs in the small pancreatic ducts, causing back pressure injury and autodigestion of the pancreatic acini drained by these obstructed channels. This process causes one form of acute pancreatitis. **Al Martini** had an episode of acute alcohol-induced pancreatitis superimposed on a more chronic alcohol-related inflammatory process in the pancreas—in other words, a chronic pancreatitis. As a result of decreased secretion of pancreatic lipase through the pancreatic ducts and into the lumen of the small intestine, dietary fat was not absorbed at a normal rate, and steatorrhea (fat-rich

stools) occurred. If abstinence from alcohol does not allow adequate recovery of the enzymatic secretory function of the pancreas, Mr. Martini will have to take a commercial preparation of pancreatic enzymes with meals that contain even minimal amounts of fat.

BIOCHEMICAL COMMENTS



The assembly of chylomicrons within the endoplasmic reticulum of the enterocyte requires the activity of microsomal triglyceride transfer protein (MTP). The protein is a dimer of two nonidentical subunits. The smaller subunit (57 kDa) is protein disulfide isomerase (PDI, see Chapter 7, section IX.A), whereas the larger subunit (97 kDa) contains the triglyceride transfer activity. MTP accelerates the transport of triglycerides, cholesterol esters, and phospholipids across membranes of subcellular organelles. The role of PDI in this complex is not known; the disulfide isomerase activity of this subunit is not needed for triglyceride transport to occur. The lack of triglyceride transfer activity leads to the disease **abetalipoproteinemia**. This disease affects both chylomicron assembly in the intestine and VLDL assembly in the liver. Both particles require a B apoprotein for their assembly (ApoB-48 for chylomicrons, ApoB-100 for VLDL), and MTP binds to the B apoproteins. For both chylomicron and VLDL assembly, a small ApoB-containing particle is first produced within the lumen of the ER. The appropriate apoB is made on the rough endoplasmic reticulum (RER) and is inserted into the ER lumen during its synthesis (see Chapter 15, section IX). As the protein is being translated, lipid (a small amount of triglyceride) begins to associate with the protein, and the lipid association is catalyzed by MTP. This leads to the generation of small ApoB-containing particles; these particles are not formed in patients with abetalipoproteinemia. Thus, it appears as though MTP activity is necessary to transfer triacylglycerol formed within the ER to the ApoB protein. The second stage of particle assembly is the fusion of the initial ApoB particle with triacylglycerol droplets within the ER. MTP also may be required for the transfer of triacylglycerol from the cytoplasm to the lumen of the ER to form this lipid droplet. These steps are depicted in Fig. 32.14.

The symptoms of abetalipoproteinemia include lipid malabsorption (and its accompanying symptoms, such as steatorrhea and vomiting), which can result in caloric deficiencies and weight loss. Because lipid-soluble vitamin distribution occurs through chylomicron circulation, signs and symptoms of deficiencies in the lipid-soluble vitamins may be seen in these patients.

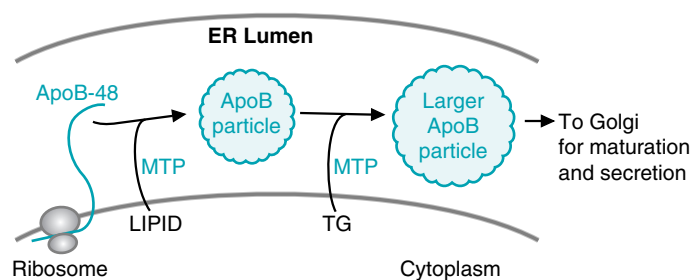


Fig. 32.14. A model of microsomal triglyceride transfer protein (MTP) action. MTP is required to transfer lipid to apoB-48 as it is synthesized, and to transfer lipid from the cytoplasm to the ER lumen.

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REVIEW QUESTIONS—CHAPTER 32

1. The most abundant component of chylomicrons is which of the following?
 - (A) ApoB-48
 - (B) Triglyceride
 - (C) Phospholipid
 - (D) Cholesterol
 - (E) Cholesterol ester
2. The conversion of nascent chylomicrons to mature chylomicrons requires which of the following?
 - (A) Bile salts
 - (B) 2-Monoacylglycerol
 - (C) Lipoprotein lipase
 - (D) High-density lipoprotein
 - (E) Lymphatic system
3. The apoproteins B-48 and B-100 are similar with respect to which of the following?
 - (A) They are synthesized from the same gene.
 - (B) They are derived by alternative splicing of the same hnRNA.
 - (C) ApoB-48 is a proteolytic product of apoB-100.
 - (D) Both are found in mature chylomicrons.
 - (E) Both are found in very-low-density lipoproteins.
4. Bile salts must reach a particular concentration within the intestinal lumen before they are effective agents for lipid digestion. This is because of which of the following?
 - (A) The bile salt concentration must be equal to the triglyceride concentration.
 - (B) The bile salt solubility in the lumen is a critical factor.
 - (C) The ability of bile salts to bind lipase is concentration dependant.
 - (D) The bile salts cannot be reabsorbed in the ileum until they reach a certain concentration.
 - (E) The bile salts do not activate lipase until they reach a particular concentration.
5. Type III hyperlipidemia is caused by a deficiency of apoprotein E. Analysis of the serum of patients with this disorder would exhibit which of the following?
 - (A) An absence of chylomicrons after eating
 - (B) Above-normal levels of VLDL after eating
 - (C) Normal triglyceride levels
 - (D) Elevated triglyceride levels
 - (E) Below-normal triglyceride levels

33 Synthesis of Fatty Acids, Triacylglycerols, and the Major Membrane Lipids

Fatty acids are synthesized mainly in the liver in humans, with dietary glucose serving as the major source of carbon. Glucose is converted through glycolysis to pyruvate, which enters the mitochondrion and forms both acetyl CoA and oxaloacetate (Fig. 33.1). These two compounds condense, forming citrate. Citrate is transported to the **cytosol**, where it is cleaved to form acetyl CoA, the source of carbon for the reactions that occur on the **fatty acid synthase complex**. The key regulatory enzyme for the process, **acetyl CoA carboxylase**, produces **malonyl CoA** from acetyl CoA.

The growing fatty acid chain, attached to the fatty acid synthase complex in the cytosol, is elongated by the sequential addition of 2-carbon units provided by malonyl CoA. **NADPH**, produced by the **pentose phosphate pathway** and the **malic enzyme**, provides **reducing equivalents**. When the growing fatty acid chain is 16 carbons in length, it is released as **palmitate**. After activation to a CoA derivative, palmitate can be elongated and desaturated to produce a series of fatty acids.

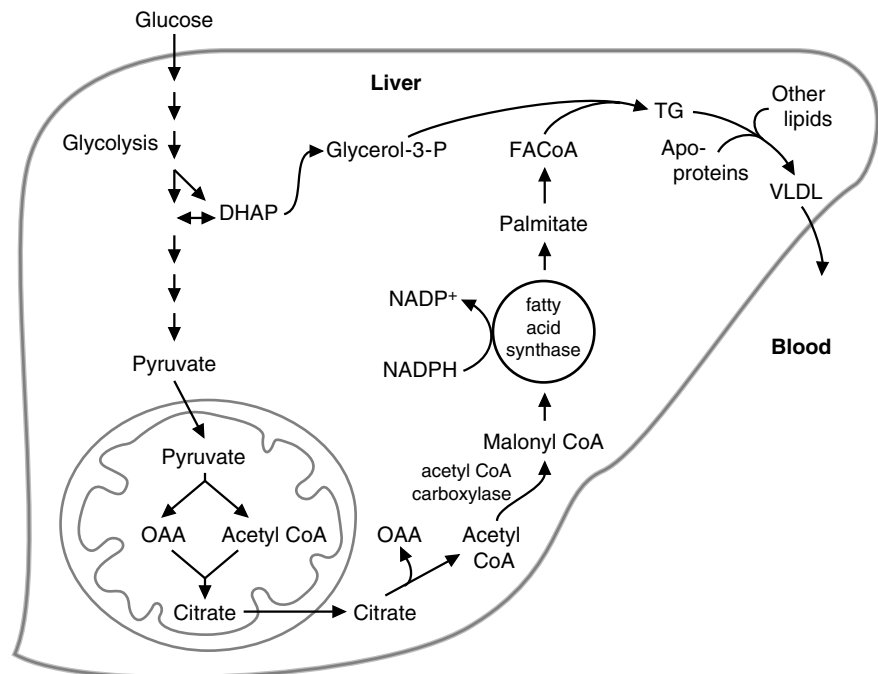


Fig. 33.1. Lipogenesis, the synthesis of triacylglycerols from glucose. In humans, the synthesis of fatty acids from glucose occurs mainly in the liver. Fatty acids (FA) are converted to triacylglycerols (TG), packaged in VLDL, and secreted into the blood. OAA = oxaloacetate.

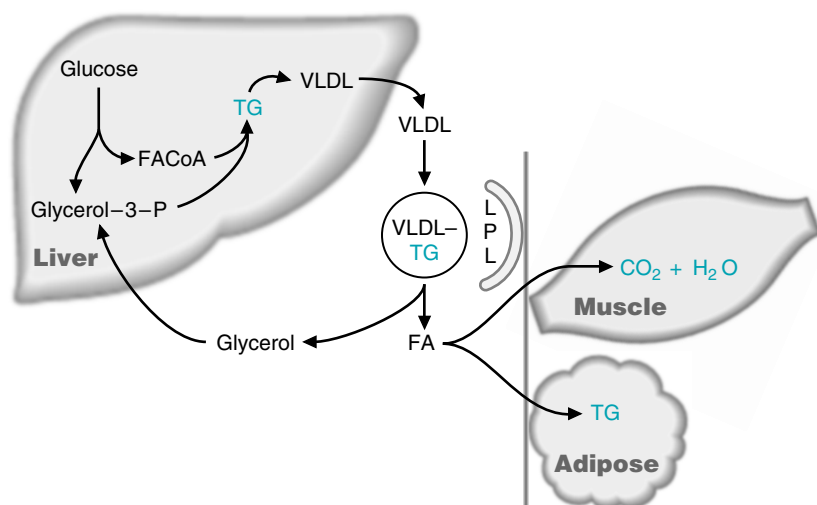


Fig. 33.2. Fate of VLDL triacylglycerol (TG). The TG of VLDL, produced in the liver, is digested by lipoprotein lipase (LPL) present on the lining cells of the capillaries in adipose and skeletal muscle tissue. Fatty acids are released and either oxidized or stored in tissues as TG. Glycerol is used by the liver and other tissues that contain glycerol kinase. FA = fatty acid (or fatty acyl group).

Fatty acids, produced in cells or obtained from the diet, are used by various tissues for the synthesis of **triacylglycerols** (the major storage form of fuel) and the **glycerophospholipids** and **sphingolipids** (the major components of cell membranes).

In the liver, triacylglycerols are produced from fatty acyl CoA and glycerol 3-phosphate. **Phosphatidic acid** serves as an intermediate in this pathway. The triacylglycerols are not stored in the liver but rather packaged with apoproteins and other lipids in **very-low-density lipoprotein (VLDL)** and secreted into the blood (see Fig. 33.1).

In the capillaries of various tissues (particularly adipose tissue, muscle, and the lactating mammary gland), **lipoprotein lipase (LPL)** digests the triacylglycerols of VLDL, forming **fatty acids** and **glycerol** (Fig. 33.2). The glycerol travels to the liver and other tissues where it is used. Some of the fatty acids are oxidized by muscle and other tissues. After a meal, however, most of the fatty acids are converted to triacylglycerols in **adipose cells**, where they are **stored**. These fatty acids are released during fasting and serve as the predominant fuel for the body.

Glycerophospholipids are also synthesized from fatty acyl CoA, which forms esters with glycerol 3-phosphate, producing phosphatidic acid. Various head groups are added to carbon 3 of the glycerol 3-phosphate moiety of phosphatidic acid, generating amphipathic compounds such as **phosphatidylcholine**, **phosphatidylinositol**, and **cardiolipin** (Fig. 33.3). In the formation of **plasmalogens** and **platelet-activating factor (PAF)**, a long-chain fatty alcohol forms an **ether** with carbon 1, replacing the fatty acyl ester (Fig. 33.4). Cleavage of phospholipids is catalyzed by **phospholipases** found in cell membranes, lysosomes, and pancreatic juice.

Sphingolipids, which are prevalent in membranes and the myelin sheath of the central nervous system, are built on **serine** rather than glycerol. In the synthesis of sphingolipids, serine and palmityl CoA condense, forming a compound that is related to **sphingosine**. Reduction of this compound, followed by addition of a second fatty acid in amide linkage, produces **ceramide**. Carbohydrate groups attach to ceramide, forming **glycolipids** such as the **cerebrosides**, **globosides**, and **gangliosides** (Fig. 33.5). The addition of **phosphocholine** to ceramide produces **sphingomyelin**. These sphingolipids are degraded by lysosomal enzymes.

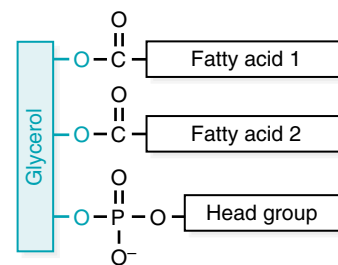


Fig. 33.3. General structure of a glycerophospholipid. The fatty acids are joined by ester bonds to the glycerol moiety. Various combinations of fatty acids may be present. The fatty acid at carbon 2 of the glycerol is usually unsaturated. The head group is the group attached to the phosphate on position 3 of the glycerol moiety. The most common head group is choline, but ethanolamine, serine, inositol, or phosphatidylglycerol also may be present. The phosphate group is negatively charged, and the head group may carry a positive charge (choline and ethanolamine), or both a positive and a negative charge (serine). The inositol may be phosphorylated and, thus, negatively charged.

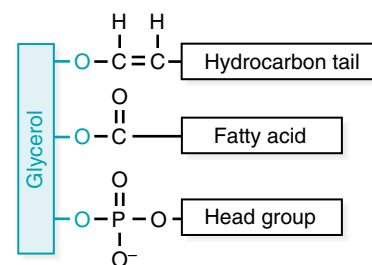


Fig. 33.4. General structure of a plasmalogen. Carbon 1 of glycerol is joined to a long-chain fatty alcohol by an ether linkage. The fatty alcohol group has a double bond between carbons 1 and 2. The head group is usually ethanolamine or choline.

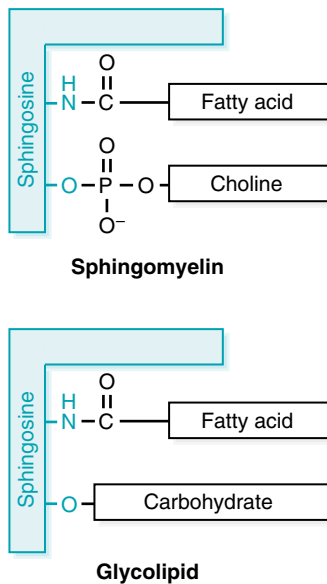


Fig. 33.5. General structures of the sphingolipids. The “backbone” is sphingosine rather than glycerol. Ceramide is sphingosine with a fatty acid joined to its amino group by an amide linkage. Sphingomyelin contains phosphocholine, whereas glycolipids contain carbohydrate groups.



The dietician did a careful analysis of **Percy Veere's** diet, which was indeed low in fat, adequate in protein, but excessive in carbohydrates, especially in refined sugars. Percy's total caloric intake averaged about 430 kilocalories (kcal) a day in excess of his isocaloric requirements. This excess carbohydrate was being converted to fats, accounting for Percy's weight gain. A new diet with a total caloric content that would prevent further gain in weight was prescribed.



THE WAITING ROOM



Percy Veere's mental depression slowly responded to antidepressant medication, to the therapy sessions with his psychiatrist, and to frequent visits from an old high school sweetheart whose husband had died several years earlier. While hospitalized for malnutrition, Mr. Veere's appetite returned. By the time of discharge, he had gained back 8 of the 22 lb he had lost and weighed 133 lb.

During the next few months, Mr. Veere developed a craving for “sweet foods” such as the candy he bought and shared with his new friend. After 6 months of this high-carbohydrate courtship, Percy had gained another 22 lb and now weighed 155 lb, just 8 lb more than he weighed when his depression began. He became concerned about the possibility that he would soon be overweight and consulted his dietitian, explaining that he had faithfully followed his low-fat diet but had “gone overboard” with carbohydrates. He asked whether it was possible to become fat without eating fat.



Cora Nari's hypertension and heart failure have been well controlled on medication, and she has lost 10 lb since she had her recent heart attack. Her fasting serum lipid profile on discharge from the hospital indicated significantly elevated serum low-density lipoprotein (LDL) cholesterol level of 175 mg/dL (recommended level for a patient with known coronary artery disease = 100 mg/dL or less), a serum triacylglycerol level of 280 mg/dL (reference range = 60–150), and a serum high-density lipoprotein (HDL) cholesterol level of 34 mg/dL (reference range > 50 for healthy women). While still in the hospital, she was asked to obtain the most recent serum lipid profiles of her older brother and her younger sister, both of whom were experiencing chest pain. Her brother's profile showed normal triacylglycerols, moderately elevated LDL cholesterol, and significantly suppressed HDL cholesterol levels. Her sister's profile showed only hypertriglyceridemia (high blood triacylglycerols).



Colleen Lakker was born 6 weeks prematurely. She appeared normal until about 30 minutes after delivery, when her respirations became rapid at 64 breaths/minute with audible respiratory grunting. The spaces between her ribs (intercostal spaces) retracted inward with each inspiration, and her lips and fingers became cyanotic from a lack of oxygen in her arterial blood. An arterial blood sample indicated a low partial pressure of oxygen (pO_2) and a slightly elevated partial pressure of carbon dioxide (pCO_2). The arterial pH was somewhat suppressed, in part from an accumulation of lactic acid secondary to the hypoxemia (a low level of oxygen in her blood). A chest x-ray showed a fine reticular granularity of the lung tissue, especially in the left lower lobe area. From these clinical data, a diagnosis of respiratory distress syndrome (RDS), also known as hyaline membrane disease, was made.

Colleen was immediately transferred to the neonatal intensive care unit, where, with intensive respiration therapy, she slowly improved.

I. FATTY ACID SYNTHESIS

Fatty acids are synthesized whenever an excess of calories is ingested. The major source of carbon for the synthesis of fatty acids is dietary carbohydrate. An excess of dietary protein also can result in an increase in fatty acid synthesis. In this case, the carbon source is amino acids that can be converted to acetyl CoA or tricarboxylic

acid (TCA) cycle intermediates (see Chapter 39). Fatty acid synthesis occurs mainly in the liver in humans, although the process also occurs in adipose tissue.

When an excess of dietary carbohydrate is consumed, glucose is converted to acetyl CoA, which provides the 2-carbon units that condense in a series of reactions on the fatty acid synthase complex, producing palmitate (see Fig. 33.1). Palmitate is then converted to other fatty acids. The fatty acid synthase complex is located in the cytosol, and, therefore, it uses cytosolic acetyl CoA.

A. Conversion of Glucose to Cytosolic Acetyl CoA

The pathway for the synthesis of cytosolic acetyl CoA from glucose begins with glycolysis, which converts glucose to pyruvate in the cytosol (Fig. 33.6). Pyruvate enters mitochondria, where it is converted to acetyl CoA by pyruvate dehydrogenase and to oxaloacetate by pyruvate carboxylase. The pathway pyruvate follows is dictated by the acetyl CoA levels in the mitochondria. When acetyl CoA levels are high, pyruvate dehydrogenase is inhibited, and pyruvate carboxylase activity is stimulated. As oxaloacetate levels increase because of the activity of pyruvate carboxylase, oxaloacetate condenses with acetyl CoA to form citrate. This condensation reduces the acetyl CoA levels, which leads to the activation of pyruvate dehydrogenase and inhibition of pyruvate carboxylase. Through such reciprocal regulation, citrate can be continuously synthesized and transported across the inner mitochondrial membrane. In the cytosol, citrate is cleaved by citrate lyase to re-form acetyl CoA and oxaloacetate. This circuitous route is required because pyruvate dehydrogenase, the enzyme that converts pyruvate to acetyl CoA, is found only in mitochondria and because acetyl CoA cannot directly cross the mitochondrial membrane.

The NADPH required for fatty acid synthesis is generated by the pentose phosphate pathway (see Chapter 29) and from recycling of the oxaloacetate produced by citrate lyase (Fig. 33.7). Oxaloacetate is converted back to pyruvate in two steps: the reduction of oxaloacetate to malate by NAD^+ -dependent malate dehydrogenase and the oxidative decarboxylation of malate to pyruvate by an NADP^+ -dependent malate dehydrogenase (malic enzyme) (Fig. 33.8). The pyruvate formed by malic enzyme is reconverted to citrate. The NADPH that is generated by malic enzyme, along with the NADPH generated by glucose 6-phosphate and gluconate 6-phosphate dehydrogenases in the pentose phosphate pathway, is used for the reduction reactions that occur on the fatty acid synthase complex (Fig. 33.9).

The generation of cytosolic acetyl CoA from pyruvate is stimulated by elevation of the insulin/glucagon ratio after a carbohydrate meal. Insulin activates pyruvate dehydrogenase by stimulating the phosphatase that dephosphorylates the enzyme to

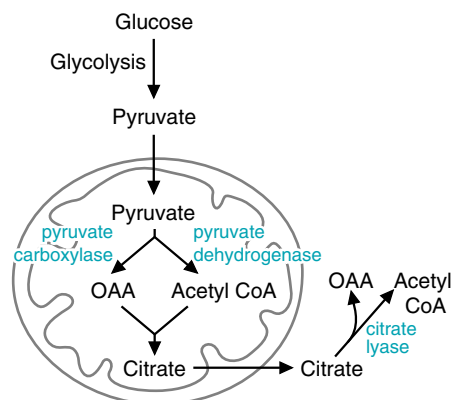


Fig. 33.6. Conversion of glucose to cytosolic acetyl CoA. OAA = oxaloacetate.

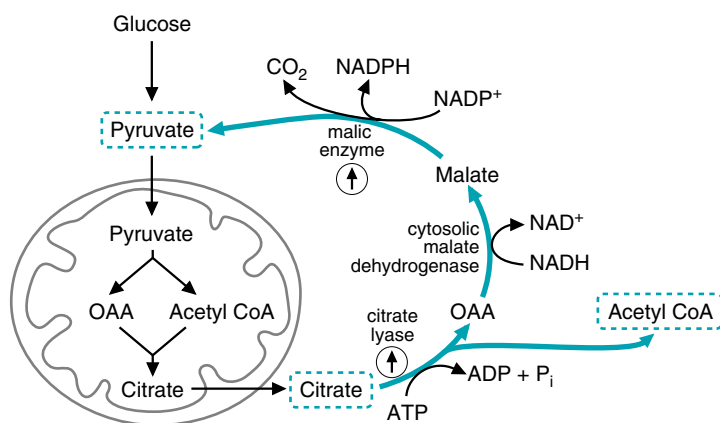


Fig. 33.7. Fate of citrate in the cytosol. Citrate lyase is also called citrate cleavage enzyme. OAA = oxaloacetate; circled $\uparrow$ = inducible enzyme.

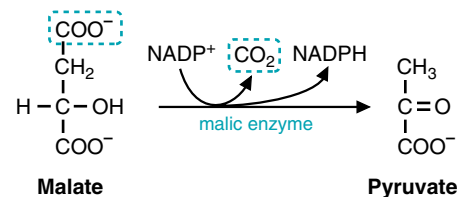


Fig. 33.8. Reaction catalyzed by malic enzyme. This enzyme is also called the decarboxylating or NADP^+ -dependent malate dehydrogenase.

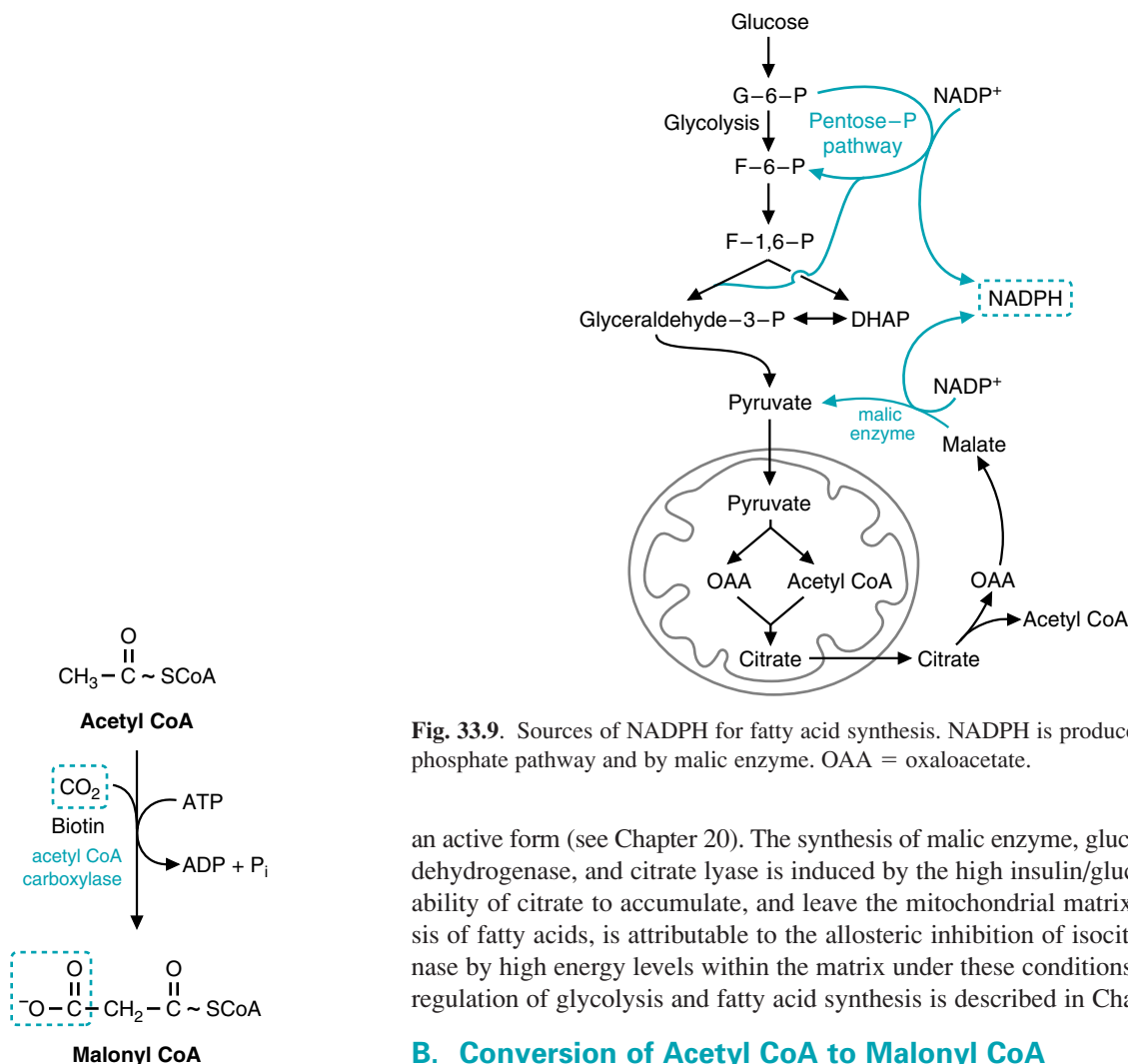


Fig. 33.10. Reaction catalyzed by acetyl CoA carboxylase. CO_2 is covalently attached to biotin, which is linked by an amide bond to the ϵ -amino group of a lysine residue of the enzyme. Hydrolysis of ATP is required for the attachment of CO_2 to biotin.



AMP is a much more sensitive indicator of low energy levels because of the adenylate kinase reaction. The $[\text{AMP}]$ to $[\text{ATP}]$ ratio is proportional to the square of the $[\text{ADP}]$ to $[\text{ATP}]$ ratio, so a fivefold change in ADP levels corresponds to a 25-fold change in AMP levels.

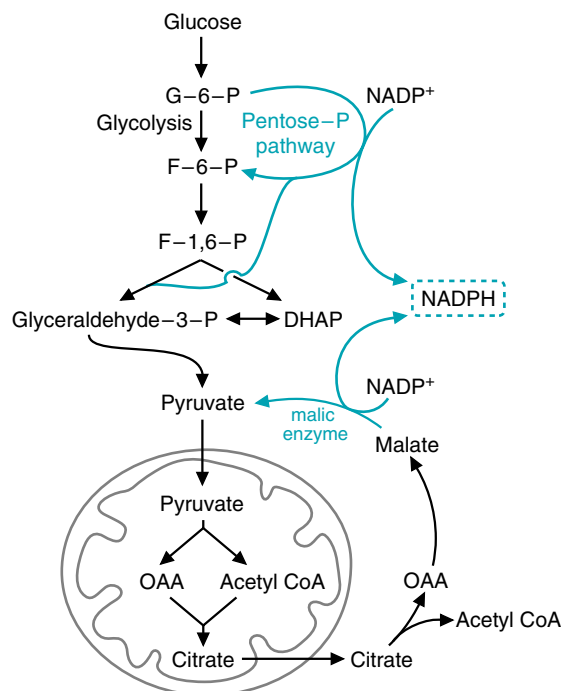


Fig. 33.9. Sources of NADPH for fatty acid synthesis. NADPH is produced by the pentose phosphate pathway and by malic enzyme. OAA = oxaloacetate.

an active form (see Chapter 20). The synthesis of malic enzyme, glucose 6-phosphate dehydrogenase, and citrate lyase is induced by the high insulin/glucagon ratio. The ability of citrate to accumulate, and leave the mitochondrial matrix for the synthesis of fatty acids, is attributable to the allosteric inhibition of isocitrate dehydrogenase by high energy levels within the matrix under these conditions. The concerted regulation of glycolysis and fatty acid synthesis is described in Chapter 36.

B. Conversion of Acetyl CoA to Malonyl CoA

Cytosolic acetyl CoA is converted to malonyl CoA, which serves as the immediate donor of the 2-carbon units that are added to the growing fatty acid chain on the fatty acid synthase complex. To synthesize malonyl CoA, acetyl CoA carboxylase adds a carboxyl group to acetyl CoA in a reaction requiring biotin and adenosine triphosphate (ATP) (Fig. 33.10).

Acetyl CoA carboxylase is the rate-limiting enzyme of fatty acid synthesis. Its activity is regulated by phosphorylation, allosteric modification, and induction/repression of its synthesis (Fig. 33.11). Citrate allosterically activates acetyl CoA carboxylase by causing the individual enzyme molecules (each composed of 4 subunits) to polymerize. Palmitoyl CoA, produced from palmitate (the endproduct of fatty acid synthase activity), inhibits acetyl CoA carboxylase. Phosphorylation by an AMP-dependent protein kinase inhibits the enzyme in the fasting state when energy levels are low. The enzyme is activated by dephosphorylation in the fed state when energy and insulin levels are high. A high insulin/glucagon ratio also results in induction of the synthesis of both acetyl CoA carboxylase and the next enzyme in the pathway, fatty acid synthase.

C. Fatty Acid Synthase Complex

As an overview, fatty acid synthase sequentially adds 2-carbon units from malonyl CoA to the growing fatty acyl chain to form palmitate. After the addition of each 2-carbon unit, the growing chain undergoes two reduction reactions that require NADPH.

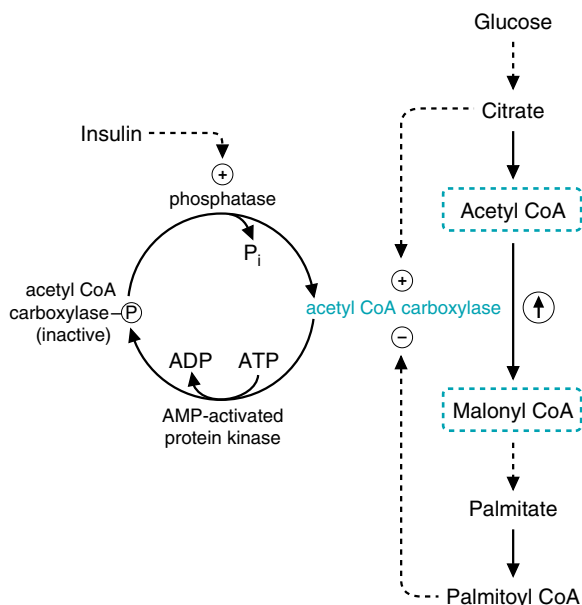


Fig. 33.11. Regulation of acetyl CoA carboxylase. This enzyme is regulated allosterically, both positively and negatively, by phosphorylation (circled P) and dephosphorylation, and by diet-induced induction (circled $\uparrow$). It is active in the dephosphorylated state when citrate causes it to polymerize. Dephosphorylation is catalyzed by an insulin-stimulated phosphatase. Low energy levels, via activation of an AMP-dependent protein kinase, cause the enzyme to be phosphorylated and inactivated. The ultimate product of fatty acid synthesis, palmitate, is converted to its CoA derivative palmitoyl CoA, which inhibits the enzyme. A high-calorie diet increases the rate of transcription of the gene for acetyl CoA carboxylase, whereas a low-calorie diet reduces transcription of this gene.

Fatty acid synthase is a large enzyme composed of two identical dimers, which each have seven catalytic activities and an acyl carrier protein (ACP) segment in a continuous polypeptide chain. The ACP segment contains a phosphopantetheine residue that is derived from the cleavage of coenzyme A (Fig. 33.12). The two dimers associate in a head-to-tail arrangement, so that the phosphopantetheinyl sulfhydryl group on one subunit and a cysteinyl sulfhydryl group on another subunit are closely aligned.

In the initial step of fatty acid synthesis, an acetyl moiety is transferred from acetyl CoA to the ACP phosphopantetheinyl sulfhydryl group of one subunit, and then to the cysteinyl sulfhydryl group of the other subunit. The malonyl moiety from malonyl CoA then attaches to the ACP phosphopantetheinyl sulfhydryl group of the first subunit. The acetyl and malonyl moieties condense, with the release of the malonyl carboxyl group as CO_2 . A 4-carbon α -keto acyl chain is now attached to the ACP phosphopantetheinyl sulfhydryl group (Fig. 33.13).

A series of three reactions reduces the 4-carbon keto group to an alcohol, removes water to form a double bond, and reduces the double bond (Fig. 33.14). NADPH provides the reducing equivalents for these reactions. The net result is that the original acetyl group is elongated by two carbons.

The 4-carbon fatty acyl chain is then transferred to the cysteinyl sulfhydryl group and subsequently condenses with a malonyl group. This sequence of reactions is repeated until the chain is 16 carbons in length. At this point, hydrolysis occurs, and palmitate is released (Fig. 33.15).

Palmitate is elongated and desaturated to produce a series of fatty acids. In the liver, palmitate and other newly synthesized fatty acids are converted to triacylglycerols that are packaged into VLDL for secretion.

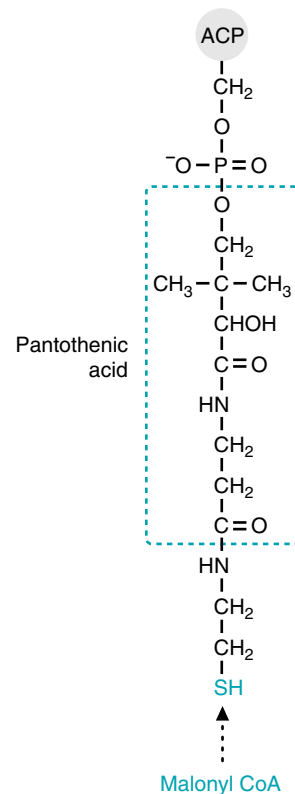


Fig. 33.12. Phosphopantetheinyl residue of the fatty acid synthase complex. The portion derived from the vitamin, pantothenic acid, is indicated. Phosphopantetheine is covalently linked to a serine residue of the acyl carrier protein (ACP) segment of the enzyme. The sulfhydryl group reacts with malonyl CoA to form a thioester.

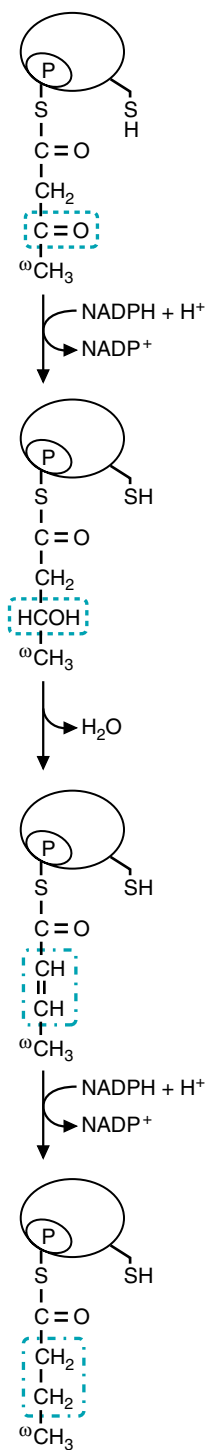


Fig. 33.14. Reduction of a β -ketoacyl group on the fatty acid synthase complex. NADPH is the reducing agent.



Where does the methyl group of the first acetyl CoA that binds to fatty acid synthase appear in palmitate, the final product?

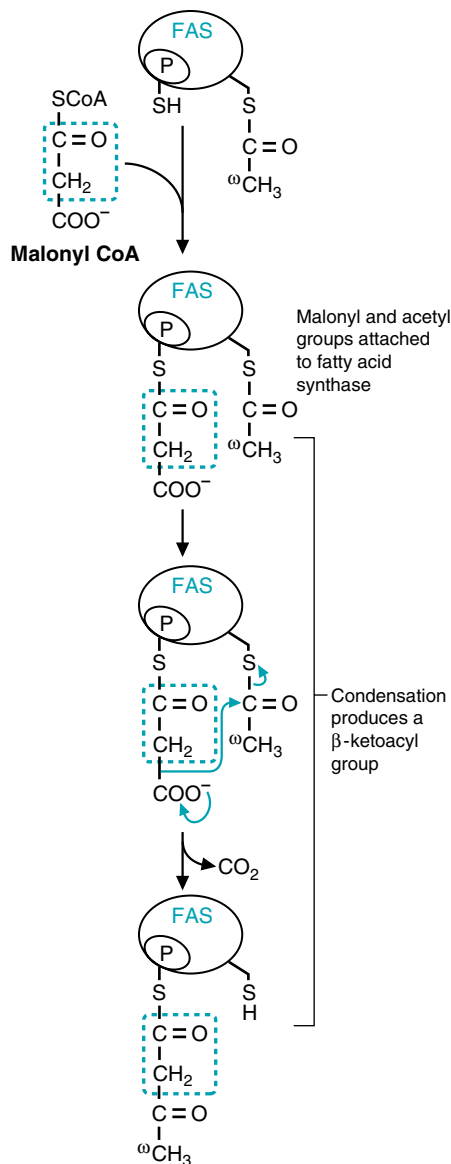


Fig. 33.13. Addition of a 2-carbon unit to an acetyl group on fatty acid synthase. The malonyl group attaches to the phosphopantetheinyl residue (P) of the ACP of the fatty acid synthase. The acetyl group, which is attached to a cysteinyl sulfhydryl group, condenses with the malonyl group. CO_2 is released, and a 3-ketoacyl group is formed. The carbon that eventually forms the ω -methyl group of palmitate is labeled ω .

In the liver, the oxidation of newly synthesized fatty acids back to acetyl CoA via the mitochondrial β -oxidation pathway is prevented by malonyl CoA. Carnitine:palmitoyltransferase I, the enzyme involved in the transport of long-chain fatty acids into mitochondria (see Chapter 23), is inhibited by malonyl CoA (Fig. 33.16). Malonyl CoA levels are elevated when acetyl CoA carboxylase is activated, and, thus, fatty acid oxidation is inhibited while fatty acid synthesis is proceeding. This inhibition prevents the occurrence of a futile cycle.

D. Elongation of Fatty Acids

After synthesis on the fatty acid synthase complex, palmitate is activated, forming palmitoyl CoA. Palmitoyl CoA and other activated long-chain fatty acids can be

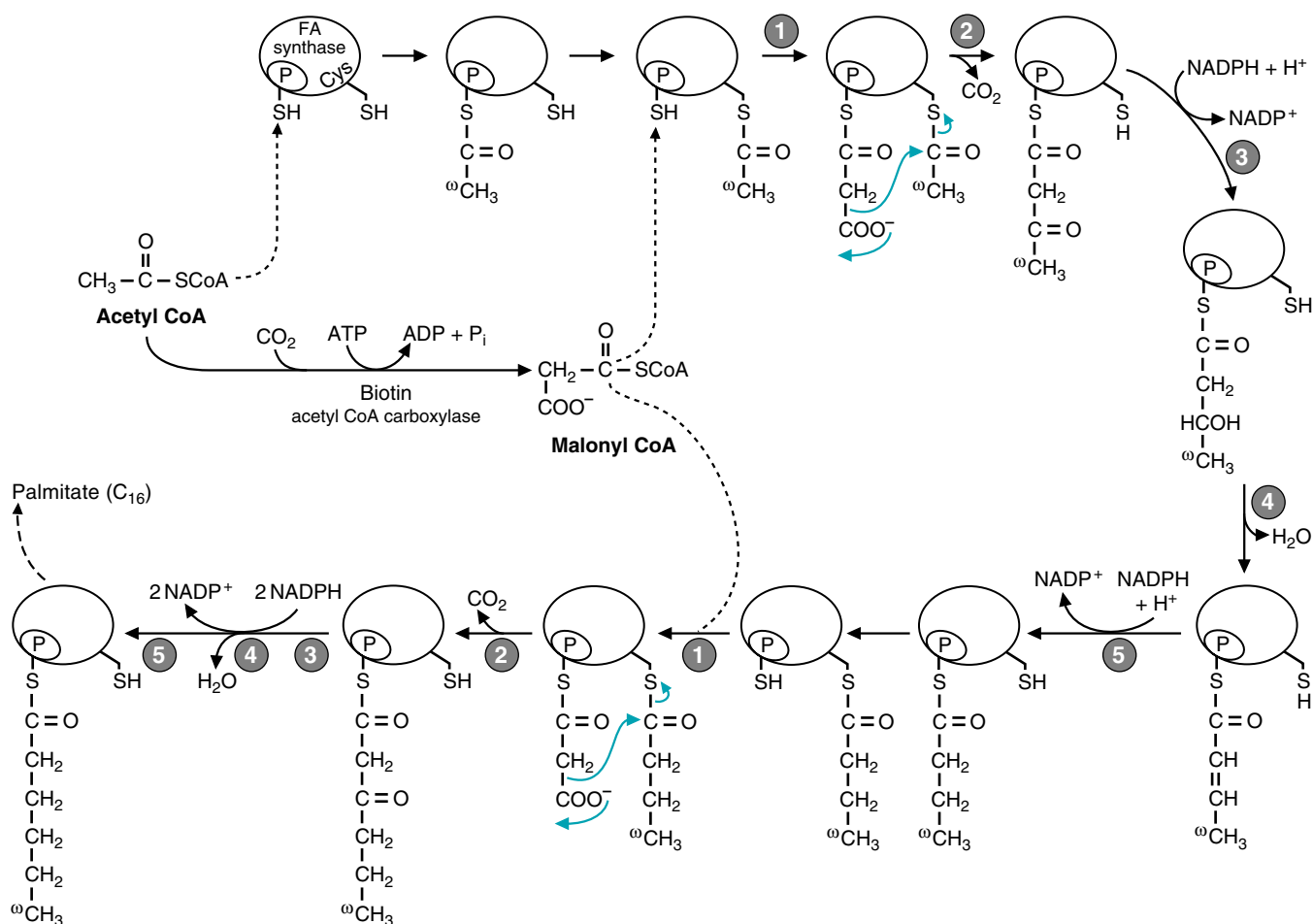


Fig. 33.15. Synthesis of palmitate on the fatty acid synthase complex. Initially, acetyl CoA adds to the synthase. It provides the ω -methyl group of palmitate. Malonyl CoA provides the 2-carbon units that are added to the growing fatty acyl chain. The addition and reduction steps are repeated until palmitate is produced. 1. Transfer of the malonyl group to the phosphopantetheinyl residue. 2. Condensation of the malonyl and fatty acyl groups. 3. Reduction of the β -ketoacyl group. 4. Dehydration. 5. Reduction of the double bond. P = a phosphopantetheinyl group attached to the fatty acid synthase complex; Cys-SH = a cysteinyl residue.

elongated, two carbons at a time, by a series of reactions that occur in the endoplasmic reticulum (Fig. 33.17). Malonyl CoA serves as the donor of the 2-carbon units, and NADPH provides the reducing equivalents. The series of elongation reactions resemble those of fatty acid synthesis except that the fatty acyl chain is attached to coenzyme A rather than to the phosphopantetheinyl residue of an ACP. The major elongation reaction that occurs in the body involves the conversion of palmitoyl CoA (C16) to stearoyl CoA (C18). Very-long-chain fatty acids (C22 to C24) are also produced, particularly in the brain.

E. Desaturation of Fatty Acids

Desaturation of fatty acids involves a process that requires molecular oxygen (O_2), NADH, and cytochrome b_5 . The reaction, which occurs in the endoplasmic reticulum, results in the oxidation of both the fatty acid and NADH (Fig. 33.18). The most common desaturation reactions involve the placement of a double bond between carbons 9 and 10 in the conversion of palmitic acid to palmitoleic acid (16:1, Δ^9) and the conversion of stearic acid to oleic acid (18:1, Δ^9). Other positions that can be desaturated in humans include carbons 4, 5, and 6.



The methyl group of acetyl CoA becomes the ω -carbon (the terminal methyl group) of palmitate. Each new 2-carbon unit is added to the carboxyl end of the growing fatty acyl chain (see Fig. 33.13).

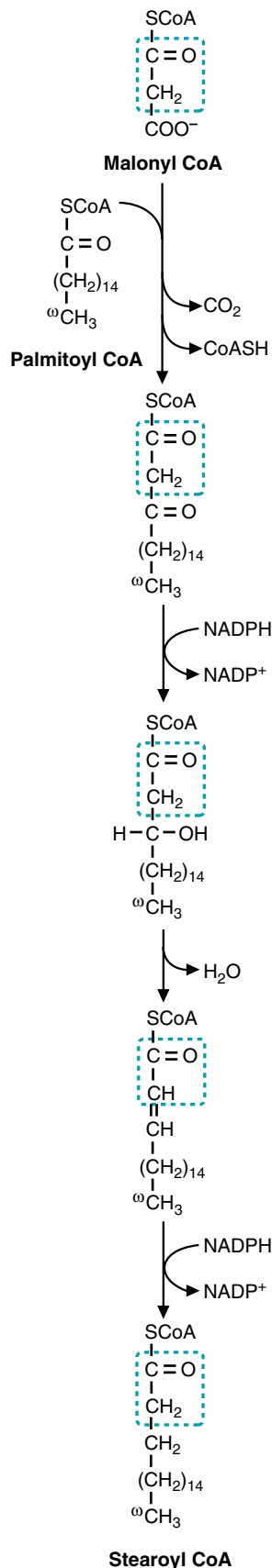


Fig. 33.17. Elongation of long-chain fatty acids in the endoplasmic reticulum.

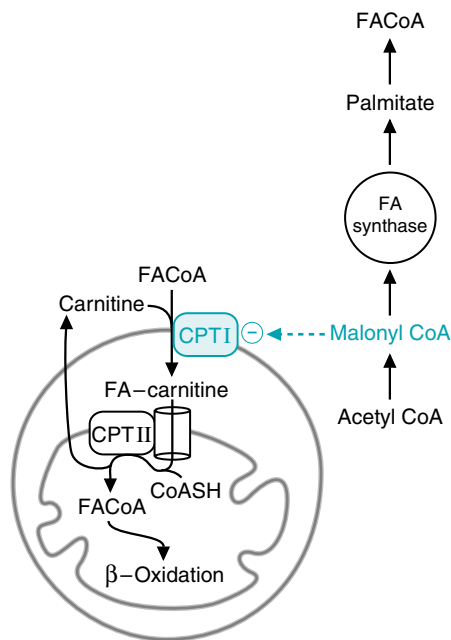


Fig. 33.16. Inhibition of carnitine:palmitoyltransferase (CPTI, also called carnitine:acyltransferase I) by malonyl CoA. During fatty acid synthesis, malonyl CoA levels are high. This compound inhibits CPTI, which is involved in the transport of long-chain fatty acids into mitochondria for β -oxidation. This mechanism prevents newly synthesized fatty acids from undergoing immediate oxidation.

Polyunsaturated fatty acids with double bonds three carbons from the methyl end (ω 3 fatty acids) and six carbons from the methyl end (ω 6 fatty acids) are required for the synthesis of eicosanoids (see Chapter 35). Because humans cannot synthesize these fatty acids *de novo* (i.e., from glucose via palmitate), they must be present in the diet or the diet must contain other fatty acids that can be converted to these fatty acids. We obtain ω 6 and ω 3 polyunsaturated fatty acids mainly from dietary plant oils that contain the ω 6 fatty acid linoleic acid (18:2, $\Delta^{9,12}$) and the ω 3 fatty acid α -linolenic acid (18:3, $\Delta^{9,12,15}$). In the body, linoleic acid can be converted by elongation and desaturation reactions to arachidonic acid (20:4, $\Delta^{5,8,11,14}$), which is used for the synthesis of the major class of human prostaglandins and other eicosanoids (Fig. 33.19). Elongation and desaturation of α -linolenic acid produces eicosapentaenoic acid (EPA; 20:5, $\Delta^{5,8,11,14,17}$), which is the precursor of a different class of eicosanoids (see Chapter 35).



Plants are able to introduce double bonds into fatty acids in the region between C10 and the ω -end and therefore can synthesize ω 3 and ω 6 polyunsaturated fatty acids. Fish oils also contain ω 3 and ω 6 fatty acids, particularly eicosapentaenoic acid (EPA; ω 3, 20:5, $\Delta^5, 8, 11, 14, 17$) and docosahexaenoic acid (DHA; ω 3, 22:6, $\Delta^4, 7, 10, 13, 16, 19$). The fish obtain these fatty acids by eating phytoplankton (plants that float in water).

Arachidonic acid is listed in some textbooks as an essential fatty acid. Although it is an ω 6 fatty acid, it is not essential in the diet if linoleic acid is present because arachidonic acid can be synthesized from dietary linoleic acid (see Fig. 33.19).



The essential fatty acid linoleic acid is required in the diet for at least three reasons: (a) It serves as a precursor of arachidonic acid from which eicosanoids are produced. (b) It covalently binds another fatty acid attached to cerebroside in the skin, forming an unusual lipid (acylglucosylceramide) that helps to make the skin impermeable to water. This function of linoleic acid may help to explain the red, scaly dermatitis and other skin problems associated with a dietary deficiency of essential fatty acids. (c) It is the precursor of C22:6 ω 3, an important neuronal fatty acid.

The other essential fatty acid, α -linolenic acid (18:3, $\Delta^9, 12, 15$), also forms eicosanoids.

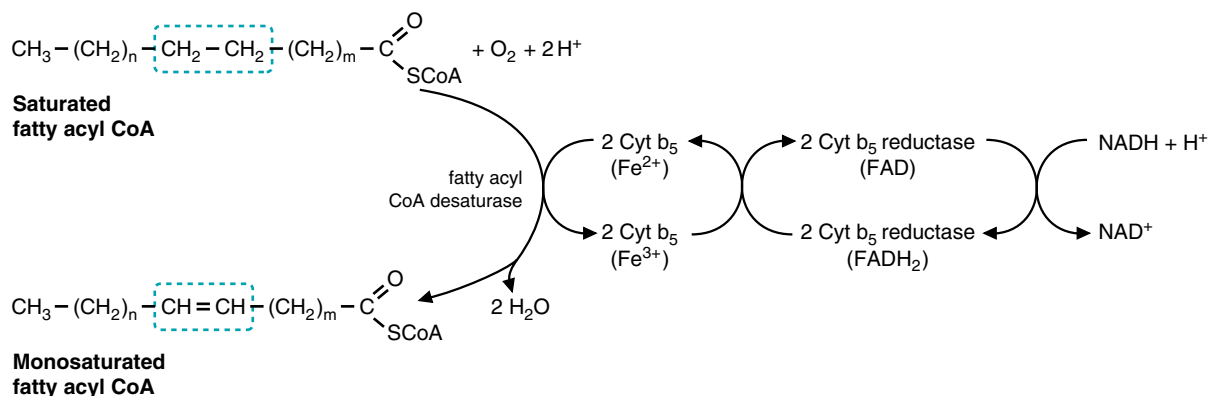


Fig. 33.18. Desaturation of fatty acids. The process occurs in the endoplasmic reticulum and uses molecular oxygen. Both the fatty acid and NADH are oxidized. Human desaturases cannot introduce double bonds between carbon 9 and the methyl end. Therefore, m is equal to or less than 7.

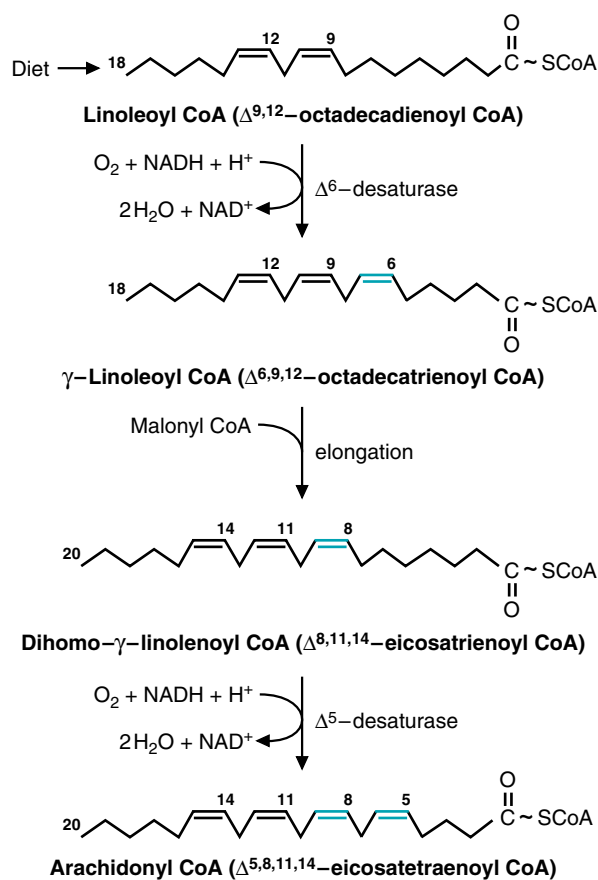


Fig. 33.19. Conversion of linoleic acid to arachidonic acid. Dietary linoleic acid (as linoleoyl CoA) is desaturated at carbon 6, elongated by 2 carbons, and then desaturated at carbon 5 to produce arachidonyl CoA.

II. SYNTHESIS OF TRIACYLGLYCEROLS AND VLDL PARTICLES

In liver and adipose tissue, triacylglycerols are produced by a pathway containing a phosphatidic acid intermediate (Fig. 33.20). Phosphatidic acid is also the precursor of the glycerolipids found in cell membranes and the blood lipoproteins.

The sources of glycerol 3-phosphate, which provides the glycerol moiety for triacylglycerol synthesis, differ in liver and adipose tissue. In liver, glycerol 3-phosphate



Recent experiments have shown functional glycerol kinase activity in muscle cells. The significance of this finding is under investigation, but it may indicate that muscle has a greater capacity for fatty acid synthesis than previously believed.

is produced from the phosphorylation of glycerol by glycerol kinase or from the reduction of dihydroxyacetone phosphate derived from glycolysis. Adipose tissue lacks glycerol kinase and can produce glycerol 3-phosphate only from glucose via dihydroxyacetone phosphate. Thus, adipose tissue can store fatty acids only when glycolysis is activated, i.e., in the fed state.

In both adipose tissue and liver, triacylglycerols are produced by a pathway in which glycerol 3-phosphate reacts with fatty acyl CoA to form phosphatidic acid. Dephosphorylation of phosphatidic acid produces diacylglycerol. Another fatty acyl CoA reacts with the diacylglycerol to form a triacylglycerol (see Fig. 33.20).

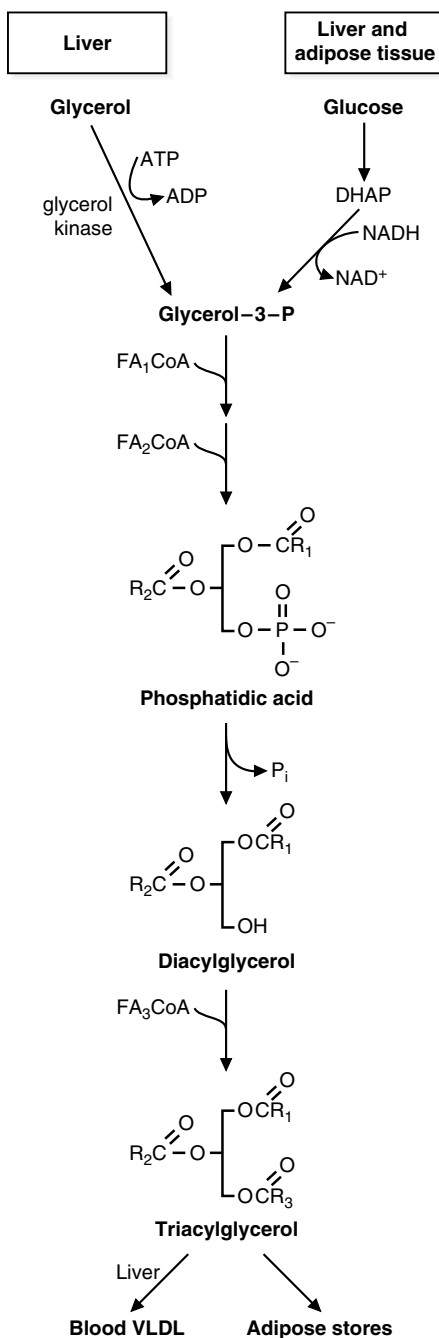
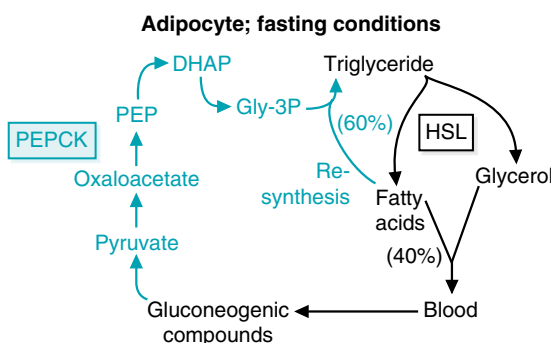


Fig. 33.20. Synthesis of triacylglycerol in liver and adipose tissue. Glycerol 3-phosphate is produced from glucose in both tissues. It is also produced from glycerol in liver, but not in adipose tissue, which lacks glycerol kinase. The steps from glycerol 3-phosphate are the same in the two tissues. FA = fatty acyl group.



Adipose tissue also undergoes glyceroneogenesis, the process of synthesizing glycerol from gluconeogenic precursors in the blood, such as alanine, aspartate, and malate. Glyceroneogenesis occurs primarily in the fasting state and is dependent on the induction of cytoplasmic PEPCK in the adipocyte. The re-synthesis of triglycerides by adipose tissue during fasting modulates the release of fatty acids in the circulation. Mice that have been engineered to not express PEPCK in adipose tissue display reduced levels of triglyceride in their adipocytes; mice that overproduce adipocyte PEPCK were obese. Thus, although activation of hormone-sensitive lipase during fasting results in the release of fatty acids from adipocytes, the release is carefully modulated through glyceroneogenesis and re-synthesis of triglycerides.



The triacylglycerol, which is produced in the smooth endoplasmic reticulum of the liver, is packaged with cholesterol, phospholipids, and proteins (synthesized in the rough endoplasmic reticulum) to form VLDL (Fig. 33.21). The microsomal triglyceride transfer protein (MTP), which is required for chylomicron assembly, is also required for VLDL assembly. The major protein of VLDL is apoB-100. There is one long apoB-100 molecule wound through the surface of each VLDL particle. ApoB-100 is encoded by the same gene as the apoB-48 of chylomicrons, but is a longer protein (see Fig. 32.11). In intestinal cells, RNA editing produces a smaller mRNA and, thus, a shorter protein, apoB-48.

VLDL is processed in the Golgi complex and secreted into the blood by the liver (Figs. 33.22 and 33.23). The fatty acid residues of the triacylglycerols ultimately are stored in the triacylglycerols of adipose cells. Note that, in comparison to chylomicrons (see Chapter 32), VLDL particles are more dense, as they contain a lower percentage of triglyceride than do the chylomicrons. Similar to chylomicrons, VLDL particles are first synthesized in a nascent form, and on entering the circulation they acquire apoproteins CII and E from HDL particles to become mature VLDL particles.



The fact that a number of different abnormal lipoprotein profiles were found in **Cora Nari** and her siblings, and that each had evidence of coronary artery disease, suggests that Cora has familial combined hyperlipidemia (FCH). This diagnostic impression is further supported by the finding that Cora's profile of lipid abnormalities appeared to change somewhat from one determination to the next, a characteristic of FCH. This hereditary disorder of lipid metabolism is believed to be quite common, with an estimated prevalence of about 1 per 100 population.

The mechanisms for FCH are incompletely understood but may involve a genetically determined increase in the production of apoprotein B-100. As a result, packaging of VLDL is increased, and blood VLDL levels may be elevated. Depending on the efficiency of lipolysis of VLDL by LPL, VLDL levels may be normal and LDL levels may be elevated, or both VLDL and LDL levels may be high. In addition, the phenotypic expression of FCH in any given family member may be determined by the degree of associated obesity, the diet, the use of specific drugs, or other factors that change over time.



Abetalipoproteinemia, which is due to a lack of MTP (microsomal triglyceride transfer protein; see Chapter 32) activity, results in an inability to assemble both chylomicrons in the intestine and VLDL particles in the liver.



Why do some alcoholics have high VLDL levels?

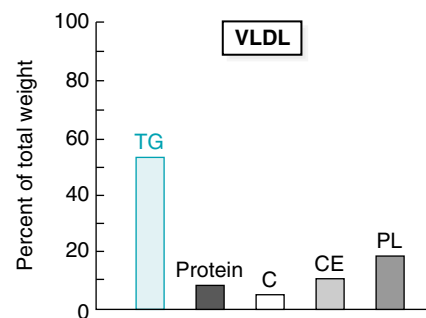


Fig. 33.21. Composition of a typical VLDL particle. The major component is triacylglycerol (TG). C = cholesterol; CE = cholesterol ester; PL = phospholipid.

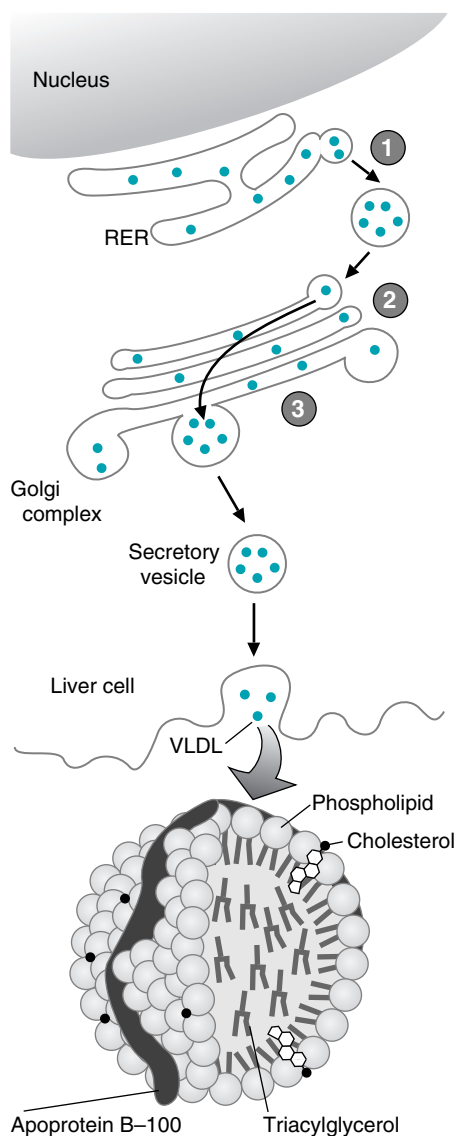


Fig. 33.23. Synthesis, processing, and secretion of VLDL. Proteins synthesized on the rough endoplasmic reticulum (RER) are packaged with triacylglycerols in the ER and Golgi complex to form VLDL. VLDL are transported to the cell membrane in secretory vesicles and secreted by endocytosis. Blue dots represent VLDL particles. An enlarged VLDL particle is depicted at the bottom of the figure.

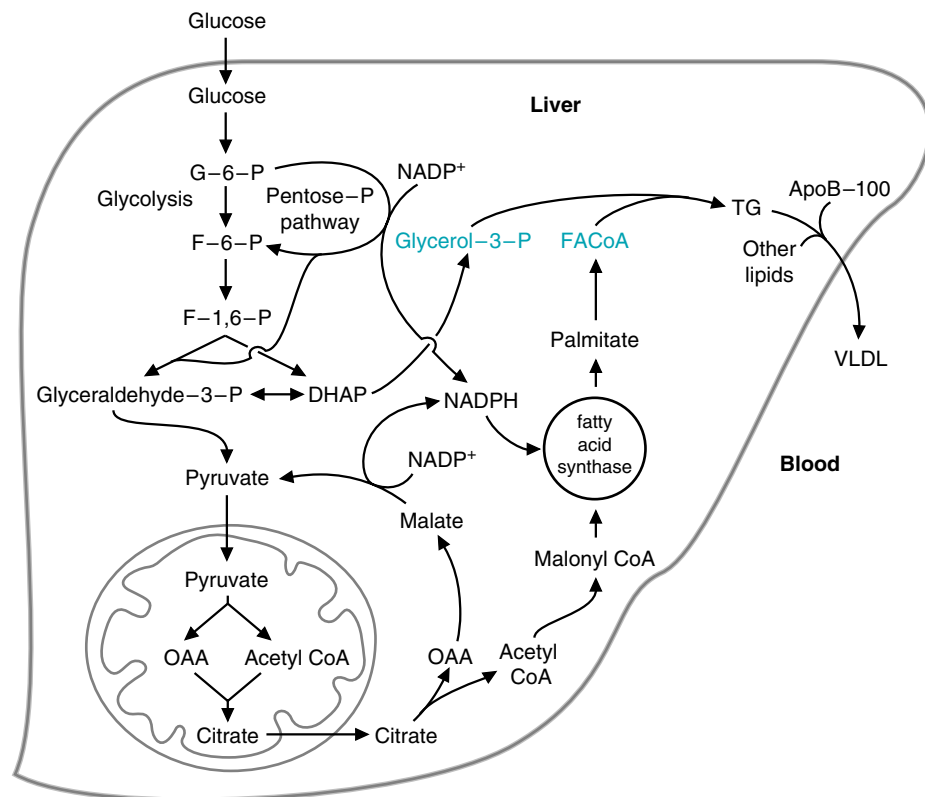


Fig. 33.22. Synthesis of VLDL from glucose in the liver. G-6-P = glucose 6-phosphate; F-6-P = fructose 6-phosphate; F-1,6-BP = fructose 1,6-bisphosphate; FA = fatty acyl group; TG = triacylglycerol.

III. FATE OF VLDL TRIACYLGLYCEROL

Lipoprotein lipase (LPL), which is attached to the basement membrane proteoglycans of capillary endothelial cells, cleaves the triacylglycerols in both VLDL and chylomicrons, forming fatty acids and glycerol. The C-II apoprotein, which these lipoproteins obtain from HDL, activates LPL. The low K_m of the muscle LPL isozyme permits muscle to use the fatty acids of chylomicrons and VLDL as a source of fuel even when the blood concentration of these lipoproteins is very low. The isozyme in adipose tissue has a high K_m and is most active after a meal, when blood levels of chylomicrons and VLDL are elevated. The fate of the VLDL particle after triglyceride has been removed by LPL is the generation of an IDL particle (intermediate-density lipoprotein), which can further lose triglyceride to become an LDL particle (low-density lipoprotein). The fate of the IDL and LDL particles is discussed in Chapter 34.



Fatty acids for VLDL synthesis in the liver may be obtained from the blood or they may be synthesized from glucose. In a healthy individual, the major source of the fatty acids of VLDL triacylglycerol is excess dietary glucose. In individuals with diabetes mellitus, fatty acids mobilized from adipose triacylglycerols in excess of the oxidative capacity of tissues are a major source of the fatty acids re-esterified in liver to VLDL triacylglycerol. These individuals frequently have elevated levels of blood triacylglycerols.



In alcoholism, NADH levels in the liver are elevated (see Chapter 25). High levels of NADH inhibit the oxidation of fatty acids. Therefore, fatty acids, mobilized from adipose tissue, are re-esterified to glycerol in the liver, forming triacylglycerols, which are packaged into VLDL and secreted into the blood. Elevated VLDL is frequently associated with chronic alcoholism. As alcohol-induced liver disease progresses, the ability to secrete the triacylglycerols is diminished, resulting in a fatty liver.

IV. STORAGE OF TRIACYLGLYCEROLS IN ADIPOSE TISSUE

After a meal, the triacylglycerol stores of adipose tissue increase (Fig. 33.24). Adipose cells synthesize LPL and secrete it into the capillaries of adipose tissue when the insulin/glucagon ratio is elevated. This enzyme digests the triacylglycerols of both chylomicrons and VLDL. The fatty acids enter adipose cells and are activated, forming fatty acyl CoA, which reacts with glycerol 3-phosphate to form triacylglycerol by the same pathway used in the liver (see Fig. 33.20). Because adipose tissue lacks glycerol kinase and cannot use the glycerol produced by LPL, the glycerol travels through the blood to the liver, which uses it for the synthesis of triacylglycerol. In adipose cells, glycerol 3-phosphate is derived from glucose.

In addition to stimulating the synthesis and release of LPL, insulin stimulates glucose metabolism in adipose cells. Insulin leads to the activation of the glycolytic enzyme phosphofructokinase-1 by an activation of PFK-2, which increases fructose 2,6-bisphosphate levels. Insulin also stimulates the dephosphorylation of pyruvate dehydrogenase, so that the pyruvate produced by glycolysis can be oxidized in the TCA cycle. Furthermore, insulin stimulates the conversion of glucose to fatty acids in adipose cells, although the liver is the major site of fatty acid synthesis in humans.

V. RELEASE OF FATTY ACIDS FROM ADIPOSE TRIACYLGLYCEROLS

During fasting, the decrease of insulin and the increase of glucagon cause cAMP levels to rise in adipose cells, stimulating lipolysis (Fig. 33.25). Protein kinase A phosphorylates hormone-sensitive lipase to produce a more active form of the enzyme. Hormone-sensitive lipase, also known as adipose triacylglycerol lipase, cleaves a fatty acid from a triacylglycerol. Subsequently, other lipases complete the process of lipolysis, and fatty acids and glycerol are released into the blood. Simultaneously, to regulate the amount of fatty acids released into circulation, triglyceride synthesis occurs along with glyceroneogenesis.



In some cases of hyperlipidemia, LPL is defective. If a blood lipid profile is performed on patients with an LPL deficiency, which lipids would be elevated?



Because the fatty acids of adipose triacylglycerols come both from chylomicrons and VLDL, we produce our major fat stores both from dietary fat (which produces chylomicrons) and dietary sugar (which produces VLDL). An excess of dietary protein also can be used to produce the fatty acids for VLDL synthesis.

The dietician carefully explained to **Percy Veere** that we can become fat from eating excess fat, excess sugar, or excess protein.

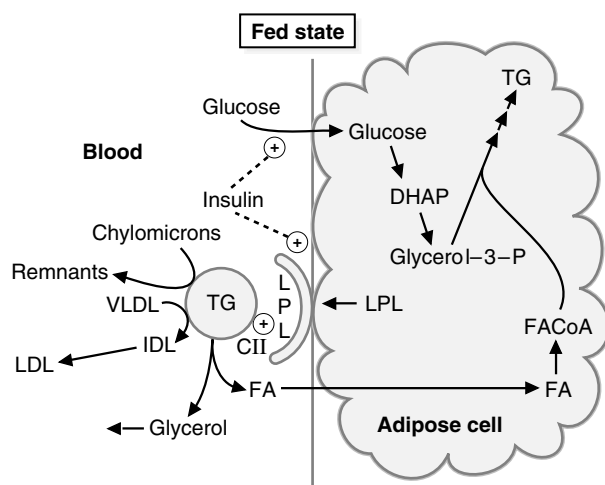


Fig. 33.24. Conversion of the fatty acid (FA) from the triacylglycerols (TG) of chylomicrons and VLDL to the TG stored in adipose cells. Note that insulin stimulates both the transport of glucose into adipose cells and the secretion of LPL from the cells. Glucose provides the glycerol 3-phosphate for TG synthesis. Insulin also stimulates the synthesis and secretion of lipoprotein lipase (LPL). Apoprotein C-II activates LPL.

A: Individuals with a defective LPL have high blood triacylglycerol levels. Their levels of chylomicrons and VLDL (which contain large amounts of triacylglycerols) are elevated because they are not digested at the normal rate by LPL.

LPL can be dissociated from capillary walls by treatment with heparin (a glycosaminoglycan). Measurements can be made on blood after heparin treatment to determine whether LPL levels are abnormal.

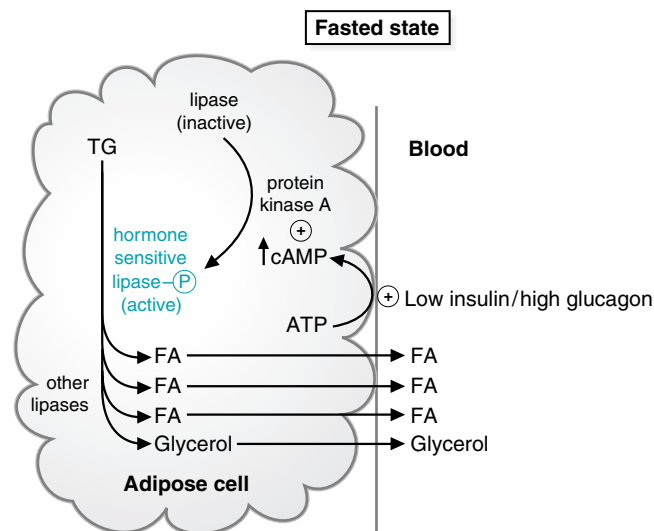


Fig. 33.25. Mobilization of adipose triacylglycerol (TG). In the fasted state, when insulin levels are low and glucagon is elevated, intracellular cAMP increases and activates protein kinase A, which phosphorylates hormone-sensitive lipase (HSL). Phosphorylated HSL is active and initiates the breakdown of adipose TG. Recall, however, that re-esterification of fatty acids does occur, along with glyceroneogenesis, in the fasted state. HSL is also called triacylglycerol lipase. FA = fatty acid.

The fatty acids, which travel in the blood complexed with albumin, enter cells of muscle and other tissues, where they are oxidized to CO_2 and water to produce energy. During prolonged fasting, acetyl CoA produced by β -oxidation of fatty acids in the liver is converted to ketone bodies, which are released into the blood. The glycerol derived from lipolysis in adipose cells is used by the liver during fasting as a source of carbon for gluconeogenesis.

VI. METABOLISM OF GLYCEROPHOSPHOLIPIDS AND SPHINGOLIPIDS

Fatty acids, obtained from the diet or synthesized from glucose, are the precursors of glycerophospholipids and of sphingolipids (Fig. 33.26). These lipids are major components of cellular membranes. Glycerophospholipids are also components of blood lipoproteins, bile, and lung surfactant. They are the source of the polyunsaturated fatty acids, particularly arachidonic acid, that serve as precursors of the eicosanoids (e.g., prostaglandins, thromboxanes, leukotrienes; see Chapter 35). Ether glycerophospholipids differ from other glycerophospholipids in that the alkyl or alkenyl chain (an alkyl chain with a double bond) is joined to carbon 1 of the glycerol moiety by an ether rather than an ester bond. Examples of ether lipids are the plasmalogens and platelet activating factor. Sphingolipids are particularly important in forming the myelin sheath surrounding nerves in the central nervous system, and in signal transduction.

In glycerolipids and ether glycerolipids, glycerol serves as the backbone to which fatty acids and other substituents are attached. Sphingosine, derived from serine, provides the backbone for sphingolipids.

A. Synthesis of Phospholipids Containing Glycerol

1. GLYCEROPHOSPHOLIPIDS

The initial steps in the synthesis of glycerophospholipids are similar to those of triacylglycerol synthesis. Glycerol 3-phosphate reacts with fatty acyl CoA to form

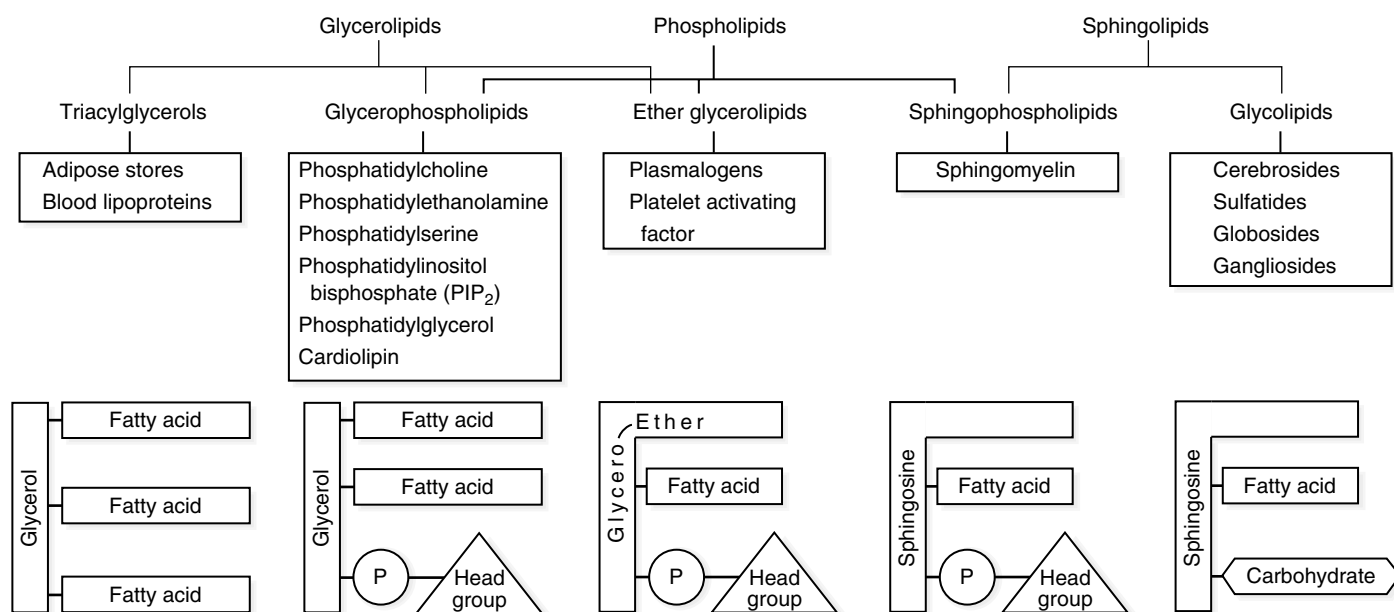


Fig. 33.26. Types of glycerolipids and sphingolipids. Glycerolipids contain glycerol, and sphingolipids contain sphingosine. The category of phospholipids overlaps both glycerolipids and sphingolipids. The head groups include choline, ethanolamine, serine, inositol, glycerol, and phosphatidylglycerol. The carbohydrates are monosaccharides (which may be sulfated), oligosaccharides, and oligosaccharides with branches of *N*-acetylneuraminic acid. P = phosphate.

phosphatidic acid. Two different mechanisms are then used to add a head group to the molecule (Fig. 33.27). A head group is a chemical group, such as choline or serine, attached to carbon 3 of a glycerol moiety that contains hydrophobic groups, usually fatty acids, at positions 1 and 2. Head groups are hydrophilic, either charged or polar.

In the first mechanism, phosphatidic acid is cleaved by a phosphatase to form diacylglycerol (DAG). DAG then reacts with an activated head group. In the synthesis of phosphatidylcholine, the head group choline is activated by combining with CTP to form CDP-choline (Fig. 33.28). Phosphocholine is then transferred to carbon 3 of DAG, and CMP is released. Phosphatidylethanolamine is produced by a similar reaction involving CDP-ethanolamine.

Various types of interconversions occur among these phospholipids (see Fig. 33.28). Phosphatidylserine is produced by a reaction in which the ethanolamine moiety of

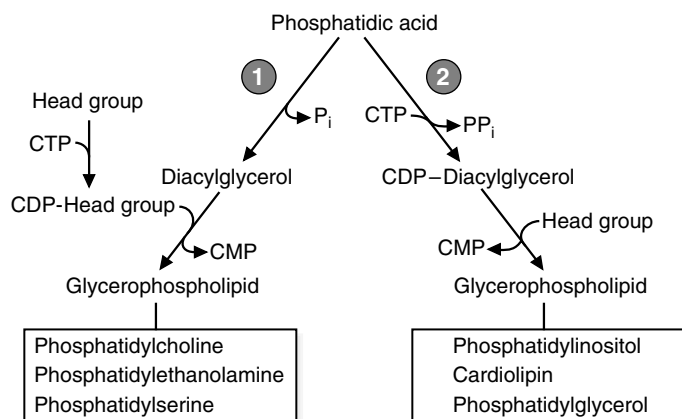


Fig. 33.27. Strategies for addition of the head group to form glycerophospholipids. In both cases, CTP is used to drive the reaction.

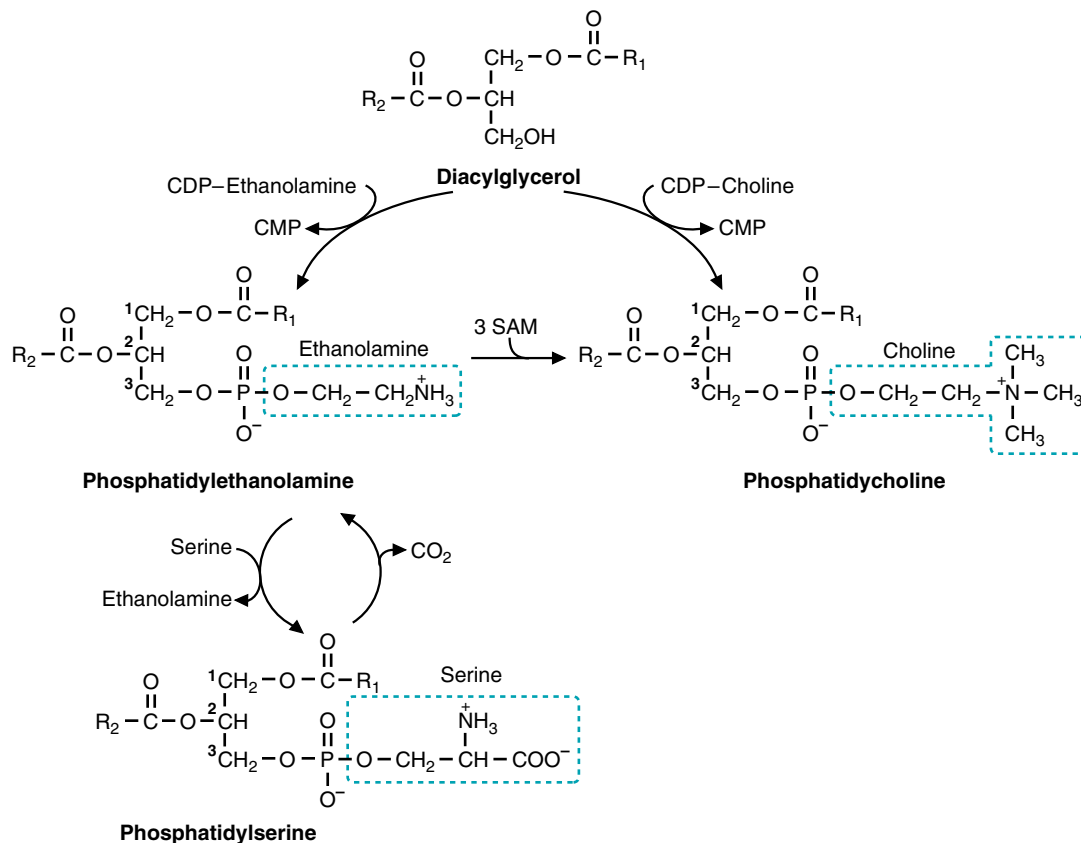


Fig. 33.28. Synthesis of phosphatidylcholine, phosphatidylethanolamine, and phosphatidylserine. The multiple pathways reflect the importance of phospholipids in membrane structure. For example, phosphatidylcholine (PC) can be synthesized from dietary choline when it is available. If choline is not available, PC can be made from dietary carbohydrate, although the amount synthesized is inadequate to prevent choline deficiency. SAM is S-adenosylmethionine, a methyl group donor for many biochemical reactions (see Chapter 40).



Phosphatidylcholine (lecithin) is not required in the diet because it can be synthesized in the body. The components of phosphatidylcholine (including choline) all can be produced, as shown in Figure 33.28. A pathway for de novo choline synthesis from glucose exists, but the rate of synthesis is inadequate to provide for the necessary amounts of choline. Thus, choline has been classified as an essential nutrient, with an AI (adequate intake) of 425 mg/day in females and 550 mg/day in males.

Because choline is widely distributed in the food supply, primarily in phosphatidylcholine (lecithin), deficiencies have not been observed in humans on a normal diet. Deficiencies may occur, however, in patients on total parental nutrition (TPN), i.e., supported solely by intravenous feeding. The fatty livers that have been observed in these patients probably result from a decreased ability to synthesize phospholipids for VLDL formation.

phosphatidylethanolamine is exchanged for serine. Phosphatidylserine can be converted back to phosphatidylethanolamine by a decarboxylation reaction. Phosphatidylethanolamine can be methylated to form phosphatidylcholine (see Chapter 40).

In the second mechanism for the synthesis of glycerolipids, phosphatidic acid reacts with CTP to form CDP-diacylglycerol (Fig. 33.29). This compound can react with phosphatidylglycerol (which itself is formed from the condensation of CDP-diacylglycerol and glycerol 3-phosphate) to produce cardiolipin or with inositol to produce phosphatidylinositol. Cardiolipin is a component of the inner mitochondrial membrane. Phosphatidylinositol can be phosphorylated to form phosphatidylinositol 4,5-bisphosphate (PIP₂), which is a component of cell membranes. In response to signals such as the binding of hormones to membrane receptors, PIP₂ can be cleaved to form the second messengers diacylglycerol and inositol triphosphate (see Chapter 11).

2. ETHER GLYCEROLIPIDS

The ether glycerolipids are synthesized from the glycolytic intermediate dihydroxyacetone phosphate (DHAP). A fatty acyl CoA reacts with carbon 1 of DHAP, forming an ester (Fig. 33.30). This fatty acyl group is exchanged for a fatty alcohol, produced by reduction of a fatty acid. Thus, the ether linkage is formed. Then the keto group on carbon 2 of the DHAP moiety is reduced and esterified to a fatty acid. Addition of the head group proceeds by a series of reactions analogous to those for synthesis of phosphatidylcholine. Formation of a double bond between carbons 1 and 2 of the alkyl group produces a plasmalogen. Ethanolamine plasmalogen is found in

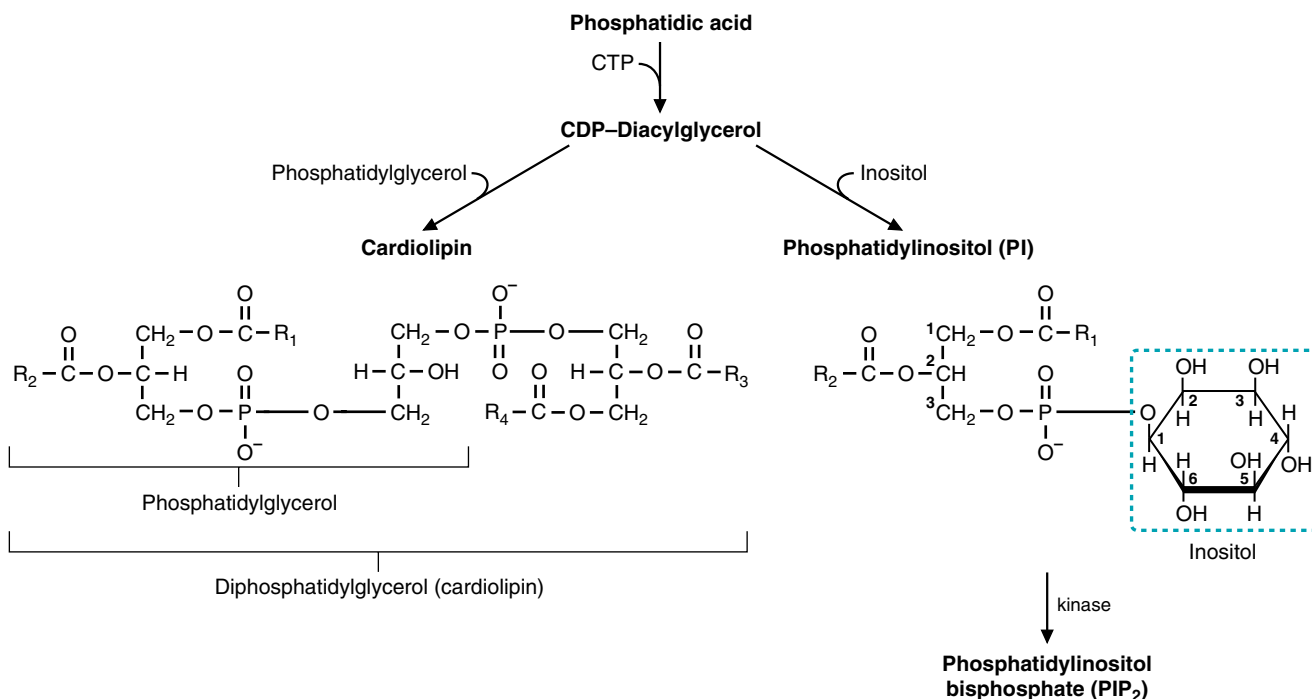
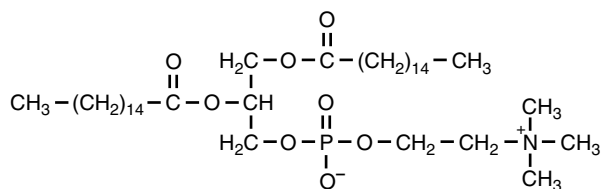


Fig. 33.29. Synthesis of cardiolipin and phosphatidylinositol.

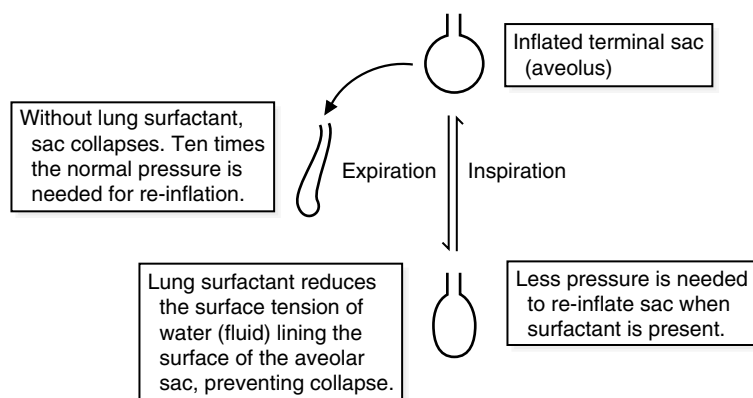


The respiratory distress syndrome (RDS) of a premature infant such as **Colleen Lakker** is, in part, related to a deficiency in the synthesis of a substance known as lung surfactant. The major constituents of surfactant are dipalmitoylphosphatidylcholine, phosphatidylglycerol, apoproteins (surfactant proteins: Sp-A,B,C), and cholesterol.



**Dipalmitoylphosphatidylcholine,
the major component of
lung surfactant**

These components of lung surfactant normally contribute to a reduction in the surface tension within the air spaces (alveoli) of the lung, preventing their collapse. The premature infant has not yet begun to produce adequate amounts of lung surfactant.



The effect of lung surfactant

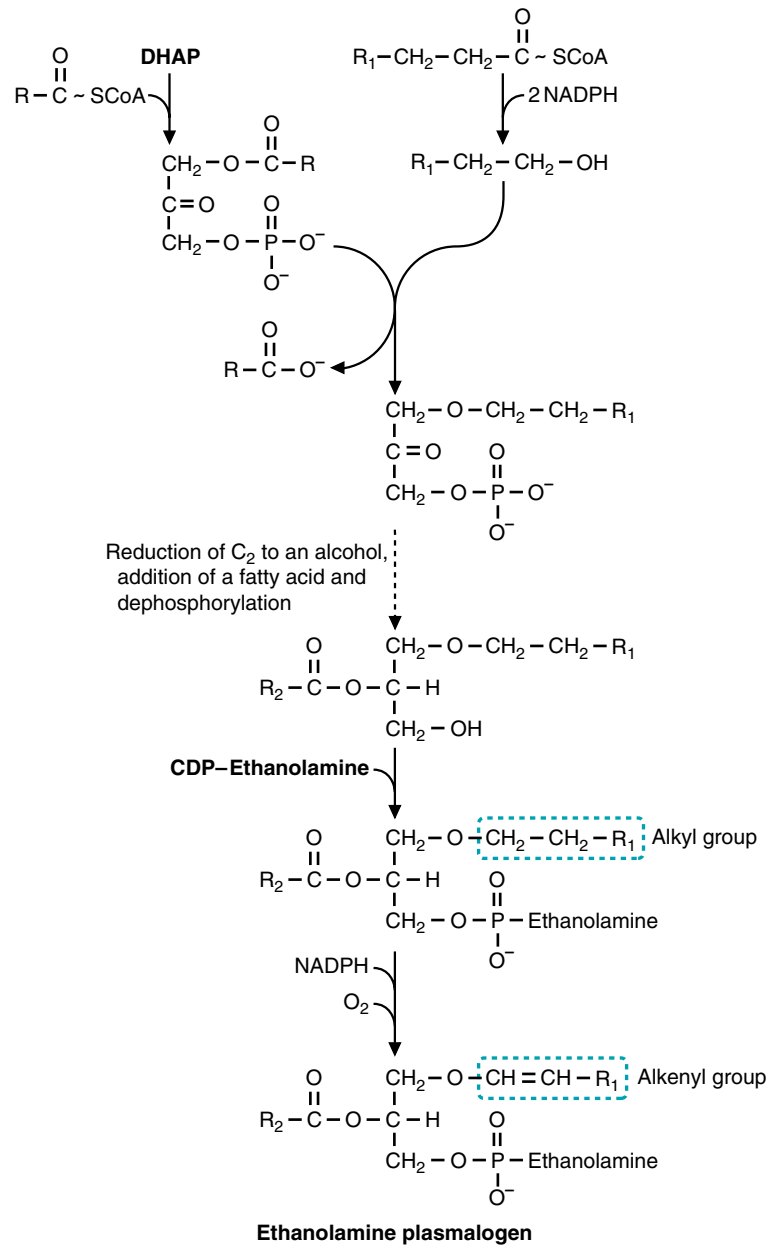


Fig. 33.30. Synthesis of a plasmalogen.



Phospholipase A₂ provides the major repair mechanism for membrane lipids damaged by oxidative free radical reactions. Arachidonic acid, which is a polyunsaturated fatty acid, can be peroxidatively cleaved in free radical reactions to malondialdehyde and other products. Phospholipase A₂ recognizes the distortion of membrane structure caused by the partially degraded fatty acid and removes it. Acyltransferases then add back a new arachidonic acid molecule.

myelin and choline plasmalogen in heart muscle. Platelet-activating factor (PAF) is similar to choline plasmalogen except that an acetyl group replaces the fatty acyl group at carbon 2 of the glycerol moiety, and the alkyl group on carbon 1 is saturated. PAF is released from phagocytic blood cells in response to various stimuli. It causes platelet aggregation, edema, and hypotension, and it is involved in the allergic response. Plasmalogen synthesis occurs within peroxisomes, and, in individuals with Zellweger's syndrome (a defect in peroxisome biogenesis), plasmalogen synthesis is compromised. If severe enough, this syndrome leads to death at an early age.

B. Degradation of Glycerophospholipids

Phospholipases located in cell membranes or in lysosomes degrade glycerophospholipids. Phospholipase A₁ removes the fatty acyl group on carbon 1 of the glycerol moiety, and phospholipase A₂ removes the fatty acid on carbon 2 (Fig. 33.31). The C2 fatty acid in cell membrane phospholipids is usually an unsaturated fatty acid,

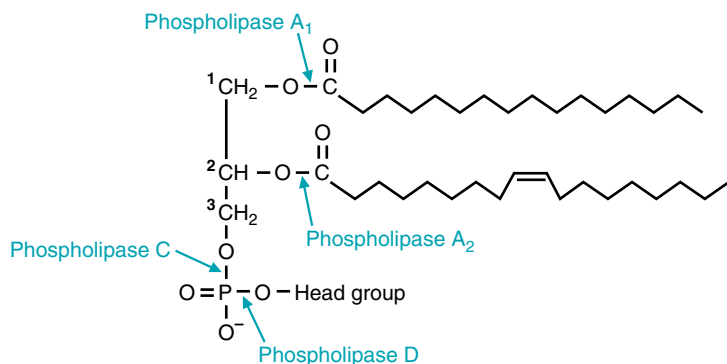


Fig. 33.31. Bonds cleaved by phospholipases.

which is frequently arachidonic acid. It is removed in response to signals for the synthesis of eicosanoids. The bond joining carbon 3 of the glycerol moiety to phosphate is cleaved by phospholipase C. Hormonal stimuli activate phospholipase C, which hydrolyzes PIP_2 to produce the second messengers DAG and inositol triphosphate (IP_3). The bond between the phosphate and the head group is cleaved by phospholipase D, producing phosphatidic acid and the free alcohol of the head group.

C. Sphingolipids

Sphingolipids serve in intercellular communication and as the antigenic determinants of the ABO blood group. Some are used as receptors by viruses and bacterial toxins, although it is unlikely that this was the purpose for which they originally evolved. Before the functions of the sphingolipids were elucidated, these compounds appeared to be inscrutable riddles. They were, therefore, named for the Sphinx of Thebes, who killed passersby that could not solve her riddle.

The synthesis of sphingolipids begins with the formation of ceramide (Fig. 33.32). Serine and palmitoyl CoA condense to form a product that is reduced. A very-long-chain fatty acid (usually containing 22 carbons) forms an amide with the amino group, a double bond is generated, and ceramide is formed.

Ceramide reacts with phosphatidylcholine to form sphingomyelin, a component of the myelin sheath (Fig. 33.33). Ceramide also reacts with UDP-sugars to form cerebrosides (which contain a single monosaccharide, usually galactose or glucose). Galactocerebroside may react with 3'-phosphoadenosine 5'-phosphosulfate (PAPS, an active sulfate donor; Figure 33.34) to form sulfatides, the major sulfolipids of the brain.

Additional sugars may be added to ceramide to form globosides, and gangliosides are produced by the addition of *N*-acetylneuraminic acid (NANA) as branches from the oligosaccharide chains (see Fig. 33.33 and Chapter 30).

Sphingolipids are degraded by lysosomal enzymes (see Chapter 30). Deficiencies of these enzymes result in a group of lysosomal storage diseases known as the sphingolipidoses.

CLINICAL COMMENTS



 If **Percy Veere** had continued to eat a hypercaloric diet rich in carbohydrates, he would have become obese. In an effort to define obesity, it has been agreed internationally that the ratio of the patient's body weight in kilograms and their height in meters squared (W/H^2) is the most useful and reproducible measure. This ratio is referred to as the body mass index or BMI. Normal men and women fall into the range of 20 to 25. Percy's current value is 21.3 and rising.

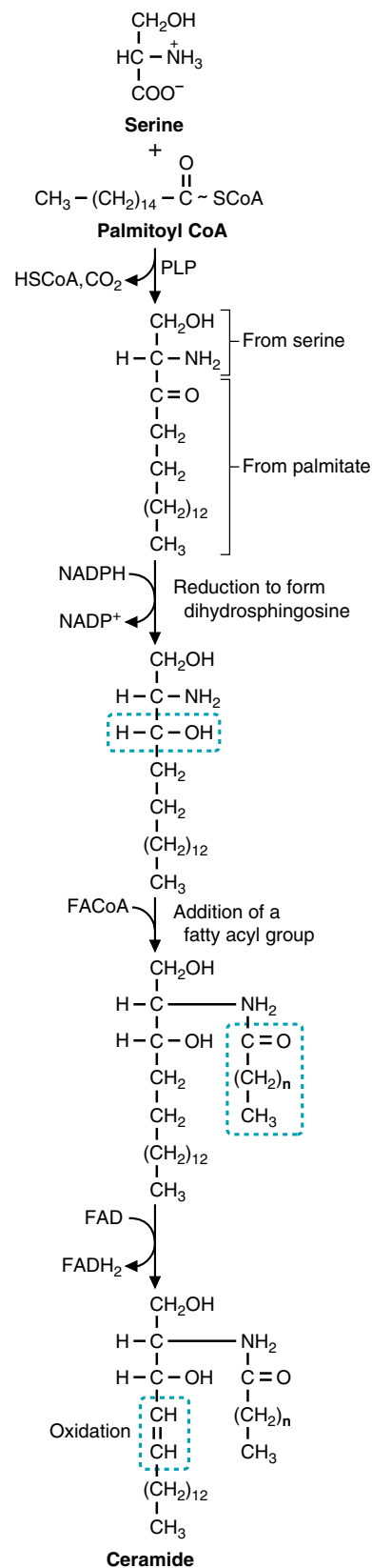


Fig. 33.32. Synthesis of ceramide. The changes that occur in each reaction are highlighted. PLP = pyridoxal phosphate.

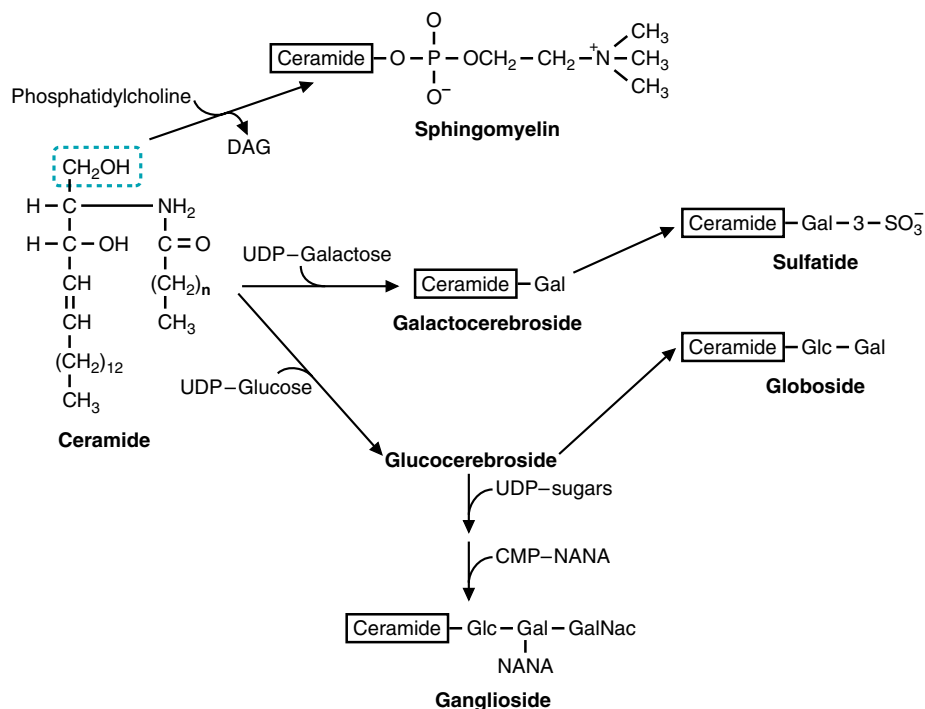


Fig. 33.33. Synthesis of sphingolipids from ceramide. Phosphocholine or sugars add to the hydroxymethyl group of ceramide (in blue) to form sphingomyelins, cerebrosides, sulfatides, globosides, and gangliosides. Gal = galactose; Glc = glucose; GalNac = *N*-acetylgalactosamine; NANA = *N*-acetylneuraminic acid.

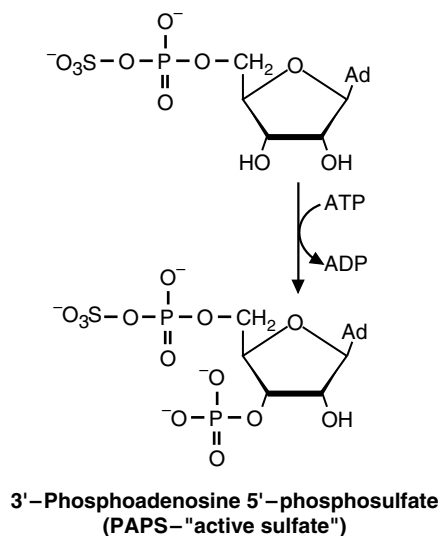


Fig. 33.34. The synthesis of 3'-phosphoadenosine 5'-phosphosulfate (PAPS), an active sulfate donor. PAPS donates sulfate groups to cerebrosides to form sulfatides and is also involved in glycosaminoglycan biosynthesis (see Chapter 49). Ad = adenosine.

Approximately 36 million people in the United States have a BMI greater than 27.8 (for men) or 27.3 (for women). At this level of obesity, which is quite close to a 20% weight increase above the "ideal" or desirable weight, an attempt at weight loss should be strongly advised. The idea that obesity is a benign condition unless accompanied by other risk factors for cardiovascular disease is disputed by several long-term, properly controlled prospective studies. These studies show that obesity

is an independent risk factor not only for heart attacks and strokes, but for the development of insulin resistance, type 2 diabetes mellitus, hypertension, and gallbladder disease.

Percy did not want to become overweight and decided to follow his new diet faithfully.



Because **Cora Nari's** lipid profile indicated an elevation in both serum triacylglycerols and LDL cholesterol, she was classified as having a combined hyperlipidemia. The dissimilarities in the lipid profiles of Cora and her two siblings, both of whom were experiencing anginal chest pain, is characteristic of the multigenic syndrome referred to as familial combined hyperlipidemia (FCH).

Approximately 1% of the North American population has FCH. It is the most common cause of coronary artery disease in the United States. In contrast to patients with familial hypercholesterolemia (FH), patients with FCH do not have fatty deposits within the skin or tendons (xanthomas) (see Chapter 34). In FCH, coronary artery disease usually appears by the fifth decade of life.

Treatment of FCH includes restriction of dietary fat. Patients who do not respond adequately to dietary therapy are treated with antilipidemic drugs. Selection of the appropriate antilipidemic drugs depends on the specific phenotypic expression of the patient's multigenic disease as manifest by their particular serum lipid profile. In Cora's case, a decrease in both serum triacylglycerols and LDL cholesterol must be achieved. If possible, her serum HDL cholesterol level should also be raised to a level above 40 mg/dL.

To accomplish these therapeutic goals, her physician initially prescribed fast-release nicotinic acid (niacin), because this agent has the potential to lower serum triacylglycerol levels and cause a reciprocal rise in serum HDL cholesterol levels, as well as to lower serum total and LDL cholesterol levels. The mechanisms suggested for niacin's triacylglycerol-lowering action include enhancement of the action of LPL, inhibition of lipolysis in adipose tissue, and a decrease in esterification of triacylglycerols in the liver (see Table 34.5). The mechanism by which niacin lowers the serum total and LDL cholesterol levels is related to the decrease in hepatic production of VLDL. When the level of VLDL in the circulation decreases, the production of its daughter particles, IDL and LDL, also decreases. Cora found niacin's side effects of flushing and itching to be intolerable, and the drug was discontinued.

Pravastatin was given instead. Pravastatin inhibits cholesterol synthesis by inhibiting hydroxymethylglutaryl CoA (HMG-CoA) reductase, the rate-limiting enzyme in the pathway (see Chapter 34). After 3 months of therapy, pravastatin decreased Cora's LDL cholesterol from a pretreatment level of 175 to 122 mg/dL (still higher than the recommended treatment goal of 100 mg/dL or less in a patient with established coronary artery disease). Her fasting serum triacylglycerol concentration was decreased from a pretreatment level of 280 to 178 mg/dL (a treatment goal for serum triacylglycerol when the pretreatment level is less than 500 mg/dL has not been established).



Colleen Lakker suffered from respiratory distress syndrome (RDS), which is a major cause of death in the newborn. RDS is preventable if prematurity can be avoided by appropriate management of high-risk pregnancy and labor. Before delivery, the obstetrician must attempt to predict and possibly treat pulmonary prematurity in utero. For example, estimation of fetal head circumference by ultrasonography, monitoring for fetal arterial oxygen saturation, and determination of the ratio of the concentrations of phosphatidylcholine (lecithin) and that of sphingomyelin in the amniotic fluid may help to identify premature infants who are predisposed to RDS (Fig. 33.35).

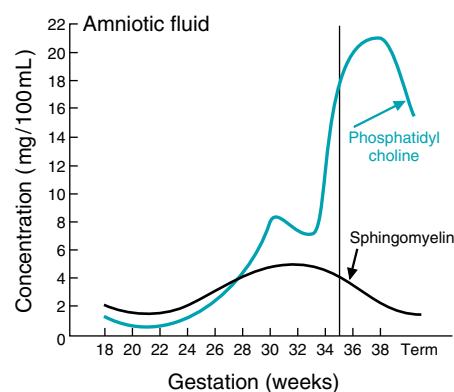


Fig. 33.35. Comparison of phosphatidylcholine and sphingomyelin in amniotic fluid. Phosphatidylcholine is the major lipid in lung surfactant. The concentration of phosphatidylcholine relative to sphingomyelin rises at 35 weeks of gestation, indicating pulmonary maturity.

The administration of synthetic corticosteroids 48 to 72 hours before delivery of a fetus of less than 33 weeks of gestation in women who have toxemia of pregnancy, diabetes mellitus, or chronic renal disease may reduce the incidence or mortality of RDS by stimulating fetal synthesis of lung surfactant.

The administration of one dose of surfactant into the trachea of the premature infant immediately after birth may transiently improve respiratory function but does not improve overall mortality. In Colleen's case, intensive therapy allowed her to survive this acute respiratory complication of prematurity.

BIOCHEMICAL COMMENTS



Biochemically, what makes people become obese? Obviously, the amount of fat an individual can store depends on the number of fat cells in the body and the amount of triacylglycerol each cell can accommodate. In obese individuals, both the number of fat cells and the size of the cells (i.e., the total storage capacity) is greater than in individuals with no history of obesity. To fill these stores, however, an individual must eat more than required to support the basal metabolic rate and physical activity.

Fat cells begin to proliferate early in life, starting in the third trimester of gestation. Proliferation essentially ceases before puberty, and thereafter fat cells change mainly in size. However, some increase in the number of fat cells can occur in adulthood if preadipocytes are induced to proliferate by growth factors and changes in the nutritional state. Weight reduction results in a decrease in the size of fat cells rather than a decrease in number. After weight loss, the amount of LPL, an enzyme involved in the transfer of fatty acids from blood triacylglycerols to the triacylglycerol stores of adipocytes, increases. In addition, the amount of mRNA for LPL also increases. All of these factors suggest that individuals who become obese, particularly those who do so early in life, will have difficulty losing weight and maintaining a lower body adipose mass.

Signals that initiate or inhibit feeding are extremely complex and include psychological and hormonal factors as well as neurotransmitter activity. These signals are integrated and relayed through the hypothalamus. Destruction of specific regions of the hypothalamus can lead to overeating and obesity or to anorexia and weight loss. Overeating and obesity are associated with damage to the ventromedial or the paraventricular nucleus, whereas weight loss and anorexia are related to damage to more lateral hypothalamic regions. Compounds that act as satiety signals have been identified in brain tissue and include leptin and glucagon-like peptide-1 (GLP-1). Appetite suppressors developed from compounds such as these may be used in the future for the treatment of obesity.

Recently it has become apparent that the adipocyte, in addition to storing triacylglycerol, secretes hormones that regulate both glucose and fat metabolism. The hormones leptin, resistin (resists insulin action), and adiponectin (also known as Acrp30) are all secreted from adipocytes under different conditions. The role of these hormones has been best understood in mouse models; unfortunately, extrapolation to the human condition has been difficult. In mice, leptin is released from adipocytes as triglyceride levels increase and signals the hypothalamus to reduce eating and to increase physical activity. Mice lacking the ability to secrete leptin (the ob mouse), or respond to leptin (the db mouse) are obese. Injecting leptin into ob mice allows them to lose weight.

The adipocytes in mice have been shown to release a hormone known as resistin. This hormone may contribute to insulin resistance in these animals. The mechanism by which resistin causes an insensitivity of cells to the actions of insulin is unknown. It is of great interest, however, that the class of drugs known as thiazolidinediones, which are given to individuals with type 2 diabetes, suppress resistin transcription, reduce resistin levels, and increase sensitivity to insulin in these patients. Addition-

ally, thiazolidinediones may upregulate adipose PEPCK, resulting in a reduced fatty acid output from the adipocyte because of increased glyceroneogenesis.

In humans, adiponectin is secreted from adipocytes in inverse proportion to their adipose mass, lean individuals secreting more adiponectin than obese individuals. This is the exact opposite of leptin secretion. The effects of adiponectin, and how it interacts with resistin and leptin, are active areas of current research.

Further complicating the issue of glucose and lipid homeostasis is the effect of nuclear receptors known as peroxisome proliferator activated receptors (PPAR). These nuclear receptors (see Chapter 10) exist in three forms; α , β/δ , and γ . PPAR γ is found in highest levels in adipocytes, and activation of the receptor leads to gene transcription, which is necessary for adipocyte differentiation and regulation of lipid metabolism. The thiazolidinediones activate PPAR γ , which leads to a decrease in circulating resistin levels. Understanding more about the physiologic regulators of PPAR γ is also an active area of research. The role of PPAR in liver is discussed in Chapter 46.

Although an increase in food intake beyond the daily requirements results in an increase in body weight and in fat stores, there is a large variation among individuals in the amount of weight gained for a given number of excess calories consumed. Both genetic and environmental factors influence the development of obesity. Studies of identical twins who were purposely overfed showed that the amount of weight gained was more similar within sets than between sets. Other studies of identical and fraternal twins, in which the members of a set were reared apart, support the conclusion that heredity plays a major role in determining body weight.

Suggested Readings

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REVIEW QUESTIONS—CHAPTER 33

1. Which of the following is involved in the synthesis of triacylglycerols in adipose tissue?
 - (A) Fatty acids obtained from chylomicrons and VLDL
 - (B) Glycerol 3-phosphate derived from blood glycerol
 - (C) 2-Monoacylglycerol as an obligatory intermediate
 - (D) Lipoprotein lipase to catalyze the formation of ester bonds
 - (E) Acetoacetyl CoA as an obligatory intermediate

2. A molecule of palmitic acid, attached to carbon 1 of the glycerol moiety of a triacylglycerol, is ingested and digested. It passes into the blood, is stored in a fat cell, and ultimately is oxidized to carbon dioxide and water in a muscle cell. Choose the molecular complex in the blood in which the palmitate residue is carried from the lumen of the gut to the surface of the gut epithelial cell.
- (A) VLDL
 - (B) Chylomicron
 - (C) Fatty acid-albumin complex
 - (D) Bile salt micelle
 - (E) LDL
3. A patient with hyperlipoproteinemia would be most likely to benefit from a low-carbohydrate diet if the lipoproteins that are elevated in blood are which of the following?
- (A) Chylomicrons
 - (B) VLDL
 - (C) HDL
 - (D) LDL
 - (E) IDL
4. Which of the following is a characteristic of sphingosine?
- (A) It is converted to ceramide by reacting with a UDP-sugar.
 - (B) It contains a glycerol moiety.
 - (C) It is synthesized from palmitoyl CoA and serine.
 - (D) It is a precursor of cardiolipin.
 - (E) It is only synthesized in neuronal cells.
5. Newly synthesized fatty acids are not immediately degraded because of which of the following?
- (A) Tissues that synthesize fatty acids do not contain the enzymes that degrade fatty acids.
 - (B) High NADPH levels inhibit β -oxidation.
 - (C) In the presence of insulin, the key fatty acid degrading enzyme is not induced.
 - (D) Newly synthesized fatty acids cannot be converted to their CoA derivatives.
 - (E) Transport of fatty acids into mitochondria is inhibited under conditions in which fatty acids are being synthesized.

34 Cholesterol Absorption, Synthesis, Metabolism, and Fate

Cholesterol is one of the most highly recognized molecules in human biology, in part because of a direct relationship between its concentrations in blood and tissues and the development of **atherosclerotic vascular disease**. Cholesterol, which is transported in the blood in **lipoproteins** because of its absolute insolubility in water, serves as a **stabilizing component of cell membranes** and as a precursor of the **bile salts** and **steroid hormones**. Precursors of cholesterol are converted to **ubiquinone**, **dolichol**, and, in the skin, to **cholecalciferol**, the **active form of vitamin D**. As a **major component of blood lipoproteins**, cholesterol can appear in its free, unesterified form in the outer shell of these macromolecules and as cholesterol esters in the lipoprotein core.

Cholesterol is obtained from the diet or synthesized by a pathway that occurs in most cells of the body, but to a greater extent in cells of the liver and intestine. The precursor for cholesterol synthesis is **acetyl CoA**, which can be produced from glucose, fatty acids, or amino acids. Two molecules of acetyl CoA form **acetoacetyl CoA**, which condenses with another molecule of acetyl CoA to form **hydroxymethylglutaryl CoA (HMG-CoA)**. Reduction of HMG-CoA produces **mevalonate**. This reaction, catalyzed by **HMG-CoA reductase**, is the major rate-limiting step of cholesterol synthesis. Mevalonate produces isoprene units that condense, eventually forming **squalene**. Cyclization of squalene produces the steroid ring system, and a number of subsequent reactions generate cholesterol. The adrenal cortex and the gonads also synthesize cholesterol in significant amounts and use it as a precursor for steroid hormone synthesis.

Cholesterol is packaged in **chylomicrons** in the intestine and in **very-low-density lipoprotein (VLDL)** in the liver. It is transported in the blood in these lipoprotein particles, which also transport triacylglycerols. As the triacylglycerols of the blood lipoproteins are digested by lipoprotein lipase, chylomicrons are converted to **chylomicron remnants**, and VLDL is converted to **intermediate-density lipoprotein (IDL)** and subsequently to **low-density lipoprotein (LDL)**. These products return to the liver, where they bind to receptors in cell membranes and are taken up by endocytosis and digested by lysosomal enzymes. LDL is also endocytosed by nonhepatic (peripheral) tissues. Cholesterol and other products of lysosomal digestion are released into the cellular pools. The liver uses this recycled cholesterol, and the cholesterol that is synthesized from acetyl CoA, to produce VLDL and to synthesize bile salts.

Intracellular cholesterol obtained from blood lipoproteins decreases the synthesis of cholesterol within cells, stimulates the storage of cholesterol as cholesterol esters, and decreases the synthesis of **LDL receptors**. LDL receptors are found on the surface of the cells and bind various classes of lipoproteins prior to endocytosis.

Although high-density lipoprotein (HDL) contains triacylglycerols and cholesterol, its function is very different from that of the chylomicrons and VLDL, which transport triacylglycerols. HDL exchanges proteins and lipids with the other lipoproteins in the blood. HDL transfers **apolipoprotein E (apoE)** and **apoC_{II}** to chylomicrons and VLDL. After digestion of the VLDL triacylglycerols, apoE and apoC_{II} are transferred back to HDL. In addition, HDL obtains cholesterol from

other lipoproteins and from cell membranes and converts it to **cholesterol esters** by the **lecithin:cholesterol acyltransferase (LCAT)** reaction. Then HDL either directly transports cholesterol and cholesterol esters to the liver or transfers cholesterol esters to other lipoproteins via the **cholesterol ester transfer protein (CETP)**. Ultimately, lipoprotein particles carry the cholesterol and cholesterol esters to the liver, where endocytosis and lysosomal digestion occur. Thus, “**reverse cholesterol transport**” (i.e., the return of cholesterol to the liver) is a major function of HDL.

Elevated levels of cholesterol in the blood are associated with the formation of **atherosclerotic plaques** that can occlude blood vessels, causing heart attacks and strokes. Although high levels of LDL cholesterol are especially atherogenic, high levels of HDL cholesterol are protective because HDL particles are involved in the process of removing cholesterol from tissues, such as the lining cells of vessels, and returning it to the liver.

Bile salts, which are produced in the liver from cholesterol obtained from the blood lipoproteins or synthesized from acetyl CoA, are secreted into the bile. They are stored in the gallbladder and released into the intestine during a meal. The bile salts emulsify dietary triacylglycerols, thus aiding in digestion. The digestive products are absorbed by intestinal epithelial cells from bile salt micelles, tiny microdroplets that contain bile salts at their water interface. After the contents of the micelles are absorbed, most of the bile salts travel to the ileum, where they are resorbed and recycled by the liver. Less than 5% of the bile salts that enter the lumen of the small intestine are eventually excreted in the feces.

Although the fecal excretion of bile salts is relatively low, it is a major means by which the body disposes of the steroid nucleus of cholesterol. Because the ring structure of cholesterol cannot be degraded in the body, it is excreted mainly in the bile as free cholesterol and bile salts.

The steroid hormones, derived from cholesterol, include the adrenal cortical hormones (e.g., cortisol, aldosterone, and the adrenal sex steroids dehydroepiandrosterone [DHEA] and androstenedione) and the gonadal hormones (e.g., the ovarian and testicular sex steroids, such as testosterone and estrogen).



THE WAITING ROOM



At his current office visit, **Ivan Applebod's** case was reviewed by his physician. Mr. Applebod has several of the major risk factors for coronary heart disease (CHD). These include a sedentary lifestyle, marked obesity, hypertension, hyperlipidemia, and early non-insulin-dependent diabetes mellitus (NIDDM). Unfortunately, he has not followed his doctor's advice with regard to a diabetic diet designed to affect a significant loss of weight, nor has he followed an aerobic exercise program. As a consequence, his weight has gone from 270 to 281 lb. After a 14-hour fast, his serum glucose is now 214 mg/dL (normal, <110), and his serum total cholesterol level is 314 mg/dL (desired level is 200 or less). His serum triacylglycerol level is 295 mg/dL (desired level is 150 or less), and his serum HDL cholesterol is 24 mg/dL (desired level is ≥ 40 for a male). His calculated serum LDL cholesterol level is 231 mg/dL (desired level for a person with two or more risk factors for CHD is 130 mg/dL or less, unless one of the risk factors is diabetes mellitus, in which case, the LDL cholesterol level should be < 100 mg/dL).



Ann Jeina was carefully followed by her physician after she survived her heart attack. Before discharge from the hospital, after a 14-hour fast, her serum triacylglycerol level was 158 mg/dL (slightly above the upper range of normal), and her HDL cholesterol level was low at 32 mg/dL (normal for women is ≥ 50). Her serum total cholesterol level was elevated at 420 mg/dL (reference range, ≤ 200 for a female with known CHD). From these values, her LDL cholesterol level was calculated to be 356 mg/dL (desirable level for a person with established heart disease is < 100).

Both of Ms. Jeina's younger brothers had "very high" serum cholesterol levels, and both had suffered heart attacks in their mid-forties. With this information, a tentative diagnosis of familial hypercholesterolemia, type IIA was made, and the patient was started on a step I diet as recommended by the National Cholesterol Education Program (NCEP) Adult Treatment Panel III. This panel recommends that decisions with regard to when dietary and drug therapy are initiated based on the serum LDL cholesterol level, as depicted in Table 34.1.

Because a Step I diet (Table 34.2) usually lowers serum total and LDL cholesterol levels by no more than 15%, it is likely that Ms. Jeina's diet will eventually have to be further restricted in cholesterol and fat and that one or more lipid-lowering drugs will have to be added to her treatment plan.



Vera Leizd is a 34-year-old woman in whom pubertal changes began at age 12, leading to the development of normal secondary sexual characteristics and the onset of menses at age 13. Her menstrual periods occurred on a monthly basis over the next 7 years, but the flow was scant. At age 20, she noted a gradual increase in the intermenstrual interval from her normal of 28 days to 32 to 38 days. The volume of her menstrual flow also gradually diminished. After 7 months, her menstrual periods ceased. She complained of increasing oiliness of her skin, the appearance of acnelike lesions on her face and upper back, and the appearance of short dark terminal hairs on the mustache and sideburn areas of her face. The amount of extremity hair also increased, and she noticed a disturbing loss of hair from her scalp.



Until recently, the concentration of LDL cholesterol could only be directly determined by sophisticated laboratory techniques not available for routine clinical use. As a consequence, the LDL cholesterol concentration in the blood was derived indirectly by using the Friedewald formula: the sum of the HDL cholesterol level and the triacylglycerol (TG) level divided by 5 (which gives an estimate of the VLDL cholesterol level) subtracted from the total cholesterol level.

$$\text{LDL cholesterol} = \text{Total cholesterol} - [\text{HDL cholesterol} + (\text{TG}/5)]$$

This equation yields inaccurate LDL cholesterol levels 15 to 20% of the time and fails completely when serum triacylglycerol levels exceed 400 mg/dL.

A recently developed test called "LDL direct" isolates LDL cholesterol by using a special immunoseparation reagent. Not only is this direct assay for LDL cholesterol more accurate than the indirect Friedewald calculation, it also is not affected by mildly to moderately elevated serum triacylglycerol levels and can be used for a patient who has not fasted. It does not require the expense of determining serum total cholesterol, HDL cholesterol, and triacylglycerol levels.

I. INTESTINAL ABSORPTION OF CHOLESTEROL

Cholesterol absorption by intestinal cells is a key regulatory point in human sterol metabolism because it ultimately determines what percentage of the 1,000 mg of biliary cholesterol produced by the liver each day and what percentage of the

Table 34.1. ATP III: LDL-C Goals and Cut Points for Therapy in Different Risk Categories

Risk Category	LDL Goal (mg/dL)	LDL level at which to initiate therapeutic lifestyle changes (mg/dL)	LDL level at which to consider drug therapy (mg/dL)
CHD or CHD risk equivalents (10-year risk $> 20\%$)	< 100	≥ 100	≥ 130 (100–129: drug optional)
2+ Risk factors (10-year risk $\leq 20\%$)	< 130	≥ 130	10-Year risk 10%–20%: ≥ 130 10-year risk $< 10\%$: ≥ 160
0–1 risk factor	< 160	≥ 160	≥ 190 (160–189: LDL-lowering drug optional)

LDL-C = low-density lipoprotein cholesterol; CHD = coronary heart disease.

Source: Executive summary of the third report of the National Cholesterol Education Programs (NCEP) Expert panel on detection, evaluation, and treatment of high blood cholesterol in Adults (Adult Treatment Panel III). Final Report, Circulation 2002;106:3145–3457.

Table 34.2: Dietary Therapy of Elevated Blood Cholesterol

Nutrient	Step I Diet	Step II Diet ^a
Cholesterol ^b	<300 mg/day	<200 mg/day
Total fat	≤30% ^b	30%
Saturated fat	8–10%	<7%
Polyunsaturated fat	≤10%	≤10%
Monounsaturated fat	≤15%	≤15%
Carbohydrates	≥55%	≥55%
Protein	~15%	~15%
Calories	To achieve and maintain desirable body weight	

Based on: NCEP. Second Report of the Adult Treatment Panel, JAMA, 1993;269(23):3015–3023.

^aThe Step II diet is applied if 3 months on the Step I diet has failed to reduce blood cholesterol to the desired level (see Table 34.1).

^bExcept for the values given in mg/day, all the values are percentage of total calories eaten daily.

300 mg of dietary cholesterol entering the gut per day is eventually absorbed into the blood. In normal subjects, approximately 55% of this intestinal pool enters the blood through the enterocyte each day. The details of cholesterol absorption from dietary sources was outlined in Chapter 32.

Although the absorption of cholesterol from the intestinal lumen is a diffusion-controlled process, there is also a mechanism to remove unwanted or excessive cholesterol and plant sterols from the enterocyte. The transport of sterols out of the enterocyte, and into the lumen, is related to the products of genes that code for the adenosine triphosphate (ATP)-binding cassette (ABC) protein family, ABC1, ABCG5, and ABCG8. These proteins couple ATP hydrolysis to the transport of unwanted or excessive cholesterol and plant sterols (phytosterols) from the enterocyte back into the gut lumen. Cholesterol cannot be metabolized to CO₂ and water and is, therefore, principally eliminated from the body in the feces as unabsorbed sterols and bile acids. ABC protein expression increases the amount of sterols present in the gut lumen, with the potential to increase elimination of the sterols into the feces. Patients with a condition known as phytosterolemia (a rare autosomal recessive disease, also known as sitosterolemia) have a defect in the function of either ABCG5 or ABCG8 in the enterocytes, thereby leading to the accumulation of cholesterol and phytosterols within these cells. These eventually reach the bloodstream, markedly elevating the level of cholesterol and phytosterol in the blood. This accounts for the increased cardiovascular morbidity in individuals with this disorder. From these experiments of nature, it is clear that agents that either amplify the expression of the ABC proteins within enterocytes, or block cholesterol absorption from the lumen, have therapeutic potential in the treatment of patients with hypercholesterolemia. Ezetimibe, now available for clinical use, is a compound that is structurally different from the sterols. Its primary action in lowering serum cholesterol levels is to block cholesterol absorption through a specific but as yet poorly characterized cholesterol absorption mechanism in the brush border of enterocytes. It also may induce ABC protein expression, but this action is relatively unimportant in reducing net cholesterol absorption. The reduction of cholesterol absorption from the intestinal lumen has been shown to reduce blood levels of LDL cholesterol.

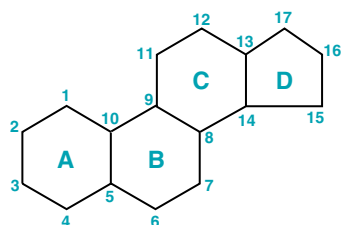


Fig. 34.1. The basic ring structure of sterols; the perhydrocyclopentanophenanthrene nucleus. Each ring is labeled either A, B, C, or D.

II. CHOLESTEROL SYNTHESIS

Cholesterol is an alicyclic compound whose basic structure includes the perhydrocyclopentanophenanthrene nucleus containing four fused rings (Figure 34.1). In its “free” form, the cholesterol molecule contains 27 carbon atoms, a simple hydroxyl group at C3, a double bond between C5 and C6, an eight-membered hydrocarbon chain attached to carbon 17 in the D ring, a methyl group (carbon 19) attached to carbon 10, and a second methyl group (carbon 18) attached to carbon 13 (Figure 34.2).

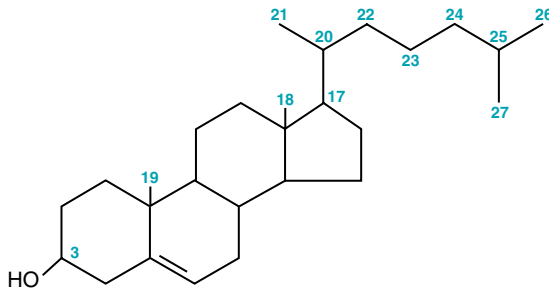


Fig. 34.2. The structure of cholesterol.

Approximately one third of plasma cholesterol exists in the free (or unesterified) form. The remaining two thirds exists as cholesterol esters in which a long-chain fatty acid (usually linoleic acid) is attached by ester linkage to the hydroxyl group at C-3 of the A ring. The proportions of free and esterified cholesterol in the blood can be measured using methods such as high-performance liquid chromatography (HPLC).

The structure of cholesterol suggests that its synthesis involves multimolecular interactions; yet all of the 27 carbons are derived from one precursor, acetyl CoA. Acetyl CoA can be obtained from several sources, including the beta oxidation of fatty acids, the oxidation of ketogenic amino acids, such as leucine and lysine, and the pyruvate dehydrogenase reaction. Carbons 1, 2, 5, 7, 9, 13, 15, 18, 19, 20, 22, 24, 26, and 27 of cholesterol are derived from the methyl group of acetyl CoA and the remaining 12 carbons of cholesterol from the carboxylate atom of acetyl CoA.

The synthesis of cholesterol requires significant reducing power, which is supplied in the form of NADPH. The latter is provided by glucose-6-phosphate dehydrogenase and 6-phosphogluconate dehydrogenase of the hexose monophosphate shunt pathway (see Chapter 29). Cholesterol synthesis occurs in the cytosol, requiring hydrolysis of high-energy thioester bonds of acetyl CoA and phosphoanhydride bonds of ATP. Its synthesis occurs in four stages.

A. Stage 1: Synthesis of Mevalonate from Acetyl CoA

The first stage of cholesterol synthesis leads to the production of the intermediate mevalonate (Fig. 34.3). *The synthesis of mevalonate is the committed, rate-limiting step in cholesterol formation.* In this cytoplasmic pathway, two molecules of acetyl CoA condense, forming acetoacetyl CoA, which then condenses with a third molecule of acetyl CoA to yield the 6-carbon compound β -hydroxy- β -methylglutaryl-CoA (HMG-CoA). The HMG-CoA synthase in this reaction is present in the cytosol and is distinct from the mitochondrial HMG-CoA synthase that catalyses HMG-CoA synthesis involved in ketone body production. *The committed step and major point of regulation of cholesterol synthesis in stage 1 involves reduction of HMG-CoA to mevalonate, a reaction catalyzed by HMG-CoA reductase, an enzyme embedded in the membrane of the endoplasmic reticulum.* HMG-CoA reductase contains eight membrane-spanning domains, and the amino terminal domain, which faces the cytoplasm, contains the enzymatic activity. The reducing equivalents for this reaction are donated by two molecules of NADPH. The regulation of the activity of HMG-CoA reductase is controlled in multiple ways.

1. TRANSCRIPTIONAL CONTROL

The rate of synthesis of HMG-CoA reductase messenger RNA (mRNA) is controlled by one of the family of sterol regulatory element binding proteins (SREBPs)(Fig. 34.4A). These transcription factors belong to the helix-loop-helix-

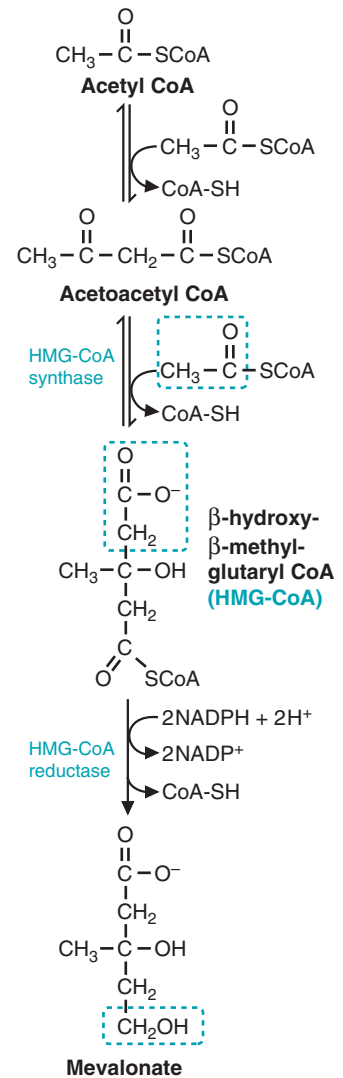


Fig. 34.3. The conversion of three molecules of acetyl-CoA to mevalonic acid.



Ann Jeina's serum total and LDL cholesterol levels improved only modestly after 3 months on a Step I diet. Three additional months on a more severe low-fat diet (Step II diet) brought little further improvement. The next therapeutic step would be to initiate lipid-lowering drug therapy (see Table 34.5).

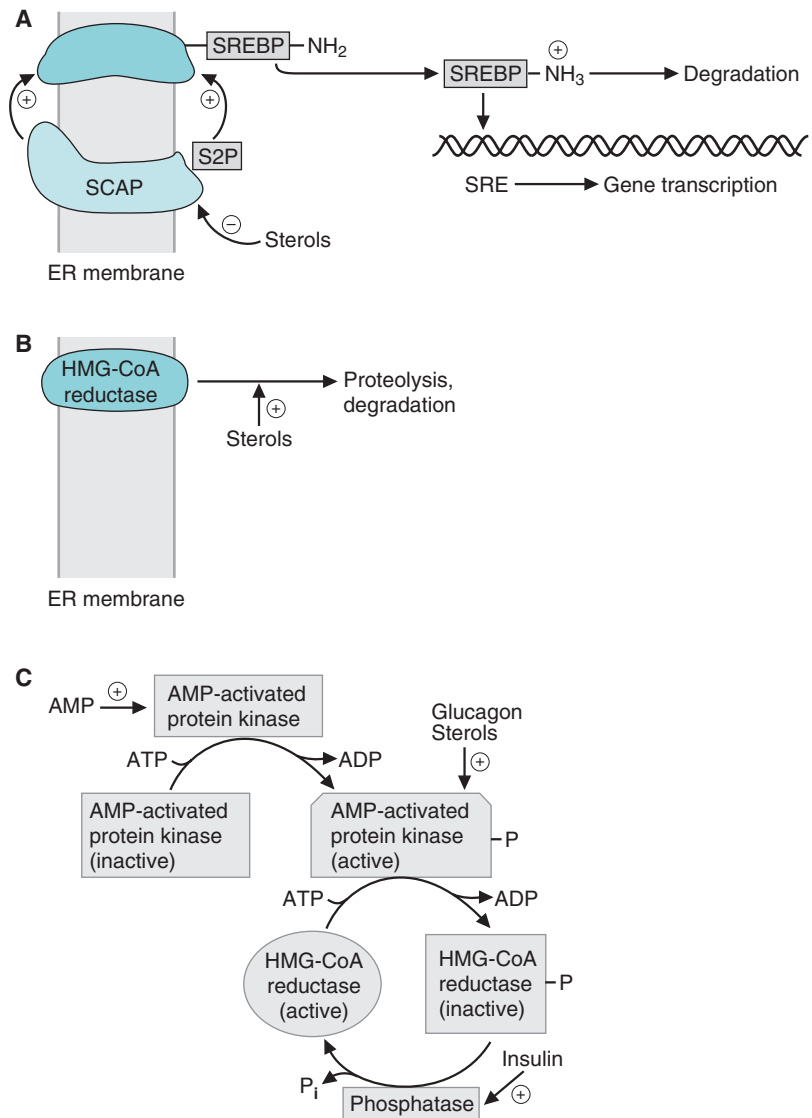


Fig. 34.4. Regulation of HMG-CoA reductase activity. See text for details. A. Transcriptional control. B. Regulation by proteolysis. C. Regulation by phosphorylation.

leucine zipper (bHLH-Zip) family of transcription factors that directly activate the expression of more than 30 genes dedicated to the synthesis and uptake of cholesterol, fatty acids, triacylglycerols, and phospholipids as well as the production of the NADPH cofactors required to synthesize these molecules.

SREBP1- α specifically enhances transcription of genes required for HMG-CoA reductase expression by binding to the sterol regulatory element (SRE) upstream of the reductase gene. When bound, the rate of transcription is increased. SREBPs, after synthesis, are integral ER proteins, and the active component of the protein is released by two proteases, SCAP (SREBP cleavage-activating protein) and S2P (site 2 protease). Once released, the active amino terminal component travels to the nucleus to bind to SREs. The soluble SREBPs are rapidly turned over and need to be continuously produced to effectively stimulate reductase mRNA transcription. When cytoplasmic sterol levels rise, the sterols bind to SCAP and inactivate it, thereby leading to a decrease in transcription of the reductase gene, and less reductase protein being produced.

2. PROTEOLYTIC DEGRADATION OF HMG-CoA REDUCTASE

Rising levels of cholesterol and bile salts in cells that synthesize these molecules also may cause a change in the oligomerization state of the membrane domain of HMG-CoA reductase, rendering the enzyme more susceptible to proteolysis (see Fig. 34.4B). This, in turn, decreases its activity. The membrane domains of HMG-CoA reductase contain sterol-sensing regions, which are similar to those in SCAP.

3. REGULATION BY COVALENT MODIFICATION

In addition to the inductive and repressive influences cited above, the activity of the reductase is also regulated by phosphorylation and dephosphorylation (see Fig. 34.4C). Elevated glucagon levels increase phosphorylation of the enzyme, thereby inactivating it, whereas hyperinsulinemia increases the activity of the reductase by activating phosphatases, which dephosphorylate the reductase. Increased levels of intracellular sterols also may increase phosphorylation of HMG-CoA reductase, thereby reducing its activity as well (feedback suppression). Thyroid hormone also increases enzyme activity, whereas glucocorticoids decrease its activity. The enzyme that phosphorylates HMG-CoA reductase is the adenosine monophosphate (AMP)-activated protein kinase, which itself is regulated by phosphorylation by the AMP-activated protein kinase kinase. Thus, cholesterol synthesis decreases when ATP levels are low and increases when ATP levels are high. This will become very clear once the further reactions of the biosynthetic pathway of cholesterol are discussed.

B. Stage 2: Conversion of Mevalonate to Two Activated Isoprenes

In the second stage of cholesterol synthesis, three phosphate groups are transferred from three molecules of ATP to mevalonate (Fig. 34.5). The purpose of these phosphate transfers is to activate both carbon 5 and the hydroxyl group on carbon 3 for further reactions in which these groups will leave the molecule. The phosphate group attached to the C-3 hydroxyl group of mevalonate in the 3-phospho-5-pyrophosphomevalonate intermediate is removed along with the carboxyl group on C-1. This produces a double bond in the 5-carbon product, Δ^3 -isopentenyl pyrophosphate, the first of two activated isoprenes necessary for the synthesis of cholesterol. The second activated isoprene is formed when Δ^3 -isopentenyl pyrophosphate is isomerized to dimethylallyl pyrophosphate (see Fig. 34.5).

C. Stage 3: Condensation of Six Activated 5-Carbon Isoprenes to Form the 30-Carbon Squalene

The next stage in the biosynthesis of cholesterol involves the head-to-tail condensation of isopentenylpyrophosphate and dimethylallyl pyrophosphate. In this reaction, one pyrophosphate group is displaced, and a 10-carbon chain, known as geranyl pyrophosphate, is generated (Fig. 34.6). (The “head” refers to the end to which pyrophosphate is joined.) Geranyl pyrophosphate then undergoes another head-to-tail condensation with isopentenyl pyrophosphate, resulting in the formation of the 15-carbon intermediate, farnesyl pyrophosphate. After this, two molecules of farnesyl pyrophosphate undergo a head-to-head fusion, and both pyrophosphate groups are removed to form squalene, a compound first isolated from the liver of sharks (genus *Squalus*). Squalene contains 30 carbons (24 in the main chain and 6 in the methyl group branches; see Fig. 34.6).



Farnesyl and geranyl groups can form covalent bonds with proteins, particularly the G proteins and certain protooncogene products involved in signal transduction. These hydrophobic groups anchor the proteins in the cell membrane.

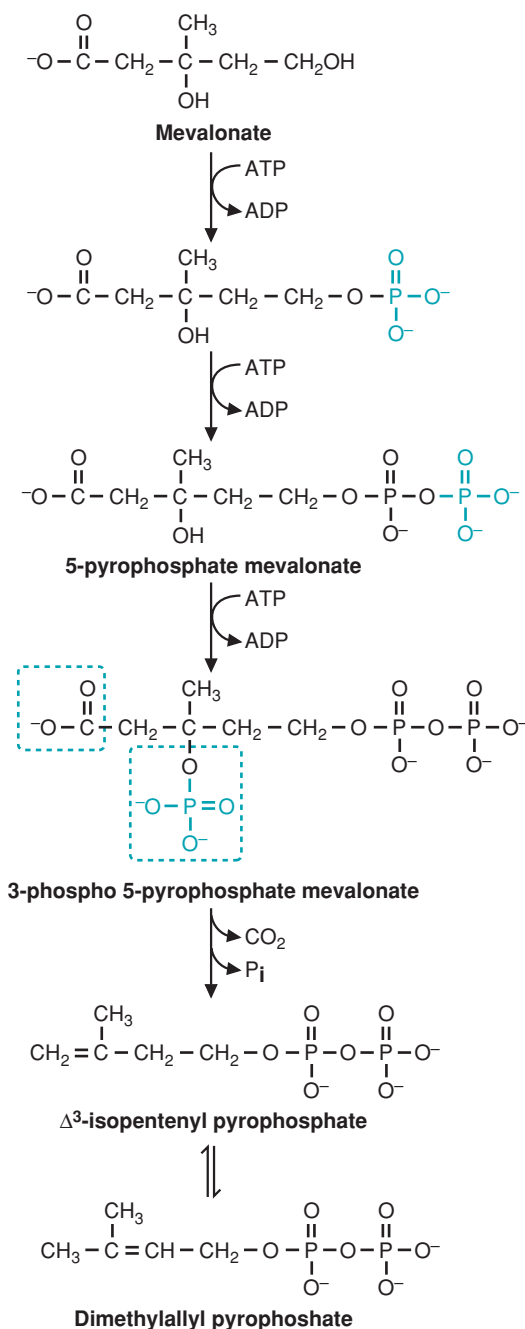


Fig. 34.5. The formation of activated isoprene units (Δ^3 -isopentenyl pyrophosphate and dimethylallyl pyrophosphate) from mevalonic acid. Note the large ATP requirement for these steps.

D. Stage 4: Conversion of Squalene to the Four-Ring Steroid Nucleus

The enzyme squalene monooxygenase adds a single oxygen atom from O_2 to the end of the squalene molecule, forming an epoxide. NADPH then reduces the other oxygen atom of O_2 to H_2O . The unsaturated carbons of the squalene 2, 3- epoxide are aligned in a way that allows conversion of the linear squalene epoxide into a cyclic structure. The cyclization leads to the formation of lanosterol, a sterol with the four-ring structure characteristic of the steroid nucleus. A series of complex

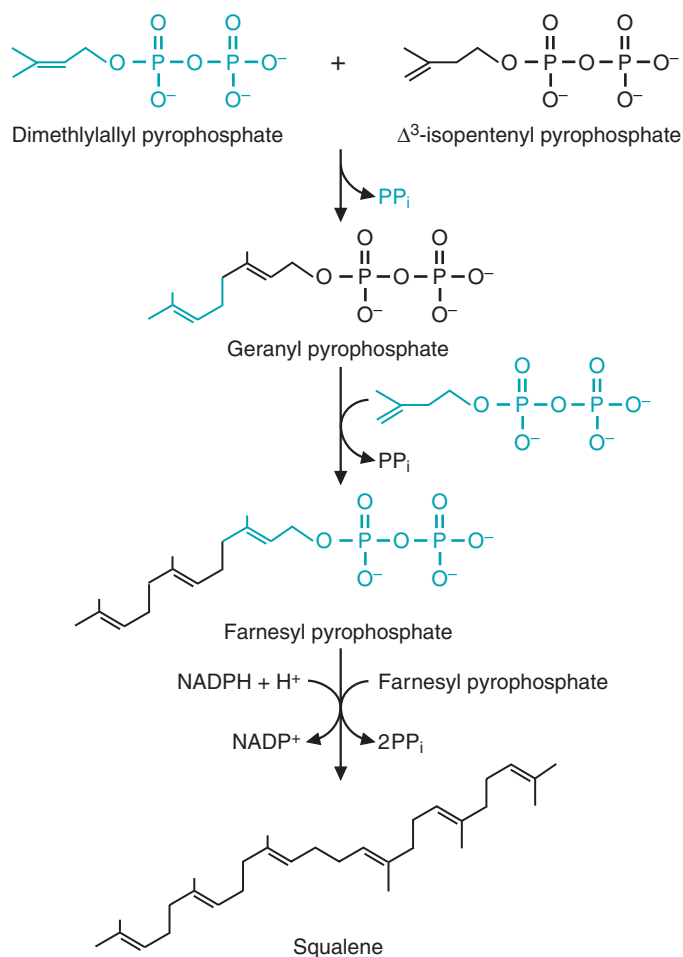


Fig. 34.6. The formation of squalene from six isoprene units. The activation of the isoprene units drives their condensation to form geranyl pyrophosphate, farnesyl pyrophosphate, and squalene.

reactions, containing many steps and elucidated in the late 1950s, leads to the formation of cholesterol (Fig. 34.7).

III. SEVERAL FATES OF CHOLESTEROL

Almost all mammalian cells are capable of producing cholesterol. Most of the biosynthesis of cholesterol, however, occurs within liver cells, although the gut, the adrenal cortex, and the gonads (as well as the placenta in pregnant women) also produce significant quantities of the sterol. Although a fraction of hepatic cholesterol is used for the synthesis of hepatic membranes, the bulk of synthesized cholesterol is secreted from the hepatocyte as one of three moieties: cholesterol esters, biliary cholesterol, or bile acids. Cholesterol ester production in the liver is catalyzed by acyl-CoA-cholesterol acyl transferase (ACAT). ACAT catalyzes the transfer of a fatty acid from coenzyme A to the hydroxyl group on carbon 3 of cholesterol (Fig. 34.8). Cholesterol esters are more hydrophobic than is free cholesterol. The liver packages some of the esterified cholesterol into the hollow core of lipoproteins, primarily VLDL. VLDL is secreted from the hepatocyte into the blood and transports the cholesterol esters (and triacylglycerols, phospholipids, apoproteins, etc.) to the tissues that require greater amounts of cholesterol than they can synthesize *de novo*. These tissues then use the cholesterol for the synthesis of membranes,

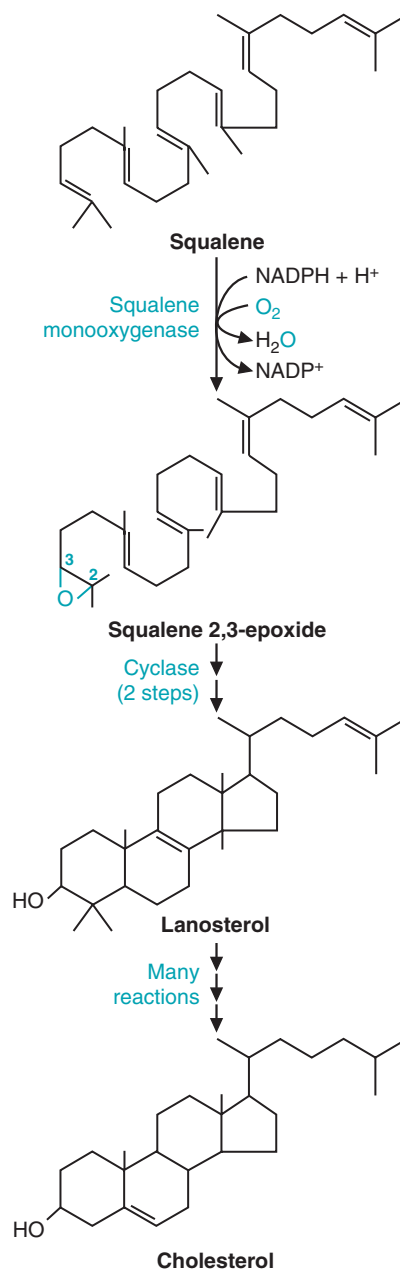


Fig. 34.7. The conversion of squalene to cholesterol. Squalene is shown in a different conformation than that of Fig. 34.6 to better indicate how the cyclization reaction occurs.

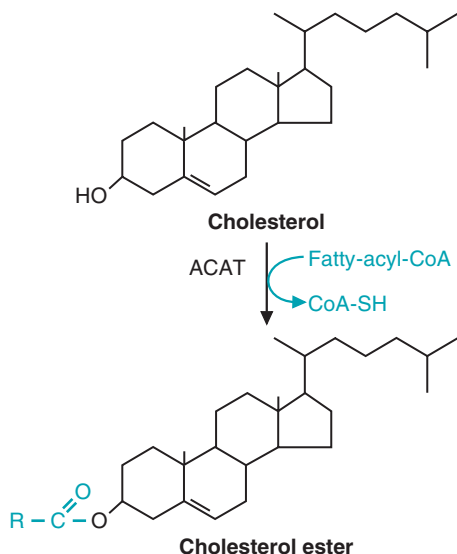


Fig. 34.8. The ACAT reaction, producing cholesterol esters. ACAT = acyl-CoA:cholesterol acyl transferase.

for the formation of steroid hormones, and for the biosynthesis of vitamin D. The residual cholesterol esters not used in these ways are stored in the liver for later use. The hepatic cholesterol pool also serves as a source of cholesterol for the synthesis of the relatively hydrophilic bile acids and their salts. These derivatives of cholesterol are highly effective detergents because they contain both polar and nonpolar regions. They are introduced in the biliary ducts of the liver. They are stored and concentrated in the gallbladder and later discharged into the gut in response to the ingestion of food. They aid in the digestion of intraluminal lipids by forming micelles with them, which increases the surface area of lipids exposed to the digestive action of intraluminal lipases. Free cholesterol also enters the gut lumen via the biliary tract (approximately 1,000 mg daily, which mixes with 300 mg dietary cholesterol to form an intestinal pool, roughly 55% of which is resorbed by the enterocytes and enters the bloodstream daily). On a low-cholesterol diet, the liver will synthesize approximately 800 mg cholesterol per day to replace bile salts and cholesterol lost from the enterohepatic circulation into the feces. Conversely, a greater intake of dietary cholesterol suppresses the rate of hepatic cholesterol synthesis (feedback repression).

IV. SYNTHESIS OF BILE SALTS

A. Conversion of Cholesterol to Cholic Acid and Chenocholic Acid

Bile salts are synthesized in the liver from cholesterol by reactions that hydroxylate the steroid nucleus and cleave the side chain. In the first reaction, an α -hydroxyl group is added to carbon 7 (on the α side of the B ring). The activity of the 7 α -hydroxylase that catalyzes this rate-limiting step is decreased by bile salts (Fig. 34.9).

In subsequent steps, the double bond in the B ring is reduced, and an additional hydroxylation may occur. Two different sets of compounds are produced. One set has α -hydroxyl groups at positions 3, 7, and 12, and produces the cholic acid series

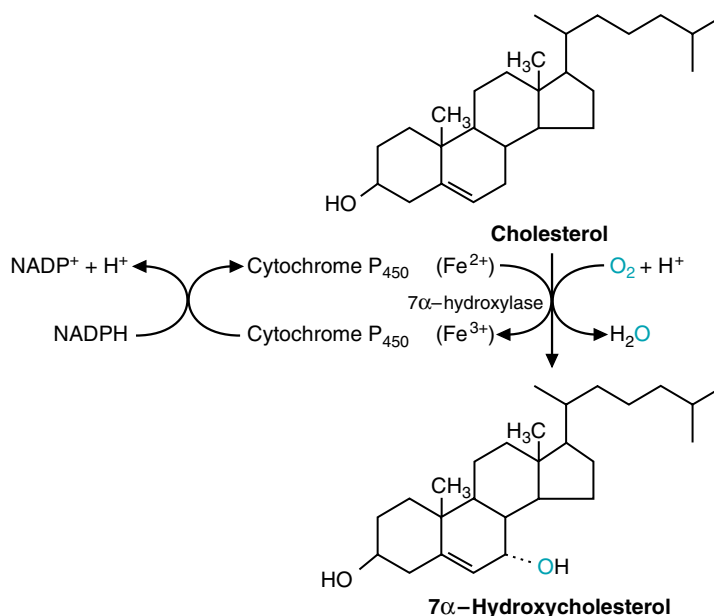


Fig. 34.9. The reaction catalyzed by 7 α -hydroxylase. An α -hydroxyl group is formed at position 7 of cholesterol. This reaction, which is inhibited by bile salts, is the rate-limiting step in bile salt synthesis.

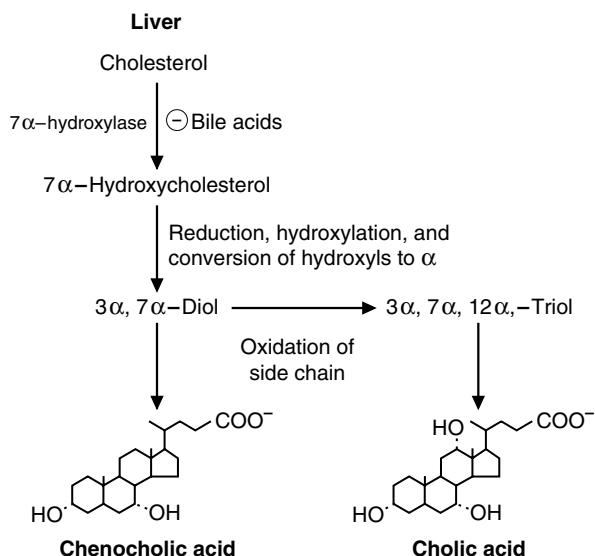


Fig. 34.10. Synthesis of bile salts. Two sets of bile salts are generated; one with α -hydroxyl groups at positions 3 and 7 (the chenocholate series), and the other with α -hydroxyls at positions 3, 7 and 12 (the cholate series).

of bile salts. The other set has α -hydroxyl groups only at positions 3 and 7 and produces the chenocholic acid series (Fig. 34.10). Three carbons are removed from the side chain by an oxidation reaction. The remaining 5-carbon fragment attached to the ring structure contains a carboxyl group (see Fig. 34.10).

The pK of the bile acids is approximately 6. Therefore, in the contents of the intestinal lumen, which normally have a pH of 6, approximately 50% of the molecules are present in the protonated form, and 50% are ionized, which forms bile salts. (The terms *bile acids* and *bile salts* are often used interchangeably, but *bile salts* actually refer to the ionized form of the molecule.)

B. Conjugation of Bile Salts

The carboxyl group at the end of the side chain of the bile salts is activated by a reaction that requires ATP and coenzyme A (CoA). The CoA derivatives can react with either glycine or taurine (which is derived from cysteine), forming amides that are known as the conjugated bile salts. In glycocholic acid and glycochenocholic acid, the bile acids are conjugated with glycine. These compounds have a pK of approximately 4, so compared to their unconjugated forms, a higher percentage of the molecules is present in the ionized form at the pH of the intestine. The taurine conjugates, taurocholic and taurochenocholic acid, have a pK of approximately 2. Therefore, compared with the glycoconjugates, an even greater percentage of the molecules of these conjugates are ionized in the lumen of the gut (Fig. 34.11).

V. FATE OF THE BILE SALTS

The bile salts are produced in the liver and secreted into the bile (Fig. 34.12). They are stored in the gallbladder and released into the intestine during a meal, where they serve as detergents that aid in the digestion of dietary lipids (see Chapter 32).

Intestinal bacteria deconjugate and dehydroxylate the bile salts, removing the glycine and taurine residues and the hydroxyl group at position 7. The bile salts that lack a hydroxyl group at position 7 are called secondary bile salts. The deconjugated and dehydroxylated bile salts are less soluble and, therefore, less readily resorbed from the intestinal lumen than the bile salts that have not been subjected to bacterial

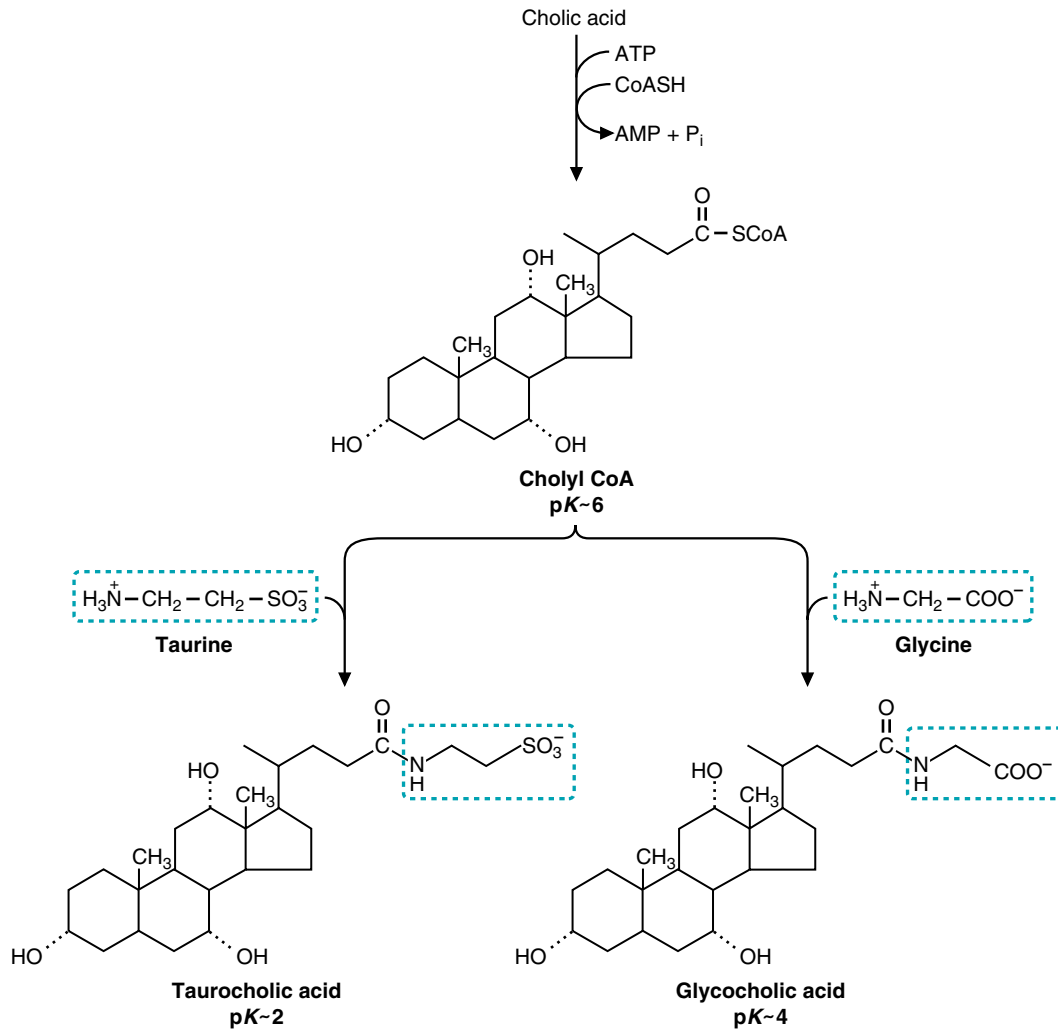


Fig. 34.11. Conjugation of bile salts. Conjugation lowers the pK of the bile salts, making them better detergents; i.e., they are more ionized in the contents of the intestinal lumen ($\text{pH} \approx 6$) than are the unconjugated bile salts ($\text{pK} \approx 6$). The reactions are the same for the chenocholic acid series of bile salts.

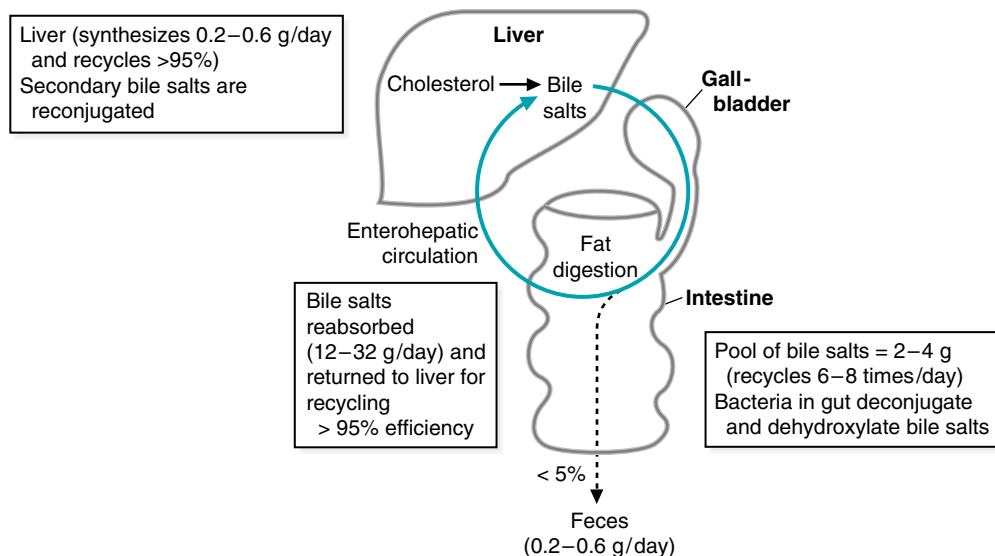


Fig. 34.12. Overview of bile salt metabolism.

action (Fig. 34.13). Lithocholic acid, a secondary bile salt that has a hydroxyl group only at position 3, is the least soluble bile salt. Its major fate is excretion.

Greater than 95% of the bile salts are resorbed in the ileum and return to the liver via the enterohepatic circulation (via the portal vein; see Fig. 34.12). The secondary bile salts may be reconstituted in the liver, but they are not rehydroxylated. The bile salts are recycled by the liver, which secretes them into the bile. This enterohepatic recirculation of bile salts is extremely efficient. Less than 5% of the bile salts entering the gut are excreted in the feces each day. Because the steroid nucleus cannot be degraded in the body, the excretion of bile salts serves as a major route for removal of the steroid nucleus and, thus, of cholesterol from the body.

VI. TRANSPORT OF CHOLESTEROL BY THE BLOOD LIPOPROTEINS

Because they are hydrophobic and essentially insoluble in the water of the blood, cholesterol and cholesterol esters, like triacylglycerols and phospholipids, must be transported through the bloodstream packaged as lipoproteins. These macromolecules are water-soluble. Each lipoprotein particle is composed of a core of hydrophobic lipids such as cholesterol esters and triacylglycerols surrounded by a shell of polar lipids (the phospholipids), which allows a hydration shell to form around the lipoprotein (see Fig. 32.9). This occurs when the positive charge of the nitrogen atom of the phospholipid (phosphatidylcholine, phosphatidylethanolamine, or phosphatidylserine) forms an ionic bond with the negatively charged hydroxyl ion of the environment. In addition, the shell contains a variety of apoproteins that also increase the water solubility of the lipoprotein. Free cholesterol molecules are dispersed throughout the lipoprotein shell to stabilize it in a way that allows it to maintain its spherical shape. The major carriers of lipids are chylomicrons (see Chapter 32), VLDL, and HDL. Metabolism of VLDL will lead to IDL and LDL. Metabolism of chylomicrons leads to chylomicron remnant formation.

Through this carrier mechanism, lipids leave their tissue of origin, enter the bloodstream, and are transported to the tissues, where their components will be either used in synthetic or oxidative process or stored for later use. The apoproteins (“apo” describes the protein within the shell of the particle in its lipid-free form) not only add to the hydrophilicity and structural stability of the particle but have other functions as well: (1) they activate certain enzymes required for normal lipoprotein metabolism and (2) they act as ligands on the surface of the lipoprotein that target specific receptors on peripheral tissues that require lipoprotein delivery for their innate cellular function.

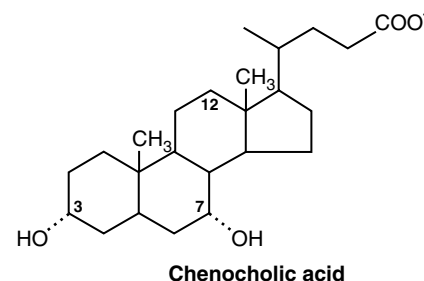
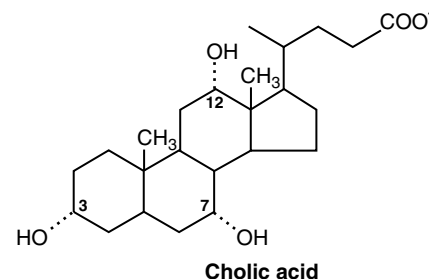
Ten principal apoproteins have been characterized. Their tissue source, molecular mass, distribution within lipoproteins, and metabolic functions are shown in Table 34.3.

The lipoproteins themselves are distributed among eight major classes. Some of their characteristics are shown in Table 34.4. Each class of lipoprotein has a specific function determined by its apolipoprotein content, its tissue of origin, and the proportion of the macromolecule made up of triacylglycerols, cholesterol esters, free cholesterol, and phospholipids (see Tables 34.3 and 34.4).

A. The Chylomicrons

Chylomicrons are the largest of the lipoproteins and the least dense because of their rich triacylglycerol content. They are synthesized from dietary lipids (the “exogenous” lipoprotein pathway) within the epithelial cells of the small intestine and then secreted into the lymphatic vessels draining the gut (see Fig. 32.13). They enter the bloodstream via the left subclavian vein. The major apoproteins of chylomicrons are apoB-48, apoC_{II}, and apoE (see Table 34.3). The apoC_{II} activates lipoprotein lipase

Primary bile salts



Secondary bile salts

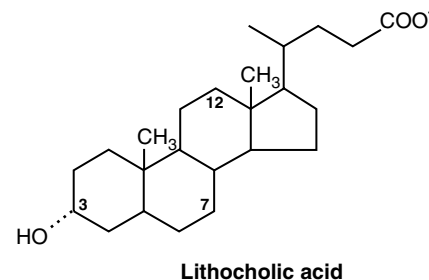
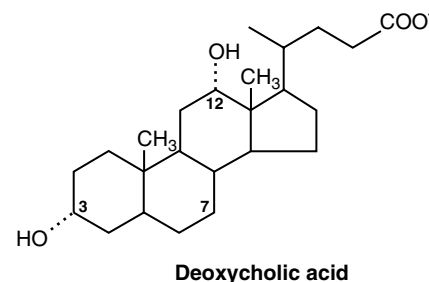


Fig. 34.13. Structures of the primary and secondary bile salts. Primary bile salts form conjugates with taurine or glycine in the liver. After secretion into the intestine, they may be deconjugated and dehydroxylated by the bacterial flora, forming secondary bile salts. Note that dehydroxylation occurs at position 7, forming the deoxy family of bile salts.

Table 34.3. CHARACTERISTICS OF THE MAJOR APOPROTEINS

Apoprotein	Primary Tissue Source	Molecular Mass (Daltons)	Lipoprotein Distribution	Metabolic Function
ApoA-1	Intestine, liver	28,016	HDL (chylomicrons)	Activates LCAT; structural component of HDL
ApoA-II	Liver	17,414	HDL (chylomicrons)	Unknown
ApoA-IV	Intestine	46,465	HDL (chylomicrons)	Unknown (may facilitate transport of other apoproteins between HDL and chylomicrons)
ApoB-48	Intestine	264,000	Chylomicrons	Assembly and secretion of chylomicrons from small bowel
ApoB-100	Liver	540,000	VLDL, IDL, LDL	VLDL assembly and secretion structured protein of VLDL, IDL, and LDL ligand for LDL receptor
ApoC-1	Liver	6,630	Chylomicrons, VLDL, IDL, HDL	Unknown; may inhibit hepatic uptake of chylomicron and VLDL remnants
ApoC-II	Liver	8,900	Chylomicrons, VLDL, IDL, HDL	Cofactor activator of lipoprotein lipase (LPL)
ApoC-III	Liver	8,800	Chylomicrons, VLDL, IDL, HDL	Inhibitor of LPL; may inhibit hepatic uptake of chylomicrons and VLDL remnants
ApoE	Liver	34,145	Chylomicron remnants, VLDL, IDL, HDL	Ligand for binding of several lipoproteins to the LDL receptor, to the LDL receptor-related protein (LRP) and possibly to a separate apo-E receptor.
Apo(a)	Liver		Lipoprotein "little" a (Lp(a))	Unknown

(LPL), an enzyme that projects into the lumen of capillaries in adipose tissue, cardiac muscle, skeletal muscle, and the acinar cells of mammary tissue. This activation allows LPL to hydrolyze the chylomicrons, leading to the release of free fatty acids derived from core triacylglycerides of the lipoprotein into these target cells. The muscle cells then oxidize the fatty acids as fuel while the adipocytes and mammary cells store them as triacylglycerols (fat) or, in the case of the lactating breast, use them for milk formation. The partially hydrolyzed chylomicrons remaining in the bloodstream (the **chylomicron remnants**), now partly depleted of their core triacylglycerols, retain their apoE and apoB48 proteins. Receptors in the plasma membranes of the liver cells bind to apoE on the surface of these remnants, allowing them to be taken up by the liver through a process of receptor-mediated endocytosis (see below).

B. Very-Low-Density Lipoproteins (VLDL)

If dietary intake of fatty acids exceeds the immediate fuel requirements of the liver, the excess fatty acids are converted to triacylglycerols, which, along with free and esterified cholesterol, phospholipids, and a variety of apoproteins (see Table 34.3),

Table 34.4. CHARACTERISTICS OF THE MAJOR LIPOPROTEINS

Lipoprotein	Density	Particle	Electrophoretic Mobility	Lipid (%)*			Function
	Range (g/mL)	Diameter (MM) range		TG	Chol	PL	
Chylomicrons	0.930	75–1200	Origin	80–95	2–7	3–9	Deliver dietary lipids
Chylomicron remnants	0.930–1.006	30–80	Slow pre β				Return dietary lipids to the liver
VLDL	0.930–1.006	30–80	Pre β	55–80	5–15	10–20	Deliver endogenous lipids
IDL	1.006–1.019	25–35	Slow pre β	20–50	20–40	15–25	Return endogenous lipids to the liver; precursor of LDL
LDL	1.019–1.063	18–25	β	5–15	40–50	20–25	Deliver cholesterol to cells
HDL ₂	1.063–1.125	9–12	α	5–10	15–25	20–30	Reverse cholesterol transport
HDL ₃	1.125–1.210	5–9	α				Reverse cholesterol transport
Lip(a)	1.050–1.120	25	Pre β				

*The remaining percent composition is composed of apoproteins.

Abbreviations: TG, Triacylglycerols; Chol, the sum of free and esterified cholesterol; PL, phospholipid; VLDL = very-low-density lipoproteins; IDL, intermediate-density lipoproteins; LDL, low-density lipoproteins; HDL, high-density lipoproteins.

including apoB-100, apoC_{II}, and apoE, are packaged to form VLDL. These particles are then secreted from the liver (the “endogenous” pathway of lipoprotein metabolism) into the bloodstream (Fig. 34.14). The density, particle size, and lipid content of VLDL particles are given in Table 34.3. These particles are then transported from the hepatic veins to capillaries in skeletal and cardiac muscle and adipose tissue, as well as lactating mammary tissues, where lipoprotein lipase is activated by apoC_{II} in the VLDL particles. The activated enzyme facilitates the hydrolysis of the triacylglycerol in VLDL, causing the release of fatty acids and glycerol from a portion of core triacylglycerols. These fatty acids are oxidized as fuel by muscle cells, used in the resynthesis of triacylglycerols in fat cells, and used for milk production in the lactating breast. The residual particles remaining in the bloodstream are called **VLDL remnants**. Approximately 50% of these remnants are taken up from the blood by liver cells through the binding of VLDL apoE to the hepatocyte plasma membrane apoE receptor followed by endocytic internalization of the VLDL remnant.

C. Intermediate-Density Lipoprotein (IDL) and Low-Density Lipoproteins (LDL)

Approximately half of the VLDL remnants are not taken up by the liver but, instead, have additional core triacylglycerols removed to form IDL, a specialized class of VLDL remnants. With the removal of additional triacylglycerols from IDL through the action of hepatic triglyceride lipase within hepatic sinusoids, LDL is generated from IDL. As seen in Table 34.4, the LDL particles are rich in cholesterol and cholesterol esters. Approximately 60% of the LDL is transported back to the liver, where its apoB-100 binds to specific apoB-100 receptors in the liver cell plasma membranes, allowing particles to be endocytosed into the hepatocyte. The remaining 40% of LDL particles are carried to extrahepatic tissues such as adrenocortical

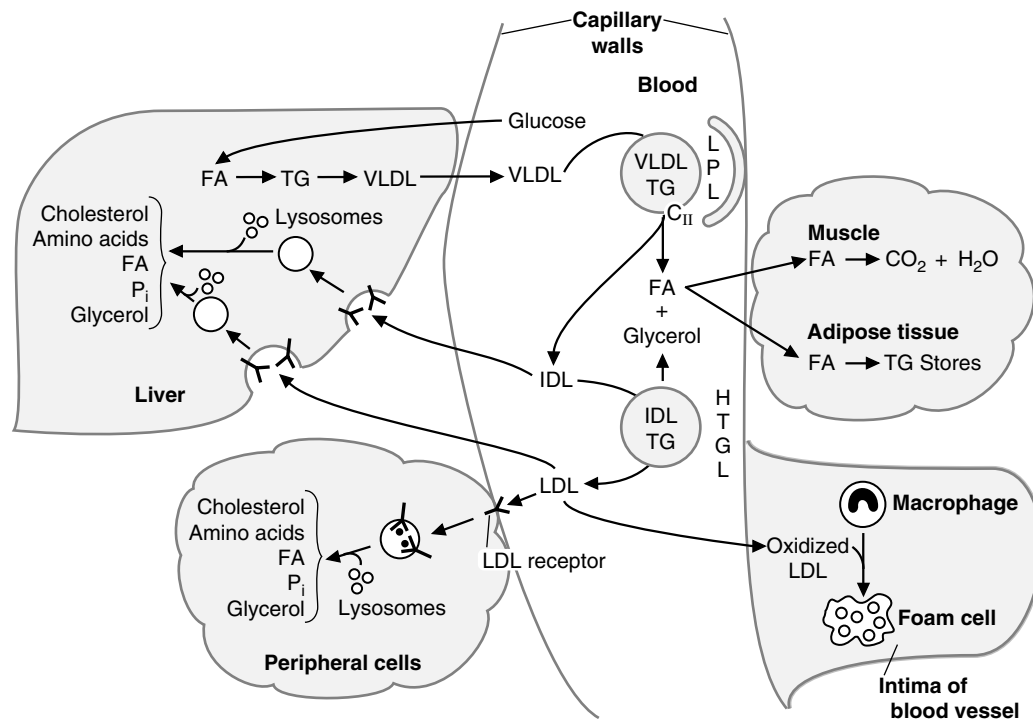


Fig. 34.14. Fate of VLDL. VLDL triacylglycerol (TG) is degraded by LPL, forming IDL. IDL can either be endocytosed by the liver through a receptor-mediated process or further digested, mainly by hepatic triacylglycerol lipase (HTGL), to form LDL. LDL may be endocytosed by receptor-mediated processes in the liver or in peripheral cells. LDL also may be oxidized and taken up by “scavenger” receptors on macrophages. The scavenger pathway plays a role in atherosclerosis. FA = fatty acids; Pi = inorganic phosphate.

and gonadal cells that also contain apoB-100 receptors, allowing them to internalize the LDL particles and use their cholesterol for the synthesis of steroid hormones. Some of the cholesterol of the internalized LDL is used for membrane synthesis and vitamin D synthesis as well. If an excess of LDL particles is present in the blood, this specific receptor-mediated uptake of LDL by hepatic and nonhepatic tissue becomes saturated. The “excess” LDL particles are now more readily available for nonspecific uptake of LDL by macrophages (scavenger cells) present near the endothelial cells of arteries. This exposure of vascular endothelial cells to high levels of LDL is believed to induce an inflammatory response by these cells, a process suggested to initiate the complex cascade of atherosclerosis discussed below.

D. High-Density Lipoprotein (HDL)

The fourth class of lipoproteins is HDL, which plays several roles in whole body lipid metabolism.

1. SYNTHESIS OF HDL

HDL particles can be created by a number of mechanisms. The first is synthesis of nascent HDL by the liver and intestine as a relatively small molecule whose shell, like that of other lipoproteins, contains phospholipids, free cholesterol, and a variety of apoproteins, predominant among which are apoA1, apoAII, apoC_I, and apoC_{II} (see Table 34.3). Very low levels of triacylglycerols or cholesterol esters are found in the hollow core of this early, or nascent, version of HDL.

A second method for HDL generation is the budding of apoproteins from chylomicrons and VLDL particles as they are digested by lipoprotein lipase. The apoproteins (particularly AI) and shells can then accumulate more lipid, as described below.

A third method for HDL generation is free apoprotein AI, which may be shed from other circulating lipoproteins. AI will acquire cholesterol and phospholipids from other lipoproteins and cell membranes, to form a nascent-like HDL particle within the circulation.

2. MATURATION OF NASCENT HDL

In the process of maturation, the nascent HDL particles accumulate phospholipids and cholesterol from cells lining the blood vessels. As the central hollow core of nascent HDL progressively fills with cholesterol esters, HDL takes on a more globular shape to eventually form the mature HDL particle. The transfer of lipids to nascent HDL does not require enzymatic activity.

3. REVERSE CHOLESTEROL TRANSPORT

A major benefit of HDL particles derives from their ability to remove cholesterol from cholesterol-laden cells and to return the cholesterol to the liver, a process known as reverse cholesterol transport. This is particularly beneficial in vascular tissue; by reducing cellular cholesterol levels in the subintimal space, the likelihood that foam cells (lipid-laden macrophages that engulf oxidized LDL-cholesterol and represent an early stage in the development of atherosclerotic plaque) will form within the blood vessel wall is reduced.

Reverse cholesterol transport requires a directional movement of cholesterol from the cell to the lipoprotein particle. Cells contain the protein ABC1 (ATP-binding cassette protein 1) which uses ATP hydrolysis to move cholesterol from the inner leaflet of the membrane to the outer leaflet. Once the cholesterol has reached the outer membrane leaflet, the HDL particle can accept it, but if the cholesterol is not modified within the HDL particle, the cholesterol can leave the particle by the same route that it entered. To trap the cholesterol within the HDL core, the HDL particle acquires the

enzyme LCAT from the circulation (LCAT is synthesized and secreted by the liver). LCAT catalyzes the transfer of a fatty acid from the 2-position of lecithin (phosphatidylcholine) in the phospholipid shell of the particle to the 3-hydroxyl group of cholesterol, forming a cholesterol ester (Fig. 34.15). The cholesterol ester migrates to the core of the HDL particle and is no longer free to return to the cell.

Elevated levels of lipoprotein-associated cholesterol in the blood, particularly that associated with LDL but also that in the more triacylglycerol-rich lipoproteins, are associated with the formation of cholesterol-rich atheromatous plaque in the vessel wall, eventually leading to diffuse atherosclerotic vascular disease resulting in acute cardiovascular events, such as a myocardial infarction, a stroke, or symptomatic peripheral vascular insufficiency. High levels of HDL in the blood, therefore, are believed to be vasculoprotective, because these high levels increase the rate of reverse cholesterol transport “away” from the blood vessels and “toward” the liver (“out of harm’s way”).

4. FATE OF HDL CHOLESTEROL

Mature HDL particles can bind to specific receptors on hepatocytes (such as the apoE receptor), but the primary means of clearance of HDL from the blood is through its uptake by the scavenger receptor SR-B1. This receptor is present on many cell types. It does not carry out endocytosis per se, but once the HDL particle



Two genetically determined disorders, familial HDL deficiency and Tangier disease, result from mutations in the ATP-binding cassette 1 (ABC 1) protein. Cholesterol-depleted HDL cannot transport free cholesterol from cells that lack the ability to express this protein. As a consequence, HDL is rapidly degraded. These disorders have established a role for ABC 1 protein in the regulation of HDL levels in the blood.



Because **Ann Jeina** continued to experience intermittent chest pain, in spite of good control of her hypertension and a 20-lb weight loss, her physician decided that a 2-drug regimen to lower her blood LDL cholesterol level must be added to the dietary measures already in place. Consequently, treatment with cholestyramine, a resin that binds some of the bile salts in the intestinal lumen, and the HMG-CoA reductase inhibitor pravastatin was initiated.

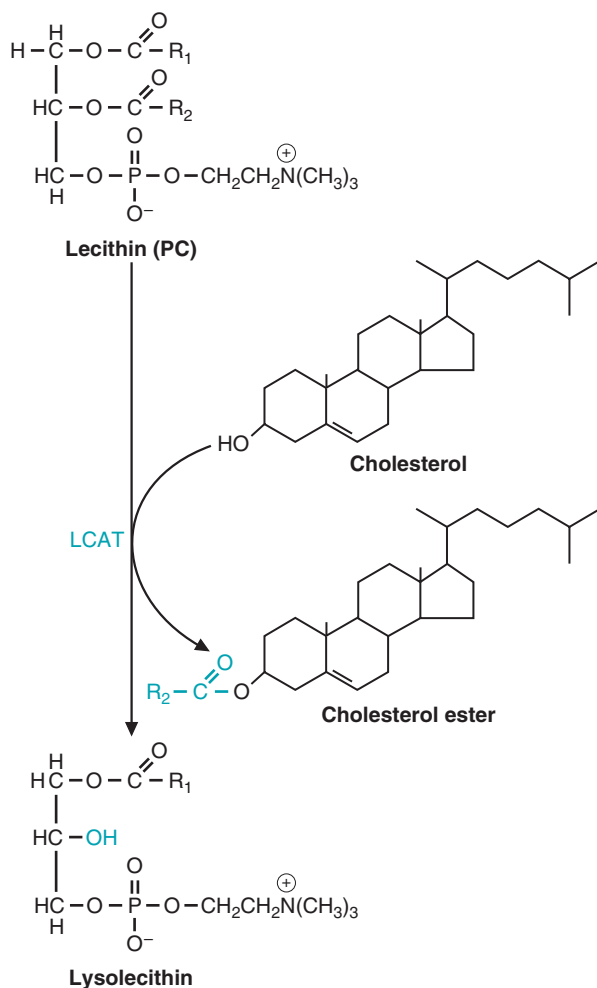


Fig. 34.15. The reaction catalyzed by LCAT. R1 = saturated fatty acid. R2 = unsaturated fatty acid.

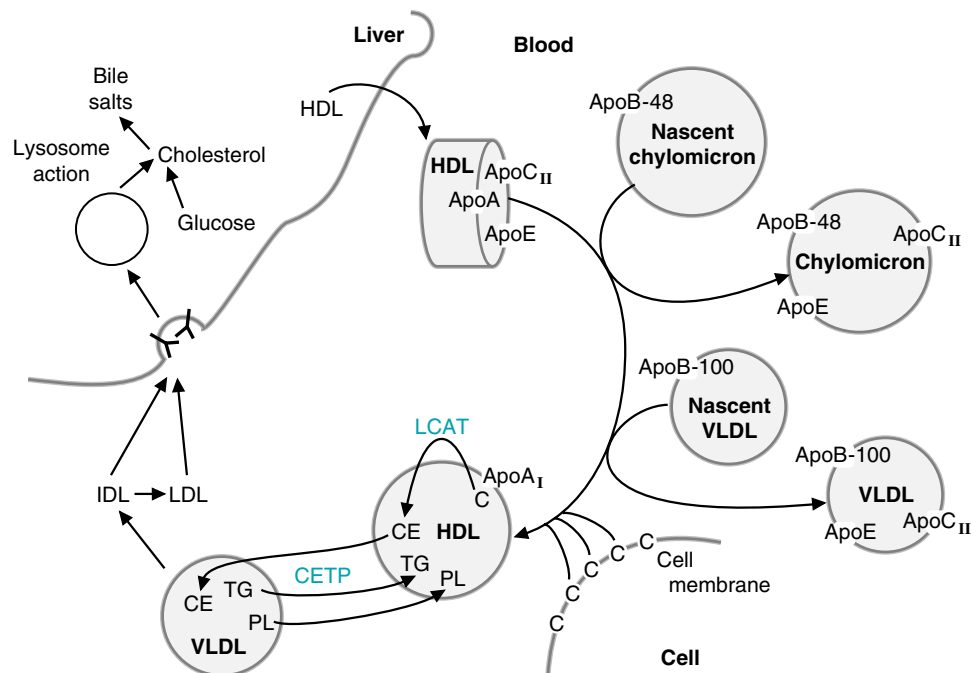


Fig. 34.16. Functions and fate of HDL. Nascent HDL is synthesized in liver and intestinal cells. It exchanges proteins with chylomicrons and VLDL. HDL picks up cholesterol (C) from cell membranes. This cholesterol is converted to cholesterol ester (CE) by the LCAT reaction. HDL transfers CE to VLDL in exchange for triacylglycerol (TG). The cholesterol ester transfer protein (CETP) mediates this exchange. PL = phospholipids.

is bound to the receptor, its cholesterol and cholesterol esters are transferred into the cells. When depleted of cholesterol and its esters, the HDL particle dissociates from the SR-B1 receptor and re-enters the circulation. SR-B1 receptors can be upregulated in certain cell types that require cholesterol for biosynthetic purposes, such as the cells that produce the steroid hormones. The SR-B1 receptors are not downregulated when cholesterol levels are high.

5. HDL INTERACTIONS WITH OTHER PARTICLES

In addition to its ability to pick up cholesterol from cell membranes, HDL also exchanges apoproteins and lipids with other lipoproteins in the blood. For example, HDL transfers apolipoprotein E (apoE) and apolipoprotein CII (apoCII) to chylomicrons and to VLDL. The apoCII stimulates the degradation of the triacylglycerols of chylomicrons and VLDL by activating LPL (Fig. 34.16). After digestion of the chylomicrons and the VLDL triacylglycerols, apoE and apoCII are transferred back to HDL. When HDL obtains free cholesterol from cell membranes, the free cholesterol is esterified at the third carbon of the A ring via the LCAT reaction (see Fig. 34.14). From this point, HDL either transports the free cholesterol and cholesterol esters directly to the liver, as described above, or by CETP to circulating triacylglycerol-rich lipoproteins such as VLDL and VLDL remnants (see Fig. 34.16). In exchange, triacylglycerols from the latter lipoproteins are transferred to HDL (Fig. 34.17). The greater the concentration of triacylglycerol-rich lipoproteins in the blood, the greater the rate of these exchanges. Thus, the CETP exchange pathway may explain the observation that whenever triacylglycerol-rich lipoproteins are present in the blood in high concentrations, the amount of cholesterol reaching the liver via cholesterol-enriched VLDL and VLDL remnants increases, and a proportional reduction in the total amount of cholesterol and cholesterol esters that are directly transferred to the liver via HDL occurs. Mature

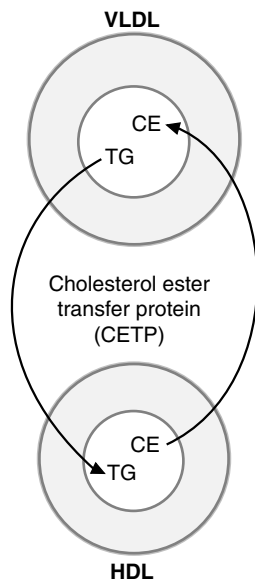


Fig. 34.17. Function of cholesterol ester transfer protein (CETP). CETP transfers cholesterol esters (CE) from HDL to VLDL in exchange for triacylglycerol (TG).

HDL particles are designated as HDL₃; after the CETP reaction and loss of cholesterol and gain of triacylglycerol, the particles become larger and are designated as HDL₂ particles (see Table 34.4).

VII. LIPOPROTEINS ENTER CELLS BY RECEPTOR-MEDIATED ENDOCYTOSIS

As stated earlier, each lipoprotein particle contains specific apoproteins on its surface that act as ligands for specific plasma membrane receptors on target tissues such as the liver, the adrenal cortex, the gonads, and other cells that require one or more of the components of the lipoproteins. With the exception of the scavenger receptor SR-B1, the interaction of ligand and receptor initiates the process of endocytosis depicted for LDL in Figure 34.18. The receptors for LDL, for example, are found in specific areas of the plasma membrane of the target cell for circulating lipoproteins. These are known as coated pits, and they contain a unique protein called clathrin. The plasma membrane in the vicinity of the receptor–LDL complex invaginates and fuses to form an endocytic vesicle. These vesicles then fuse with lysosomes, acidic subcellular vesicles that contain a number of degradative enzymes. The cholesterol esters of LDL are hydrolyzed to form free cholesterol, which is rapidly reesterified through the action of ACAT. This rapid reesterification is necessary to avoid the damaging effect of high levels of free cholesterol on cellular membranes. The newly esterified cholesterol contains primarily oleate or palmitoleate (monounsaturated fatty acids), unlike those of the cholesterol esters in LDL, which are rich in linoleate, a polyunsaturated fatty acid.

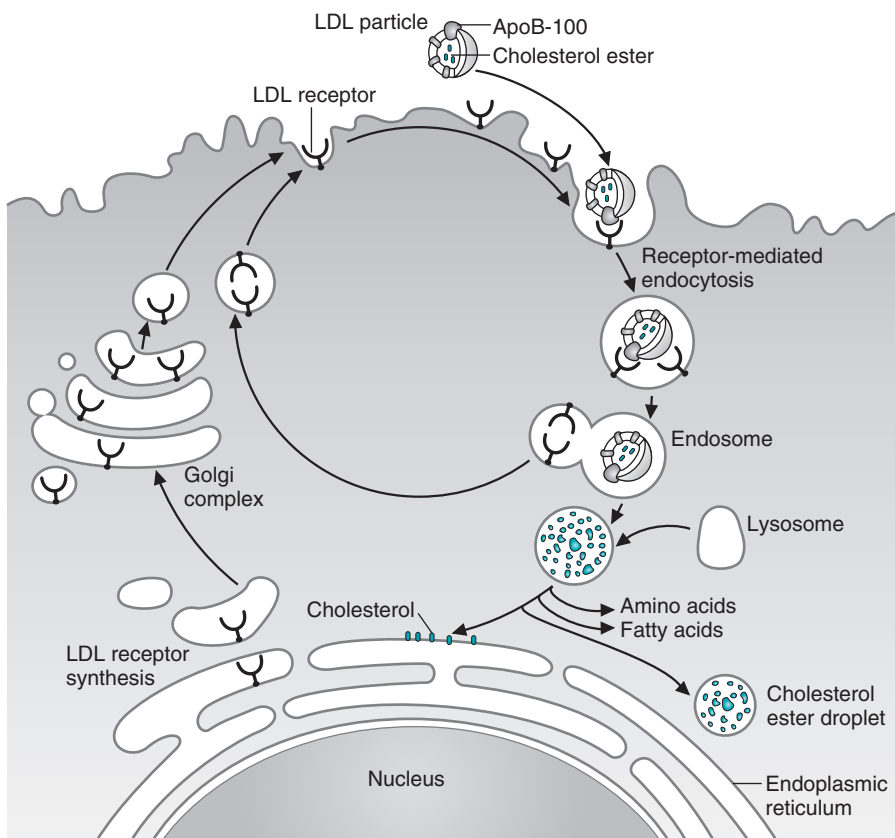


Fig. 34.18. Cholesterol uptake by receptor-mediated endocytosis.

As is true for the synthesis and activity of HMG CoA reductase, the synthesis of the LDL receptor itself is subject to feedback inhibition by increasing levels of cholesterol within the cell. One probable mechanism for this feedback regulation involves one or more of the SREBP described earlier. These proteins or the cofactors that are required for the full expression of genes that code for the LDL receptor are also capable of sensing the concentration of sterols within the cell. When sterol levels are high, the process that leads to the binding of the SREBP to the sterol regulatory element of these genes is suppressed (see Fig. 34.4). The rate of synthesis from mRNA for the LDL receptor is diminished under these circumstances. This, in turn, appropriately reduces the amount of cholesterol that can enter these cholesterol-rich cells by receptor-mediated endocytosis (downregulation of receptor synthesis). When the intracellular levels of cholesterol decrease, these processes are reversed, and cells act to increase their cholesterol levels. Both synthesis of cholesterol from acetyl CoA and synthesis of LDL receptors are stimulated. An increased number of receptors (upregulation of receptor synthesis) results in an increased uptake of LDL cholesterol from the blood, with a subsequent reduction of LDL-cholesterol levels. At the same time, the cellular cholesterol pool is replenished.

VIII. LIPOPROTEIN RECEPTORS

The best-characterized lipoprotein receptor, the LDL receptor, specifically recognizes apoB-100 and apo E. Therefore, this receptor binds VLDL, IDL, and chylomicron remnants in addition to LDL. The binding reaction is characterized by its saturability and occurs with high affinity and a narrow range of specificity. Other receptors, such as the LDL receptor-related proteins (LRP) and the macrophage scavenger receptor (notably types SR-A1 and SR-A2, which are located primarily near the endothelial surface of vascular endothelial cells), have broad specificity and bind many other ligands in addition to the blood lipoproteins.

A. The LDL Receptor

The LDL receptor has a mosaic structure encoded by a gene that was assembled by a process known as exon shuffling. It is composed of six different regions (Fig. 34.19). The first region, at the amino terminus, contains the LDL-binding region, a cysteine-rich sequence of 40 residues. Acidic side chains in this region bind ionic calcium. When these side chains are protonated, calcium is released from its binding sites. This release leads to conformational changes that allow the LDL to disconnect from its receptor docking site. Disulfide bonds, formed from the cysteine residues, have a stabilizing influence on the structural integrity of this portion of the receptor.

The second region of the receptor contains domains that are homologous with epidermal growth factor (EGF) as well as a complex consisting of six repeats that resemble the blades of the transducin beta subunit forming a propeller-like moiety.

The third region of the LDL receptor contains a chain of N-linked oligosaccharides, whereas the fourth region contains a domain that is rich in serine and threonine and contains O-linked sugars. This region may have a role in physically extending the receptor away from the membrane so that the LDL-binding region is accessible to the LDL molecule.

The fifth region contains 22 hydrophobic residues constituting the membrane-spanning unit of the receptor, whereas the sixth region extends into the cytosol, where it regulates the interaction between the C-terminal domain of the LDL receptor and the clathrin-containing coated pit where the process of receptor-mediated endocytosis is initiated.

The number of LDL receptors, the binding of LDL to its receptors, and the postreceptor binding process can be diminished for a variety of reasons, all of which

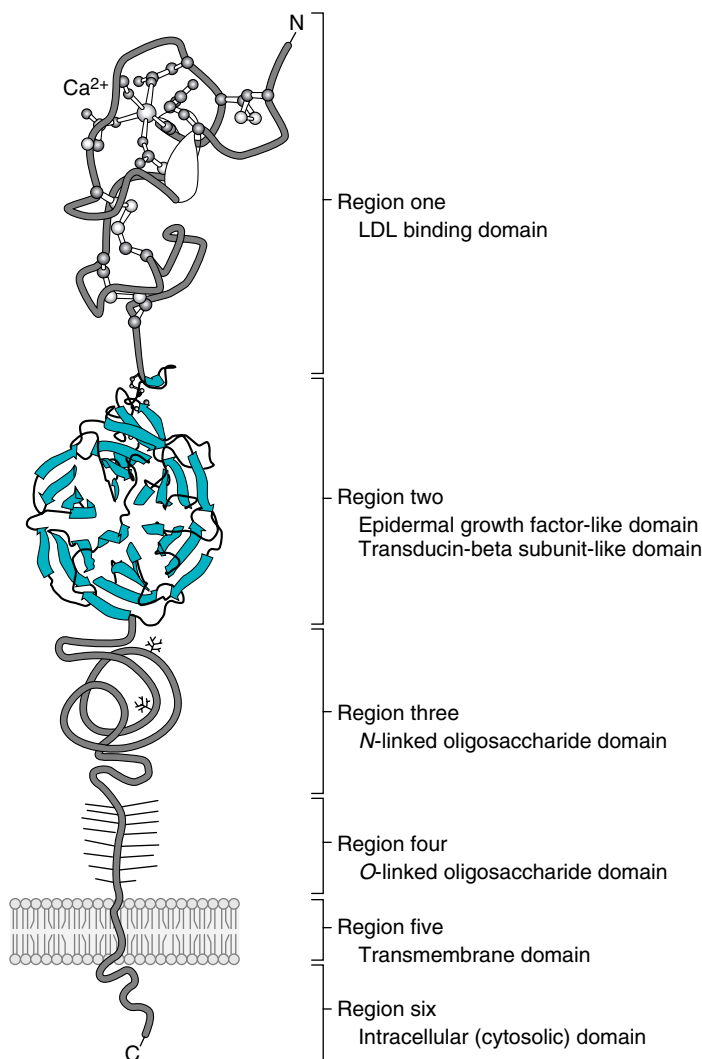


Fig. 34.19. Structure of the LDL receptor. The protein has six major regions.

may lead to an accumulation of LDL cholesterol in the blood and premature atherosclerosis. These abnormalities can result from mutations in one (heterozygous—seen in approximately 1 in 500 people) or both (homozygous—seen in about 1 in 1 million people) alleles for the LDL receptor (familial hypercholesterolemia). Heterozygotes produce approximately half of the normal complement of LDL receptors, whereas the homozygotes produce almost no LDL receptor protein (receptor negative familial hypercholesterolemia). The latter have serum total cholesterol levels in the range of 500 to 800 mg/dL. In a subset of patients with familial hypercholesterolemia, the LDL receptor is normally synthesized and transported to the cell surface, but an amino acid substitution or deletion leads to changes in the protein's structure such that LDL-binding is impaired. As a result, cholesterol does not enter the target cell from the bloodstream and, therefore, cannot feedback negatively on cholesterol biosynthesis in the cell. As a result, the serum cholesterol level rises. A third form of familial hypercholesterolemia involves a genetic defect in the transport or migration mechanism that normally delivers the LDL receptor from its point of synthesis within the cell to its proper location in that cell's plasma membrane. Genetic mutations can lead to yet another form of familial hypercholesterolemia in which a structural change occurs in the carboxy terminus of the LDL receptor.



Ann Jeina's blood lipid levels (in mg/dL) were:

Triacylglycerol	158
Total cholesterol	420
HDL cholesterol	32
LDL cholesterol	356

She was diagnosed as having familial hypercholesterolemia (FH) type IIA, which is caused by genetic defects in the gene that encodes the LDL receptor (see Biochemical Comments). As a result of the receptor defect, LDL cannot readily be taken up by cells, and its concentration in the blood is elevated.

LDL particles contain a high percentage, by weight, of cholesterol and cholesterol esters, more than other blood lipoproteins. However, LDL triacylglycerol levels are low because LDL is produced by digestion of the triacylglycerols of VLDL and IDL. Therefore, individuals with a type IIA hyperlipoproteinemia have very high blood cholesterol levels, but their levels of triacylglycerols may be in or near the normal range (see Table 34.4).



Ivan Applebod's blood lipid levels were:

Triacylglycerol	295
Total cholesterol	314
HDL cholesterol	24
LDL cholesterol	231

The elevated serum levels of LDL cholesterol found in patients such as **Ivan Applebod** who have type 2 diabetes mellitus is multifactorial. One of the mechanisms responsible for this increase involves the presence of chronically elevated levels of glucose in the blood of poorly controlled diabetics. This prolonged hyperglycemia increases the rate of nonenzymatic attachment of glucose to various proteins in the body, a process referred to as glycation or glycosylation of proteins.

Glycation may adversely affect the structure or the function of the protein involved. For example, glycation of the LDL receptor and of proteins in the LDL particle may interfere with the normal “fit” of LDL particles with their specific receptors. As a result, less circulating LDL is internalized into cells by receptor-mediated endocytosis, and the serum LDL cholesterol level rises.

Although such patients are able to place structurally normal LDL receptors in the plasma membrane of the cell, they are unable to internalize the LDL–LDL receptor complex because they cannot properly translocate the complex into the clathrin-containing coated pits. The spectrum of mutations of the LDL receptor gene is shown in Figure 34.20.

B. LDL Receptor-Related Protein (LRP)

LRP is structurally related to the LDL receptor but recognizes a broader spectrum of ligands. In addition to lipoproteins, it binds the blood proteins α_2 -macroglobulin (a protein that inhibits blood proteases) and tissue plasminogen activator (TPA) and its inhibitors. The LRP receptor recognizes the apoE of lipoproteins and binds remnants produced by the digestion of the triacylglycerols of chylomicrons and VLDL by LPL. Thus, one of its functions is believed to be the clearance of these remnants from the blood. The LRP receptor is abundant in the cell membranes of the liver, brain, and placenta. In contrast to the LDL receptor, synthesis of the LRP receptor is not significantly affected by an increase in the intracellular concentration of cholesterol. However, insulin causes the number of these receptors on the cell surface to increase, consistent with the need for removal of chylomicron remnants that otherwise would accumulate after eating a meal.

C. Macrophage Scavenger Receptor

Some cells, particularly the phagocytic macrophages, have nonspecific receptors known as “scavenger” receptors that bind various types of molecules, including oxidatively modified LDL particles. There are a number of different types of scavenger receptors. SR-B1 is used primarily for HDL binding, whereas the scavenger receptors expressed on macrophages are SR-A1 and SR-A2. Modification of LDL frequently involves oxidative damage, particularly of polyunsaturated fatty acyl groups (see Chapter 24). In contrast to the LDL receptors, the scavenger receptors

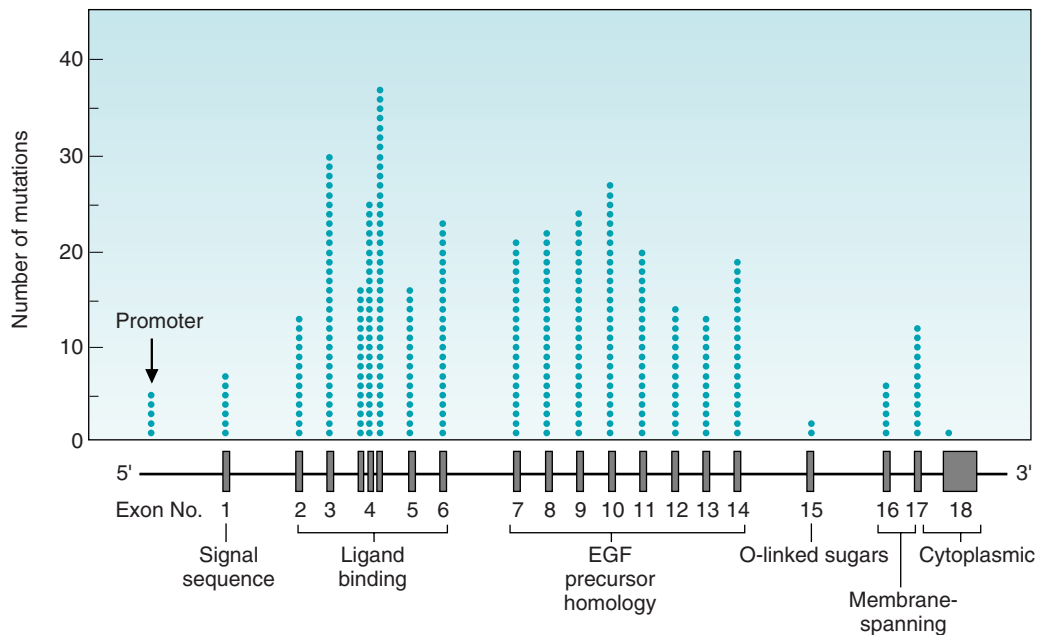


Fig. 34.20. Location of 353 point mutations and small deletions/insertions (<25 bp) in the LDL receptor gene in individuals with familial hypercholesterolemia (FH). Exons are shown as vertical boxes and introns as the lines connecting them. The figure was obtained from Goldstein JL, Hobbs HH, Brown MS. Familial hypercholesterolemia. In: Scriver CR, Beaudet AL, Sly WS, Valle D, et al., eds. *The Metabolic and Molecular Bases of Inherited Disease*. 8th Ed., vol III. New York: McGraw-Hill, 2001:2863–2913.

are not subject to downregulation. The continued presence of scavenger receptors in the cell membrane allows the cells to take up oxidatively modified LDL long after intracellular cholesterol levels are elevated. When the macrophages become engorged with lipid, they are called foam cells. An accumulation of these foam cells in the subendothelial space of blood vessels form the earliest gross evidence of a developing atherosclerotic plaque known as a fatty streak.

The processes that cause oxidation of LDL involve superoxide radicals, nitric oxide, hydrogen peroxide, and other oxidants (see Chapter 24). Antioxidants, such as vitamin E, ascorbic acid (vitamin C), and carotenoids, may be involved in protecting LDL from oxidation.

IX. ANATOMIC AND BIOCHEMICAL ASPECTS OF ATHEROSCLEROSIS

The normal artery is composed of three distinct layers (Fig. 34.21). That which is closest to the lumen of the vessel, the intima, is lined by a monolayer of endothelial cells that are bathed by the circulating blood. Just beneath these specialized cells lies the subintimal extracellular matrix, in which some vascular smooth muscle cells are embedded (the subintimal space). The middle layer, known as the tunica media, is separated from the intima by the internal elastic lamina. The tunica media contains lamellae of smooth muscle cells surrounded by an elastin- and collagen-rich matrix. The external elastic lamina forms the border between the tunica media and the outermost layer, the adventitia. This layer contains nerve fibers and mast cells. It is the origin of the vasa vasorum, which supply blood to the outer two thirds of the tunica media.

The initial step in the development of an atherosclerotic lesion within the wall of an artery is the formation of a fatty streak. The latter represents an accumulation of lipid-laden macrophages or foam cells in the subintimal space. These fatty streaks are visible as a yellow-white linear streak that bulges slightly into the lumen of the vessel. These streaks are initiated when one or more known “vascular risk factors for atherosclerosis,” all of which have the potential to injure the vascular endothelial cells, reach a critical threshold at the site of future lesions. Examples of such risk factors include elevated intra-arterial pressure (arterial hypertension), elevated circulating levels of various lipids such as LDL, chylomicron remnants, and VLDL remnants, or low levels of circulating HDL, cigarette smoking, chronic elevations in blood glucose levels, high circulating levels of the vasoconstricting octapeptide angiotensin II, and others. The resulting insult to

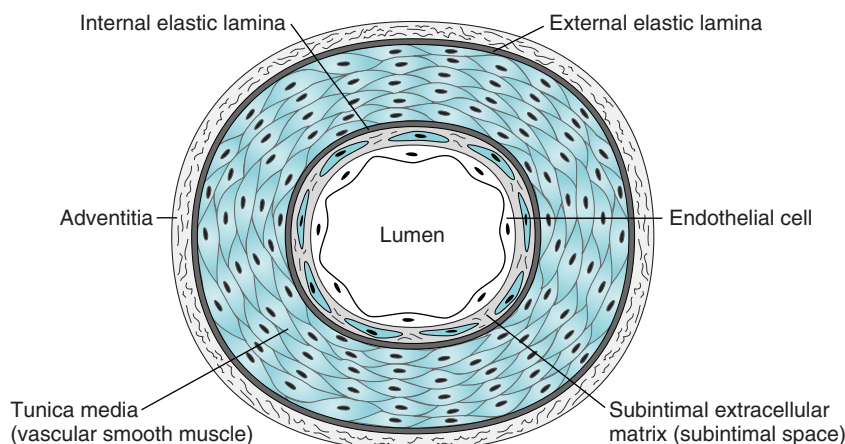


Fig. 34.21. The different layers of the arterial wall.



In addition to dietary therapy, aimed at reducing her blood cholesterol levels, **Ann Jeina** was treated with pravastatin, an HMG-CoA reductase inhibitor. The HMG-CoA reductase inhibitors decrease the rate of synthesis of cholesterol in cells. As cellular cholesterol levels decrease, the synthesis of LDL receptors increases. As the number of receptors rises on the cell surface, the uptake of LDL is increased. Consequently, the blood level of LDL cholesterol decreases.



HDL is considered to be the “good cholesterol,” because it accepts free cholesterol from peripheral tissues, such as cells in the walls of blood vessels. This cholesterol is converted to cholesterol ester, part of which is transferred to VLDL by CETP, and returned to the liver by IDL and LDL. The remainder of the cholesterol is transferred directly as part of the HDL molecule to the liver. The liver reutilizes the cholesterol in the synthesis of VLDL, converts it to bile salts, or excretes it directly into the bile. HDL therefore tends to lower blood cholesterol levels. Lower blood cholesterol levels correlate with a lower rates of death of atherosclerosis.



In patients such as **Ann Jeina** and **Ivan Applebod**, who have elevated levels of VLDL or LDL, HDL levels are often low. These patients are predisposed to atherosclerosis and suffer from a high incidence of heart attacks and strokes.

Exercise and estrogen administration both increase HDL levels. This is one of the reasons exercise is often recommended to aid in the prevention or treatment of heart disease, and estrogen replacement therapy (ERT) is often prescribed for postmenopausal women. Before menopause, the incidence of heart attacks is relatively low in women, but it rises after menopause and increases to the level found in men by the age of 65 or 70 years. Moderate consumption of ethanol (alcohol) has also been correlated with increased HDL levels. Recent studies suggest that the beneficial amount of ethanol may be quite low, about two small glasses of wine a day, and that beneficial effects ascribed to ethanol may result from other components of wine and alcoholic beverages. In spite of the evidence that postmenopausal estrogen replacement therapy decreases circulating levels of LDL and increases HDL levels, recent data suggest that ERT may actually increase the rate of atherosclerotic vascular disease in these women. As a result, the accepted indications for ERT are now limited to intolerable “hot flashes” or vaginal dryness.

endothelial cells may trigger these cells to secrete adhesion molecules that bind to circulating monocytes and markedly slow their rate of movement past the endothelium. When sufficiently slowed, these monocytic cells accumulate and have access to the physical spaces that exist between endothelial cells. This accumulation of monocytic cells resembles the classical inflammatory response to injury. These changes have led to the suggestion that atherosclerosis is, in fact, an inflammatory disorder and, therefore, is one that might be prevented or attenuated through the use of anti-inflammatory agents such as acetylsalicylic acid (e.g., aspirin) and HMG CoA reductase inhibitors (statins), which have been shown to suppress the inflammatory cascade as well as to inhibit the action of HMG CoA reductase.

The monocytic cells are transformed into macrophages that migrate through the spaces between endothelial cells. They enter the subintimal space under the influence of chemoattractant cytokines (e.g., chemokine macrophage chemoattractant protein I) secreted by vascular cells in response to exposure to oxidatively modified fatty acids within the lipoproteins.

The macrophages can replicate and exhibit augmented expression of receptors that recognize oxidatively modified lipoproteins. Unlike the classic LDL receptors on liver and many nonhepatic cells, these macrophage-bound receptors are high-capacity, low-specificity receptors (scavenger receptors). They bind to and internalize oxidatively modified fatty acids within LDLs to become subintimal foam cells as described previously. As these foam cells accumulate, they deform the overlying endothelium, causing microscopic separations between endothelial cells, exposing these foam cells and underlying extracellular matrix to the blood. These exposed areas serve as sites for platelet adhesion and aggregation. Activated platelets secrete cytokines that perpetuate this process and increase the potential for thrombus (clot) formation locally. As the evolving plaque matures, a fibrous cap forms over its expanding “roof,” which now bulges into the vascular lumen, thereby partially occluding it. Vascular smooth muscle cells now migrate from the tunica media to the subintimal space and secrete additional plaque matrix material. The smooth muscle cells also secrete metalloproteinases that thin the fibrous cap near its “elbow” at the periphery of the plaque. This thinning progresses until the fibrous cap ruptures, allowing the plaque contents to physically contact the procoagulant elements present within the circulation. This leads to acute thrombus formation. If this thrombus completely occludes the remaining lumen of the vessel, an infarction of tissues distal to the occlusion (i.e., an acute myocardial infarction) may occur (Fig. 34.22). Most plaques that rupture also contain focal areas of calcification, which appears to result from the induction of the same cluster of genes as those that promote the formation of bone. The inducers for this process include oxidized sterols as well as transforming growth factor beta (TGF- β) derived from certain vascular cells.

Finally, high intraluminal shear forces develop in these thinning or eroded areas of the plaque’s fibrous cap, inducing macrophages to secrete additional metalloproteinases that further degrade the arterial-fibrous cap matrix. This contributes further to plaque rupture and thrombus formation (see Fig. 34.22). The consequence is a macrovascular ischemic event such as an acute myocardial infarction (AMI) or an acute cerebrovascular accident (CVA).



Lipoprotein(a) is essentially an LDL particle that is covalently bound to apoprotein(a). It is called “lipoprotein little a” to avoid confusion with the apoprotein A found in HDL. The structure of apoprotein(a) is very similar to that of plasminogen, a precursor of the protease plasmin that degrades fibrin, a major component of blood clots. Lipoprotein(a), however, cannot be converted to active plasmin. There are reports that high concentrations of lipoprotein(a) correlate with an increased risk of coronary artery disease, even in patients in whom the lipid profile is otherwise normal.

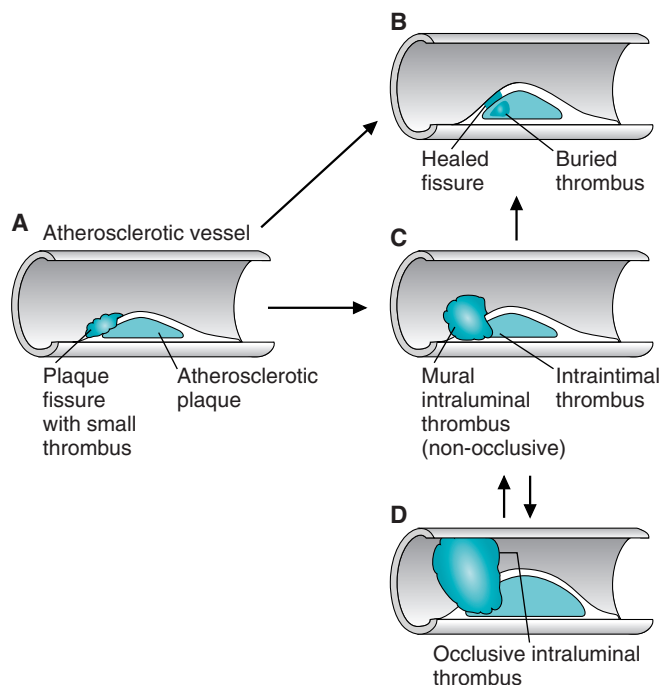


Fig. 34.22. Evolution of an atherosclerotic plaque. Plaque capsule eroded near the “elbow” of plaque creating an early plaque fissure (A), which may heal as plaque increases in size (B) or may grow as thrombus expands, having an intraluminal portion and an intraintimal portion (C). If the fissure is not properly sealed, the thrombus may grow and completely occlude the vessel lumen (D), causing an acute infarction of tissues downstream of the vessel occlusion.

X. STEROID HORMONES

Cholesterol is the precursor of all five classes of steroid hormones: glucocorticoids, mineralcorticoids, androgens, estrogens, and progestins. These hormones are synthesized in the adrenal cortex, ovaries, testes, and ovarian corpus luteum. Steroid hormones are transported through the blood from their sites of synthesis to their target organs, where, because of their hydrophobicity, they cross the cell membrane and bind to specific receptors in either the cytoplasm or nucleus. The bound receptors then bind to DNA to regulate gene transcription (see Chapter 16, section III.C.2, and Fig. 16.13). Because of their hydrophobicity, steroid hormones must be complexed with a serum protein. Serum albumin can act as a nonspecific carrier for the steroid hormones, but there are specific carriers as well. The cholesterol used for steroid hormone synthesis is either synthesized in the tissues from acetyl CoA, extracted from intracellular cholesterol ester pools, or taken up by the cell in the form of cholesterol-containing lipoproteins (either internalized by the LDL-receptor, or absorbed by the SR-B1 receptor). In general, glucocorticoids and progestins contain 21 carbons, androgens contain 19 carbons, and estrogens contain 18 carbons. The specific complement of enzymes present in the cells of an organ determines which hormones the organ can synthesize.

The oxidative reactions that lead to the synthesis and secretion of glucocorticoids such as cortisol are stimulated by adrenal corticotrophic hormone (ACTH). The role of cortisol as a stress-released hormone is discussed in Chapter 43.

Mineralocorticoids such as aldosterone are also synthesized in the adrenal cortex and are secreted in response to angiotensin II or III, rising potassium levels in the blood, or hyponatremia (low levels of sodium ions in the blood). Aldosterone stimulates sodium reuptake in the kidney, sweat glands, salivary glands, and other



Vera Leizd consulted her gynecologist, who confirmed that her problems were probably the result of an excess production of androgens (virilization) and ordered blood and urine studies to determine whether Vera's adrenal cortices or her ovaries were causing her virilizing syndrome.

tissues, with a resultant increase in extracellular fluid volume and eventually in blood pressure. The angiotensins are produced in response to a reduction in extracellular fluid volume, which may occur as a result of such things as excessive sweating, persistent vomiting without sufficient rehydration, or bleeding without adequate replacement of blood.

Androgens such as testosterone are synthesized in the Leydig cells of the testes and to a lesser extent in the ovary and are secreted in response to luteinizing hormone (LH). In males, testosterone is commonly converted to dihydrotestosterone, a higher-affinity form of the hormone, within specific target tissues. This active form of the hormone stimulates the production of sperm proteins in Sertoli cells and the development of secondary sex characteristics.

Estrogens such as 17- β -estradiol are synthesized in the ovarian follicle and the corpus luteum, from which their secretion is stimulated by follicle-stimulating hormone (FSH). In the female, 17 β -estradiol feeds back negatively on the synthesis and secretion of the pituitary gonadotropins, such as FSH. Estrogen and progesterone prepare the uterine endometrium for implantation of the fertilized ovum, and among other actions promotes differentiation of the mammary gland.

Progestogens such as progesterone are synthesized in the corpus luteum, and their secretion is stimulated by LH. As mentioned, in concert with estradiol, progesterone prepares the uterine endometrium for implantation of the fertilized ovum and acts as a differentiation factor in mammary gland development.

The biosynthesis of glucocorticoids and mineralocorticoids (in the adrenal cortex), and that of sex steroids (in the adrenal cortex and gonads), requires four cytochrome P450 enzymes (see Chapter 24). These monooxygenases are involved in the transfer of electrons from NADPH through electron transfer protein intermediates to molecular oxygen, which then oxidizes a variety of the ring carbons of cholesterol.

Cholesterol is converted to progesterone in the first two steps of synthesis of all steroid hormones. Cytochrome P450_{SCC} side-chain cleavage enzyme (previously referred to as cholesterol desmolase) is located in the mitochondrial inner membrane and removes six carbons from the side chain of cholesterol, forming pregnenolone, which has 21 carbons (Fig. 34.23). The next step, the conversion of pregnenolone to progesterone, is catalyzed by 3 β -hydroxysteroid dehydrogenase, an enzyme that is not a member of the cytochrome P450 family. Other steroid hormones are produced from progesterone by reactions that involve members of the P450 family. As the synthesis of the steroid hormones is discussed, notice how certain enzymes are used in more than one pathway. Defects in such enzymes will lead to multiple abnormalities in steroid synthesis, which, in turn, results in a variety of abnormal phenotypes.



Cytochrome P450_{C11}, another enzyme located in the mitochondrial membrane, catalyzes β -hydroxylation at C11. Hydroxylations at C17 and C21 are catalyzed by two enzymes located in the membranes of the endoplasmic reticulum (P450_{C17} for 17 α -hydroxylation and P450_{C21} for 21-hydroxylation).

A. Synthesis of Cortisol

The adrenocortical biosynthetic pathway that leads to cortisol synthesis occurs in the middle layer of the adrenal cortex known as the zona fasciculata. Free cholesterol is transported by an intracellular carrier protein to the inner mitochondrial membrane of cells (Fig. 34.24), where the side chain is cleaved to form pregnenolone. Pregnenolone returns to the cytosol, where it forms progesterone.

In the membranes of the endoplasmic reticulum, the enzyme P450_{C17} catalyzes the hydroxylation of C17 of progesterone or pregnenolone and can also catalyze the cleavage of the 2-carbon side chain of these compounds at C17 (a C17-C20 lyase activity). These two separate functions of the same enzyme allow further steroid synthesis to proceed along two separate pathways: the 17-hydroxylated steroids that retain their side chains are precursors of cortisol (C21), whereas those from which the side chain was cleaved (C19 steroids) are precursors of androgens (male sex hormones) and estrogens (female sex hormones).

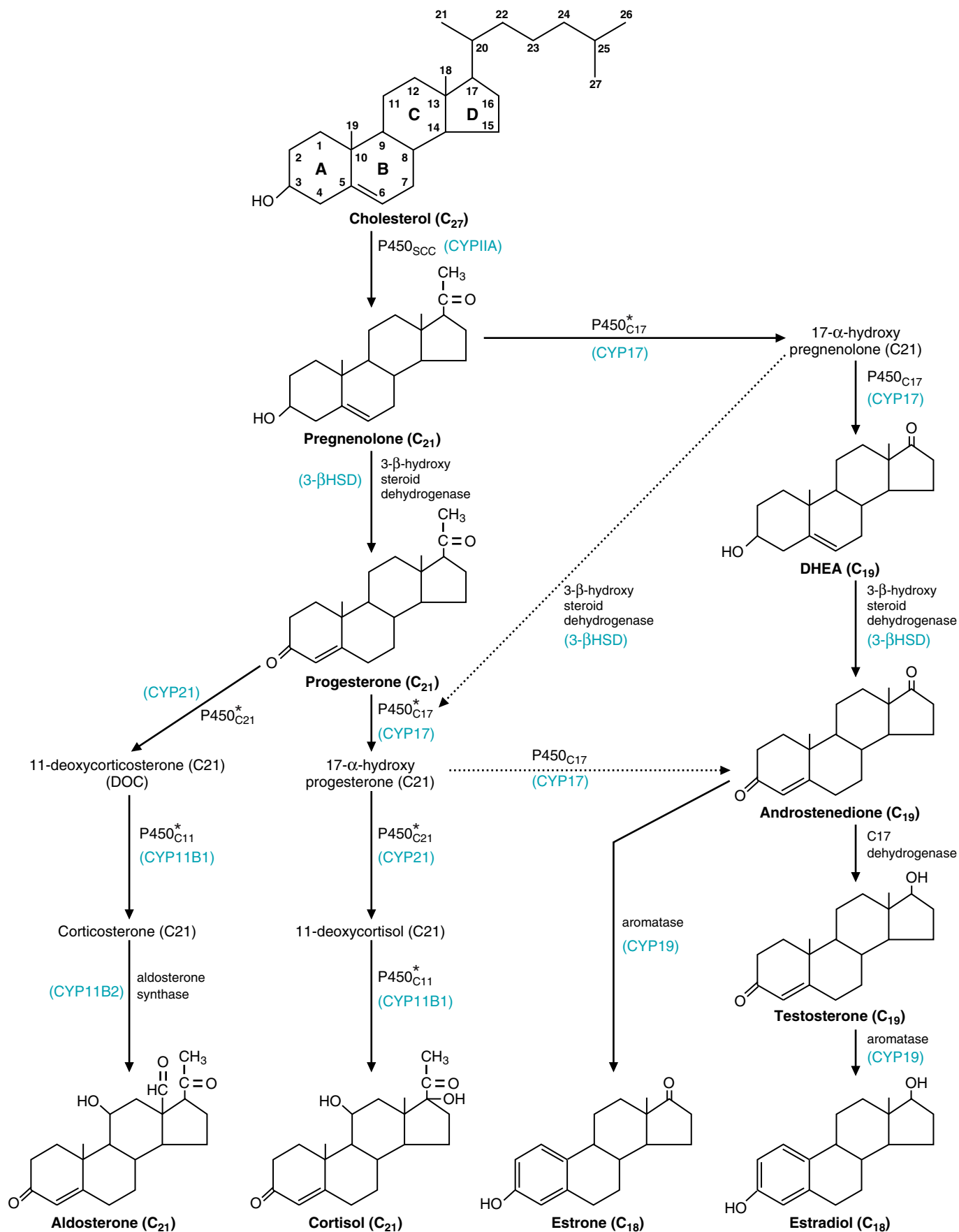


Fig. 34.23. Synthesis of the steroid hormones. The rings of the precursor, cholesterol, are lettered. Dihydrotestosterone is produced from testosterone by reduction of the carbon–carbon double bond in ring A. Structural changes between the precursor and final hormone are noted in blue. DHEA = dehydroepiandrosterone. The dashed lines indicate alternative pathways to the major pathways indicated. The starred enzymes are those that may be defective in the condition congenital adrenal hyperplasia.

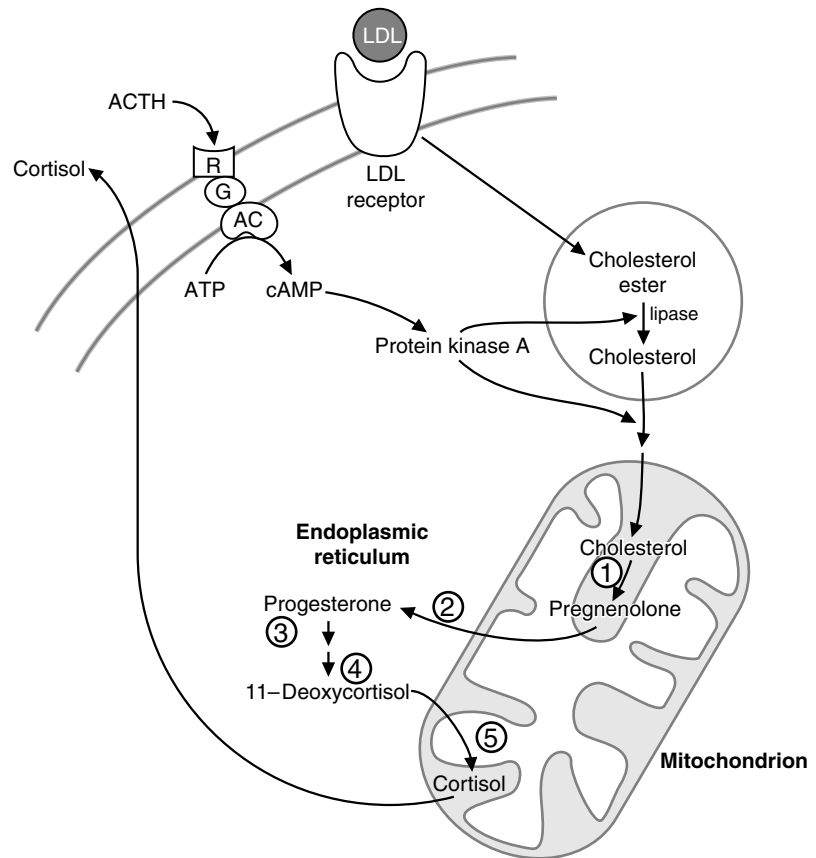


Fig. 34.24. Cellular route for cortisol synthesis. Cholesterol is synthesized from acetyl CoA or derived from low-density lipoprotein (LDL), which is endocytosed and digested by lysosomal enzymes. Cholesterol is stored in cells of the adrenal cortex as cholesterol esters. ACTH signals the cell to convert cholesterol to cortisol. 1 = cholesterol desmolase (involved in side chain cleavage); 2 = 3β -hydroxysteroid dehydrogenase; 3 = 17α -hydroxylase; 4 = 21 -hydroxylase; 5 = 11β -hydroxylase.

In the pathway of cortisol synthesis, the 17 -hydroxylation of progesterone yields 17α -hydroxyprogesterone, which, along with progesterone, is transported to the smooth endoplasmic reticulum. There the membrane-bound $P450_{C21}$ (21α -hydroxylase) enzyme catalyzes the hydroxylation of C21 of 17α -hydroxyprogesterone to form 11 -deoxycortisol (and of progesterone to form deoxycorticosterone [DOC], a precursor of the mineralocorticoid, aldosterone; see Fig. 34.23).

The final step in cortisol synthesis requires transport of 11 -deoxycortisol back to the inner membrane of the mitochondria, where $P450_{C11}$ (11β -hydroxylase) receives electrons from electron transport protein intermediates (adrenodoxin, which when oxidized is reduced by adrenodoxin reductase). The enzyme then transfers these reducing equivalents by way of oxygen to 11 -deoxycortisol for hydroxylation at C11 to form cortisol. The rate of biosynthesis of cortisol and other adrenal steroids is dependent on stimulation of the adrenal cortical cells by adrenocorticotrophic hormone (ACTH).

B. Synthesis of Aldosterone

The synthesis of the potent mineralocorticoid aldosterone in the zona glomerulosa of the adrenal cortex also begins with the conversion of cholesterol to progesterone (see Figs. 34.23 and 34.24). Progesterone is then hydroxylated at C21,

a reaction catalyzed by P450_{C21}, to yield DOC. The P450_{C11} enzyme system then catalyzes the reactions that convert DOC to corticosterone. The terminal steps in aldosterone synthesis, catalyzed by the P450 aldosterone system, involve the oxidation of corticosterone to 18-hydroxycorticosterone, which is oxidized to aldosterone.

The primary stimulus for aldosterone production is the octapeptide angiotensin II, although hyperkalemia (greater than normal levels of potassium in the blood) or hyponatremia (less than normal levels of sodium in the blood) may directly stimulate aldosterone synthesis as well. ACTH has a permissive action in aldosterone production. It allows cells to respond optimally to their primary stimulus, angiotensin II.

C. Synthesis of the Adrenal Androgens

Adrenal androgen biosynthesis proceeds from cleavage of the 2-carbon side chain of 17-hydroxypregnenolone at C17 to form the 19-carbon adrenal androgen dehydroepiandrosterone (DHEA) and its sulfate derivative (DHEAS) in the zona reticulosa of the adrenal cortex (see Fig. 34.23). These compounds, which are weak androgens, represent a significant percentage of the total steroid production by the normal adrenal cortex, and are the major androgens synthesized in the adrenal gland.

Androstenedione, another weak adrenal androgen, is produced when the 2-carbon side chain is cleaved from 17 α -hydroxypregesterone by the C17-C20 lyase activity of P450_{C17}. This androgen is converted to testosterone primarily in extra-adrenal tissues. Although the adrenal cortex makes very little estrogen, the weak adrenal androgens may be converted to estrogens in the peripheral tissues, particularly in adipose tissue (Fig. 34.25).

D. Synthesis of Testosterone

Luteinizing hormone (LH) from the anterior pituitary stimulates the synthesis of testosterone and other androgens by the Leydig cells of the human testicle. In many ways, the pathways leading to androgen synthesis in the testicle are similar to those described for the adrenal cortex. In the human testicle, the predominant pathway leading to testosterone synthesis is through pregnenolone to 17 α -hydroxypregnenolone to DHEA (the Δ^5 pathway), and then from DHEA to androstenedione, and from androstenedione to testosterone (see Fig. 34.23). As for all steroids, the rate-limiting step in testosterone production is the conversion of cholesterol to pregnenolone. LH controls the rate of side-chain cleavage from cholesterol at carbon 21 to form pregnenolone, and thus regulates the rate of



Hyperplasia or tumors of the adrenal cortex that produce excess aldosterone result in a condition known as primary aldosteronism, which is characterized by enhanced sodium and water retention, resulting in hypertension.



Although aldosterone is the major mineralocorticoid in humans, excessive production of a weaker mineralocorticoid, DOC, which occurs in patients with a deficiency of the 11-hydroxylase (the P450_{C11} enzyme), may lead to clinical signs and symptoms of mineralocorticoid excess even though aldosterone secretion is suppressed in these patients.



Androstenedione can be purchased at health food stores under the name Andros. It is touted to improve athletic performance through its ability to be converted to testosterone. Its use has been banned by most major sports, although in 1998 it was a legal supplement in baseball. During that year, the drug received a lot of publicity, as the supplement had been used by a player who broke the major league home run record.



Congenital adrenal hyperplasia (CAH) is a group of diseases caused by a genetically determined deficiency in a variety of enzymes required for cortisol synthesis. The most common deficiency is that of 21- α hydroxylase, the activity of which is necessary to convert progesterone to 11-deoxycorticosterone and 17- α hydroxy progesterone to 11-deoxycortisol. Thus, this deficiency reduces both aldosterone and cortisol production, without affecting androgen production. If the enzyme deficiency is severe, the precursors for aldosterone and cortisol production are shunted to androgen synthesis, producing an overabundance of androgens, which leads to prenatal masculinization in females and postnatal virilization of males. Another enzyme deficiency in this group of diseases is that of 11- β hydroxylase, which results in the accumulation of 11-deoxycorticosterone. An excess of this mineralocorticoid leads to hypertension (through binding of 11-deoxycorticosterone to the aldosterone receptor). In this form of CAH, 11-deoxycortisol also accumulates, but its biologic activity is minimal, and no specific clinical signs and symptoms result. The androgen pathway is unaffected, and the increased ACTH levels may increase the levels of adrenal androgens in the blood. A third possible enzyme deficiency is that of 17- α hydroxylase. A defect in 17- α hydroxylase leads to aldosterone excess and hypertension; however, because adrenal androgen synthesis requires this enzyme, no virilization occurs in these patients.

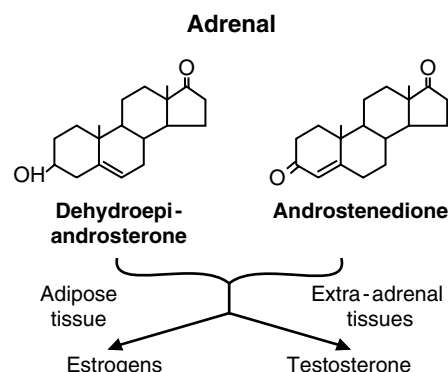


Fig. 34.25. Adrenal androgens. These weak androgens are converted to testosterone or estrogens in other tissues.



Biologically, the most potent circulating androgen is testosterone. Approximately 50% of the testosterone in the blood in a normal woman is produced equally in the ovaries and in the adrenal cortices. The remaining half is derived from ovarian and adrenal androstenedione, which, after secretion into the blood, is converted to testosterone in adipose tissue, muscle, liver, and skin. The adrenal cortex, however, is the major source of the relatively weak androgen dehydroepiandrosterone (DHEA). The serum concentration of its stable metabolite, DHEAS, is used as a measure of adrenal androgen production in hyperandrogenic patients with diffuse excessive growth of secondary sexual hair, e.g., facial hair as well as that in the axillae, the suprapubic area, the chest, and the upper extremities.



The results of the blood tests on **Vera Leizd** showed that her level of testosterone was normal but that her serum dehydroepiandrosterone sulfate (DHEAS) level was significantly elevated. Which tissue was the most likely source of the androgens that caused Vera's hirsutism (a male pattern of secondary sexual hair growth)?



Ergosterol is the provitamin of vitamin D_2 , which differs from 7-dehydrocholesterol and vitamin D_3 , respectively, only by having a double bond between C22 and C23 and a methyl group at C24. Vitamin D_2 is the constituent in many commercial vitamin preparations and in irradiated milk and bread. The antirachitic potencies of D_2 and D_3 in humans are equal, but both must be converted to 25-(OH)-cholecalciferol and eventually to the active form calcitriol ($1,25-(OH)_2D_3$) for biologic activity.



Rickets is a disorder of young children caused by a deficiency of vitamin D. Low levels of calcium and phosphorus in the blood are associated with skeletal deformities in these patients.

testosterone synthesis. In its target cells, the double bond in ring A of testosterone is reduced through the action of 5- α reductase, forming the active hormone dihydrotestosterone (DHT).

E. Synthesis of Estrogens and Progesterone

Ovarian production of estrogens, progestins (compounds related to progesterone), and androgens requires the activity of the cytochrome P450 family of oxidative enzymes used for the synthesis of other steroid hormones. Ovarian estrogens are C18 steroids with a phenolic hydroxyl group at C3 and either a hydroxyl group (estradiol) or a ketone group (estrone) at C17. Although the major steroid-producing compartments of the ovary (the granulosa cell, the theca cell, the stromal cell, and the cells of the corpus luteum) have all of the enzyme systems required for the synthesis of multiple steroids, the granulosa cells secrete primarily estrogens, the thecal and stromal cells secrete primarily androgens, and the cells of the corpus luteum secrete primarily progesterone.

The ovarian granulosa cell, in response to stimulation by follicle-stimulating hormone (FSH) from the anterior pituitary gland and through the catalytic activity of P450 aromatase, converts testosterone to estradiol, the predominant and most potent of the ovarian estrogens (see Fig. 34.23). Similarly, androstenedione is converted to estrone in the ovary, although the major site of estrone production from androstenedione occurs in extraovarian tissues, principally skeletal muscle and adipose tissue.

XI. VITAMIN D SYNTHESIS

Vitamin D is unique in that it can be either obtained from the diet (as vitamin D_2 or D_3) or synthesized from a cholesterol precursor, a process that requires reactions in the skin, liver, and intestine. The calciferols, including several forms of vitamin D, are a family of steroids that affect calcium homeostasis (Fig. 34.26). Cholecalciferol (vitamin D_3) requires ultraviolet light for its production from 7-dehydrocholesterol present in cutaneous tissues (skin) in animals and from ergosterol in plants. This irradiation cleaves the carbon-carbon bond at C9-C10 to open the B ring to form cholecalciferol, an inactive precursor of $1,25-(OH)_2$ -cholecalciferol (calcitriol). Calcitriol is the most potent biologically active form of vitamin D (see Fig. 34.26).

The formation of calcitriol from cholecalciferol begins in the liver and ends in the kidney, where the pathway is regulated. In this activation process, carbon 25 of vitamin D_2 or D_3 is hydroxylated in the microsomes of the liver to form 25-hydroxycholecalciferol (calcidiol). Calcidiol circulates to the kidney bound to vitamin D-binding globulin (transcalciferin). In the proximal convoluted tubule of the kidney, a mixed function oxidase, which requires molecular O_2 and NADPH as cofactors, hydroxylates carbon 1 on the A ring to form calcitriol. This step is tightly regulated and is the rate-limiting step in the production of the active hormone.

$1,25-(OH)_2D_3$ (calcitriol) is approximately 100 times more potent than $25-(OH)D_3$ in its actions, yet $25-(OH)D_3$ is present in the blood in a concentration that may be 100 times greater, which suggests that it may play some role in calcium and phosphorus homeostasis.

The biologically active forms of vitamin D are sterol hormones and, like other steroids, diffuse passively through the plasma membrane. In the intestine, bone, and kidney, the sterol then moves into the nucleus and binds to specific vitamin D_3 receptors. This complex activates genes that encode proteins mediating the action of active vitamin D_3 . In the intestinal mucosal cell, for example, transcription of genes encoding calcium-transporting proteins is activated. These proteins are capable of carrying Ca^{2+} (and phosphorus) absorbed from the gut lumen across the cell, making it available for eventual passage into the circulation.

CLINICAL COMMENTS



Ann Jeina is typical of patients with essentially normal serum triacylglycerol levels and elevated serum total cholesterol levels that are repeatedly in the upper 1% of the general population (e.g., 325–500 mg/dL). When similar lipid abnormalities are present in other family members in a pattern of autosomal dominant inheritance and no secondary causes for these lipid alterations (e.g., hypothyroidism) are present, the entity referred to as “familial hypercholesterolemia (FH), type IIA” is the most likely cause of this hereditary dyslipidemia.

FH is a genetic disorder caused by an abnormality in one or more alleles responsible for the formation or the functional integrity of high-affinity LDL receptors on the plasma membrane of cells that normally initiate the internalization of circulating LDL and other blood lipoproteins. Heterozygotes for FH (1 in 500 of the population) have roughly one half of the normal complement or functional capacity of such receptors, whereas homozygotes (1 in 1 million of the population) have essentially no functional LDL receptors. The rare patient with the homozygous form of FH has a more extreme elevation of serum total and LDL cholesterol than does the heterozygote and, as a result, has a more profound predisposition to premature coronary artery disease.

Chronic hypercholesterolemia not only may cause the deposition of lipid within vascular tissues leading to atherosclerosis but also may cause the deposition of lipid within the skin and eye. When this occurs in the medial aspect of the upper and lower eyelids, it is referred to as xanthelasma. Similar deposits known as xanthomas may occur in the iris of the eye (arcus lipidalis) as well as the tendons of the hands (“knucklepads”) and Achilles tendons.

Although therapy aimed at inserting competent LDL receptor genes into the cells of patients with homozygous FH is undergoing clinical trials, the current approach in the heterozygote is to attempt to increase the rate of synthesis of LDL receptors in cells pharmacologically.

Ann Jeina was treated with cholestyramine, a resin that binds some of the bile salts in the intestine, causing these resin-bound salts to be carried into the feces rather than recycled to the liver. The liver must now synthesize more bile salts, which lowers the intrahepatic free cholesterol pool. As a result, hepatic LDL receptor synthesis is induced, and more circulating LDL is taken up by the liver.

HMG-CoA reductase inhibitors, such as pravastatin, also stimulate the synthesis of additional LDL receptors but do so by inhibiting HMG-CoA reductase, the rate-limiting enzyme for cholesterol synthesis. The subsequent decline in the intracellular free cholesterol pool also stimulates the synthesis of additional LDL receptors. These additional receptors reduce circulating LDL-cholesterol levels by increasing receptor-mediated endocytosis of LDL particles.

A combination of strict dietary and dual pharmacologic therapy, aimed at decreasing the cholesterol levels of the body, is usually quite effective in cor-



Ann Jeina was treated with a statin (pravastatin) and cholestyramine, a bile acid sequestrant. With the introduction of the cholesterol absorption blocker ezetimibe, the use of cholestyramine with its high level of gastrointestinal side effects may decline. Ezetimibe reduces the percentage of absorption of free cholesterol present in the lumen of the gut and hence the amount of cholesterol available to the enterocyte to package into chylomicrons. This, in turn, reduces the amount of cholesterol returning to the liver in chylomicron remnants. The net result is a reduction in the cholesterol pool in hepatocytes. The latter induces the synthesis of an increased number of LDL receptors by the liver cells. As a consequence, the capacity of the liver to increase hepatic uptake of LDL from the circulation leads to a decrease in serum LDL levels.

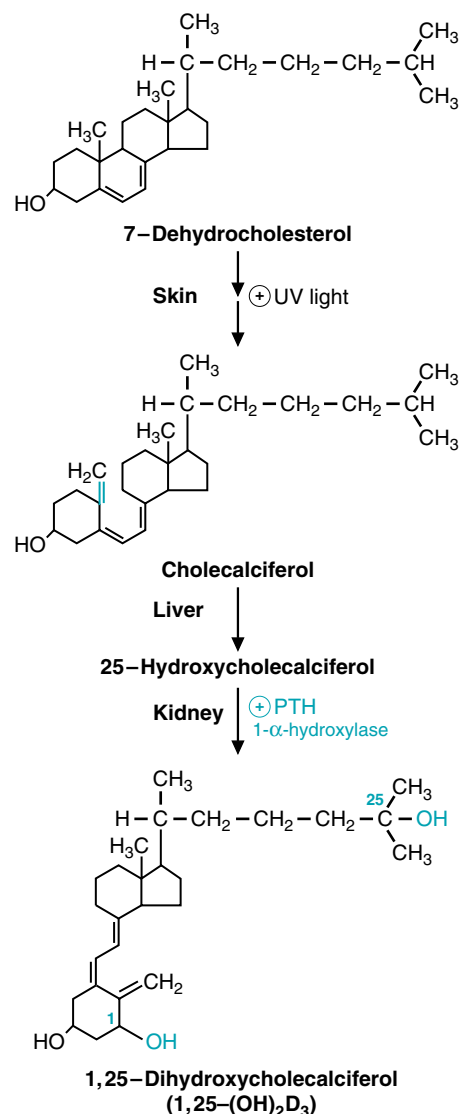


Fig. 34.26. Synthesis of active vitamin D. (1, 25-di (OH)₂D₃) is produced from 7-dehydrocholesterol, a precursor of cholesterol. In the skin, ultraviolet (UV) light produces cholecalciferol, which is hydroxylated at the 25-position in the liver and the 1-position in the kidney to form the active hormone.



Vera Leizd's hirsutism was most likely the result of a problem in her adrenal cortex that caused excessive production of DHEA.

recting the lipid abnormality and, hopefully, the associated risk of atherosclerotic cardiovascular disease in patients with heterozygous familial hypercholesterolemia.



Low-density lipoprotein cholesterol is the primary target of cholesterol-lowering therapy because both epidemiologic and experimental evidence strongly suggest a benefit of lowering serum LDL cholesterol in the prevention of atherosclerotic cardiovascular disease. Similar evidence for raising subnormal levels of serum HDL cholesterol is less conclusive but adequate to support such efforts, particularly in high-risk patients, such as **Ivan Applebod**, who have multiple cardiovascular risk factors. The first-line therapy in this attempt is non-pharmacologic and includes such measures as increasing aerobic exercise, weight loss in overweight patients, avoidance of excessive alcohol intake, reducing the intake of refined sugars, and cessation of smoking. If these efforts fail, drug therapy to raise serum HDL cholesterol levels must be considered.

So far, Mr. Applebod has failed in his attempts to diet and exercise. His LDL cholesterol level is 231 mg/dL. According to Table 34.1, he is a candidate for more stringent dietary therapy and for drug treatment. He could be given an HMG CoA reductase inhibitor such as pravastatin and, perhaps, a bile salt-binding resin such as cholestyramine. Other lipid-lowering drugs such as the fibric acid derivatives and ezetimibe, which also decrease triacylglycerol levels and potentially increase HDL levels, should be considered (Table 34.5).



Vera Leizd was born with a normal female genotype and phenotype, had normal female sexual development, spontaneous onset of puberty, and regular, although somewhat scanty, menses until the age of 20. At that point, she developed secondary amenorrhea (cessation of menses) and evidence of male hormone excess with early virilization (masculinization).

The differential diagnosis included an ovarian versus an adrenocortical source of the excess androgenic steroids. A screening test to determine whether

Table 34.5. Mechanism(s) of Action and Efficacy of Lipid-Lowering Agents

Agent	Mechanism of Action	Percentage change in serum lipid level (monotherapy)			
		Total cholesterol	LDL-cholesterol	HDL cholesterol	Triacylglycerols
Statins	Inhibits HMG-CoA reductase activity	↓15–60%	↓ 20–60%	↑ 5–15%	↓10–40%
Bile acid resins	Increase fecal excretion of bile salts	↓15–20%	↓10–25%	↑ 3–5%	Variable, depending on pretreatment level of triacylglycerols (may increase)
Niacin	Activates LPL; reduces hepatic production of VLDL; reduces catabolism of HDL	↓ 22–25%	↓10–25%	↑ 15–35%	↓ 20–50%
Fibrates	Antagonizes PPAR- α causing an increase in LPL activity, a decrease in apoprotein C-III production, and an increase in apoprotein A-I production.	↓12–15%	Variable, depending on pretreatment levels of other lipids	↑ 5–15%	↓ 20–50%
Ezetimibe	Reduces intestinal absorption of free cholesterol from the gut lumen	↓10–15%	↓ 15–20%	↑ 1–3%	↓ 5–8% if triacylglycerols are high pretreatment

Abbreviations: LPL, lipoprotein lipase; LDL, low-density lipoprotein; HDL, high-density lipoprotein; triacylglycerols, triglycerides; PPAR, peroxisome proliferators-activated receptor (the Table is adapted from *Circulation* 2002; 106:3145–3457).

the adrenal cortex or the ovary is the source of excess male hormone involves the measurement of the concentration of dehydroepiandrosterone sulfate (DHEAS) in the patient's plasma, because the adrenal cortex makes most of the DHEA, and the ovary makes little or none. Vera's plasma DHEAS level was moderately elevated, identifying her adrenal cortices as the likely source of her virilizing syndrome.

If the excess production of androgens is not the result of an adrenal tumor, but the result of a defect in the pathway for cortisol production, the simple treatment is to administer glucocorticoids by mouth. The rationale for such treatment can be better understood by reviewing Fig. 34.23. If **Vera Leizd** has a genetically determined partial deficiency in the $P450_{C11}$ enzyme system needed to convert 11-deoxycortisol to cortisol, her blood cortisol levels would fall. By virtue of the normal positive feedback mechanism, a subnormal level of cortisol in the blood would induce the anterior pituitary to make more ACTH. The latter would not only stimulate the cortisol pathway to increase cortisol synthesis to normal but, in the process, would also induce increased production of adrenal androgens such as DHEA and DHEAS. The increased levels of the adrenal androgens (although relatively weak androgens) would cause varying degrees of virilization, depending on the severity of the enzyme deficiency. The administration of a glucocorticoid by mouth would suppress the high level of secretion of ACTH from the anterior pituitary gland that occurs in response to the reduced levels of cortisol secreted from the adrenal cortex. Treatment with prednisone (a synthetic glucocorticoid), therefore, will prevent the ACTH-induced overproduction of adrenal androgens. However, when ACTH secretion returns to normal, endogenous cortisol synthesis falls below normal. The administered prednisone brings the net glucocorticoid activity in the blood back to physiologic levels. Vera's adrenal androgen levels in the blood returned to normal after several weeks of therapy with prednisone (a synthetic glucocorticoid). As a result, her menses eventually resumed, and her virilizing features slowly resolved.

Because Vera's symptoms began in adult life, her genetically determined adrenal hyperplasia is referred to as a "nonclassic" or "atypical" form of the disorder. A more severe enzyme deficiency leads to the "classic" disease, which is associated with excessive fetal adrenal androgen production in utero and, therefore, manifests itself at birth, often with ambiguous external genitalia and virilizing features in the female neonate.

BIOCHEMICAL COMMENTS



Defects in the LDL receptor gene are responsible for the elevated blood levels of LDL, and thus of cholesterol, in FH. Over 300 mutations have been found in the LDL receptor gene, affecting all stages in the production and functioning of the receptor.

The LDL receptor gene, which contains 18 exons and is 45 kilobases (kb) in length, is located on the short arm of chromosome 19. The exons share sequences for the C9 component of complement (a blood protein involved in the immune response), and the *N*-linked oligosaccharide domain is homologous to the genes for the precursor of EGF and also for three proteases of the blood clotting system, Factors IX and X and protein C (see Chapter 45). The LDL receptor gene encodes a glycoprotein that contains 839 amino acids.

Heterozygotes for FH have one normal and one mutant allele. Their cells produce approximately half the normal amount of receptor and take up LDL at about half the normal rate. Homozygotes have two mutant alleles, which may either be identical or differ. They produce very little functional receptor.

The genetic mutations are mainly deletions, but insertions or duplications also occur, as well as missense and nonsense point mutations (see Fig. 34.20). Four classes of mutations have been identified. The first class involves “null” alleles that either direct the synthesis of no protein at all or a protein that cannot be precipitated by antibodies to the LDL receptor. In the second class, the alleles encode proteins, but they cannot be transported to the cell surface. The third class of mutant alleles encodes proteins that reach the cell surface but cannot bind LDL normally. Finally, the fourth class encodes proteins that reach the surface and bind LDL but fail to cluster and internalize the LDL particles. The result of each of these mutations is that blood levels of LDL are elevated because cells cannot take up these particles at a normal rate.

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REVIEW QUESTIONS—CHAPTER 34

- Which of the following steps in the biosynthesis of cholesterol is the committed rate-limiting step?
 - The condensation of acetoacetyl-CoA with a molecule of acetyl-CoA to yield β -hydroxy- β methylglutaryl-CoA (HMG-CoA)
 - The reduction of HMG-CoA to mevalonate
 - The conversion of mevalonate to two activated isoprenes
 - The formation of farnesyl pyrophosphate
 - Condensation of six activated isoprene units to form squalene
- Considering the final steps in cholesterol biosynthesis, when squalene is eventually converted to lanosterol, which of the following statements is correct?
 - All of the sterols have three fused rings (the steroid nucleus) and are alcohols with a hydroxyl group at C-3.
 - The action of squalene monooxygenase oxidizes carbon 14 of the squalene chain, forming an epoxide.
 - Squalene monooxygenase is considered a mixed function oxidase because it catalyzes a reaction in which only one of the oxygen atoms of O_2 is incorporated into the organic substrate.
 - Squalene monooxygenase uses reduced flavin nucleotides (e.g., FAD(2H)) as the cosubstrate in the reaction.
 - Squalene is joined at carbons 1 and 30 to form the fused ring structure of sterols.
- Of the major risk factors for the development of atherosclerotic cardiovascular disease (ASCVD) such as sedentary lifestyle, obesity, cigarette smoking, diabetes mellitus, hypertension, and hyperlipidemia, which one, if present, is the only risk factor in a given patient without a history of having had a myocardial infarction that requires that the therapeutic goal for the serum LDL cholesterol level be $< 100\text{mg/dL}$?
 - Obesity
 - Cigarette smoking
 - Diabetes mellitus
 - Hypertension
 - Sedentary lifestyle

4. Which one of the following apoproteins acts as a cofactor activator of the enzyme lipoprotein lipase (LPL)?
- (A) ApoC-III
 - (B) ApoC-II
 - (C) ApoB-100
 - (D) ApoB-48
 - (E) ApoE
5. Which one of the following sequences places the lipoproteins in the order of most dense to least dense?
- (A) HDL/VLDL/chylomicrons/LDL
 - (B) HDL/LDL/VLDL/chylomicrons
 - (C) LDL/chylomicrons/HDL/VLDL
 - (D) VLDL/chylomicrons/LDL/HDL
 - (E) LDL/chylomicrons/VLDL/HDL

35 Metabolism of the Eicosanoids

The **eicosanoids**, which include the **prostaglandins** (PG), **thromboxanes** (TX), and **leukotrienes** (LT), are among the most potent regulators of cellular function in nature and are produced by almost every cell in the body. They act mainly as “local” hormones, affecting the cells that produce them or neighboring cells of a different type.

Eicosanoids participate in many processes in the body, particularly the **inflammatory response** that occurs after **infection** or **injury**. The inflammatory response is the sum of the body’s efforts to destroy invading organisms and to repair damage. It includes the control of bleeding through the formation of blood clots. In the process of protecting the body from a variety of insults, the inflammatory response can produce **symptoms** such as **pain**, **swelling**, and **fever**. An exaggerated or inappropriate expression of the normal inflammatory response may occur in individuals who have **allergic** or **hypersensitivity** reactions.

In addition to participating in the inflammatory response, eicosanoids also regulate **smooth muscle contraction** (particularly in the intestine and uterus). They increase **water** and **sodium excretion** by the kidney and are involved in **regulating blood pressure**. They frequently serve as modulators; some eicosanoids stimulate and others inhibit the same process. For example, some serve as constrictors and others as dilators of blood vessels. They are also involved in regulating **bronchoconstriction** and **bronchodilation**.

Eicosanoids are derived from polyunsaturated fatty acids containing 20 carbon atoms, which are found in cell membranes esterified to membrane phospholipids. Arachidonic acid, derived from the diet or synthesized from linoleate, is the compound from which most of the eicosanoids are produced in the body. Compounds that serve as signals for eicosanoid production bind to cell membrane receptors and activate phospholipases that cleave the polyunsaturated fatty acids from cell membrane phospholipids (Fig. 35.1).

Arachidonic acid is enzymatically metabolized by three major pathways. The two pathways that have been most thoroughly studied are the **cyclooxygenase** pathway (which produces prostaglandins and thromboxanes) and the **lipoxygenase** pathway (which produces leukotrienes). The **cytochrome P450** pathway generates eicosanoids with less well-defined physiologic functions. **Isoprostanes** are a relatively new class of eicosanoids derived from nonenzymatic free radical–catalyzed reactions. The isoprostanes are similar to prostaglandins in structure and may play a role in inflammatory responses and free radical–mediated tissue injury. In brain tissue, arachidonic acid can be coupled to ethanolamine to generate **anandamide**. This compound can bind and activate cannabinoid receptors with actions similar to those of Δ^9 -tetrahydrocannabinol (THC).

Many eicosanoids have very **short half-lives**, in the range of a few minutes or less. They are rapidly inactivated and excreted.

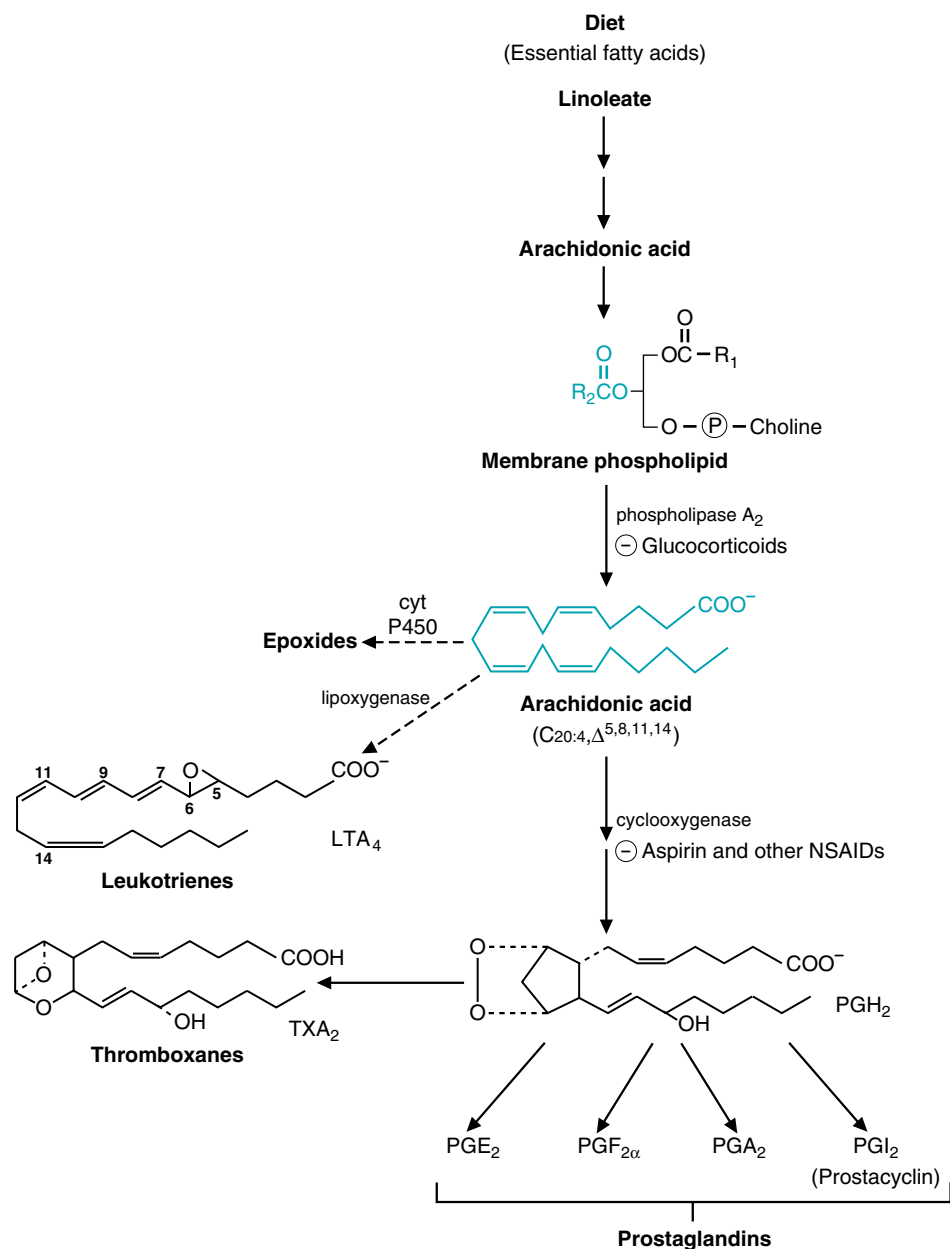


Fig. 35.1. Overview of eicosanoid metabolism. Eicosanoids are produced from fatty acids released from membrane phospholipids. In humans, arachidonic acid is the major precursor of the eicosanoids, which include the prostaglandins, leukotrienes, and thromboxanes. The circled minus sign = inhibits; cyt = cytochrome.



THE WAITING ROOM



Since her admission to the hospital for an acute myocardial infarction, **Ann Jeina** has been taking the bile salt sequestrant cholestyramine and the HMG-CoA reductase inhibitor pravastatin to lower her blood cholesterol levels (see Chapter 34). She also takes 160 mg acetylsalicylic acid (ASA; aspirin) each day. At her most recent visit to her cardiologist, she asked whether she should



“Eicosa” is the Greek word for the number 20. Eicosanoids are synthesized from polyunsaturated fatty acids with 20 carbon atoms.



The composition of the diet affects the fatty acid content of membrane phospholipids. Individuals with a high content of saturated fatty acids in their diets have a high content of saturated fatty acids in their membrane lipids. Likewise, individuals with a high content of polyunsaturated fatty acids in their diets have a high content of polyunsaturated fatty acids in their membrane lipids.



Dietary deficiencies of essential fatty acids are rare. However, some cases have been reported in patients receiving total parenteral nutrition (TPN). Although the most obvious symptom is a red scaly dermatitis, deficiencies of essential fatty acids also result in a decreased availability of precursors for eicosanoid synthesis.

continue to take aspirin because she no longer has any chest pain. She was told that the use of aspirin in her case was not to alleviate pain but to reduce the risk of a second heart attack and that she should continue to take this drug for the remainder of her life unless a complication, such as gastrointestinal bleeding, occurred as a result of its use.



Emma Wheezer has done well with regard to her respiratory function since her earlier hospitalization for an acute asthmatic attack. She has been maintained on two puffs of triamcinolone acetonide, a potent inhaled corticosteroid, three times per day, and has not required systemic steroids for months. The glucose intolerance precipitated by high intravenous and oral doses of the synthetic glucocorticoid dexamethasone during her earlier hospitalization resolved after this drug was discontinued. She has come to her doctor now because she is concerned that the low-grade fever and cough she has developed over the last 36 hours may trigger another acute asthma attack.

I. SOURCE OF THE EICOSANOIDS

The most abundant and therefore the most common precursor of the eicosanoids is arachidonic acid (eicosatetraenoic acid, ω 6,20:4, Δ 5,8,11,14), a polyunsaturated fatty acid with 20 carbons and 4 double bonds (see Fig. 35.1). It is esterified to phospholipids located in the lipid bilayer that constitutes the plasma membrane of the cell. Because arachidonic acid cannot be synthesized *de novo* in the body, the diet must contain arachidonic acid or other fatty acids from which arachidonic acid can be produced. The major dietary precursor for arachidonic acid synthesis is the essential fatty acid linoleate, which is present in plant oils (see Chapter 33).

The arachidonic acid present in membrane phospholipids is released from the lipid bilayer as a consequence of the activation of membrane-bound phospholipase A₂ or C (see Fig. 33.31 and Fig. 35.2). This activation occurs when a variety of stimuli (agonists), such as histamine and the cytokines, interact with a specific plasma membrane receptor on the target cell surface. Phospholipase A₂ is specific for the sn-2 position of phosphoacylglycerols, the site of attachment of arachidonic acid to the glycerol moiety. Phospholipase C hydrolyzes phosphorylated inositol

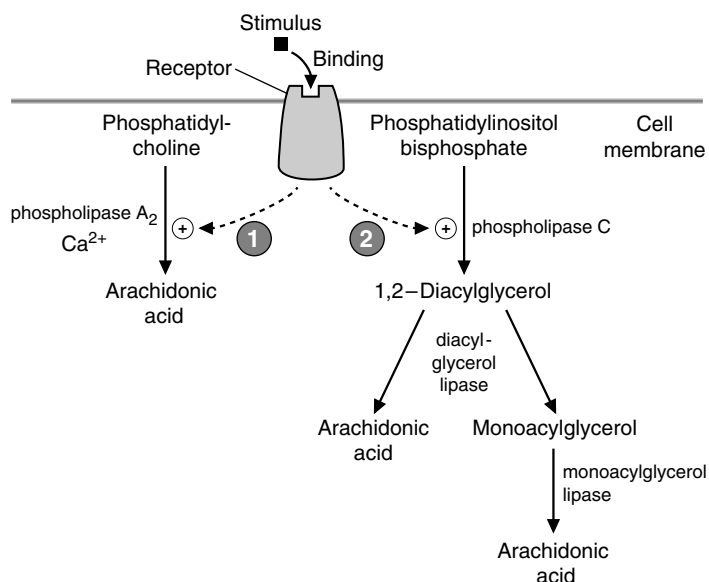


Fig. 35.2. Release of arachidonic acid from membrane lipids. The binding of a stimulus to its receptor activates pathway 1 or 2.

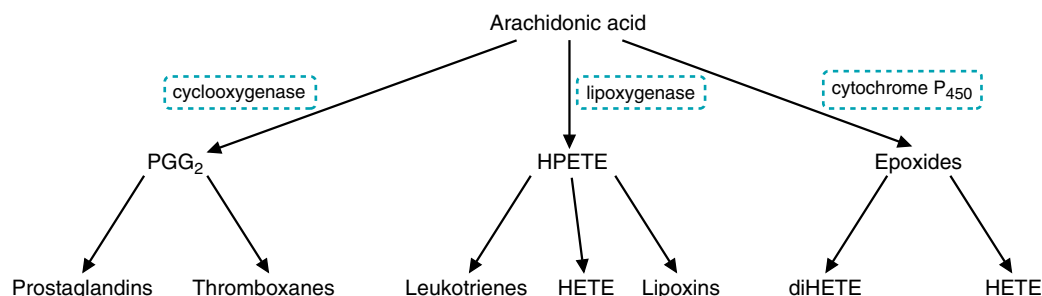


Fig. 35.3. Pathways for the metabolism of arachidonic acid.

from the inositol glycerophospholipids, generating a diacylglycerol containing arachidonic acid. This arachidonic acid is subsequently released by the action of other lipases.

II. PATHWAYS FOR EICOSANOID SYNTHESIS

After arachidonic acid is released into the cytosol, it is converted to eicosanoids by a variety of enzymes with activities that vary among tissues. This variation explains why some cells, such as those in the vascular endothelium, synthesize prostaglandins E_2 and I_2 (PGE_2 and PGI_2) whereas cells, such as platelets, synthesize primarily thromboxane A_2 (TXA_2) and 12-hydroxyeicosatetraenoic acid (12-HETE).

Three major pathways for the metabolism of arachidonic acid occur in various tissues (Fig. 35.3). The first of these, the cyclooxygenase pathway, leads to the synthesis of prostaglandins and thromboxanes. The second, the lipoxygenase pathway, yields the leukotrienes, HETEs, and lipoxins. The third pathway, catalyzed by the cytochrome P450 system, is responsible for the synthesis of the epoxides, HETEs, and diHETEs.

A. Cyclooxygenase Pathway: Synthesis of the Prostaglandins and Thromboxanes

1. STRUCTURES OF THE PROSTAGLANDINS

Prostaglandins are fatty acids containing 20 carbon atoms, including an internal 5-carbon ring. In addition to this ring, each of the biologically active prostaglandins has a hydroxyl group at carbon 15, a double bond between carbons 13 and 14, and various substituents on the ring (Fig. 35.4)

The nomenclature for the prostaglandins (PGs) involves the assignment of a capital letter (PGE), an Arabic numeral subscript (PGE_1), and, for the PGF family, a Greek letter subscript (e.g., $PGF_{2\alpha}$). The capital letter, in this case “F,” refers to the ring substituents shown in figure 35.5.

The subscript that follows the capital letter (PGF_1) refers to the PG series 1, 2, or 3, determined by the number of unsaturated bonds present in the linear portion of the hydrocarbon chain (Fig. 35.6). It does not include double bonds in the internal ring. Prostaglandins of the 1-series have one double bond (between carbons 13 and 14). The 2-series has two double bonds (between carbons 13 and 14, and 5 and 6), and the 3-series has three double bonds (between carbons 13 and 14, 5 and 6, and 17 and 18). The double bonds between carbons 13 and 14 are *trans*; the others are *cis*.

The Greek letter subscript, found only in the F series, refers to the position of the hydroxyl group at carbon 9. This hydroxyl group primarily exists in the α position, where it lies below the plane of the ring, as does the hydroxyl group at carbon 11 (see Figs. 35.4 and 35.5).



Inflammation involving the mucosal and smooth muscle layers of the respiratory tract plays a major role in the development of acute asthmatic bronchospasm in patients such as **Emma Wheezer**. Dexamethasone and other potent glucocorticoids are capable of preventing or suppressing this inflammation.

In part, the glucocorticoids act by inhibiting the recruitment of leukocytes and monocytes–macrophages into affected areas. They also limit the ability of these cells to elaborate a variety of chemotactic factors and other substances, such as certain eicosanoids, which enhance the inflammatory process. Glucocorticoids, for example, suppress the transcription and translation of the inducible form of the cyclooxygenase enzyme, COX-2. Glucocorticoids also induce the synthesis of a protein or family of proteins (lipocortins or macrocortins) that inhibit the activity of phospholipase A_2 . As a result, the synthesis of prostaglandins and leukotrienes is decreased, and the inflammatory response in bronchial tissues is reduced (see Figs. 35.1 and 35.2).

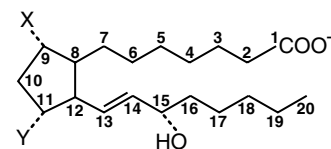


Fig. 35.4. Structural features common to the biologically active prostaglandins. These compounds have 20 carbons, with a carboxyl group at carbon 1. Carbons 8 through 12 form a five-membered ring with substituents (usually a hydroxyl or keto group) at carbons 9 (X) and 11 (Y). Carbon 15 contains a hydroxyl group, and a *trans* double bond connects carbons 13 and 14. Double bonds also may be present between carbons 5 and 6 and between carbons 17 and 18 (see Fig. 35.6).

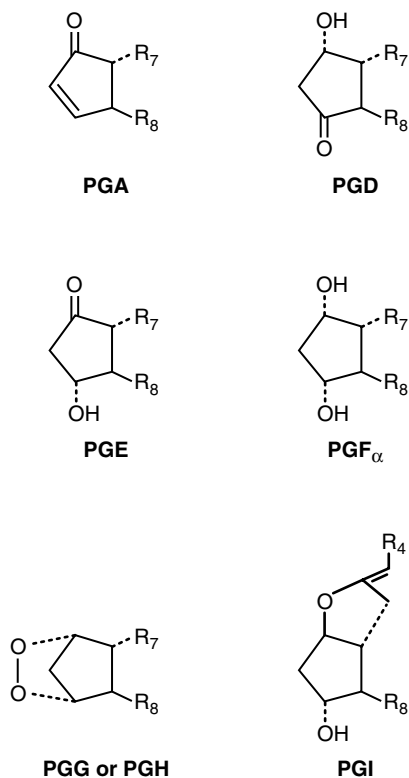


Fig. 35.5. Ring substituents of the prostaglandins (PG). The letter after PG denotes the configuration of the ring and its substituents. R₄, R₇, and R₈ represent the non-ring portions of the molecule. R₄ contains four carbons (including the carboxyl group). R₇ and R₈ contain seven and eight carbons, respectively, and correspond to the 1-, 2-, or 3-series shown in Figure 35.6. Note that the prostacyclins (PGI) contain two rings.

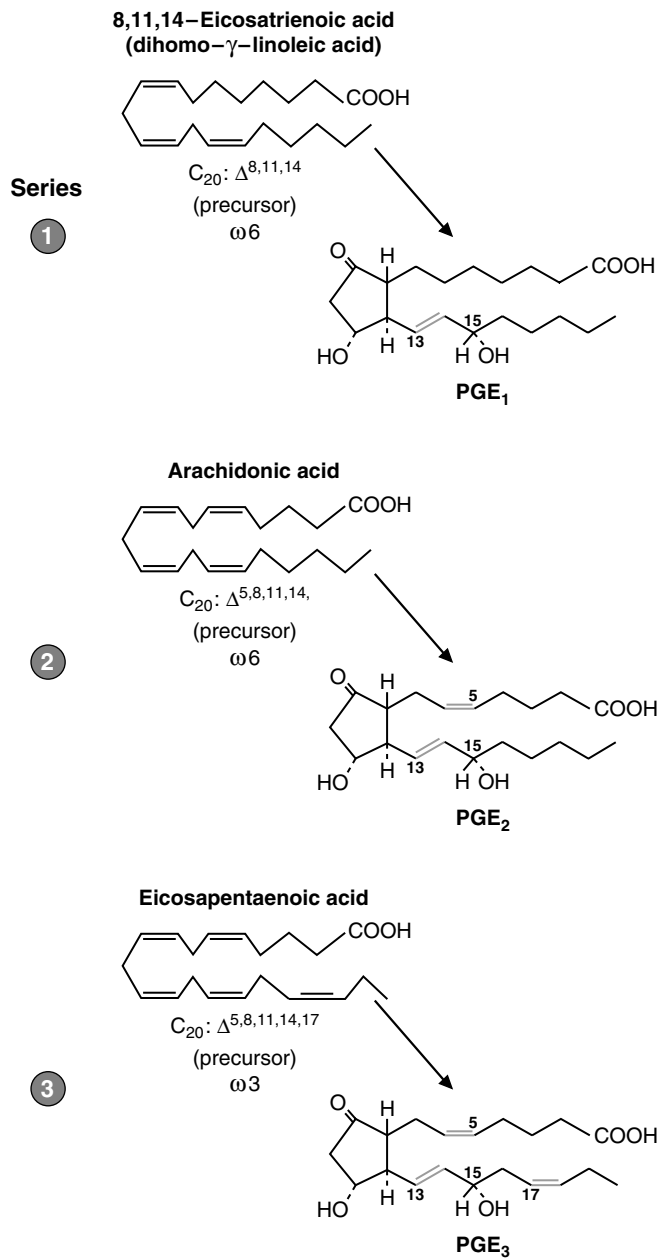


Fig. 35.6. Prostaglandins of the 1-, 2-, and 3-series and their precursors. The numeral (as a subscript in the name of the compound) refers to the number of double bonds in the non-ring portion of the prostaglandin. *Trans* double bonds are at position 13, and *cis* double bonds at positions 5 and 17. The hydroxyl group at carbon 15 is required for biologic activity.

2. STRUCTURE OF THE THROMBOXANES

The thromboxanes, derived from arachidonic acid via the cyclooxygenase pathway, closely resemble the prostaglandins in structure except that they contain a 6-membered ring that includes an oxygen atom (Fig. 35.7). The most common thromboxane, TXA₂, contains an additional oxygen atom attached both to carbon 9 and carbon 11 of the ring.



The thromboxanes were named for their action in producing blood clots (thrombi).

3. BIOSYNTHESIS OF THE PROSTAGLANDINS AND THROMBOXANES

Only the biosynthesis of those prostaglandins derived from arachidonic acid (e.g., the 2-series, such as PGE_2 , PGI_2 , TXA_2) are described, because those derived from eicosatrienoic acid (the 1-series) or from eicosapentaenoic acid (the 3-series) are present in very small amounts in humans on a normal diet (see Fig. 35.6).

The biochemical reactions that lead to the synthesis of prostaglandins and thromboxanes are illustrated in figure 35.8. The initial step, which is catalyzed by a cyclooxygenase, forms the five-membered ring and adds four atoms of oxygen (two between carbons 9 and 11, and two at carbon 15) to form the unstable endoperoxide, PGG_2 . The hydroperoxy group at carbon 15 is quickly reduced to a hydroxyl group by a peroxidase to form another endoperoxide, PGH_2 .

The next step is tissue specific (see Fig. 35.8). Depending on the type of cell involved, PGH_2 may be reduced to PGE_2 or PGD_2 by specific isomerases (PGE synthase and PGD synthase). PGE_2 may be further reduced by PGE 9-ketoreductase to form $\text{PGF}_{2\alpha}$. $\text{PGF}_{2\alpha}$ also may be formed directly from PGH_2 by the action of an endoperoxide reductase. Some of the major functions of the prostaglandins are listed in Table 35.1.

PGH_2 may be converted to the thromboxane TXA_2 , a reaction catalyzed by TXA synthase (see Fig. 35.8). This enzyme is present in high concentration in platelets. In the vascular endothelium, however, PGH_2 is converted to the prostaglandin PGI_2

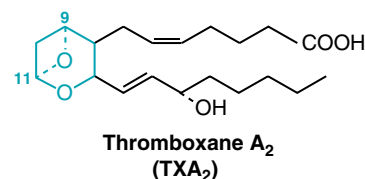


Fig. 35.7. The thromboxane ring. In contrast to the prostaglandins, which have a five-membered carbon ring, thromboxanes have a six-membered ring (shown in blue) containing an oxygen atom. Substituents are attached to the ring at carbons 9 and 11. In the case of TXA_2 (shown above), an oxygen atom connects carbons 9 and 11.

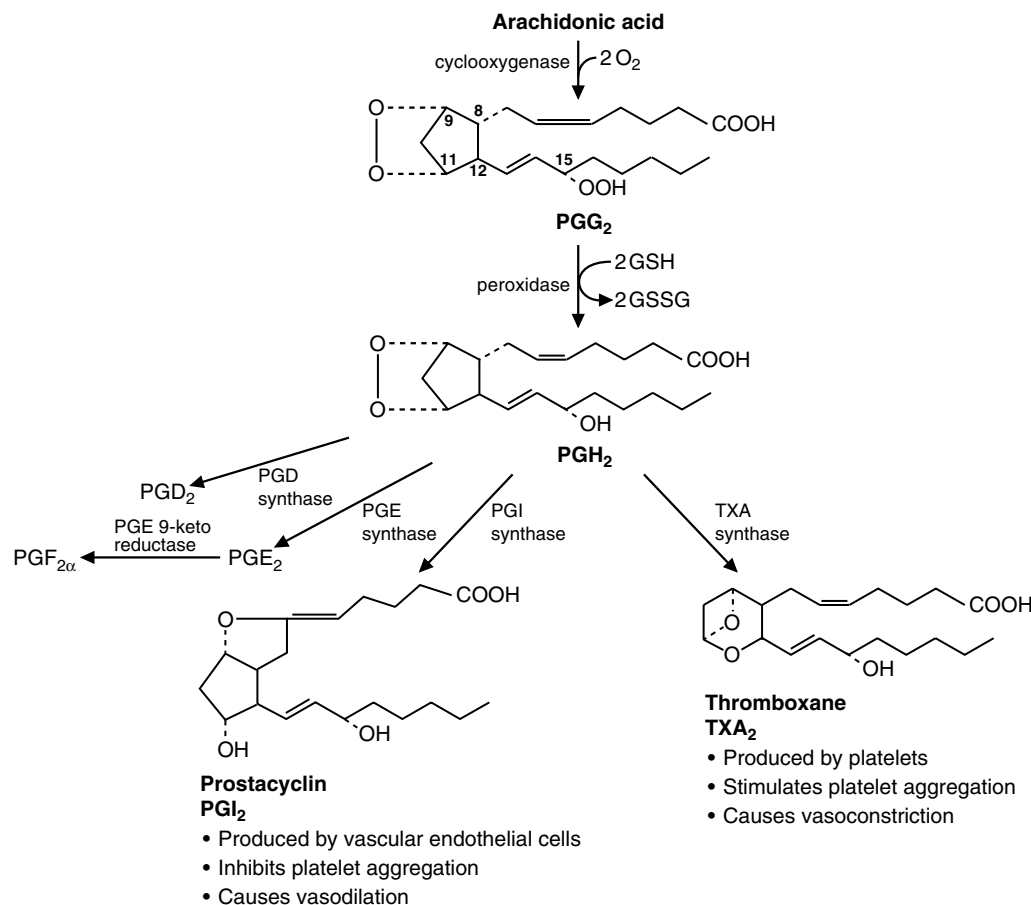


Fig. 35.8. Formation of prostaglandins (including the prostacyclin PGI_2) and thromboxane TXA_2 from arachidonic acid. The conversion of arachidonic acid to PGH_2 is catalyzed by a membrane-bound enzyme, prostaglandin endoperoxide synthase, which has cyclooxygenase and peroxidase activities. The reducing agent is glutathione (GSH), which is oxidized to GSSG.



The predominant eicosanoid in platelets is TXA_2 , a potent vasoconstrictor and a stimulator of platelet aggregation. The latter action initiates thrombus formation at sites of vascular injury as well as in the vicinity of a ruptured atherosclerotic plaque in the lumen of vessels such as the coronary arteries. Such thrombi may cause sudden total occlusion of the vascular lumen, causing acute ischemic damage to tissues distal to the block (i.e., acute myocardial infarction).

Aspirin, by covalently acetylating the active site of cyclooxygenase, blocks the production of TXA_2 from its major precursor, arachidonic acid. By causing this mild hemostatic defect, low-dose aspirin has been shown to be effective in prevention of acute myocardial infarction (see Clinical Comments). For **Ivan Applebod** (who has symptoms of coronary heart disease), aspirin is used to prevent a first heart attack (primary prevention). For **Ann Jeina** and **Cora Nari** (who already have had heart attacks), aspirin is used to prevent a second heart attack (secondary prevention).



Diets that include cold water fish (e.g., salmon, mackerel, brook trout, herring), with a high content of polyunsaturated fatty acids, eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA) (see Chapter 33), result in a high content of these fatty acids in membrane phospholipids. It has been suggested that such diets are effective in preventing heart disease, in part because they lead to formation of more TXA_3 relative to TXA_2 . TXA_3 is less effective in stimulating platelet aggregation than its counterpart in the 2-series, TXA_2 .

Table 35.1. Some Functions of the Prostaglandins

PGI_2 , PGE_2 , PGD_2

Increase

Vasodilation
cAMP

Decrease

Platelet aggregation
Leukocyte aggregation
 IL-1^a and IL-2
T-cell proliferation
Lymphocyte migration

$\text{PGF}_{2\alpha}$

Increases

Vasoconstriction
Bronchoconstriction
Smooth muscle contraction

^a IL = interleukin, a cytokine that augments the activity of many cells in the immune system.

Table 35.2. Some Functions of Thromboxane A_2

Increases

Vasoconstriction
Platelet aggregation
Lymphocyte proliferation
Bronchoconstriction

(prostacyclin) by PGI synthase (see Fig. 35.8). TXA_2 and PGI_2 have important antagonistic biologic effects on vasomotor and smooth muscle tone and on platelet aggregation. Some of the known functions of the thromboxanes are listed in Table 35.2.

In the 1990s, the cyclooxygenase enzyme was found to exist as two distinct isoforms, designated COX-1 and COX-2. COX-1 is regarded as a constitutive form of the enzyme, widely expressed in almost all tissues, and involved in the production of prostaglandins and thromboxanes for “normal” physiologic functions. COX-2 is an inducible form of the enzyme regulated by a variety of cytokines and growth factors. COX-2 mRNA and protein levels are usually low in most healthy tissue, but are expressed at high levels in inflamed tissue.

Because of the importance of prostaglandins in mediating the inflammatory response, drugs that block prostaglandin production should provide relief from pain. The cyclooxygenase enzyme is inhibited by all nonsteroidal anti-inflammatory drugs (NSAIDs) such as aspirin (acetylsalicylic acid). Aspirin transfers an acetyl group to the enzyme, irreversibly inactivating it (Fig. 35.9). Other NSAIDs (e.g., acetaminophen, ibuprofen) act as reversible inhibitors of cyclooxygenase. Acetaminophen is the major ingredient in Tylenol, and ibuprofen is the major ingredient in other NSAIDs such as Motrin, Nuprin, and Advil (see Fig. 35.9). Although having some relative selectivity for inhibiting either COX-1 or COX-2,

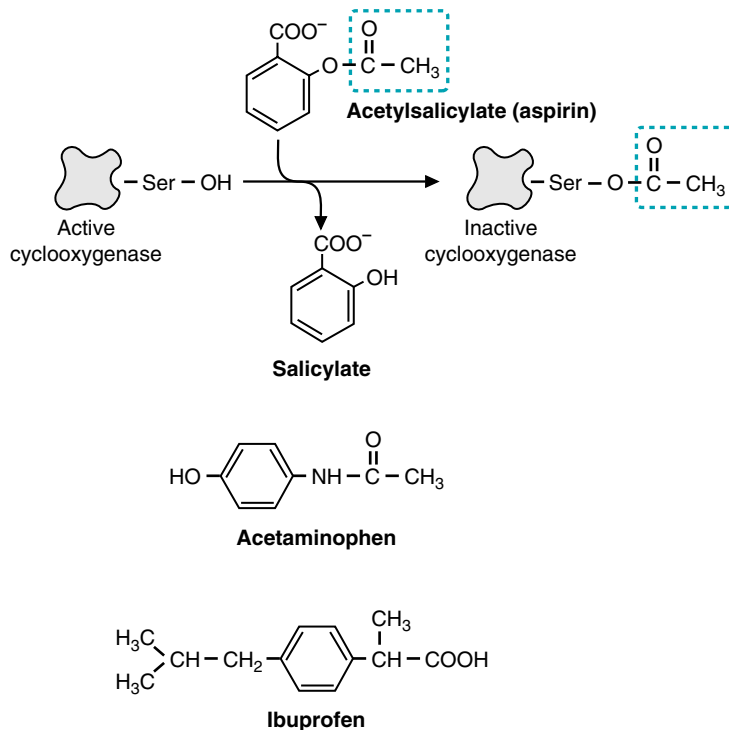


Fig. 35.9. Action of aspirin and other nonsteroidal anti-inflammatory drugs.

NSAIDs block the activity of both isoforms. These findings have provided the impetus for the development of selective COX-2 inhibitors, which are proposed to act as potent anti-inflammatory agents by inhibiting COX-2 activity, without the gastrointestinal and anti-platelet side effects commonly associated with NSAID use. These NSAID adverse effects are thought to be a result of COX-1 inhibition. Examples of these newer selective COX-2 inhibitors are celecoxib (Celebrex) and rofecoxib (Vioxx).

4. INACTIVATION OF THE PROSTAGLANDINS AND THROMBOXANES

Prostaglandins and thromboxanes are rapidly inactivated. Their half-lives ($t_{1/2}$) range from seconds to minutes. The prostaglandins are inactivated by oxidation of the 15-hydroxy group, critical for their activity, to a ketone. The double bond at carbon 13 is reduced. Subsequently, both β - and ω -oxidation of the non-ring portions occur, producing dicarboxylic acids that are excreted in the urine. Active TXA_2 is rapidly metabolized to TXB_2 by cleavage of the oxygen bridge between carbons 9 and 11 to form two hydroxyl groups. TXB_2 has no biologic activity.

B. Lipoxygenase Pathway: Synthesis of the Leukotrienes, HETE, and Lipoxins

In addition to serving as a substrate for the cyclooxygenase pathway, arachidonic acid also acts as a substrate for the lipoxygenase pathway. The lipoxygenase enzymes catalyze the incorporation of an oxygen molecule onto a carbon of one of several double bonds of arachidonic acid, forming a hydroperoxy ($-\text{OOH}$) group at these positions. With this oxygenation, the double bond isomerizes to a position one carbon removed from the hydroperoxy group and is transformed from the *cis* to the *trans* configuration (Fig. 35.10). The unstable hydroperoxy group is then converted to the more stable hydroxy group.

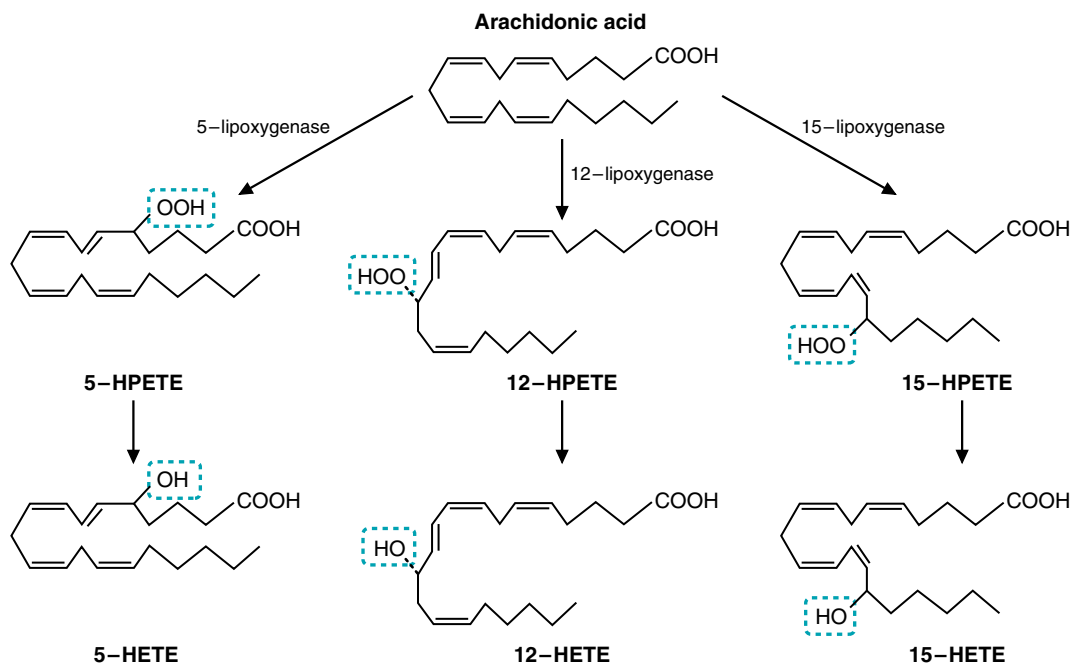


Fig. 35.10. Action of lipoxygenases in the formation of HPETEs and HETEs. Lipoxygenases add hydroperoxy groups at position 5, 12, or 15 with rearrangement of the double bond. HPETEs are unstable and are rapidly reduced to form HETEs or converted to leukotrienes and lipoxins (see Figs. 35.11 and 35.12).



Leukotrienes were so named because they are synthesized in leukocytes and contain the typical triene structure, i.e., three double bonds in series (in this case, at positions 7, 9, and 11) (see Fig. 35.11).

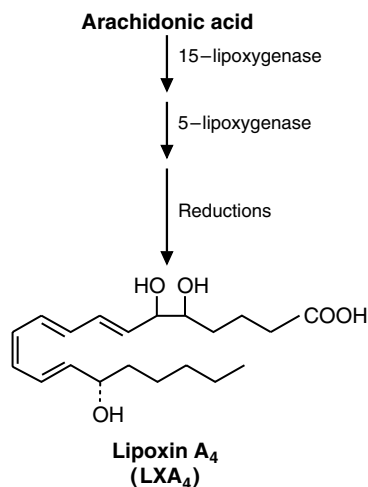


Fig. 35.12. Formation of the lipoxins. These compounds contain three hydroxyl groups.

Lipoxygenases may act at carbons 5, 12, or 15. The type of lipoxygenase varies from tissue to tissue. For example, polymorphonuclear leukocytes contain primarily 5-lipoxygenase, platelets are rich in 12-lipoxygenase, and eosinophilic leukocytes contain primarily 15-lipoxygenase.

1. LEUKOTRIENE SYNTHESIS

As shown in figure 35.11, the synthesis of the leukotrienes begins with the formation of hydroperoxyeicosatetraenoic acids (HPETEs). This product is either reduced to the corresponding hydroxy metabolites, HETEs (see Fig. 35.10), or it is metabolized to form leukotrienes or lipoxins (see Figs. 35.11 and 35.12). The major leukotrienes are produced by 5-lipoxygenase.

In leukocytes and mast cells, 5-HPETE is converted to an epoxide, leukotriene A₄ (LTA₄). The subscript number 4 refers to the presence of four double bonds in the leukotriene. Three of the double bonds (7, 9, 11) are conjugated, that is, they form a triene.

Other functional leukotrienes are formed from LTA₄ by one of two pathways. In the first, LTA₄ is converted to LTB₄, a 5,12-dihydroxy derivative. The second metabolic pathway involves the addition of reduced glutathione to carbon 6 to form LTC₄, a reaction catalyzed by glutathione *S*-transferase. Glutamate is removed from the glutathione moiety of LTC₄ through the action of γ -glutamyl transpeptidase to form LTD₄. A dipeptidase then cleaves the glycine residue from LTD₄ to form LTE₄ (see Fig. 35.11). The major functions of some of the leukotrienes are listed in Table 35.3.

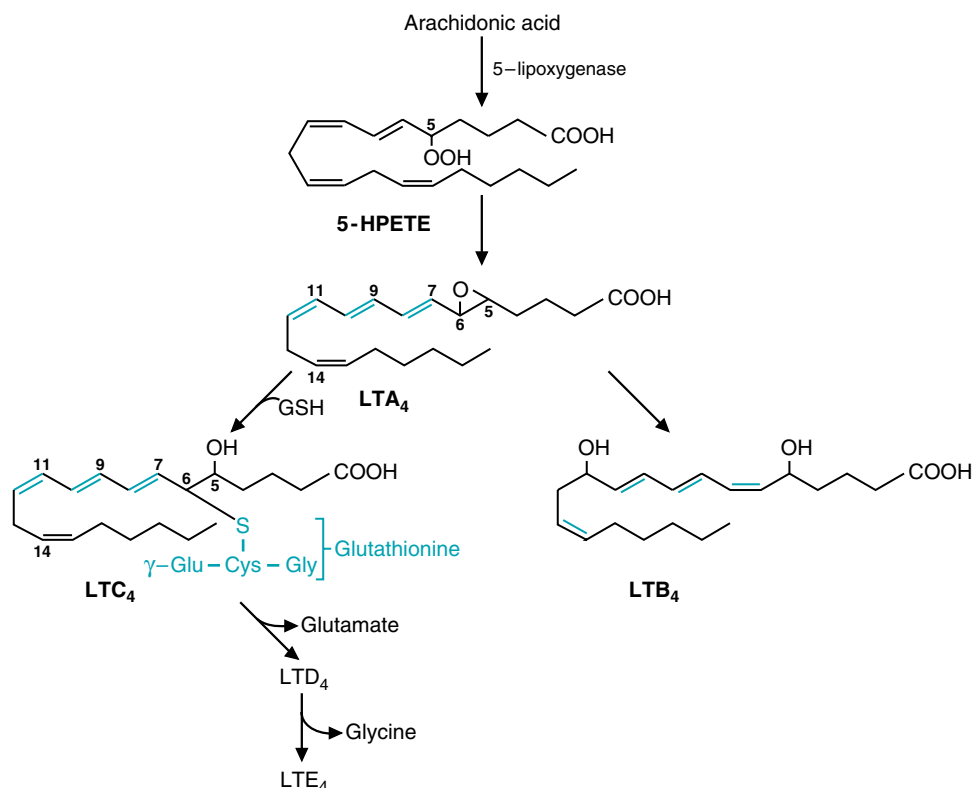


Fig. 35.11. Formation of leukotrienes and their glutathione (GSH) derivatives. HPETE = hydroperoxyeicosatetraenoic acids; LT = leukotriene (thus, LTA₄ is leukotriene A₄).

2. LIPOXIN SYNTHESIS AND ACTIONS

The lipoxins are formed through the action of 15-lipoxygenase followed by the action of 5-lipoxygenase on arachidonic acid. A series of reductions of the resultant hydroperoxy groups leads to the formation of trihydroxy derivatives of arachidonic acid known as the lipoxins (see Fig. 35.12). Lipoxins induce chemotaxis and stimulate superoxide anion production in leukocytes.

C. Cytochrome P450 Pathway: Synthesis and Actions of Epoxides, HETEs, and diHETES

A third mechanism for the oxygenation of arachidonic acid involves the cytochrome P450 pathway. The activity of the monooxygenases in this microsomal system yields epoxides, certain forms of HETEs (e.g., ω -hydroxy derivatives), and diol forms (diHETEs) (Fig. 35.13). The biologic activities of these compounds include actions in ocular, vascular, endocrine, and renal systems. Some of these actions are attributed to inhibition of Na^+ , K^+ -ATPase. The physiologic role of these compounds remains to be fully characterized.

D. Isoprostane Synthesis

Isoprostanes are derived from arachidonic acid by lipid peroxidation, initiated by free radicals. There is no enzymatic mechanism for their production. Arachidonic acid, while still a component of a phospholipid, undergoes free radical damage, and then phospholipase A2 removes the altered eicosanoid from the phospholipid and releases it into circulation (Fig. 35.14). The level of isoprostanes in the urine can be used as a measure of the oxidative stress of a patient and is a useful biologic marker for patients undergoing oxidative stress caused by a variety of disorders. Surprisingly, these altered arachidonic acid molecules also have biologic activity when measured on cultured cells; it is not known whether intracellular levels reach high enough concentrations to elicit these biologic effects. The best studied isoprostane is similar to $\text{PGF}_{2\alpha}$, and this molecule has similar effects on cultured cells as does $\text{PGF}_{2\alpha}$ (see Table 35.1).

E. Endocannabinoid Synthesis

Endocannabinoids are endogenous ligands for the cannabinoid receptor, with effects primarily in the nervous system. Anandamide was the first such ligand isolated and identified. Anandamide is synthesized in neurons from phosphatidylethanolamine, as outlined in figure 35.15. The biosynthetic pathway is unique in transferring an arachidonic acid group from the 2-position to the free amino group on ethanolamine, and then using a unique phospholipase D to cleave the modified ethanolamine from the phospholipid. The synthesis of anandamide is regulated, in part, by agonists that cause calcium influx into nerve cells. Once anandamide is released, it acts as a retrograde messenger, binding to receptors on the presynaptic membrane that alter ion fluxes such that neurotransmitter release from the presynaptic neuron can be increased and an analgesic effect obtained. Anandamide is degraded by the enzyme fatty acid amide hydrolase, which splits anandamide to arachidonic acid and ethanolamine. The hydrolase enzyme is the target of drug research, because inhibiting the action of this enzyme will prolong the analgesic effects induced by anandamide.

III. MECHANISM OF ACTION OF THE EICOSANOIDS

The eicosanoids have a wide variety of physiologic effects, which are generally initiated through an interaction of the eicosanoid with a specific receptor on the plasma membrane of a target cell (Table 35.4). This eicosanoid-receptor binding either activates the adenylate cyclase-cAMP-protein kinase A system (PGE, PGD,

Table 35.3. Some Functions of Leukotrienes

LTB₄
Increases
Vascular permeability
T-cell proliferation
Leukocyte aggregation
INF- γ
IL-1
IL-2
LTC₄ and LTD₄
Increase
Bronchoconstriction
Vascular permeability
INF- γ

Abbreviations: INF = interferon; IL = interleukin.

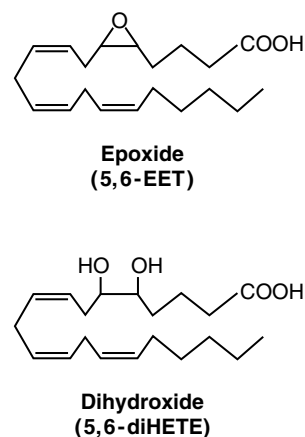


Fig. 35.13. Examples of compounds produced from arachidonic acid by the cytochrome P450 pathway.



Recall that oxidative stress results from reactive oxygen or nitrogen species that are not fully neutralized by anti-oxidants (see Chapter 24). The increase in reactive oxygen and nitrogen species results in protein, lipid, and nucleic acid modifications. Many neurodegenerative diseases, such as Alzheimer disease, Parkinson disease, and Lou Gehrig disease, have in common increased oxidative stress.

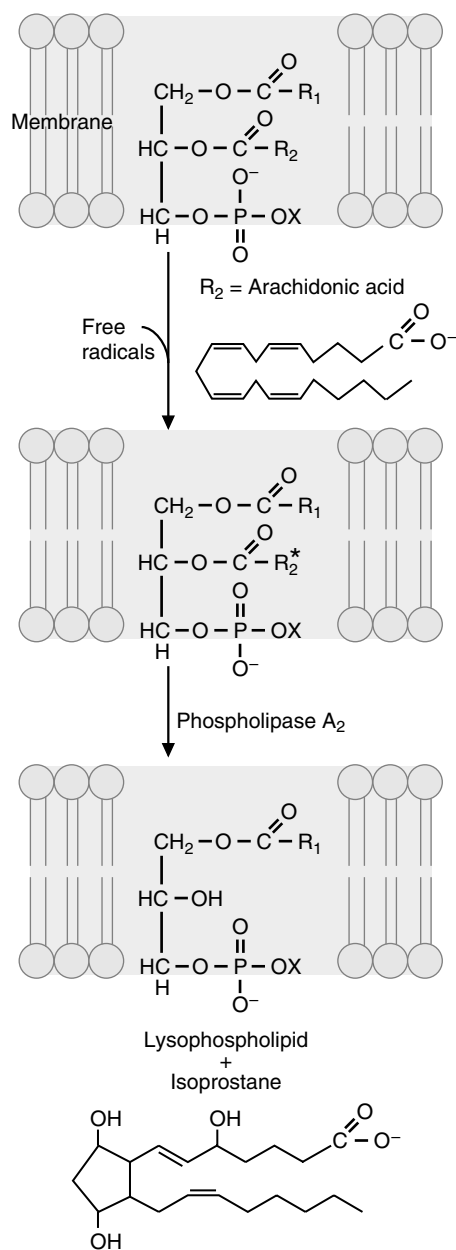


Fig. 35.14. Generation of an isoprostane. Radical damage to a phospholipid on the arachidonic acid residue at position 2 generates an isoprostane, which is then removed from the damaged phospholipid by phospholipase A₂. The example of an isoprostane shown in this figure is just one of many that can be produced.

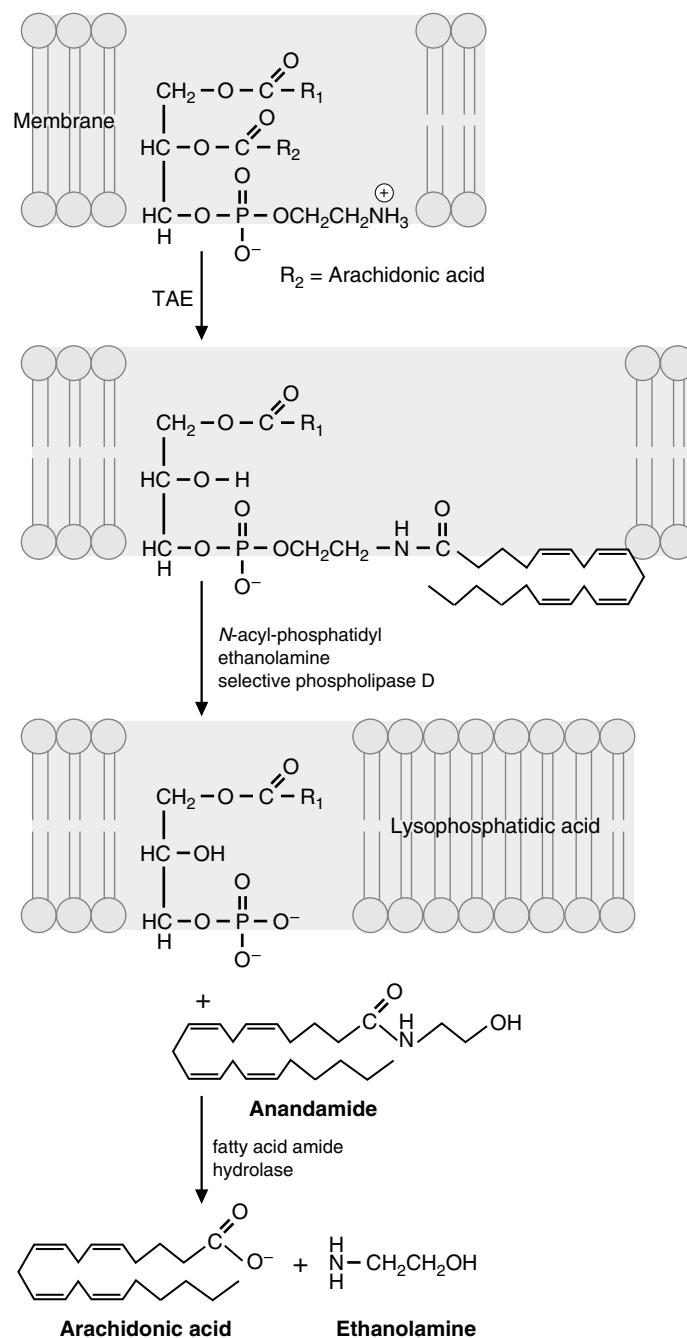


Fig. 35.15. Anandamide synthesis and degradation. TAE = transacylase.

Table 35.4. Prostaglandin and Thromboxane Receptors

Receptor	Type Ligand	G-protein coupled	cAMP response	Calcium response
DP	PGD series	Yes	Increase	None
EP1	PGE series	Yes	None	Increase
EP2	PGE series	Yes	Increase	None
EP3	PGE series	Yes	Decrease	None
EP4	PGE series	Yes	Increase	None
FP	PGF series	Yes	None	Increase
IP	PGI series	Yes	Increase	None
TP	Thromboxane A	Yes	None	Increase

and PGI series) or causes an increase in the level of calcium in the cytosol of target cells (PGF_{2α}, TXA₂, the endoperoxides, and the leukotrienes).

In some systems, the eicosanoids appear to modulate the degree of activation of adenylate cyclase in response to other stimuli. In these instances, the eicosanoid may bind to a regulatory subunit of the GTP-binding proteins (G proteins) within the plasma membrane of the target cell (see Chapter 11). If the eicosanoid binds to the stimulatory subunit, the effect of the stimulus is amplified. Conversely, if the eicosanoid binds to the inhibitory subunit, the cellular response to the stimulus is reduced. Through these influences on the activation of adenylate cyclase, eicosanoids contribute to the regulation of cell function.

Some of the biologic effects of certain eicosanoids occur as a result of a paracrine or autocrine action. One paracrine action is the contraction of vascular smooth muscle cells caused by TXA₂ released from circulating platelets (vasoconstriction). An autocrine action of eicosanoids is exemplified by platelet aggregation induced by TXA₂ produced by the platelets themselves.

The eicosanoids influence the cellular function of almost every tissue of the body. Certain organ systems are affected to a greater degree than others.

CLINICAL COMMENTS



In the presence of aspirin, cyclooxygenase is irreversibly inactivated by acetylation. New cyclooxygenase molecules are not produced in platelets, because these cells have no nuclei and, therefore, cannot synthesize new mRNA. Thus, the inhibition of cyclooxygenase by aspirin persists for the lifespan of the platelet (7–10 days). When aspirin is taken daily at doses between 81 and 325 mg, new platelets are affected as they are generated. Higher doses do not improve efficacy but do increase side effects, such as gastrointestinal bleeding and easy bruising.

Patients with established or suspected atherosclerotic coronary disease, such as **Ann Jeina**, **Cora Nari**, and **Ivan Applebod**, benefit from the action of low-dose aspirin (approximately 162 mg/day), which produces a mild defect in hemostasis. This action of aspirin helps to prevent thrombus formation in the area of an atherosclerotic plaque at critical sites in the vascular tree.



Corticosteroids reduce inflammation, in part, through their inhibitory effect on phospholipase A2. In addition, suppression of COX-2 induction is now thought to be a primary anti-inflammatory mechanism of action for glucocorticoids. Despite the unquestionable value of glucocorticoid therapy in a variety of diseases associated with acute inflammation of tissues, the suppression of the inflammatory response with pharmacologic doses of corticosteroids is potentially hazardous. The sudden appearance of temporary glucose intolerance when **Emma Wheezer** was treated with large doses of dexamethasone, a gluconeogenic steroid (glucocorticoid), is just one of the many potential adverse effects of this class of drugs when given systemically in pharmacologic doses over an extended period. The inhaled steroids, conversely, have far fewer systemic side effects because their absorption across the bronchial mucosa into the circulation is very limited. This property allows them to be used over prolonged periods in the treatment of asthma. The fact that inhalation allows direct delivery of the agent to the primary site of inflammation adds to their effectiveness in the treatment of these patients.

BIOCHEMICAL COMMENTS



Inflammation is the response of the body to infection or injury, directed at destroying the infectious agents and repairing the damaged areas. It involves an increase of the blood supply to the affected region by means of



Although our knowledge of the spectrum of biologic actions of the endogenous eicosanoids is incomplete, several actions are well-enough established to allow their application in a variety of clinical situations or diseases. For example, drugs that are analogs of PGE₁ and PGE₂ suppress gastric ulceration, in part by inhibiting secretion of hydrochloric acid in the mucosal cells of the stomach. Analogs of PGE₁ are used in the treatment of sexual impotence. Men with certain forms of sexual impotence can self-inject this agent into the corpus cavernosum of the penis to induce immediate but temporary penile tumescence. The erection lasts for 1 to 3 hours. The stimulatory action of PGE₂ and PGF_{2α} on uterine muscle contraction has led to the use of these prostaglandins to induce labor and to control postpartum bleeding. PGE₁ is also used as palliative therapy in neonates with congenital heart defects to maintain patency of the ductus arteriosus until surgery can be performed. Analogs of PGI₂ have been shown to be effective in the treatment of primary pulmonary hypertension.



Although uncomfortable, the pain, swelling, and fever that are part of the inflammatory response serve as an important warning sign that the host is threatened and that some specific counteractions must be taken against the offending agent or process. Although the use of anti-inflammatory drugs may bring welcome symptomatic relief, their use may, in part, diminish the effectiveness of the host's response to the inciting agent.

vasodilation. The capillaries become more permeable so that fluid, large molecules, and white blood cells can cross, leaving the blood and entering the tissue. White blood cells (particularly neutrophils and monocytes) move by chemotaxis to the injured site. Redness (rubor), heat (calor), swelling (tumor), and pain (dolor) are associated with the inflammatory process. Redness and heat are caused by the increased blood flow. Swelling is the result of the increased movement of fluid and white blood cells into the area of inflammation. Pain is caused by the release of chemical compounds and the compression of nerves in the vicinity of the inflammatory process.

The chemical mediators of inflammation usually are produced by activation of complement (a family of blood proteins that are cleaved to form active fragments) or of the blood clotting cascade (see Chapter 45). These processes cause the release of histamine from mast cells, and the production of kinins by cleavage of kininogens. Among their other effects, both histamine and kinins increase vascular permeability. They stimulate the synthesis of eicosanoids that act on the motility and metabolism of white blood cells and cause the aggregation of platelets to arrest bleeding. Some of the prostaglandins act on thermoregulatory centers of the brain, producing fever. Cytokines are also released that stimulate the proliferation of cells involved in the immune response.

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REVIEW QUESTIONS—CHAPTER 35

- In humans, prostaglandins are primarily derived from which of the following?
 - Glucose
 - Acetyl CoA
 - Arachidonic acid
 - Oleic acid
 - Leukotrienes
- Aspirin will inhibit which of the following reaction pathways?
 - Arachidonic acid → thromboxanes
 - Arachidonic acid → leukotrienes
 - Arachidonic acid → phospholipids
 - Linoleic acid → arachidonic acid
 - Acetyl CoA → linoleic acid

3. Which of the following drugs leads to the covalent modification, and inactivation, of both the COX-1 and COX-2 enzymes?
 - (A) Aspirin
 - (B) Tylenol
 - (C) Celebrex
 - (D) Vioxx
 - (E) Advil

4. Thromboxane A₂, which is found in high levels in platelets, aids in wound repair through induction of which of the following activities?
 - (A) Inhibits COX-2 gene expression
 - (B) Inhibits COX-1 gene expression
 - (C) Vasoconstriction
 - (D) Vasodilation
 - (E) Bronchodilation

5. Certain prostaglandins, when binding to their receptor, induce an increase in intracellular calcium levels. The signal that leads to the elevation of intracellular calcium is initiated by which of the following enzymes?
 - (A) Protein kinase A
 - (B) Phospholipase C
 - (C) Phospholipase A₂
 - (D) Protein kinase C
 - (E) Cyclooxygenase

36 Integration of Carbohydrate and Lipid Metabolism

The purpose of this chapter is to summarize and integrate the major pathways for the utilization of carbohydrates and fats as fuels. We will concentrate on reviewing the regulatory mechanisms that determine the flux of metabolites in the fed and fasting states, integrating the pathways that were described separately under carbohydrate and lipid metabolism. The next section of the book covers the mechanisms by which the pathways of nitrogen metabolism are coordinated with fat and carbohydrate metabolism.

*For the species to survive, it is necessary for us to store excess food when we eat and to use these stores when we are fasting. Regulatory mechanisms direct compounds through the pathways of metabolism involved in the storage and utilization of fuels. These mechanisms are controlled by **hormones**, by the **concentration of available fuels**, and by the **energy needs** of the body.*

*Changes in hormone levels, in the concentration of fuels, and in energy requirements affect the activity of key enzymes in the major pathways of metabolism. Intracellular **enzymes** are generally regulated by **activation and inhibition**, by **phosphorylation and dephosphorylation** (or other covalent modifications), by **induction and repression** of synthesis, and by **degradation**. Activation and inhibition of enzymes cause immediate changes in metabolism. Phosphorylation and dephosphorylation of enzymes affect metabolism slightly less rapidly. Induction and repression of enzyme synthesis are much slower processes, usually affecting metabolic flux over a period of hours. Degradation of enzymes decreases the amount available to catalyze reactions.*

*The pathways of metabolism have **multiple control points** and **multiple regulators** at each control point. The function of these complex mechanisms is to produce a graded response to a stimulus and to provide sensitivity to multiple stimuli so that an exact balance is maintained between flux through a given step (or series of steps) and the need or use for the product. Pyruvate dehydrogenase is an example of an enzyme that has multiple regulatory mechanisms. Regardless of insulin levels, the enzyme cannot become fully activated in the presence of products and absence of substrates.*

*The major hormones that regulate the pathways of fuel metabolism are **insulin** and **glucagon**. In liver, all effects of glucagon are reversed by insulin, and some of the pathways that insulin activates are inhibited by glucagon. Thus, the pathways of carbohydrate and lipid metabolism are generally regulated by changes in the **insulin/glucagon ratio**.*

*Although **glycogen** is a critical storage form of fuel because blood glucose levels must be carefully maintained, adipose **triacylglycerols** are quantitatively the major fuel store in the human. After a meal, both dietary glucose and fat are stored in adipose tissue as triacylglycerol. This fuel is released during fasting, when it provides the main source of energy for the tissues of the body. The length of time we can survive without food depends mainly on the size of our bodies' fat stores.*



THE WAITING ROOM



Within 2 months of her surgery to remove a benign insulin-secreting β cell tumor of the pancreas, **Bea Selmass** was again jogging lightly. She had lost the 8 lb that she had gained in the 6 weeks before her surgery. Because her hypoglycemic episodes were no longer occurring, she had no need to eat frequent carbohydrate snacks to prevent the adrenergic and neuroglycopenic symptoms that she had experienced when her tumor was secreting excessive amounts of insulin.



A few months after her last hospitalization, **Di Abietes** was once again brought to the hospital emergency room in diabetic ketoacidosis (DKA). Blood samples for glucose and electrolytes were drawn repeatedly during the first 24 hours. The hospital laboratory reported that the serum in each of these specimens appeared opalescent rather than having its normal clear or transparent appearance. This opalescence results from light scattering caused by the presence of elevated levels of triacylglycerol-rich lipoproteins in the blood.



When **Ann Sulin** initially presented with type 2 diabetes mellitus at age 39, she was approximately 30 lb above her ideal weight. Her high serum glucose levels were accompanied by abnormalities in her 14-hour fasting lipid profile. Her serum total cholesterol, low-density lipoprotein (LDL) cholesterol, and triacylglycerol levels were elevated, and her serum high-density lipoprotein (HDL) cholesterol level was below the normal range.

I. REGULATION OF CARBOHYDRATE AND LIPID METABOLISM IN THE FED STATE

A. Mechanisms That Affect Glycogen and Triacylglycerol Synthesis in Liver

After a meal, the liver synthesizes glycogen and triacylglycerol. The level of glycogen stored in the liver can increase from approximately 80 g after an overnight fast to a limit of approximately 200–300 g. Although the liver synthesizes triacylglycerol, it does not store this fuel but rather packages it in very-low-density lipoprotein (VLDL) and secretes it into the blood. The fatty acids of the VLDL triacylglycerols secreted from the liver are stored as adipose triacylglycerols. Adipose tissue has an almost infinite capacity to store fat, limited mainly by the ability of the heart to pump blood through the capillaries of the expanding adipose mass. Although we store fat throughout our bodies, it tends to accumulate in places where it does not interfere too much with our mobility: in the abdomen, hips, thighs, and buttocks.

Both the synthesis of liver glycogen and the conversion by the liver of dietary glucose to triacylglycerol (lipogenesis) are regulated by mechanisms involving key enzymes in these pathways.

1. GLUCOKINASE

After a meal, glucose can be converted to glycogen or to triacylglycerol in the liver. For both processes, glucose is first converted to glucose 6-phosphate by glucokinase, a liver enzyme that has a high K_m for glucose (Fig. 36.1). Because of the enzyme's low affinity for glucose, this enzyme is most active in the fed state, when the concentration of glucose is particularly high because the hepatic portal vein

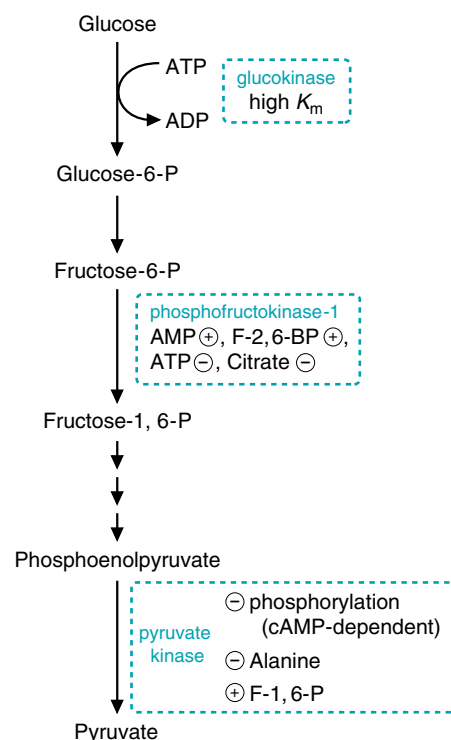


Fig. 36.1. Regulation of glucokinase, PFK-1, and pyruvate kinase in the liver.

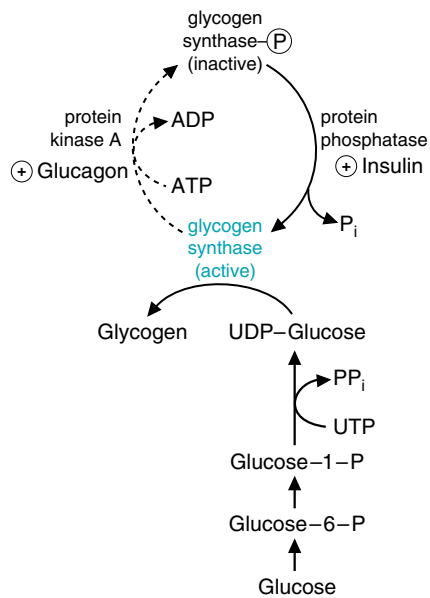


Fig. 36.2. Regulation of glycogen synthase. This enzyme is phosphorylated by a series of kinases, which are initiated by the cAMP-dependent protein kinase, under fasting conditions. It is dephosphorylated and active after a meal, and glycogen is stored. Circled P = phosphate; a circled + sign = activated by; a circled - sign = inhibited by.

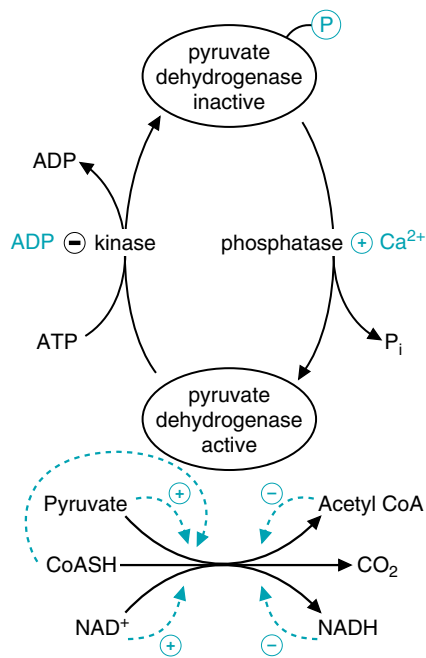


Fig. 36.3. Regulation of pyruvate dehydrogenase (PDH). A kinase associated with the PDH complex phosphorylates and inactivates the enzyme.

carries digestive products directly from the intestine to the liver. Synthesis of glucokinase is also induced by insulin (which is elevated after a meal) and repressed by glucagon (which is elevated during fasting). In keeping with the liver's function in maintaining blood glucose levels, this system is set up such that the liver can only metabolize glucose when sugar levels are high, and not when sugar levels are low.

2. GLYCOGEN SYNTHASE

In the conversion of glucose 6-phosphate to glycogen, the key regulatory enzyme is glycogen synthase. This enzyme is activated by the dephosphorylation that occurs when insulin is elevated and glucagon is decreased (Fig. 36.2) and by the increased level of glucose.

3. PHOSPHOFRUCTOKINASE-1 AND PYRUVATE KINASE

For lipogenesis, glucose 6-phosphate is converted through glycolysis to pyruvate. Key enzymes that regulate this pathway in the liver are phosphofructokinase-1 (PFK-1) and pyruvate kinase. PFK-1 is allosterically activated in the fed state by fructose 2,6-bisphosphate and adenosine monophosphate (AMP) (see Fig. 36.1). Phosphofructokinase-2, the enzyme that produces the activator fructose 2,6-bisphosphate, is dephosphorylated and active after a meal (see Chapter 22). Pyruvate kinase is also activated by dephosphorylation, which is stimulated by the increase of the insulin/glucagon ratio in the fed state (see Fig. 36.1).

4. PYRUVATE DEHYDROGENASE AND PYRUVATE CARBOXYLASE

The conversion of pyruvate to fatty acids requires a source of acetyl CoA in the cytosol. Pyruvate can only be converted to acetyl CoA in mitochondria, so it enters mitochondria and forms acetyl CoA through the pyruvate dehydrogenase (PDH) reaction. This enzyme is dephosphorylated and most active when its supply of substrates and adenosine diphosphate (ADP) is high, its products are used, and insulin is present (Fig. 36.3).

Pyruvate is also converted to oxaloacetate. The enzyme that catalyzes this reaction, pyruvate carboxylase, is activated by acetyl CoA. Because acetyl CoA cannot directly cross the mitochondrial membrane to form fatty acids in the cytosol, it condenses with oxaloacetate, producing citrate. The citrate that is not required for tri-carboxylic acid (TCA) cycle activity crosses the membrane and enters the cytosol.

As fatty acids are produced under conditions of high energy, the high NADH/NAD⁺ ratio in the mitochondria inhibits isocitrate dehydrogenase, which leads to citrate accumulation within the mitochondrial matrix. As the citrate accumulates, it is transported out into the cytosol to donate carbons for fatty acid synthesis.

5. CITRATE LYASE, MALIC ENZYME, AND GLUCOSE 6-PHOSPHATE DEHYDROGENASE

In the cytosol, citrate is cleaved by citrate lyase, an inducible enzyme, to form oxaloacetate and acetyl CoA (Fig. 36.4). The acetyl CoA is used for fatty acid biosynthesis and for cholesterol synthesis, pathways that are activated by insulin. Oxaloacetate is recycled to pyruvate via cytosolic malate dehydrogenase and malic enzyme, which is inducible. Malic enzyme generates NADPH for the reactions of the fatty acid synthase complex. NADPH is also produced by the two enzymes of the pentose phosphate pathway (see Chapter 29), glucose 6-phosphate dehydrogenase and 6-phosphogluconate dehydrogenase. Glucose 6-phosphate dehydrogenase is also induced by insulin.

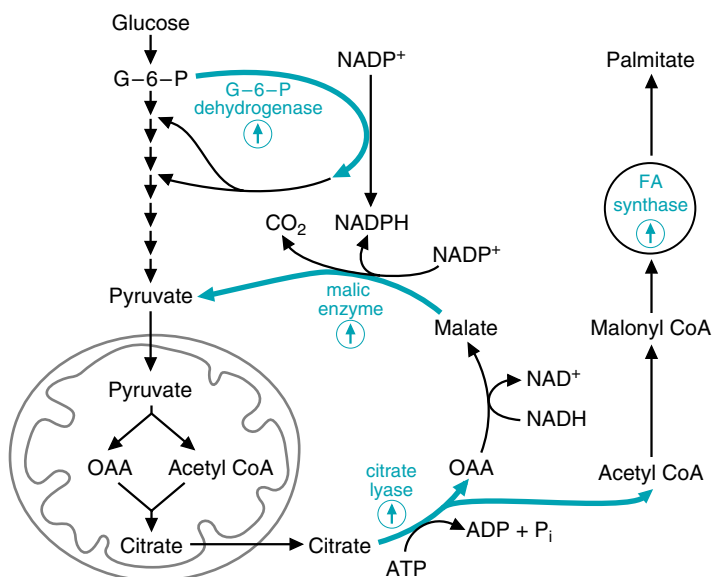


Fig. 36.4. Regulation of citrate lyase, malic enzyme, glucose 6-phosphate dehydrogenase, and fatty acid synthase. Citrate lyase, which provides acetyl CoA for fatty acid biosynthesis, and fatty acid synthase, the enzymes that provide NADPH (malic enzyme, glucose 6-phosphate dehydrogenase), as well as fatty acid synthase, are inducible (circled ↑).

6. ACETYL CoA CARBOXYLASE

Acetyl CoA is converted to malonyl CoA, which provides the 2-carbon units for elongation of the growing fatty acyl chain on the fatty acid synthase complex. Acetyl CoA carboxylase, the enzyme that catalyzes the conversion of acetyl CoA to malonyl CoA, is controlled by three of the major mechanisms that regulate enzyme activity (Fig. 36.5). It is activated by citrate, which causes the enzyme to

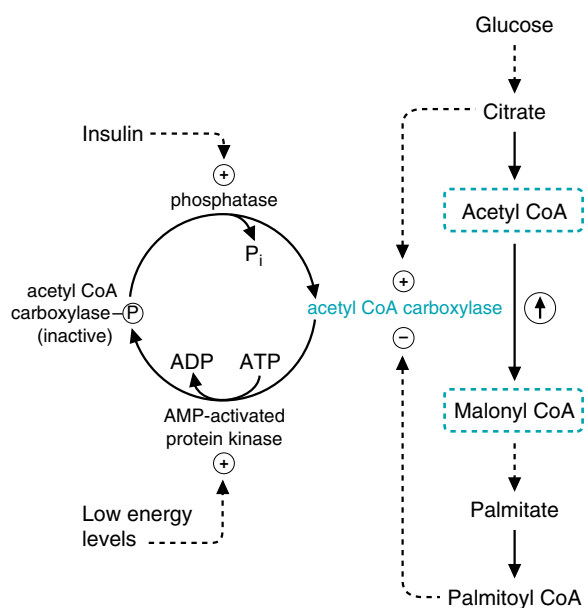


Fig. 36.5. Regulation of acetyl CoA carboxylase (AcC). AcC is regulated by activation and inhibition, by phosphorylation (mediated by the AMP-activated protein kinase) and dephosphorylation (via an insulin-stimulated phosphatase), and by induction and repression. It is active in the fed state.

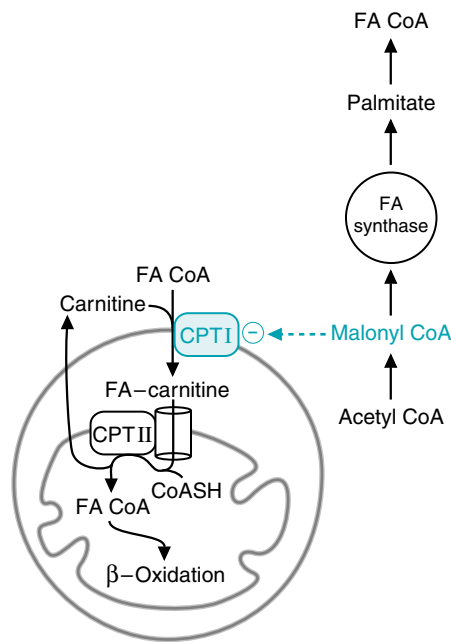


Fig. 36.6. Inhibition of transport of fatty acids (FA) into mitochondria by malonyl CoA. In the fed state, malonyl CoA (the substrate for fatty acid synthesis produced by acetyl CoA carboxylase) is elevated. It inhibits CPTI, preventing the transport of long-chain fatty acids into mitochondria. Therefore, substrate is not available for β -oxidation and ketone body synthesis.

Q: Why does **Ann Sulin** have a hypertriglyceridemia?

 **Di Abietes** has insulin-dependent diabetes mellitus (type 1), a disease associated with a severe deficiency or absence of insulin production by the β cells of the pancreas. One of the effects of insulin is to stimulate production of LPL. Because of low insulin levels, **Di Abietes** tends to have low levels of this enzyme. Hydrolysis of the triacylglycerols in chylomicrons and in VLDL is decreased, and hypertriglyceridemia results.

polymerize, and inhibited by long-chain fatty acyl CoA. A phosphatase stimulated by insulin activates the enzyme by dephosphorylation. The third means by which this enzyme is regulated is induction: the quantity of the enzyme increases in the fed state.

Malonyl CoA, the product of the acetyl CoA carboxylase reaction, provides the carbons for the synthesis of palmitate on the fatty acid synthase complex. Malonyl CoA also inhibits carnitine:palmitoyltransferase I (CPTI, also known as carnitine:acyltransferase I), the enzyme that prepares long-chain fatty acyl CoA for transport into mitochondria (Fig. 36.6). In the fed state, when acetyl CoA carboxylase is active and malonyl CoA levels are elevated, newly synthesized fatty acids are converted to triacylglycerols for storage, rather than being transported into mitochondria for oxidation and ketone body formation.

7. FATTY ACID SYNTHASE COMPLEX

In a well-fed individual, the quantity of the fatty acid synthase complex is increased (see Fig. 36.4). The genes that produce this enzyme complex are induced by increases in the insulin/glucagon ratio. The amount of the complex increases slowly after a few days of a high-carbohydrate diet.

Glucose 6-phosphate dehydrogenase, which generates NADPH in the pentose phosphate pathway, and malic enzyme, which produces NADPH, are also induced by the increase of insulin.

The palmitate produced by the synthase complex is converted to palmitoyl CoA and elongated and desaturated to form other fatty acyl CoA molecules, which are converted to triacylglycerols. These triacylglycerols are packaged and secreted into the blood as VLDL.

B. Mechanisms That Affect the Fate of Chylomicrons and VLDL

The lipoprotein triacylglycerols in chylomicrons and VLDL are hydrolyzed to fatty acids and glycerol by lipoprotein lipase (LPL), an enzyme attached to endothelial cells of capillaries in muscle and adipose tissue. The enzyme found in muscle, particularly heart muscle, has a low K_m for these blood lipoproteins. Therefore, it acts even when these lipoproteins are present at very low concentrations in the blood. The fatty acids enter muscle cells and are oxidized for energy. The enzyme found in adipose tissue has a higher K_m and is most active after a meal when blood lipoprotein levels are elevated.

C. Mechanisms That Affect Triacylglycerol Storage in Adipose Tissue

Insulin stimulates adipose cells to synthesize and secrete LPL, which hydrolyzes the chylomicron and VLDL triacylglycerols. The C_{II} apoprotein, donated to chylomicrons and VLDL by HDL, activates LPL (Fig. 36.7).

Fatty acids released from chylomicrons and VLDL by LPL are stored as triacylglycerols in adipose cells. The glycerol released by LPL is not used by adipose cells because they lack glycerol kinase. Glycerol can be used by liver cells, however, because these cells do contain glycerol kinase. In the fed state, liver cells convert glycerol to the glycerol moiety of the triacylglycerols of VLDL, which is secreted from the liver to distribute the newly synthesized triglycerides to the tissues.

Insulin causes the number of glucose transporters in adipose cell membranes to increase. Glucose enters these cells, and is oxidized, producing energy and providing the glycerol 3-phosphate moiety for triacylglycerol synthesis (via the dihydroxyacetone phosphate intermediate of glycolysis).

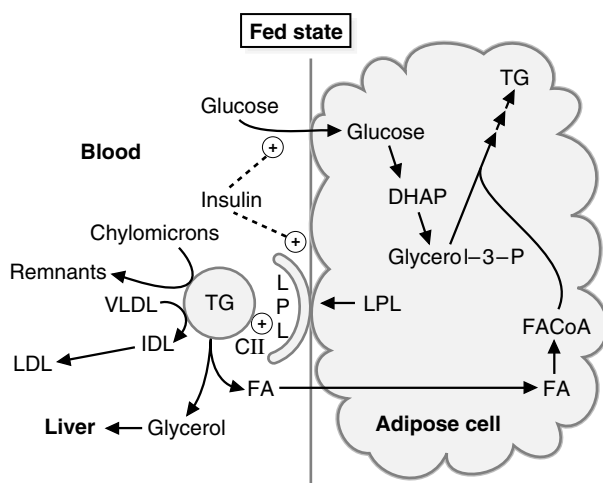


Fig. 36.7. Regulation of the storage of triacylglycerols (TG) in adipose tissue. Insulin stimulates the secretion of LPL from adipose cells and the transport of glucose into these cells. ApoC_{II} activates LPL. FA = fatty acids.

II. REGULATION OF CARBOHYDRATE AND LIPID METABOLISM DURING FASTING

A. Mechanisms in Liver That Serve to Maintain Blood Glucose Levels

During fasting, the insulin/glucagon ratio decreases. Liver glycogen is degraded to produce blood glucose because enzymes of glycogen degradation are activated by cAMP-directed phosphorylation (Fig. 36.8). Glucagon stimulates adenylate cyclase to produce cAMP, which activates protein kinase A. Protein kinase A phosphorylates phosphorylase kinase, which then phosphorylates and activates glycogen phosphorylase. Protein kinase A also phosphorylates but, in this case, inactivates glycogen synthase.

Gluconeogenesis is stimulated because the synthesis of phosphoenolpyruvate carboxykinase, fructose 1,6-bisphosphatase, and glucose 6-phosphatase is induced and because there is an increased availability of precursors. Fructose 1,6-bisphosphatase is also activated because the levels of its inhibitor, fructose 2,6-bisphosphate, are low (Fig. 36.9). During fasting, the activity of the corresponding enzymes of glycolysis is decreased.

B. Mechanisms That Affect Lipolysis in Adipose Tissue

During fasting, as blood insulin levels fall and glucagon levels rise, the level of cAMP rises in adipose cells. Consequently, protein kinase A is activated and causes phosphorylation of hormone-sensitive lipase (HSL). The phosphorylated form of this enzyme is active and cleaves fatty acids from triacylglycerols (Fig. 36.10). Other hormones (e.g., epinephrine, adrenocorticotrophic hormone [ACTH], growth hormone) also activate this enzyme (see Chapter 43). Glyceroneogenesis and resynthesis of triglyceride by the adipocyte regulates the rate of release of fatty acids during fasting.

C. Mechanisms That Affect Ketone Body Production by the Liver

As fatty acids are released from adipose tissue during fasting, they travel in the blood complexed with albumin. These fatty acids are oxidized by various tissues, particularly muscle. In the liver, fatty acids are transported into mitochondria

A: **Ann Sulin** has type 2 diabetes mellitus. She produces insulin, but her adipose tissue is resistant to its actions. Therefore, her adipose tissue does not produce as much LPL as a normal person, which is one of the reasons why VLDL and chylomicrons remain elevated in her blood.

B: Twenty to thirty percent of patients with an insulinoma gain weight as part of their syndrome. **Bea Selmass** gained 8 lb in the 6 weeks before her surgery. Although she was primed by her high insulin levels both to store and to use fuel more efficiently, she would not have gained weight if she had not consumed more calories than she required for her daily energy expenditure during her illness (see Chapter 1). Bea consumed extra carbohydrate calories, mostly as hard candies and table sugar, to avoid the symptoms of hypoglycemia.

C: Induction of enzyme synthesis requires activation of transcription factors. One of the factors that is activated is CREB, which stands for cAMP response element binding protein. CREB is phosphorylated and activated by the cAMP-dependent protein kinase, which itself is activated on glucagon or epinephrine stimulation. A second set of transcription factors activated are the C/EBPs (CCAAT/ enhancer binding proteins), although the regulation of their activity is still under investigation.

D: **Di Abietes** suffers from hyperglycemia because her insulin levels tend to be low and her glucagon levels tend to be high. Her muscle and adipose cells do not take glucose up at a normal rate, and she produces glucose by glycogenolysis and gluconeogenesis. As a result, her blood glucose levels are elevated.

Ann Sulin is in a similar metabolic state. However, in this case, she produces insulin, but her tissues are resistant to its actions.

Q: Would **Ann Sulin's** serum triacylglycerol level be elevated?



Because the adipose tissue of individuals with type 2 diabetes mellitus is relatively resistant to insulin's inhibition of HSL and its stimulation of LPL, **Ann Sulin's** serum triacylglycerol level would be elevated for the same reasons as those that caused the hypertriglyceridemia in **Di Abietes**.

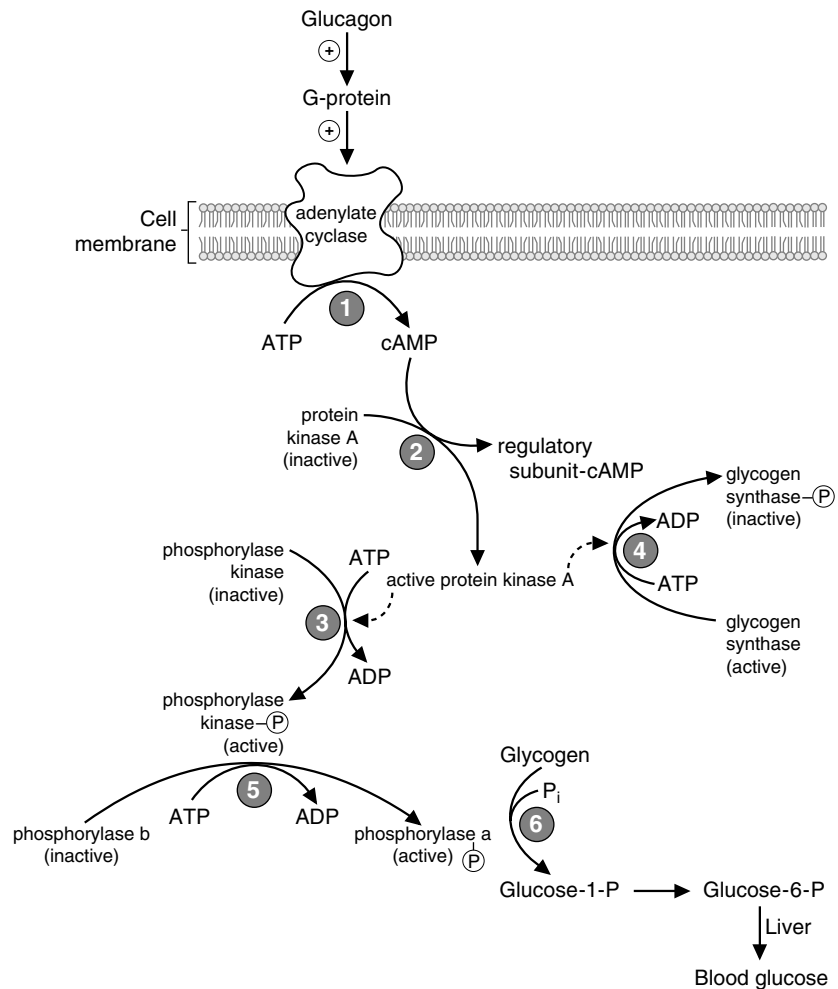


Fig. 36.8. Regulation of the enzymes of glycogen degradation in the liver. 1. Glucagon (or epinephrine) binds to its cell membrane receptor, initially activating a G protein, which activates adenylate cyclase. 2. As cAMP levels rise, inhibitory subunits are removed from protein kinase A, which now phosphorylates phosphorylase kinase (step 3). 4. The cAMP-dependent protein kinase also phosphorylates glycogen synthase, inactivating the enzyme. 5. Phosphorylated phosphorylase kinase phosphorylates glycogen phosphorylase. 6. Phosphorylated glycogen phosphorylase catalyzes the phosphorolysis of glycogen, producing glucose 1-phosphate. These events occur during fasting and produce glucose to maintain a relatively constant level of blood glucose.



Insulin normally inhibits lipolysis by decreasing the lipolytic activity of HSL in the adipocyte. Individuals such as **Di Abietes**, who have a deficiency of insulin, have an increase in lipolysis and a subsequent increase in the concentration of free fatty acids in the blood. The liver, in turn, uses some of these fatty acids to synthesize triacylglycerols, which then are used in the hepatic production of VLDL. VLDL is not stored in the liver but is secreted into the blood, raising its serum concentration. **Di** also has low levels of LPL because of decreased insulin levels. Her hypertriglyceridemia is the result, therefore, of both overproduction of VLDL by the liver and decreased breakdown of VLDL triacylglycerol for storage in adipose cells.

The serum begins to appear cloudy when the triacylglycerol level reaches 200 mg/dL. As the triacylglycerol level increases still further, the degree of serum opalescence increases proportionately.

because acetyl CoA carboxylase is inactive, malonyl CoA levels are low, and CPTI (carnitine:acyltransferase I) is active (see Fig. 36.6). Acetyl CoA, produced by β -oxidation, is converted to ketone bodies. Ketone bodies are used as an energy source by many tissues (Table 36.1) to spare the use of glucose and the necessity of degrading muscle protein to provide the precursors for gluconeogenesis. The high levels of acetyl CoA in the liver (derived from fat oxidation) inhibit pyruvate dehydrogenase (which prevents pyruvate from being converted to acetyl CoA) and activate pyruvate carboxylase, which produces oxaloacetate for gluconeogenesis. The oxaloacetate does not condense with acetyl CoA to form citrate for two reasons. The first is that under these conditions (a high rate of fat oxidation in the liver mitochondria), energy levels in the mitochondrial matrix are high; that is, there are high levels of NADH and ATP present. The high NADH level inhibits isocitrate dehydrogenase. As a result, citrate accumulates and inhibits citrate synthase from producing more citrate. The second reason that citrate synthesis is depressed is that the

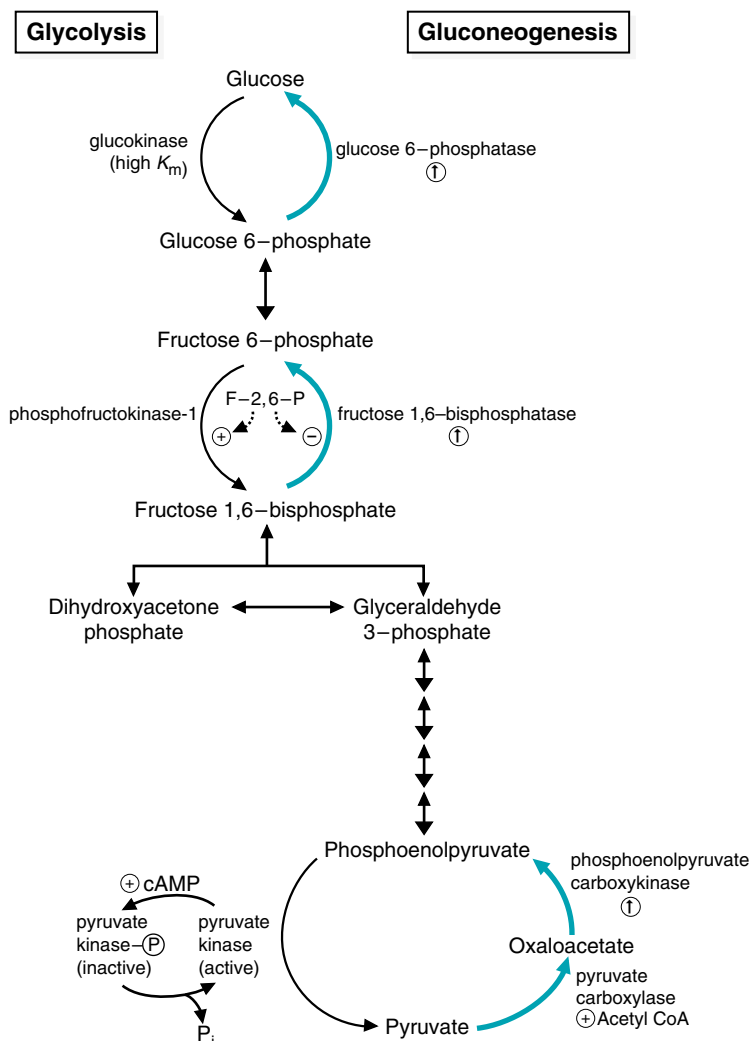


Fig. 36.9. Regulation of gluconeogenesis and glycolysis during fasting. The gluconeogenic enzymes phosphoenolpyruvate carboxykinase, fructose 1,6-bisphosphatase, and glucose 6-phosphatase are induced. Fructose 1,6-bisphosphatase is also active because, during fasting, the level of its inhibitor, fructose 2,6-bisphosphate, is low. The corresponding enzymes of glycolysis are not very active during fasting. The rate of glucokinase is low because it has a high K_m for glucose and the glucose concentration is low. Phosphofructokinase-1 is not very active because the concentration of its activator fructose 2,6-bisphosphate is low. Pyruvate kinase is inactivated by cAMP-mediated phosphorylation.

Table 36.1. Fuel Utilization by Various Tissues during Starvation (Fasting)

Tissue	Glucose	Fatty Acids	Ketone Bodies
Nervous system	++	—	++
Skeletal muscle	—	++	++
Heart muscle	—	++	++
Liver	—	++	—
Intestinal epithelial cells	—	—	++
Kidney	—	+	+

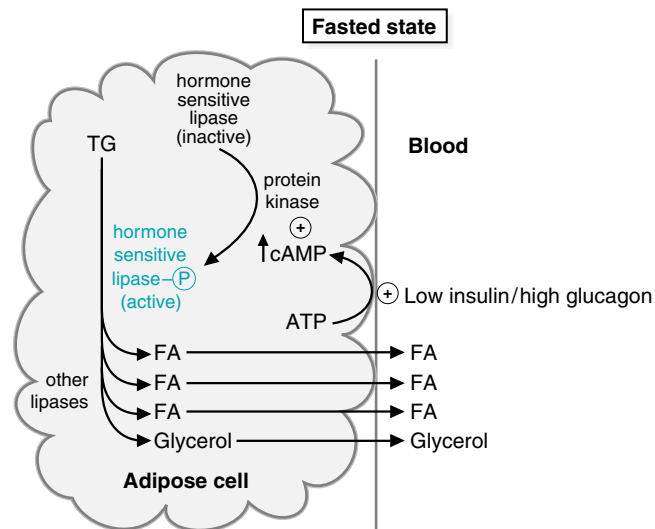


Fig. 36.10. Regulation of hormone-sensitive lipase (HSL) in adipose tissue. During fasting, the glucagon/insulin ratio rises, causing cAMP levels to be elevated. Protein kinase A is activated and phosphorylates HSL, activating this enzyme. HSL-P initiates the mobilization of adipose triacylglycerol by removing a fatty acid (FA). Other lipases then act, producing fatty acids and glycerol. Insulin stimulates the phosphatase that inactivates HSL in the fed state.

high NADH/NAD⁺ ratio also diverts oxaloacetate into malate, such that the malate can exit the mitochondria (via the malate/aspartate shuttle) for use in gluconeogenesis.

D. Regulation of the Use of Glucose and Fatty Acids by Muscle

During exercise, the fuel that is used initially by muscle cells is muscle glycogen. As exercise continues and the blood supply to the tissue increases, glucose is taken up from the blood and oxidized. Liver glycogenolysis and gluconeogenesis replenish the blood glucose supply. However, as insulin levels drop, the concentration of GLUT4 transporters in the membrane is reduced, thereby reducing glucose entry from the circulation into the muscle.

Thus, as fatty acids become available because of increased lipolysis of adipose triacylglycerols, the exercising muscle begins to oxidize fatty acids. β -Oxidation produces NADH and acetyl CoA, which slow the flow of carbon from glucose through the reaction catalyzed by pyruvate dehydrogenase (see Fig. 36.3). Thus, the oxidation of fatty acids provides a major portion of the increased demand for ATP generation and spares blood glucose.



Muscle GLUT4 transporters also can be induced by AMP levels (see Chapter 47), through the actions of the AMP-activated protein kinase. Thus, if energy levels are very low, and AMP levels increase, glucose can still be transported from the circulation into the muscle to provide energy. This most often occurs during exercise.

III. THE IMPORTANCE OF AMP AND FRUCTOSE 2,6-BISPHOSPHATE

The switch between catabolic and anabolic pathways is often regulated by the levels of AMP and fructose 2,6-bisphosphate in cells, particularly the liver. It is logical for AMP to be a critical regulator. As a cell uses ATP in energy-requiring pathways, the levels of AMP accumulate more rapidly than that of ADP because of the adenylate kinase reaction ($2 \text{ ADP} \rightarrow \text{ATP and AMP}$). The rise in AMP levels then signals that more energy is required (usually through allosteric binding sites on enzymes and the activation of the AMP-activated protein kinase), and the cell will switch to the activation of catabolic pathways. As AMP levels drop, and ATP levels rise, the anabolic pathways are now activated to store the excess energy.

The levels of fructose 2,6-bisphosphate are also critical in regulating glycolysis versus gluconeogenesis in the liver. Under conditions of high blood glucose, and insulin release, fructose 2,6 bisphosphate levels will be high because PFK-2 will be in its activated state. The fructose 2,6 bisphosphate activates PFK-1, and inhibits fructose 2,6 bisphosphatase, thereby allowing glycolysis to proceed. When blood glucose levels are low, and glucagon is released, PFK-2 is phosphorylated by the cAMP-dependent protein kinase and is inhibited, thereby lowering fructose 2,6-bisphosphate levels and inhibiting glycolysis, while favoring gluconeogenesis.

CLINICAL COMMENTS



Bea Selmass's younger sister was very concerned that Bea's pancreatic tumor might be genetically determined or potentially malignant, so she accompanied Bea to her second postoperative visit to the endocrinologist. The doctor explained that insulinomas may be familial in up to 20% of cases and that in 10% of patients with insulinomas, additional secretory neoplasms may occur in the anterior pituitary or the parathyroid glands (a genetically determined syndrome known as multiple endocrine neoplasia, type I or, simply, MEN I). Bea's tumor showed no evidence of malignancy, and the histologic slides, although not always definitive, showed a benign-appearing process. The doctor was careful to explain, however, that close observation for recurrent hypoglycemia and for the signs and symptoms suggestive of other facets of MEN I would be necessary for the remainder of Bea's and her immediate family's life.



Diabetes mellitus is a well-accepted risk factor for the development of coronary artery disease; the risk is three to four times higher in the diabetic than in the nondiabetic population. Although chronically elevated serum levels of chylomicrons and VLDL may contribute to this atherogenic predisposition, the premature vascular disease seen in **Di Abietes** and other patients with type 1 diabetes mellitus, as well as **Ann Sulin** and other patients with type 2 diabetes mellitus, is also related to other abnormalities in lipid metabolism. Among these are the increase in glycation (nonenzymatic attachment of glucose molecules to proteins) of LDL apoproteins as well as glycation of the proteins of the LDL receptor, which occurs when serum glucose levels are chronically elevated. These glycations interfere with the normal interaction or "fit" of the circulating LDL particles with their specific receptors on cell membranes. As a consequence, the rate of uptake of circulating LDL by the normal target cells is diminished. The LDL particles, therefore, remain in the circulation and eventually bind nonspecifically to "scavenger" receptors located on macrophages adjacent to the endothelial surfaces of blood vessels, one of the early steps in the process of atherogenesis.



Because **Di Abietes** produces very little insulin, she is prone to developing ketoacidosis. When insulin levels are low, HSL of adipose tissue is very active, resulting in increased lipolysis. The fatty acids that are released travel to the liver, where they are converted to the triacylglycerols of VLDL. They also undergo β -oxidation and conversion to ketone bodies. If Di does not take exogenous insulin or if her insulin levels decrease abruptly for some physiologic reason, she may develop a ketoacidosis (DKA). In fact, she has had repeated bouts of DKA.

For reasons that are not as well understood, individuals with type 2 diabetes mellitus, such as **Ann Sulin**, do not tend to develop ketoacidosis. One possible explanation is that the insulin resistance is tissue-specific; the insulin sensitivity of adipocytes may be greater than that of muscle and liver. Such a tissue-specific sensitivity would lead to less fatty acids being released from adipocytes in type 2 diabetes than in type 1 diabetes, although in both cases the release of fatty acids would be greater than that of an individual without the disease.

BIOCHEMICAL COMMENTS



All of the material in this chapter was presented previously. However, because this information is so critical for understanding biochemistry in a way that will allow it to be used in interpreting clinical situations, it was summarized in this chapter. In addition, the information previously presented under carbohydrate metabolism was integrated with lipid metabolism. We have, for the most part, left out the role of allosteric modifiers and other regulatory mechanisms that finely coordinate these processes to an exquisite level. Because such details may be important for specific clinical situations, we hope this summary will serve as a framework to which the details can be fitted as students advance in their clinical studies.

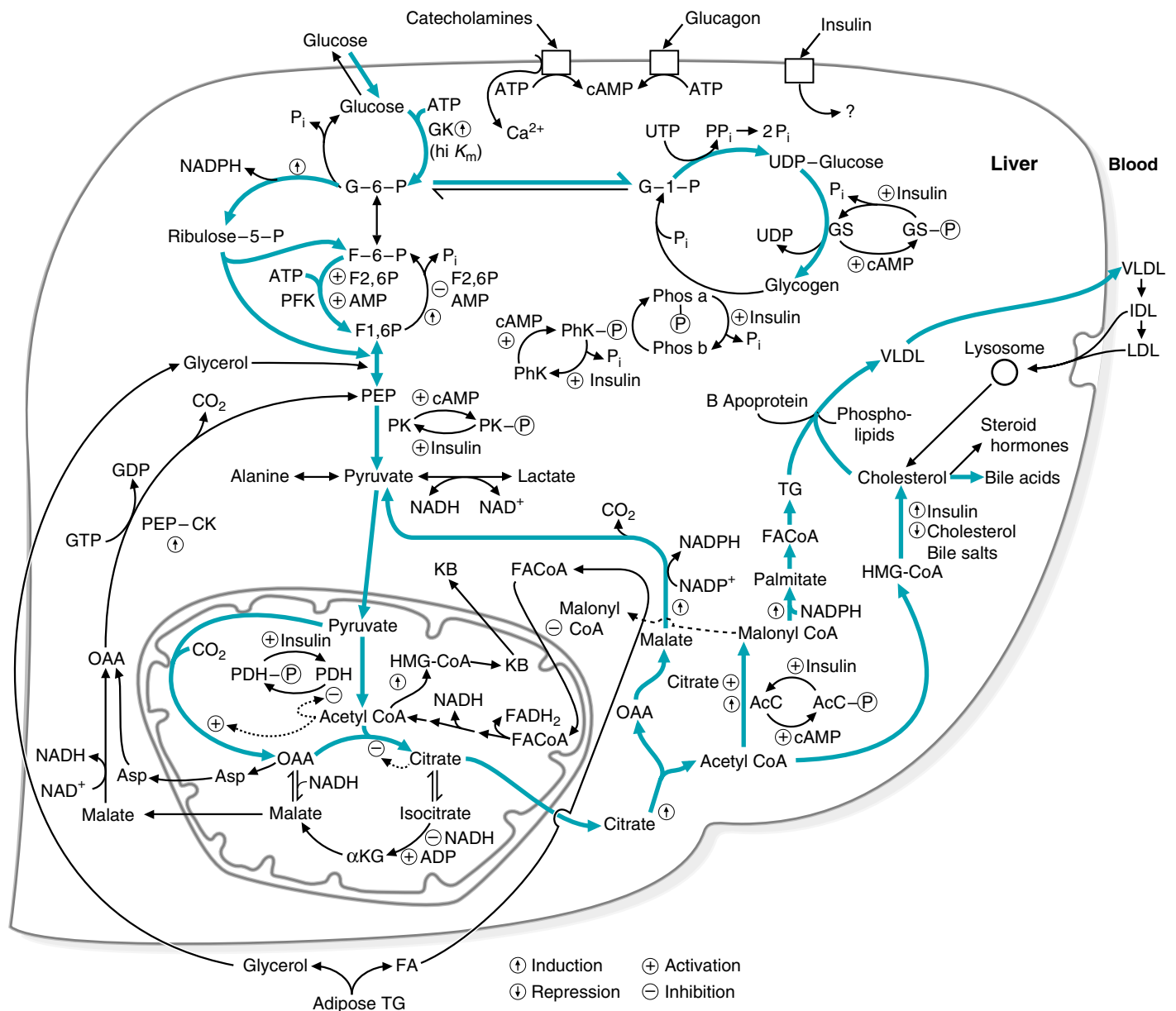


Fig. 36.11. Regulation of carbohydrate and lipid metabolism in the liver. Solid blue arrows indicate the flow of metabolites in the fed state. Solid black arrows indicate the flow during fasting. G = glucose; GK = glucokinase; F = fructose; PFK = phosphofructokinase-1; PEP = phosphoenolpyruvate; PK = pyruvate kinase; OAA = oxaloacetate; α KG = α -ketoglutarate; GS = glycogen synthase; Phos = glycogen phosphorylase; PhK = phosphorylase kinase; AcC = acetyl CoA carboxylase; FA = fatty acid or fatty acyl group; TG = triacylglycerol; circled P = phosphate group.

Figure 36.11 is a comprehensive figure, and Tables 36.2 and 36.3 provide a list of the major regulatory enzymes of carbohydrate and lipid metabolism in the liver, an order to the events that occur, and the mechanisms by which they are controlled. This figure and tables should help students to integrate this mass of material.

Now that many of the details of the pathways have been presented, it would be worthwhile to re-read the first three chapters of this book. A student who understands biochemistry within the context of fuel metabolism is in a very good position to solve clinical problems that involve metabolic derangements.

Table 36.2. Flowchart of Changes in Liver Metabolism

When blood sugar increases: Insulin is released, which leads to the <i>dephosphorylation</i> of:	When blood sugar decreases: Glucagon is released, which leads to the <i>phosphorylation</i> of:
• PFK-2 (now active) • Pyruvate kinase (now active) • Glycogen synthase (now active) • Phosphorylase kinase (now inactive) • Glycogen phosphorylase (now inactive) • Pyruvate dehydrogenase (now active) • Acetyl CoA Carboxylase (now active)	• PFK-2 (now inactive) • Pyruvate kinase (now inactive) • Glycogen synthase (now inactive) • Phosphorylase kinase (now active) • Glycogen phosphorylase (now active) • Pyruvate dehydrogenase (now inactive) • Acetyl CoA Carboxylase (now inactive)
Which leads to <i>active</i>	Which leads to <i>active</i>
• Glycolysis • Fatty acid synthesis • Glycogen synthesis	• Glycogenolysis • Fatty acid oxidation • Gluconeogenesis

Table 36.3. Regulation of Liver Enzymes Involved in Glycogen, Blood Glucose, and Triacylglycerol Synthesis and Degradation

LIVER ENZYMES REGULATED BY ACTIVATION/INHIBITION		
Enzyme	Activated By	State in Which Active
Phosphofructokinase-1	Fructose-2,6-bisP, AMP	Fed
Pyruvate carboxylase	Acetyl CoA	Fed and fasting
Acetyl CoA carboxylase	Citrate	Fed
Carnitine: palmitoyltransferase I	Loss of inhibitor (malonyl CoA)	Fasting
LIVER ENZYMES REGULATED BY PHOSPHORYLATION/DEPHOSPHORYLATION		
Enzyme	Active Form	State in Which Active
Glycogen synthase	Dephosphorylated	Fed
Phosphorylase kinase	Phosphorylated	Fasting
Glycogen phosphorylase	Phosphorylated	Fasting
Phosphofructokinase-2/F-2, 6-bisphosphatase (acts as a kinase, increasing fructose- 2,6-bisP levels)	Dephosphorylated	Fed
Phosphofructokinase-2/F-2, 6-bisphosphatase (acts as a phosphatase, decreasing fructose- 2,6-bisP levels)	Phosphorylated	Fasting
Pyruvate kinase	Dephosphorylated	Fed
Pyruvate dehydrogenase	Dephosphorylated	Fed
Acetyl CoA carboxylase	Dephosphorylated	Fed
LIVER ENZYMES REGULATED BY INDUCTION/REPRESSION		
Enzyme	State in Which Induced	Process Affected
Glucokinase	Fed	Glucose → TG
Citrate lyase	Fed	Glucose → TG
Acetyl CoA carboxylase	Fed	Glucose → TG
Fatty acid synthase	Fed	Glucose → TG
Malic enzyme	Fed	Production of NADPH
Glucose-6-P dehydrogenase	Fed	Production of NADPH
Glucose 6-phosphatase	Fasted	Production of blood glucose
Fructose 1, 6-bisphosphatase	Fasted	Production of blood glucose
Phosphoenolpyruvate carboxykinase	Fasted	Production of blood glucose

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REVIEW QUESTIONS—CHAPTER 36

- A 20-year-old woman with diabetes mellitus was admitted to the hospital in a semiconscious state with fever, nausea, and vomiting. Her breath smelled of acetone. A urine sample was strongly positive for ketone bodies. Which of the following statements is correct concerning this woman?
 - A blood glucose test would probably show that her blood glucose level was well below 80 mg/dL.
 - An insulin injection would decrease her ketone body production.
 - She should be given a glucose infusion to regain consciousness.
 - Glucagon should be administered to stimulate glycogenolysis and gluconeogenesis in the liver.
 - The acetone was produced by decarboxylation of the ketone body β -hydroxybutyrate.
- A woman was told by her physician to go on a low-fat diet. She decided to continue to consume the same number of calories by increasing her carbohydrate intake while decreasing her fat intake. Which of the following blood lipoprotein levels would be decreased as a consequence of her diet?
 - VLDL
 - IDL
 - HDL
 - Chylomicrons
 - HDL
- Assume that an individual has been eating excess calories daily such that they will gain weight. Under which of the following conditions would the person gain weight most rapidly?
 - If all the excess calories were due to carbohydrate
 - If all the excess calories were due to triacylglycerol
 - If all the excess calories were split 50%–50% between carbohydrate and triacylglycerol
 - If all the excess calories were split 25%–75% between carbohydrate and triacylglycerol
 - It makes no difference what form the excess calories are in
- A chronic alcoholic has been admitted to the hospital because of a severe hypoglycemic episode brought about by excessive alcohol consumption for the past 5 days. A blood lipid analysis indicates much higher than expected VLDL levels. The elevated VLDL is attributable to which of the following underlying causes?
 - Alcohol-induced inhibition of lipoprotein lipase
 - Elevated NADH levels in the liver
 - Alcohol-induced transcription of the apoB-100 gene
 - NADH activation of phosphoenolpyruvate carboxykinase
 - Acetaldehyde induction of enzymes on the endoplasmic reticulum

5. Certain patients with abetalipoproteinemia frequently have difficulties in maintaining blood volume; their blood has trouble clotting. This symptom is attributable to which of the following?
- (A) Inability to produce chylomicrons
 - (B) Inability to produce VLDL
 - (C) Inability to synthesize clotting factors
 - (D) Inability to synthesize fatty acids
 - (E) Inability to absorb short-chain fatty acids

Nitrogen Metabolism

Dietary proteins are the primary source of the nitrogen that is metabolized by the body (Fig. 1). Amino acids, produced by digestion of dietary proteins, are absorbed through intestinal epithelial cells and enter the blood. Various cells take up these amino acids, which enter the cellular pools. They are used for the synthesis of proteins and other nitrogen-containing compounds or they are oxidized for energy.

Protein synthesis, the translation of mRNA on ribosomes (see Chapter 15), is a dynamic process. Within the body, proteins are constantly being synthesized and degraded, partially draining and then refilling the cellular amino acid pools.

Compounds derived from amino acids include cellular proteins, hormones, neurotransmitters, creatine phosphate, the heme of hemoglobin and the cytochromes, the skin pigment melanin, and the purine and pyrimidine bases of nucleotides and nucleic acids. In fact, all of the nitrogen-containing compounds of the body are synthesized from amino acids. Many of these pathways are outlined in the following chapters of the book.

In addition to serving as the precursors for the nitrogen-containing compounds of the body and as the building blocks for protein synthesis, amino acids are also a source of energy. Amino acids are directly oxidized or they are converted to glucose and then oxidized or stored as glycogen. They also may be converted to fatty acids and stored as adipose triacylglycerols. Glycogen and triacylglycerols are oxidized during periods of fasting. The liver is the major site of amino acid oxidation. However, most tissues can oxidize the branched-chain amino acids (leucine, isoleucine, and valine).

Before the carbon skeletons of amino acids are oxidized, the nitrogen must be removed. Amino acid nitrogen forms ammonia, which is toxic to the body. In the liver, ammonia and the amino groups from amino acids are converted to urea, which is nontoxic, water-soluble, and readily excreted in the urine. The process by which urea is produced is known as the urea cycle. The liver is the organ responsible for producing urea. Branched-chain amino acids can be oxidized in many tissues, but the nitrogen must always travel to the liver for disposal.

Although urea is the major nitrogenous excretory product of the body, nitrogen is also excreted in other compounds (Table 1). Uric acid is the degradation product

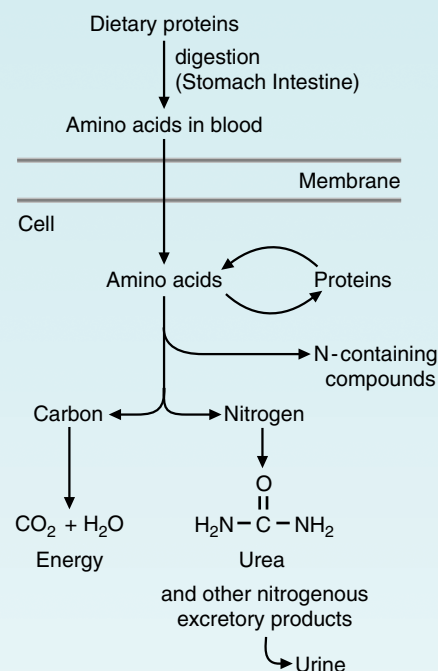


Fig. 1. Summary of amino acid metabolism. Dietary proteins are digested to amino acids in the stomach and intestine, which are absorbed by the intestinal epithelium, transferred to the circulation, and taken up by cells. Amino acids are used to synthesize proteins and other nitrogen-containing compounds. The carbon skeletons of amino acids are also oxidized for energy, and the nitrogen is converted to urea and other nitrogenous excretory products.

Table 1. Major Nitrogenous Urinary Excretory Products

Amount excreted in urine/day ^a	
Urea ^b	12–20 g urea nitrogen (12,000–20,000 mg)
NH ₄ ⁺	140–1,500 mg ammonia nitrogen
Creatinine	Males: 14–26 mg/kg
	Females: 11–20 mg/kg
Uric acid	250–750 mg

^aThe amounts are expressed in units generally reported by clinical laboratories. Note that the amounts for creatinine and uric acid are for the whole compound, whereas those for urea and ammonia are for the nitrogen content.

^bUnder normal circumstances, approximately 90% of the nitrogen excreted in the urine is in the form of urea. The exact amounts of each component vary, however, depending on dietary protein intake and physiologic state. For instance, NH₄⁺ excretion increases during an acidosis because the kidney secretes ammonia to bind protons in the urine.



The healthy human adult is in nitrogen balance; in other words, the amount of nitrogen excreted each day (mainly in the urine) equals the amount consumed (mainly as dietary protein). Negative nitrogen balance occurs when the amount of nitrogen excreted is greater than the amount consumed, and positive nitrogen balance occurs when the amount excreted is less than that consumed (see Chapter 1).

Table 2. Amino Acids Synthesized in the Body^a

From Glucose	From an Essential Amino Acid
Serine Glycine Cysteine ^b Alanine Aspartate Asparagine Glutamate Glutamine Proline Arginine	Tyrosine (from phenylalanine)

^aThese amino acids are called “nonessential” or “dispensable,” terms that refer to dietary requirements. Of course, within the body, they are necessary. We cannot survive without them.

^bAlthough the carbons of cysteine can be derived from glucose, its sulfur is obtained from the essential amino acid methionine.

of the purine bases, creatinine is produced from creatine phosphate, and ammonia is released from glutamine, particularly by the kidney, where it helps to buffer the urine by reacting with protons to form ammonium ions (NH_4^+). These compounds are excreted mainly in the urine, but substantial amounts are also lost in the feces and through the skin. Small amounts of nitrogen-containing metabolites are formed from the degradation of neurotransmitters, hormones, and other specialized amino acid products and excreted in the urine. Some of these degradation products, such as bilirubin (formed from the degradation of heme), are excreted mainly in the feces.

Eleven of the 20 amino acids used to form proteins are synthesized in the body if an adequate amount is not present in the diet (Table 2). Ten of these amino acids can be produced from glucose; the 11th, tyrosine, is synthesized from the essential amino acid phenylalanine. It should be noted that cysteine, one of the 10 amino acids produced from glucose, obtains its sulfur atom from the essential amino acid methionine.

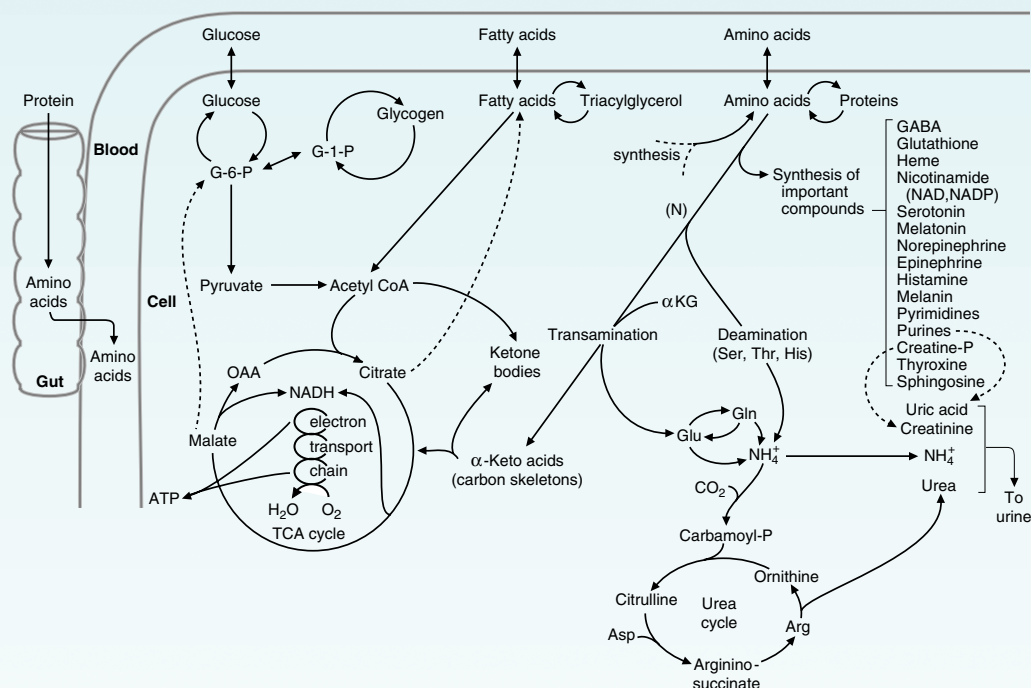


Fig. 2. Overview of nitrogen metabolism. The metabolism of nitrogen-containing compounds is shown on the right and that of glucose and fatty acids is shown on the left. This figure shows a hypothetical, composite cell. No single cell type has all of these pathways. Many of the pathways shown are described in the next few chapters. α KG = α -ketoglutarate; OAA = oxaloacetate; G-6-P = glucose 6-phosphate; G-1-P = glucose 1-phosphate.

Nine amino acids are essential in the human. “Essential” means that the carbon skeleton cannot be synthesized and, therefore, these amino acids are required in the diet (Table 3). The essential amino acids are also called the indispensable amino acids. Arginine is essential during periods of growth; in adults it is no longer considered essential.

After nitrogen is removed from amino acids, the carbon skeletons are oxidized (Fig. 2). Most of the carbons are converted to pyruvate, intermediates of the tricarboxylic acid (TCA) cycle, or to acetyl CoA. In the liver, particularly during fasting, these carbons may be converted to glucose or to ketone bodies and released into the blood. Other tissues then oxidize the glucose and ketone bodies. Ultimately, the carbons of the amino acids are converted to CO_2 and H_2O .

Table 3. Amino Acids Essential in the Diet

Lysine
Isoleucine
Leucine
Threonine
Valine
Tryptophan
Phenylalanine
Methionine
Histidine
Arginine (not required by the adult, but required for growth)

37 Protein Digestion and Amino Acid Absorption

Proteolytic enzymes (also called **proteases**) break down dietary proteins into their constituent amino acids in the **stomach** and the **intestine**. Many of these digestive proteases are synthesized as larger, inactive forms known as **zymogens**. After zymogens are secreted into the digestive tract, they are cleaved to produce the active proteases.

In the stomach, **pepsin** begins the digestion of proteins by hydrolyzing them to smaller polypeptides. The contents of the stomach pass into the small intestine, where enzymes produced by the exocrine pancreas act. The pancreatic proteases (**trypsin**, **chymotrypsin**, **elastase**, and the **carboxypeptidases**) cleave the polypeptides into oligopeptides and amino acids.

Further cleavage of the oligopeptides to amino acids is accomplished by enzymes produced by the intestinal epithelial cells. These enzymes include **aminopeptidases** located on the brush border and other peptidases located within the cells. Ultimately, the amino acids produced by protein digestion are absorbed through the **intestinal epithelial cells** and enter the **blood**.

A large number of overlapping transport systems exist for amino acids in cells. Some systems contain **facilitative transporters**, whereas others express **sodium-linked transporters**, which allow the **active transport** of amino acids into cells. Defects in amino acid transport can lead to disease.

Proteins are also continually synthesized and degraded (**turnover**) in cells. A wide variety of proteases exist in cells to carry out this activity. **Lysosomal proteases** (**cathepsins**) degrade proteins that enter lysosomes. Cytoplasmic proteins targeted for turnover are covalently linked to the small protein **ubiquitin**, which then interacts with a large protein complex, the **proteasome**, to degrade the protein in an adenosine triphosphate (ATP)-dependent process. The amino acids released from proteins during turnover can then be used for the synthesis of new proteins or for energy generation.



THE WAITING ROOM



Sissy Fibrosa, a young child with cystic fibrosis, has had repeated bouts of bronchitis caused by *Pseudomonas aeruginosa*. With each of these infections, her response to aerosolized antibiotics has been good. However, her malabsorption of food continues, resulting in foul-smelling, glistening, bulky stools. Her growth records show a slow decline. She is now in the 24th percentile for height and the 20th percentile for weight. She is often listless and irritable, and she tires easily. When her pediatrician discovered that her levels of the serum proteins

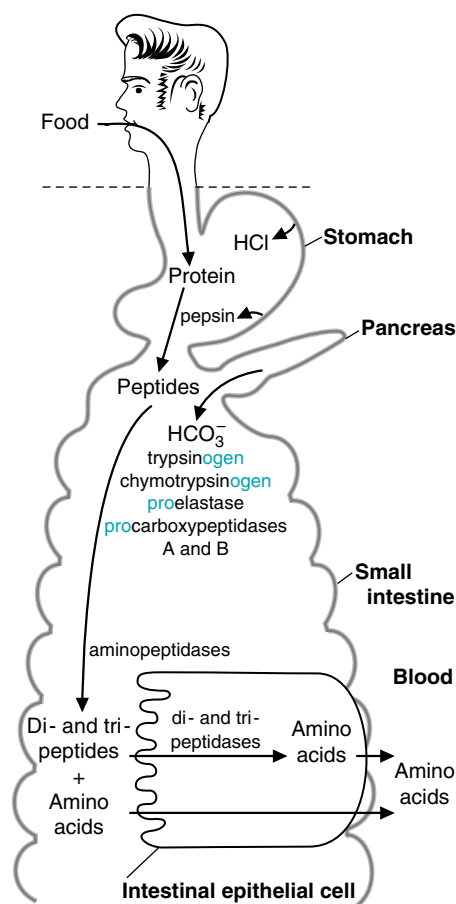


Fig. 37.1. Digestion of proteins. The proteolytic enzymes, pepsin, trypsin, chymotrypsin, elastase, and the carboxypeptidases, are produced as zymogens (the [pro] and [ogen] accompanying the enzyme name) that are activated by cleavage after they enter the gastrointestinal lumen (see Fig. 37.2).



Kwashiorkor, a common problem of children in Third World countries, is caused by a deficiency of protein in a diet that is adequate in calories. Children with kwashiorkor suffer from muscle wasting and a decreased concentration of plasma proteins, particularly albumin. The result is an increase in interstitial fluid that causes edema and a distended abdomen that make the children appear “plump” (see Chapter 44). The muscle wasting is caused by the lack of essential amino acids in the diet; existing proteins must be broken down to produce these amino acids for new protein synthesis.

These problems may be compounded by a decreased ability to produce digestive enzymes and new intestinal epithelial cells because of a decreased availability of amino acids for the synthesis of new proteins.

albumin, transferrin, and thyroid hormone binding prealbumin (transthyretin) were low to low-normal (indicating protein malnutrition), Sissy was given enteric-coated microspheres of pancreatic enzymes. Almost immediately, the character of Sissy’s stools became more normal and she began gaining weight. In the next 6 months, her growth curves showed improvement, and she seemed brighter, more active, and less irritable.



For the first few months after a painful episode of renal colic, during which he passed a kidney stone (see Chapter 6), **Cal Kulis** had faithfully maintained a high daily fluid intake and had taken the medication required to increase the pH of his urine. Because he has cystinuria, these measures were necessary to increase the solubility of the large amounts of cystine present in his urine and, thereby, to prevent further formation of kidney stones (calculi). With time, however, he became increasingly complacent about his preventive program. After failing to take his medication for a month, he experienced another severe episode of renal colic with grossly bloody urine. Fortunately, he passed the stone spontaneously, after which he vowed to faithfully comply with therapy.

His mother heard that some dietary amino acids were not absorbed in patients with cystinuria and asked whether any dietary changes would reduce Cal’s chances of developing additional renal stones.

I. PROTEIN DIGESTION

The digestion of proteins begins in the stomach and is completed in the intestine (Fig. 37.1). The enzymes that digest proteins are produced as inactive precursors (zymogens) that are larger than the active enzymes. The inactive zymogens are secreted from the cells in which they are synthesized and enter the lumen of the digestive tract, where they are cleaved to smaller forms that have proteolytic activity (Fig. 37.2). These active enzymes have different specificities; no single enzyme can completely digest a protein. However, by acting in concert, they can digest dietary proteins to amino acids and small peptides, which are cleaved by peptidases associated with intestinal epithelial cells

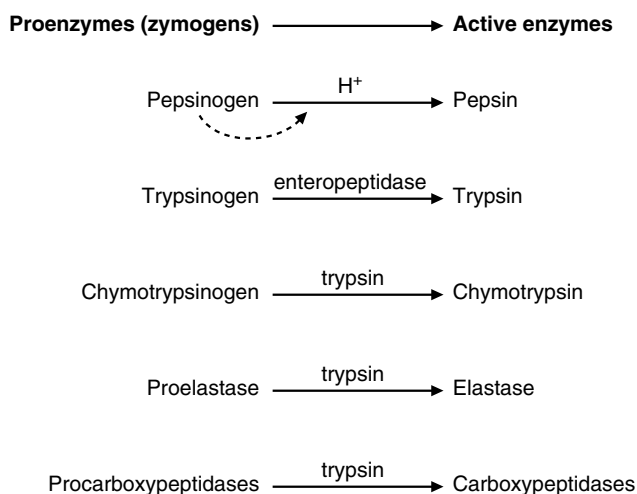


Fig. 37.2. Activation of the gastric and pancreatic zymogens. Pepsinogen catalyzes its own cleavage as the pH of the stomach drops. Trypsinogen is cleaved by enteropeptidase in the intestine to form the active protease trypsin. Trypsin then plays a key role by catalyzing the cleavage and activation of the other pancreatic zymogens.

A. Digestion of Proteins in the Stomach

Pepsinogen is secreted by the chief cells of the stomach. The gastric parietal cells secrete HCl. The acid in the stomach lumen alters the conformation of pepsinogen so that it can cleave itself, producing the active protease pepsin. Thus, the activation of pepsinogen is autocatalytic.

Dietary proteins are denatured by the acid in the stomach. This serves to inactivate the proteins and partially unfolds them such that they are better substrates for proteases. However, at the low pH of the stomach, pepsin is not denatured and acts as an endopeptidase, cleaving peptide bonds at various points within the protein chain. Although pepsin has a fairly broad specificity, it tends to cleave peptide bonds in which the carboxyl group is provided by an aromatic or acidic amino acid (Fig. 37.3). Smaller peptides and some free amino acids are produced.

B. Digestion of Proteins by Enzymes from the Pancreas

As the gastric contents empty into the intestine, they encounter the secretions from the exocrine pancreas. One of these secretions is bicarbonate, which, in addition to neutralizing the stomach acid, raises the pH such that the pancreatic proteases, which are also present in pancreatic secretions, can be active. As secreted, these pancreatic proteases are in the inactive proenzyme form (zymogens). Because the active forms of these enzymes can digest each other, it is important for their zymogen forms all to be activated within a short span of time. This feat is accomplished by the cleavage of trypsinogen to the active enzyme trypsin, which then cleaves the other pancreatic zymogens, producing their active forms (see Fig. 37.2).

The zymogen trypsinogen is cleaved to form trypsin by enteropeptidase (a protease, formerly called enterokinase) secreted by the brush-border cells of the small intestine. Trypsin catalyzes the cleavages that convert chymotrypsinogen to the active enzyme chymotrypsin, proelastase to elastase, and the procarboxypeptidases to the carboxypeptidases. Thus, trypsin plays a central role in digestion because it both cleaves dietary proteins and activates other digestive proteases produced by the pancreas.

Trypsin, chymotrypsin, and elastase are serine proteases (see Chapter 9) that act as endopeptidases. Trypsin is the most specific of these enzymes, cleaving peptide bonds in which the carboxyl (carbonyl) group is provided by lysine or arginine (see Fig. 37.3). Chymotrypsin is less specific but favors residues that contain hydrophobic or acidic amino acids. Elastase cleaves not only elastin (for which it was named) but also other proteins at bonds in which the carboxyl group is contributed by amino acid residues with small side chains (alanine, glycine, or serine). The actions of these pancreatic endopeptidases continue the digestion of dietary proteins begun by pepsin in the stomach.



Elastase is also found in neutrophils, white blood cells that have the job of engulfing and destroying invading bacteria. Neutrophils frequently act in the lung, and elastase is sometimes released into the lung as the neutrophils work. In normal individuals, the released elastase is blocked from destroying lung cells by the action of circulating α -1-antitrypsin, a protease inhibitor synthesized and secreted by the liver. Certain individuals have a genetic mutation that leads to the production of an inactive α -1-antitrypsin protein (α -1-antitrypsin deficiency). The lack of this enzyme activity leads to the development of emphysema caused by proteolytic destruction of lung cells, which results in a reduction in the expansion/contraction capability of the lungs.



Patients with cystic fibrosis, such as **Sissy Fibrosa**, have a genetically determined defect in the function of the chloride channels. In the pancreatic secretory ducts, which carry pancreatic enzymes into the lumen of the small intestines, this defect causes inspissation (drying and thickening) of pancreatic exocrine secretions, eventually leading to obstruction of these ducts. One result of this problem is the inability of pancreatic enzymes to enter the intestinal lumen to digest dietary proteins.

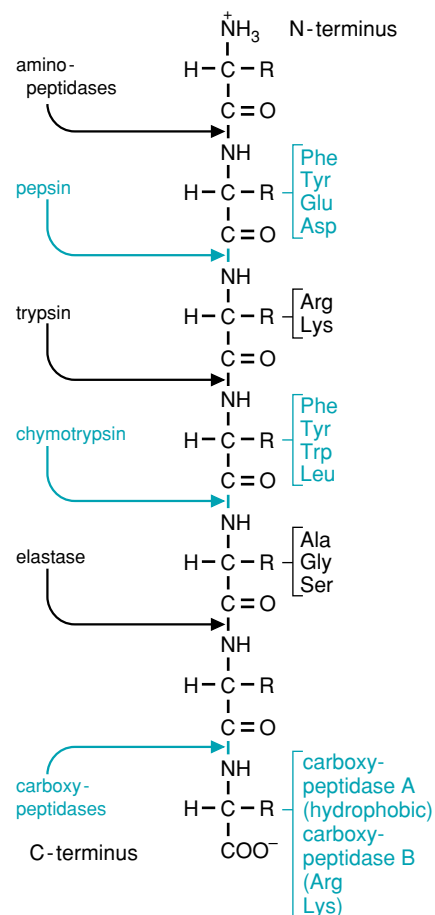


Fig. 37.3. Action of the digestive proteases. Pepsin, trypsin, chymotrypsin, and elastase are endopeptidases; they hydrolyze peptide bonds within chains. The others are exopeptidases; aminopeptidases remove the amino acid at the N-terminus and the carboxypeptidases remove the amino acid at the C terminus. For each proteolytic enzyme, the amino acid residues involved in the peptide bond that is cleaved are listed beside the R group to the right of the enzyme name.



Recall that the exocrine pancreas, in addition to secreting proteolytic zymogens, also secretes amylase for starch digestion and lipase and co-lipase for dietary triacylglycerol digestion.



The pancreas synthesizes and stores the zymogens in secretory granules. The pancreas also synthesizes a secretory trypsin inhibitor. The need for the inhibitor is to block any trypsin activity that may occur from accidental trypsinogen activation. If the inhibitor were not present, trypsinogen activation would lead to the activation of all of the zymogens in the pancreas, which would lead to the digestion of intracellular pancreatic proteins. Such episodes can lead to pancreatitis.



Why do patients with cystinuria and Hartnup disease have a hyperaminoaciduria without an associated hyperaminoacidemia?



Hartnup disease is another genetically determined and relatively rare autosomal recessive disorder. It is caused by a defect in the transport of neutral amino acids across both intestinal and renal epithelial cells. The signs and symptoms are, in part, caused by a deficiency of essential amino acids (see Clinical Comments). Cystinuria and Hartnup disease involve defects in two different transport proteins. In each case, the defect is present both in intestinal cells, causing malabsorption of the amino acids from the digestive products in the intestinal lumen and in kidney tubular cells, causing a decreased resorption of these amino acids from the glomerular filtrate.



Trace amounts of polypeptides pass into the blood. They may be transported through intestinal epithelial cells, probably by pinocytosis, or they may slip between the cells that line the gut wall. This process is particularly troublesome for premature infants, because it can lead to allergies caused by proteins in their food.

The smaller peptides formed by the action of trypsin, chymotrypsin, and elastase are attacked by exopeptidases, which are proteases that cleave one amino acid at a time from the end of the chain. Procarboxypeptidases, zymogens produced by the pancreas, are converted by trypsin to the active carboxypeptidases. These exopeptidases remove amino acids from the carboxyl ends of peptide chains. Carboxypeptidase A preferentially releases hydrophobic amino acids, whereas carboxypeptidase B releases basic amino acids (arginine and lysine).

C. Digestion of Proteins by Enzymes from Intestinal Cells

Exopeptidases produced by intestinal epithelial cells act within the brush border and also within the cell. Aminopeptidases, located on the brush border, cleave one amino acid at a time from the amino end of peptides. Intracellular peptidases act on small peptides that are absorbed by the cells.

The concerted action of the proteolytic enzymes produced by cells of the stomach, pancreas, and intestine cleaves dietary proteins to amino acids. The digestive enzymes digest themselves as well as dietary protein. They also digest the intestinal cells that are regularly sloughed off into the lumen. These cells are replaced by cells that mature from precursor cells in the duodenal crypts. The amount of protein that is digested and absorbed each day from digestive juices and cells released into the intestinal lumen may be equal to, or greater than, the amount of protein consumed in the diet (50–100 g).

II. ABSORPTION OF AMINO ACIDS

Amino acids are absorbed from the intestinal lumen through secondary active Na^+ -dependent transport systems and through facilitated diffusion.

A. Cotransport of Na^+ and Amino Acids

Amino acids are absorbed from the lumen of the small intestine principally by semispecific Na^+ -dependent transport proteins in the luminal membrane of the intestinal cell brush border, similar to that already seen for carbohydrate transport (Fig. 37.4). The cotransport of Na^+ and the amino acid from the outside of the apical membrane to the inside of the cell is driven by the low intracellular Na^+ concentration. Low intracellular Na^+ results from the pumping of Na^+ out of the cell by a Na^+, K^+ -ATPase on the serosal membrane. Thus, the primary transport mechanism is the creation of a sodium gradient; the secondary transport process is the coupling of amino acids to the influx of sodium. This mechanism allows the cells to concentrate amino acids from the intestinal lumen. The amino acids are then transported out of the cell into the interstitial fluid principally by facilitated transporters in the serosal membrane (see Fig. 37.4).

At least six different Na^+ -dependent amino acid carriers are located in the apical brush border membrane of the epithelial cells. These carriers have an overlapping specificity for different amino acids. One carrier preferentially transports neutral amino acids, another transports proline and hydroxyproline, a third preferentially transports acidic amino acids, and a fourth transports basic amino acids (lysine, arginine, the urea cycle intermediate ornithine) and cystine (two cysteine residues linked by a disulfide bond). In addition to these Na^+ -dependent carriers, some amino acids are transported across the luminal membrane by facilitated transport carriers. Most amino acids are transported by more than one transport system.

As with glucose transport, the Na^+ -dependent carriers of the apical membrane of the intestinal epithelial cells are also present in the renal epithelium. However, different isozymes are present in the cell membranes of other tissues. Conversely, the facilitated transport carriers in the serosal membrane of the intestinal epithelia are similar to those found in other cell types in the body. During starvation, the intestinal epithelia, like these other cells, take up amino acids from the blood to use as an energy source. Thus, amino acid transport across the serosal membrane is bidirectional.

Table 37.1. A Partial Listing of Amino Acid Transport Systems^a

System Name	Sodium-dependent?	Specificity	Tissues Expressed
A	Yes	Small amino acids (ala, ser, gln)	Many
ASC	Yes	Small amino acids (ala, ser, cys)	Many
N	Yes	Gln, asn, his	Liver
L	No	Branched and aromatic amino acids	Many
B ^{0,+}	Yes	Basic amino acids	Intestine (brush border) ^b
ATB ^o	Yes	Zwitterionic amino acids (monoamino, monocarboxylic acid amino acids)	Intestine and kidney ^c
X _{AG} ⁻	Yes	Anionic amino acids	Intestine (brush border)
Imino	Yes	Pro, hypro, gly	Intestine (brush border)

^a Not all transport systems are listed.^b This system is most likely defective in cystinuria.^c This system is most likely defective in Hartnup disease.

B. Transport of Amino Acids into Cells

Amino acids that enter the blood are transported across cell membranes of the various tissues principally by Na⁺-dependent cotransporters and, to a lesser extent, by facilitated transporters (Table 37.1). In this respect, amino acid transport differs from glucose transport, which is Na⁺-dependent transport in the intestinal and renal epithelium but facilitated transport in other cell types. The Na⁺ dependence of amino acid transport in liver, muscle, and other tissues allows these cells to concentrate amino acids from the blood. These transport proteins have a different genetic basis, amino acid composition, and somewhat different specificity than those in the luminal membrane of intestinal epithelia. They also differ somewhat between tissues. For instance, the N system for glutamine uptake is present in the liver but either not present in other tissues or present as an isoform with different properties. There is also some overlap in specificity of the transport proteins, with most amino acids being transported by more than one carrier.

III. PROTEIN TURNOVER AND REPLENISHMENT OF THE INTRACELLULAR AMINO ACID POOL

The amino acid pool within cells is generated both from dietary amino acids and from the degradation of existing proteins within the cell. All proteins within cells have a half-life ($t_{1/2}$), a time at which 50% of the protein that was synthesized at a particular time will have been degraded. Some proteins are inherently short-lived, with half-lives of 5 to 20 minutes. Other proteins are present for extended periods, with half-lives of many hours, or even days. Thus, proteins are continuously being



Cal Kulis and other patients with cystinuria have a genetically determined defect in the transport of cystine and the basic amino acids, lysine, arginine, and ornithine, across the brush-border membranes of cells in both their small intestine and renal tubules. However, they do not appear to have any symptoms of amino acid deficiency, in part because the amino acids cysteine (which is oxidized in blood and urine to form the disulfide cystine) and arginine can be synthesized in the body (i.e., they are “nonessential” amino acids). Ornithine (an amino acid that is not found in proteins but serves as an intermediate of the urea cycle) can also be synthesized. The most serious problem for these patients is the insolubility of cystine, which can form kidney stones that may lodge in the ureter, causing bleeding and severe pain.



Patients with cystinuria and Hartnup disease have defective transport proteins in both the intestine and the kidney. These patients do not absorb the affected amino acids at a normal rate from the digestive products in the intestinal lumen. They also do not readily resorb these amino acids from the glomerular filtrate into the blood. Therefore, they do not have a hyperaminoacidemia (a high concentration in the blood). Normally, only a few percent of the amino acids that enter the glomerular filtrate are excreted in the urine; most are resorbed. In these diseases, much larger amounts of the affected amino acids are excreted in the urine, resulting in a hyper-aminoaciduria.

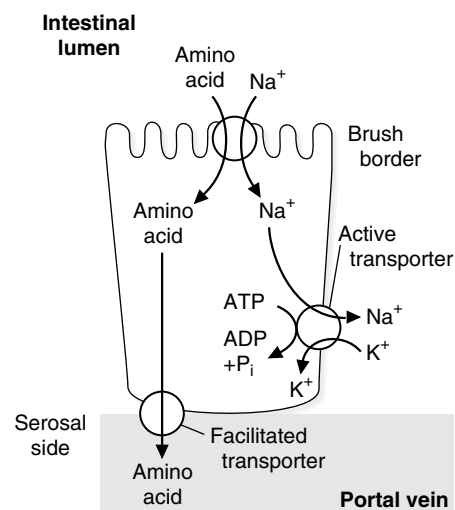


Fig. 37.4. Transepithelial amino acid transport. Na⁺-dependent carriers transport both Na⁺ and an amino acid into the intestinal epithelial cell from the intestinal lumen. Na⁺ is pumped out on the serosal side (across the basolateral membrane) in exchange for K⁺ by the Na⁺,K⁺-ATPase. On the serosal side, the amino acid is carried by a facilitated transporter down its concentration gradient into the blood. This process is an example of secondary active transport.



Protein turnover is quite extensive. For example, red blood cells have a lifespan of 120 days. Every day 3×10^{11} (300,000 million) red blood cells die and are phagocytosed. The hemoglobin in these cells is degraded to amino acids by lysosomal proteases, and these amino acids are reutilized. Approximately 6 lb hemoglobin is recycled in this way every year. As the aged cells are dying, newly generated reticulocytes are synthesizing hemoglobin in preparation for their conversion into new red blood cells, which replace the dying cells.



Adults cannot increase the amount of muscle or other body proteins by eating an excess amount of protein. If dietary protein is consumed in excess of our needs, it is converted to glycogen and triacylglycerols, which are then stored.

synthesized and degraded in the body, using a variety of enzyme systems to do so (Table 37.2). Examples of proteins that undergo extensive synthesis and degradation are hemoglobin, muscle proteins, digestive enzymes, and the proteins of cells sloughed off from the gastrointestinal tract. Hemoglobin is produced in reticulocytes and reconverted to amino acids by the phagocytic cells that remove mature red blood cells from the circulation on a daily basis. Muscle protein is degraded during periods of fasting, and the amino acids are used for gluconeogenesis. After ingestion of protein in the diet, muscle protein is resynthesized.

A large amount of protein is recycled daily in the form of digestive enzymes, which are themselves degraded by digestive proteases. In addition, approximately one fourth of the cells lining the walls of the gastrointestinal tract are lost each day and replaced by newly synthesized cells. As cells leave the gastrointestinal wall, their proteins and other components are digested by enzymes in the lumen of the gut, and the products are absorbed. Only approximately 6% (roughly 10 g) of the protein that enters the digestive tract (including dietary proteins, digestive enzymes, and the proteins in sloughed-off cells) is excreted in the feces each day. The remainder is recycled.

Proteins are also recycled within cells. The differences in amino acid composition of the various proteins of the body, the vast range in turnover times ($t_{1/2}$), and the recycling of amino acids are all important factors that help to determine the requirements for specific amino acids and total protein in the diet. The synthesis of many enzymes is induced in response to physiologic demand (such as fasting or feeding). These enzymes are continuously being degraded. Intracellular proteins are also damaged by oxidation and other modifications that limit their function. Mechanisms for intracellular degradation of unnecessary or damaged proteins involve lysosomes and the ubiquitin/proteasome system.

A. Lysosomal Protein Turnover

Lysosomes participate in the process of autophagy, in which intracellular components are surrounded by membranes that fuse with lysosomes, and endocytosis (see Chapter 10). Autophagy is a complex regulated process in which cytoplasm is sequestered into vesicles and delivered to the lysosomes. Within the lysosomes, the cathepsin family of proteases degrades the ingested proteins to individual amino acids. The recycled amino acids can then leave the lysosome and rejoin the intracellular amino acid pool. Although the details of how autophagy is induced are still not known, starvation of a cell is a trigger to induce this process. This will allow old proteins to be recycled and the newly released amino acids used for new protein synthesis, to enable the cell to survive starvation conditions.

Table 37.2. Proteases Involved in Protein Turnover/Degradation

Classification	Mechanism	Role
Cathepsins	Cysteine proteases	Lysosomal enzymes
Caspases	Cysteine proteases, which cleave after aspartate	Apoptosis; activated from pro-caspases (see Chapter 18)
Matrix metalloproteinases	Require zinc for catalysis	Model extracellular matrix components; regulated by TIMPs (tissue inhibitors of matrix metalloproteinases)
Proteasome	Large complex that degrades ubiquitin-tagged proteins	Protein turnover
Serine proteases	Active site serine in a catalytic triad with histidine and aspartic acid	Digestion and blood clotting; activated usually from zymogens (see Chapter 45)
Calpains	Calcium-dependent cysteine proteases	Many different cellular roles

B. The Ubiquitin-Proteasome Pathway

Ubiquitin is a small protein (76 amino acids) that is highly conserved. Its amino acid sequence in yeast and humans differs by only three residues. Ubiquitin targets intracellular proteins for degradation by covalently binding to the ϵ -amino group of lysine residues. This is accomplished by a three-enzyme system that adds ubiquitin to proteins targeted for degradation. Oftentimes, the target protein is polyubiquitinated, in which additional ubiquitin molecules are added to previous ubiquitin molecules, forming a long ubiquitin tail on the target protein. After polyubiquitinylation is complete, the targeted protein is released from the three-enzyme complex.

A protease complex, known as the proteasome, then degrades the targeted protein, releasing intact ubiquitin that can again mark other proteins for degradation (Figure 37.5). The basic proteasome is a cylindrical 20S protein complex with multiple internal proteolytic sites. ATP hydrolysis is used both to unfold the tagged protein and to push the protein into the core of the cylinder. The complex is regulated by cap protein complexes, which bind the ubiquitinated protein (a step that requires ATP) and deliver them to the complex. After the target protein is degraded, the ubiquitin is released intact and recycled. The resultant amino acids join the intracellular pool of free amino acids. Different cap complexes alter the specificity of the proteasome. For example, the PA700 cap is required for ubiquitinated proteins, whereas the PA28 cap targets only short peptides to the complex.



Another protein modification, which occurs through a three-enzyme complex similar to that required for ubiquitin addition, is SUMOylation. SUMO stands for small ubiquitin-like modifier, and when proteins are tagged with SUMO their activities are altered (either positively or negatively, depending on the protein). SUMOylation presents yet another means of fine-tuning existing regulatory systems.



Many proteins that contain regions rich in the amino acids proline (P), glutamate (E), serine (S), and threonine (T) have short half-lives. These regions are known as PEST sequences, based on the one-letter abbreviations used for these amino acids. Most of the amino acids that contain PEST sequences are hydrolyzed by the ubiquitin-proteasome system.

CLINICAL COMMENTS



Sissy Fibrosa's growth and weight curves were both subnormal until her pediatrician added pancreatic enzyme supplements to her treatment plan. These supplements digest dietary protein, releasing essential and other amino acids from the dietary protein that are then absorbed by the endothelial cells of Sissy's small intestine, through which they are transported into the blood. A discernable improvement in Sissy's body weight and growth curves was noted within months of the start of this therapy.

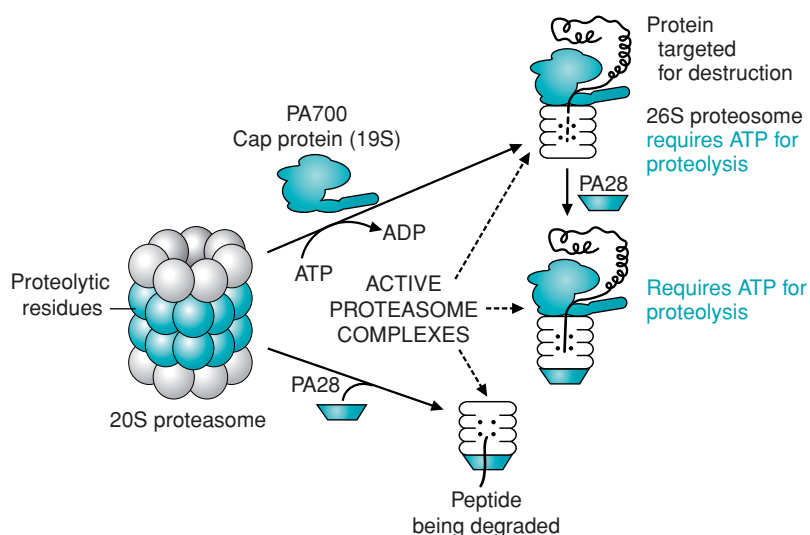


Fig 37.5. The proteasome and cap proteins. The cap proteins (PA700 and PA28) regulate the activity of this proteolytic complex by recruiting to the complex the substrates for proteolysis. The ATP requirement is to unfold and denature the proteins targeted for destruction, although the PA28 protease complex does not require ATP.

Apart from the proportions of essential amino acids present in various foods, the quality of a dietary protein is also determined by the rate at which it is digested and, in a more general way, by its capacity to contribute to the growth of the infant. In this regard, the proteins in foods of animal origin are more digestible than are those derived from plants. For example, the digestibility of proteins in eggs is approximately 97%; that for meats, poultry, and fish is 85 to 100%, and that from wheat, soybeans, and other legumes ranges from 75 to 90%.

The official daily dietary “protein requirement” accepted by the U.S. and Canadian governments is 0.8 g of protein per kilogram of desirable body weight for adults (approximately 56 g for an adult male and 44 g for an adult female). On an average weight basis, the requirement per kilogram is much greater for infants and children. This fact underscores the importance of improving **Sissy Fibrosa’s** protein digestion to optimize her potential for normal growth and development.



In patients with cystinuria, such as **Cal Kulis**, the inability to normally absorb cystine and basic amino acids from the gut and the increased loss of these amino acids in the urine would be expected to cause a deficiency of these compounds in the blood. However, because three of these amino acids can be synthesized in the body (i.e., they are nonessential amino acids), their concentrations in the plasma remain normal, and clinical manifestations of a deficiency state do not develop. It is not clear why symptoms related to a lysine deficiency have not been observed.

In the disorder that was first observed in the Hartnup family and bears their name, the intestinal and renal transport defect involves the neutral amino acids (monoamine, monocarboxylic acids), including a number of the essential amino acids (isoleucine, leucine, phenylalanine, threonine, tryptophan, and valine) as well as certain nonessential amino acids (alanine, serine, and tyrosine). A reduction in the availability of these essential amino acids would be expected to cause a variety of clinical disorders. Yet children with the Hartnup disorder identified by routine newborn urine screening almost always remain clinically normal.

However, some patients with the Hartnup biochemical phenotype eventually develop pellagra-like manifestations, which usually include a photosensitivity rash, ataxia, and neuropsychiatric symptoms. Pellagra results from a dietary deficiency of the vitamin niacin or the essential amino acid tryptophan, which are both precursors for the nicotinamide moiety of NAD and NADP. In asymptomatic patients, the transport abnormality may be incomplete and so subtle as to allow no phenotypic expression of Hartnup disease. These patients also may be capable of absorbing some small peptides that contain the neutral amino acids.

The only rational treatment of patients having pellagra-like symptoms is the administration of niacin (nicotinic acid) in oral doses up to 300 mg/day. Although the rash, ataxia, and neuropsychiatric manifestations of niacin deficiency may disappear, the hyperaminoaciduria and intestinal transport defect do not respond to this therapy. In addition to niacin, a high-protein diet may benefit some patients.

BIOCHEMICAL COMMENTS



The γ -glutamyl cycle is necessary for the synthesis of glutathione, a compound that protects cells from oxidative damage (see Chapter 24). When originally discovered, the cycle was thought to be important in amino acid transport, but its involvement in such transport is now thought to be limited to salvage of cysteine. The enzymes of the cycle are present in many tissues, although certain tissues lack one or more of the enzymes of the cycle.

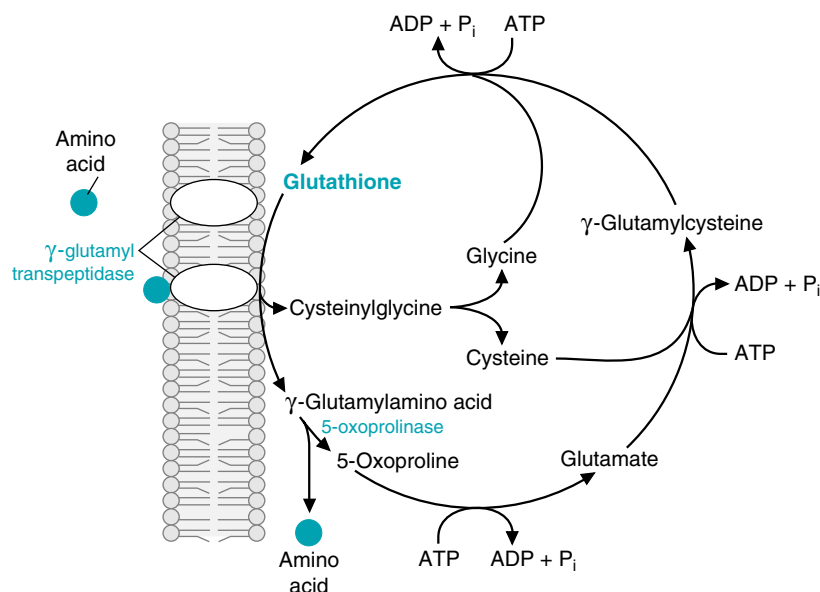


Fig. 37.6. γ -Glutamyl cycle. In cells of the intestine and kidney, amino acids can be transported across the cell membrane by reacting with glutathione (γ -glutamyl-cysteinyl-glycine) to form a γ -glutamyl amino acid. The amino acid is released into the cell, and glutathione is resynthesized. However, the major role of this cycle is glutathione synthesis, and many tissues lack the transpeptidase and 5-oxo-prolinase activities.

The entire cycle is presented in Figure 37.6. In this case, the extracellular amino acid reacts with glutathione (γ -glutamyl-cysteinyl-glycine) in a reaction catalyzed by a transpeptidase present in the cell membrane. A γ -glutamyl amino acid is formed, which travels across the cell membrane and releases the amino acid into the cell. The other products of these two reactions are reconverted to glutathione.

The reactions converting glutamate to glutathione in the γ -glutamyl cycle are the same reactions required for the synthesis of glutathione. The enzymes for glutathione synthesis, but not the transpeptidase, are found in most tissues. The oxo-prolinase is also missing from many tissues, such that the major role of this pathway is one of glutathione synthesis from glutamate, cysteine, and glycine. The transpeptidase is the only protease in the cell that can break the γ -glutamyl linkage in glutathione. Glutathione is also involved in reducing compounds such as hydrogen peroxide (see Chapter 21). It also protects cells, in particular erythrocytes, from oxidative damage, through formation of oxidized glutathione, two glutathione residues connected by a disulfide bond.

Suggested References

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REVIEW QUESTIONS—CHAPTER 37

1. An individual with a deficiency in the conversion of trypsinogen to trypsin would be expected to experience a more detrimental effect on protein digestion than an individual who was defective in any of the other digestive proteases. This is due to which of the following?
 - (A) Trypsin has a greater and wider range of substrates to act on.
 - (B) Trypsin activates pepsinogen, so digestion can begin in the stomach.
 - (C) Trypsin activates the other zymogens that are secreted by the pancreas.
 - (D) Trypsin activates enteropeptidase, which is needed to activate the other pancreatic zymogens.
 - (E) Trypsin inhibits intestinal motility, so the substrates can be hydrolyzed for longer periods.
2. An individual has been shown to have a deficiency in an intestinal epithelial cell amino acid transport system for leucine. However, the individual shows no symptoms of amino acid deficiency. This could be due to which of the following?
 - (A) The body synthesizes leucine to compensate for the transport defect.
 - (B) The kidney reabsorbs leucine and sends it to other tissues.
 - (C) There are multiple transport systems for leucine.
 - (D) Isoleucine takes the place of leucine in proteins.
 - (E) Leucine is not necessary for bulk protein synthesis.
3. Kwashiorkor can result from which of the following?
 - (A) Consuming a calorie-deficient diet that is also deficient in protein
 - (B) Consuming a calorie-adequate diet that is deficient in carbohydrates
 - (C) Consuming a calorie-adequate diet that is deficient in fatty acids
 - (D) Consuming a calorie-adequate diet that is deficient in proteins
 - (E) Consuming a calorie-deficient diet that is primarily proteins
4. Which of the following enzymes is activated through an autocatalytic process?
 - (A) Enteropeptidase
 - (B) Trypsinogen
 - (C) Pepsinogen
 - (D) Aminopeptidase
 - (E) Proelastase
5. Children with kwashiorkor usually have a fatty liver. This is due to which of the following?
 - (A) The high fat content of their diet
 - (B) The high carbohydrate content of their diet
 - (C) The high protein content of their diet
 - (D) The lack of substrates for gluconeogenesis in the liver
 - (E) The lack of substrates for protein synthesis in the liver
 - (F) The lack of substrates for glycogen synthesis in the liver

38 Fate of Amino Acid Nitrogen: Urea Cycle

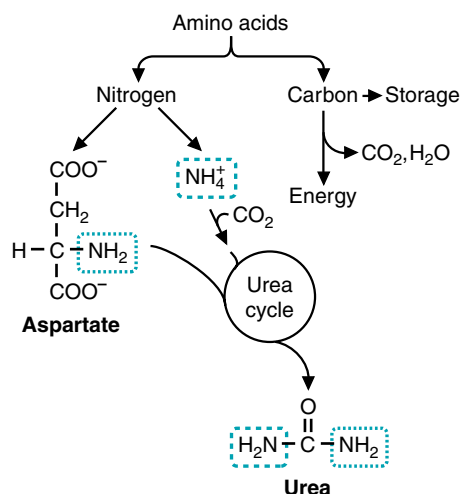


Fig 38.1. Fate of amino acid carbons and nitrogen. Amino acid carbon can be used either for energy storage (glycogen, fatty acids) or for energy. Amino acid nitrogen is used for urea synthesis. One nitrogen of urea comes from NH_4^+ , the other from aspartate.

In comparison with carbohydrate and lipid metabolism, the metabolism of amino acids is complex. We must be concerned not only with the fate of the carbon atoms of the amino acids but also with the fate of the **nitrogen**. During their metabolism, amino acids travel in the blood from one tissue to another. Ultimately, most of the nitrogen is converted to **urea** in the **liver** and the carbons are oxidized to CO_2 and H_2O by a number of tissues (Fig. 38.1).

After a meal that contains protein, amino acids released by digestion (see Chapter 37) pass from the gut through the hepatic portal vein to the liver (see Fig. 38.2A). In a normal diet containing 60 to 100 g protein, most of the amino acids are used for the synthesis of proteins in the liver and in other tissues. Excess amino acids may be converted to glucose or triacylglycerol.

During fasting, muscle protein is cleaved to amino acids. Some of the amino acids are partially oxidized to produce energy (see Fig. 38.2B). Portions of these amino acids are converted to **alanine** and **glutamine**, which, along with other amino acids, are released into the blood. Glutamine is oxidized by various tissues, including the **lymphocytes**, **gut**, and **kidney**, which convert some of the carbons and nitrogen to alanine. Alanine and other amino acids travel to the liver, where the carbons are converted to glucose and ketone bodies and the nitrogen is converted to urea, which is excreted by the kidneys. Glucose, produced by gluconeogenesis, is subsequently oxidized to CO_2 and H_2O by many tissues, and ketone bodies are oxidized by tissues such as muscle and kidney.

Several enzymes are important in the process of interconverting amino acids and in removing nitrogen so that the carbon skeletons can be oxidized. These include **dehydratases**, **transaminases**, **glutamate dehydrogenase**, **glutaminase**, and **deaminases**.

The conversion of amino acid nitrogen to urea occurs mainly in the liver. Urea is formed in the urea cycle from NH_4^+ , bicarbonate, and the nitrogen of **aspartate** (see Fig. 38.1). Initially, NH_4^+ , bicarbonate, and ATP react to produce **carbamoyl phosphate**, which reacts with **ornithine** to form **citrulline**. Aspartate then reacts with citrulline to form **argininosuccinate**, which releases **fumarate**, forming **arginine**. Finally, **arginase** cleaves arginine to release urea and regenerate ornithine. The cycle is regulated in a **feed-forward** manner, such that when amino acid degradation is occurring, the rate of the cycle is increased.



THE WAITING ROOM



Percy Veere and his high school friend decided to take a Caribbean cruise, during which they sampled the cuisine of many of the islands on their itinerary. One month after their return to the United States, Percy complained of

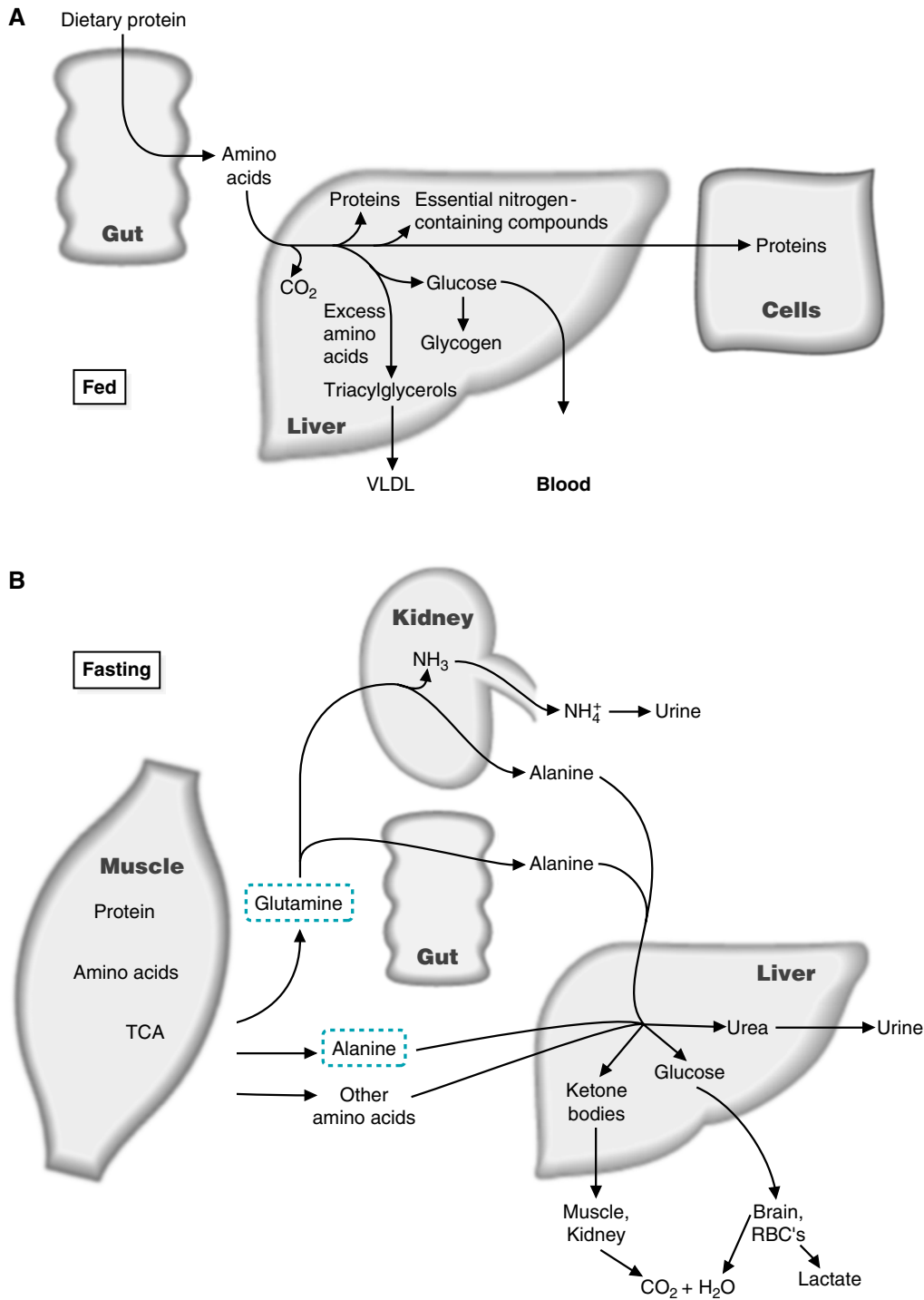


Fig 38.2. Roles of various tissues in amino acid metabolism. **A.** In the fed state, amino acids released by digestion of dietary proteins travel through the hepatic portal vein to the liver, where they are used for the synthesis of proteins, particularly the blood proteins, such as serum albumin. Excess amino acids are converted to glucose or to triacylglycerols. The latter are then packaged and secreted in VLDL. The glucose produced from amino acids in the fed state is stored as glycogen or released into the blood if blood glucose levels are low. Amino acids that pass through the liver are converted to proteins in cells of other tissues. **B.** During fasting, amino acids are released from muscle protein. Some enter the blood directly. Others are partially oxidized and the nitrogen stored in the form of alanine and glutamine, which enter the blood. In the kidney, glutamine releases ammonia into the urine and is converted to alanine and serine. In the cells of the gut, glutamine is converted to alanine. Alanine (the major gluconeogenic amino acid) and other amino acids enter the liver, where their nitrogen is converted to urea, which is excreted in the urine, and their carbons to glucose and ketone bodies, which are oxidized by various tissues for energy.

severe malaise, loss of appetite, nausea, vomiting, and arthralgias (joint pains). He had a low-grade fever and noted a persistent and increasing pain in the area of his liver. His friend noted a yellow discoloration of the whites of Percy's eyes and skin. Percy's urine turned the color of iced tea, and his stool became a light-clay color. His doctor found his liver to be enlarged and tender. Liver function tests were ordered.

Serologic testing for viral hepatitis type B, C, and D were nonreactive, but fecal studies showed "shedding" of hepatitis virus type A. Tests for antibodies to antigens of the hepatitis A virus (anti-HAV) in the serum were positive for the immunoglobulin M type.

A diagnosis of acute viral hepatitis type A was made, probably contracted from virus-contaminated food Percy had eaten while on his cruise. His physician explained that there was no specific treatment for type A viral hepatitis but recommended symptomatic and supportive care and prevention of transmission to others by the fecal–oral route. Percy took acetaminophen 3 to 4 times a day for fever and arthralgias throughout his illness.

I. FATE OF AMINO ACID NITROGEN

A. Transamination Reactions

Transamination is the major process for removing nitrogen from amino acids. In most instances, the nitrogen is transferred as an amino group from the original amino acid to α -ketoglutarate, forming glutamate, whereas the original amino acid is converted to its corresponding α -keto acid (Fig. 38.3). For example, the amino acid aspartate can be transaminated to form its corresponding α -keto acid, oxaloacetate. In the process, the amino group is transferred to α -ketoglutarate, which is converted to its corresponding amino acid, glutamate.

All amino acids except lysine and threonine undergo transamination reactions. The enzymes catalyzing these reactions are known as transaminases or aminotransferases. For most of these reactions, α -ketoglutarate and glutamate serve as one of the α -keto acid–amino acid pairs. Pyridoxal phosphate is the cofactor, and the mechanism of the reaction is indicated in Figure 38.4.

Overall, in a transamination reaction, an amino group from one amino acid becomes the amino group of a second amino acid. Because these reactions are readily reversible, they can be used to remove nitrogen from amino acids or to transfer nitrogen to α -keto acids to form amino acids. Thus, they are involved both in amino acid degradation and in amino acid synthesis.

B. Removal of Amino Acid Nitrogen as Ammonia

Cells in the body and bacteria in the gut release the nitrogen of certain amino acids as ammonia or ammonium ion (NH_4^+) (Fig. 38.5). Because these two forms of nitrogen can be interconverted, the terms are sometimes used interchangeably. Ammonium ion releases a proton to form ammonia by a reaction with a pK of 9.3 (Fig. 38.6). Therefore, at physiologic pH, the equilibrium favors NH_4^+ by a factor of approximately 100/1 (see Chapter 4, the Henderson-Hasselbalch equation). However, it is important to note that NH_3 is also present in the body, because this is the form that can cross cell membranes. For example, NH_3 passes into the urine from kidney tubule cells and decreases the acidity of the urine by binding protons, forming NH_4^+ . Once the NH_4^+ is formed, the compound can no longer freely diffuse across membranes.

Glutamate can be oxidatively deaminated by a reaction catalyzed by glutamate dehydrogenase that produces ammonium ion and α -ketoglutarate (Fig. 38.7). Either NAD^+ or NADP^+ can serve as the cofactor. This reaction, which occurs in the mitochondria of most cells, is readily reversible; it can incorporate ammonia

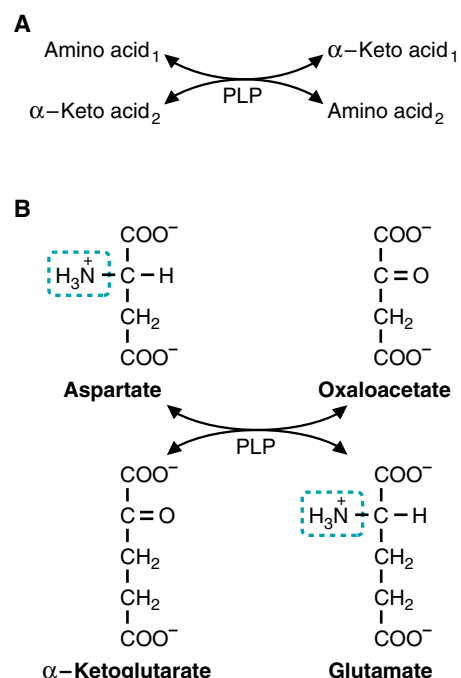


Fig. 38.3. Transamination. The amino group from one amino acid is transferred to another. Pairs of amino acids and their corresponding α -keto acids are involved in these reactions. α -ketoglutarate and glutamate are usually one of the pairs. The reactions, which are readily reversible, use pyridoxal phosphate (PLP) as a cofactor. The enzymes are called transaminases or aminotransferases. A. A generalized reaction. B. The aspartate transaminase reaction.



Whereas pyridoxal phosphate is used primarily for reactions involving amino acids, it is also required for the glycogen phosphorylase reaction, in which it acts as a general acid/base catalyst.

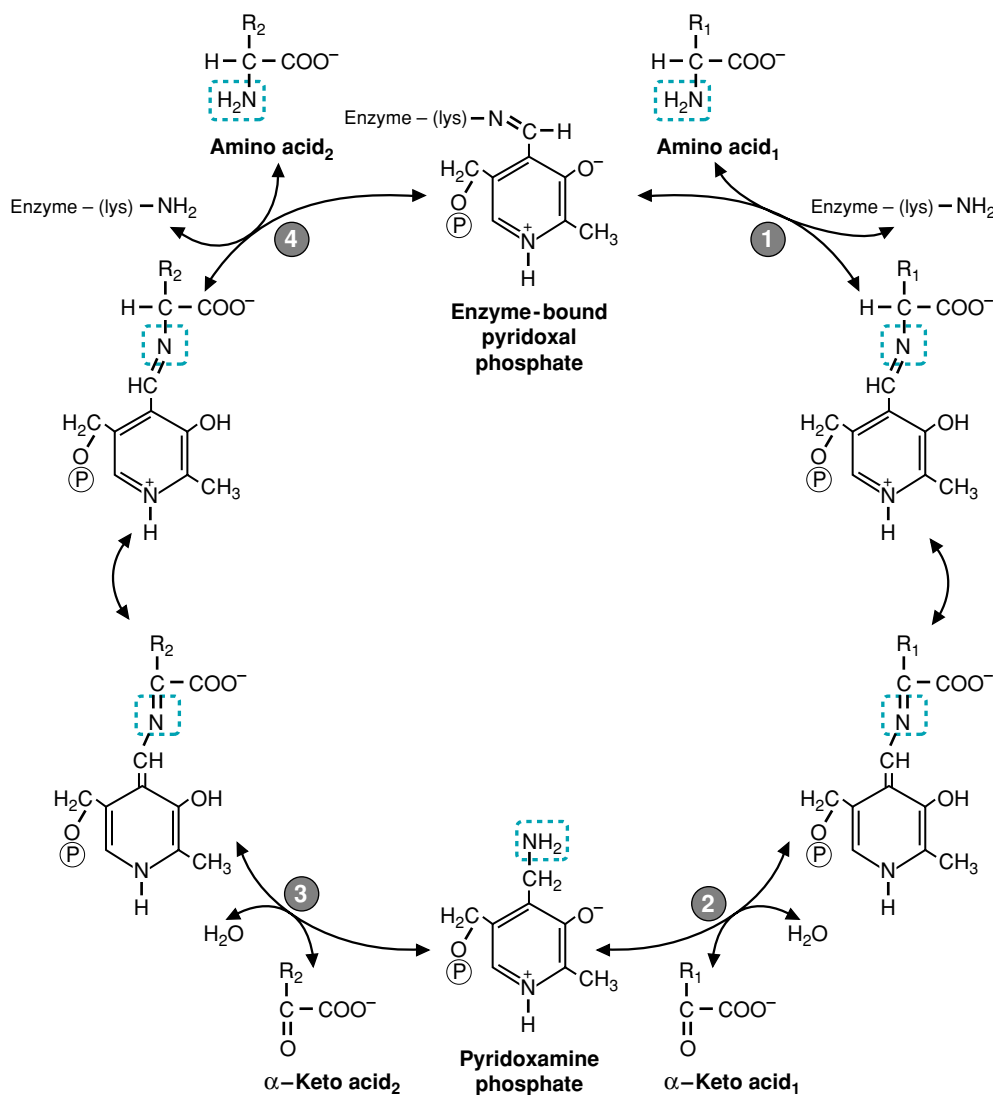


Fig. 38.4. Function of pyridoxal phosphate (PLP) in transamination reactions. The order in which the reactions occur is 1 to 4. Pyridoxal phosphate (enzyme-bound) reacts with amino acid₁, forming a Schiff base (a carbon–nitrogen double bond). After a shift of the double bond, α -keto acid₁ is released through hydrolysis of the Schiff base, and pyridoxamine phosphate is produced. Pyridoxamine phosphate then forms a Schiff base with α -keto acid₂. After the double bond shifts, amino acid₂ is released through hydrolysis of the Schiff base and enzyme-bound pyridoxal phosphate is regenerated. The net result is that the amino group from amino acid₁ is transferred to amino acid₂.

into glutamate or release ammonia from glutamate. Glutamate can collect nitrogen from other amino acids as a consequence of transamination reactions and then release ammonia through the glutamate dehydrogenase reaction. This process provides one source of the ammonia that enters the urea cycle.

In addition to glutamate, a number of amino acids release their nitrogen as NH_4^+ (see Fig. 38.5). Histidine may be directly deaminated to form NH_4^+ and urocanate. The deaminations of serine and threonine are dehydration reactions that require pyridoxal phosphate and are catalyzed by serine dehydratase. Serine forms pyruvate, and threonine forms α -ketobutyrate. In both cases, NH_4^+ is released.

Glutamine and asparagine contain R group amides that may be released as NH_4^+ by deamidation. Asparagine is deamidated by asparaginase, yielding aspartate and NH_4^+ . Glutaminase acts on glutamine, forming glutamate and NH_4^+ . The glutaminase



Glutamate dehydrogenase is one of three mammalian enzymes that can “fix” ammonia into organic molecules. The other two are glutamine synthetase and carbamoyl phosphate synthetase I.

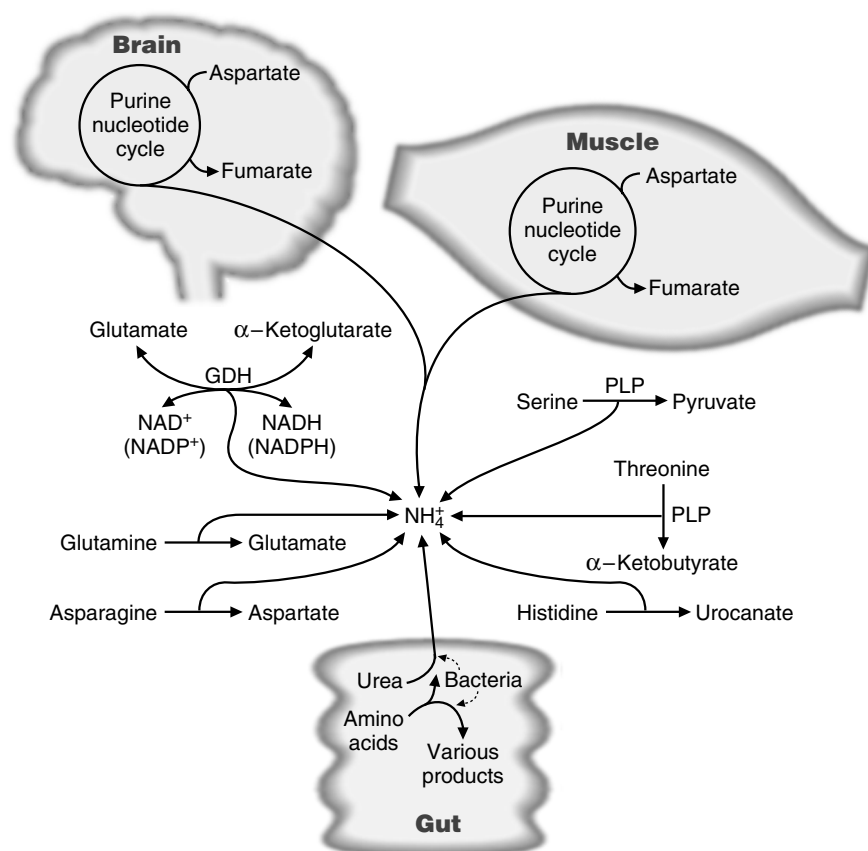


Fig. 38.5. Summary of the sources of NH_4^+ for the urea cycle. All of the reactions are irreversible except glutamate dehydrogenase (GDH). Only the dehydratase reactions, which produce NH_4^+ from serine and threonine, require pyridoxal phosphate as a cofactor. The reactions that are not shown occurring in the muscle or the gut can all occur in the liver, where the NH_4^+ generated can be converted to urea. The purine nucleotide cycle of the brain and muscle is further described in Chapter 41.

reaction is particularly important in the kidney, where the ammonium ion produced is excreted directly into the urine, where it forms salts with metabolic acids, facilitating their removal in the urine.

In muscle and brain, but not in liver, the purine nucleotide cycle allows NH_4^+ to be released from amino acids (see Fig. 38.5). Nitrogen is collected by glutamate from other amino acids by means of transamination reactions. Glutamate then transfers its amino group to oxaloacetate to form aspartate, which supplies nitrogen to the purine nucleotide cycle (see Chapter 41). The reactions of the cycle release fumarate and NH_4^+ . The ammonium ion formed can leave the muscle in the form of glutamine.

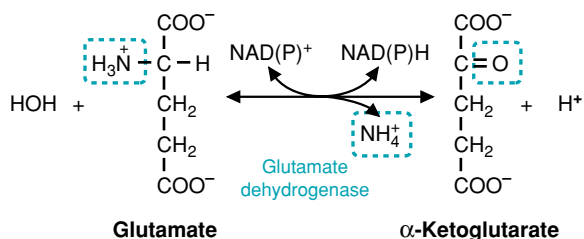


Fig. 38.7. Reaction catalyzed by glutamate dehydrogenase. This reaction is readily reversible and can use either NAD^+ or NADP^+ as a cofactor. The oxygen on α -ketoglutarate is derived from H_2O .



Percy Veere's laboratory studies showed that his serum alanine transaminase (ALT) level was 294 units/L (reference range = 5–30), and his serum aspartate transaminase (AST) level was 268 units/L (reference range = 10–30). His serum alkaline phosphatase level was 284 units/L (reference range for an adult male = 40–125), and his serum total bilirubin was 9.6 mg/dL (reference range = 0.2–1.0). Bilirubin is a degradation product of heme, as described in Chapter 44.

Cellular enzymes such as AST, ALT, and alkaline phosphatase leak into the blood through the membranes of hepatic cells that have been damaged as a result of the inflammatory process. In acute viral hepatitis, the serum ALT level is often elevated to a greater extent than the serum AST level. Alkaline phosphatase, which is present on membranes between liver cells and the bile duct, is also elevated in the blood in acute viral hepatitis.

The rise in serum total bilirubin occurs as a result of the inability of the infected liver to conjugate bilirubin and of a partial or complete occlusion of the hepatic biliary drainage ducts caused by inflammatory swelling within the liver. In fulminant hepatic failure, the serum bilirubin level may exceed 20 mg/dL, a poor prognostic sign.

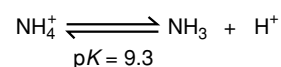


Fig. 38.6. Formation of ammonia from ammonium ion. Because the pK is 9.3, at physiologic pH, the concentration of NH_4^+ is almost 100 times that of NH_3 .



Vitamin B_6 deficiency symptoms include dermatitis, a microcytic, hypochromic anemia, weakness, irritability, and, in some cases, convulsions. Xanthurenic acid (a degradation product of tryptophan) and other compounds appear in the urine because of an inability to completely metabolize amino acids. A decreased ability to synthesize heme from glycine may cause the microcytic anemia (see Chapter 44), and decreased decarboxylation of amino acids to form neurotransmitters (see Chapter 48) may explain the convulsions.

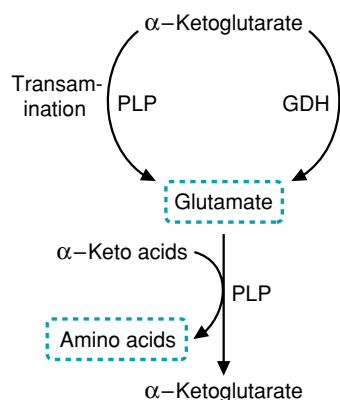
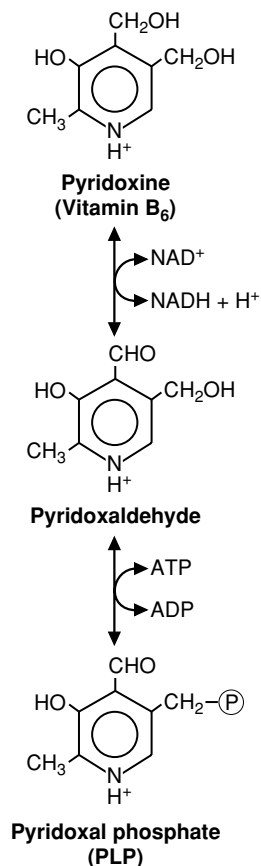


Fig. 38.8. Role of glutamate in amino acid synthesis. Glutamate transfers nitrogen by means of transamination reactions to α -keto acids to form amino acids. This nitrogen is either obtained by glutamate from transamination of other amino acids or from NH_4^+ by means of the glutamate dehydrogenase (GDH) reaction. PLP = pyridoxal phosphate.



Pyridoxal phosphate is derived from pyridoxine (vitamin B_6). Pyridoxal phosphate is the cofactor not only for transamination reactions but also for decarboxylations and a number of other reactions involving amino acids.



In summary, NH_4^+ that enters the urea cycle is produced in the body by deamination or deamidation of amino acids (see Fig. 38.5). A significant amount of NH_4^+ is also produced by bacteria that live in the lumen of the intestinal tract. This ammonium ion enters the hepatic portal vein and travels to the liver.

C. Role of Glutamate in the Metabolism of Amino Acid Nitrogen

Glutamate plays a pivotal role in the metabolism of amino acids. It is involved in both synthesis and degradation.

Glutamate provides nitrogen for amino acid synthesis (Fig. 38.8). In this process, glutamate obtains its nitrogen either from other amino acids by transamination reactions or from NH_4^+ by the glutamate dehydrogenase reaction. Transamination reactions then serve to transfer amino groups from glutamate to α -keto acids to produce their corresponding amino acids.

When amino acids are degraded and urea is formed, glutamate collects nitrogen from other amino acids by transamination reactions. Some of this nitrogen is released as ammonia by the glutamate dehydrogenase reaction, but much larger amounts of ammonia are produced from the other sources shown in Figure 38.5. NH_4^+ is one of the two forms in which nitrogen enters the urea cycle (Fig. 38.9).

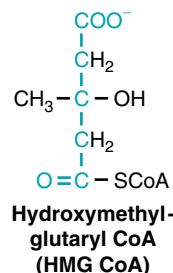
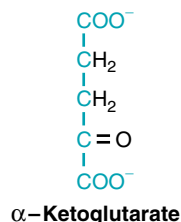
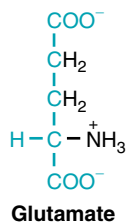
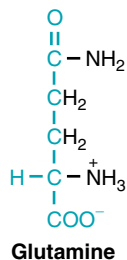
The second form of nitrogen for urea synthesis is provided by aspartate (see Fig. 38.9). Glutamate can be the source of the nitrogen. Glutamate transfers its amino group to oxaloacetate, and aspartate and α -ketoglutarate are formed.

D. Role of Alanine and Glutamine in Transporting Amino Acid Nitrogen to the Liver

Protein turnover and amino acid degradation occur in all tissues; however, the urea cycle enzymes are primarily active in the liver (the intestine expresses low levels of activity of these enzymes; see Chapter 42). Thus, a mechanism needs to be in place to transport amino acid nitrogen to the liver. Alanine and glutamine are the major carriers of nitrogen in the blood. Alanine is primarily exported by the muscle. Because the muscle is metabolizing glucose through glycolysis, pyruvate is available in the muscle. The pyruvate is transaminated by glutamate to form alanine, which travels to the liver (Fig. 38.10). The glutamate is formed by transamination of an amino acid that is being degraded. On arriving at the liver, alanine is transaminated to pyruvate, and the nitrogen will be used for urea synthesis. The pyruvate formed is used for gluconeogenesis and the glucose exported to the muscle for use as energy. This cycle of moving carbons and nitrogen between the muscle and liver is known as the glucose/alanine cycle.



Compounds that contain “glut” in their name have five carbons in a straight chain. At each end of the chain, the carbon is part of a carboxyl group. In glutamine, the carboxyl group has formed an amide, and in hydroxymethylglutaryl CoA (HMG-CoA), it has formed a thioester with coenzyme A.



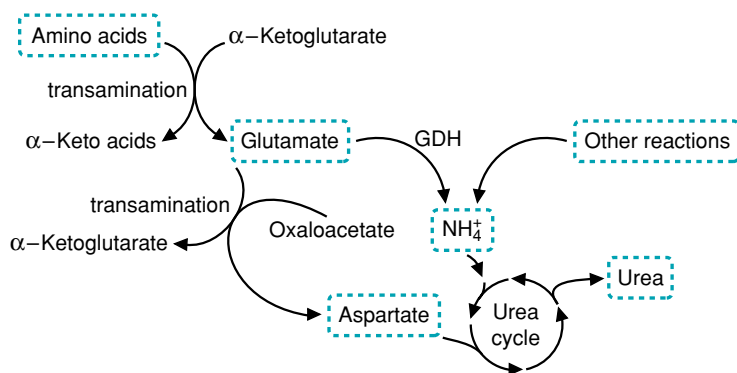


Fig. 38.9. Role of glutamate in urea production. Glutamate collects nitrogen from other amino acids by transamination reactions. This nitrogen can be released as NH_4^+ by glutamate dehydrogenase (GDH). NH_4^+ is also produced by other reactions (see Fig. 38.5). NH_4^+ provides one of the nitrogens for urea synthesis. The other nitrogen comes from aspartate and is obtained from glutamate by transamination of oxaloacetate.

Glutamine is synthesized from glutamate by the fixation of ammonia, requiring energy (adenosine triphosphate [ATP]) and the enzyme glutamine synthetase (Fig. 38.11), which is a cytoplasmic enzyme found in all cells. Under conditions of rapid amino acid degradation within a tissue, such that ammonia levels increase, the glutamate that has been formed from transamination reactions will accept another nitrogen molecule to form glutamine. The glutamine travels to the liver, kidney, or intestines, where glutaminase (see Fig. 38.11) will remove the amide nitrogen to form glutamate plus ammonia. In the kidney, the release of ammonia, and the formation of ammonium ion, serves to form salts with metabolic acids in the urine. In the intestine, the glutamine is used as a fuel (see Chapter 42). In the liver, the ammonia is used for urea biosynthesis.



Glutamine synthetase in liver is located in cells surrounding the portal vein. Its major role is to convert any ammonia that has escaped from urea production into glutamine, such that free ammonia does not leave the liver and enter the circulation.

II. UREA CYCLE

The normal human adult is in nitrogen balance; that is, the amount of nitrogen ingested each day, mainly in the form of dietary protein, is equal to the amount of nitrogen excreted. The major nitrogenous excretory product is urea, which exits from the body in the urine. This innocuous compound, produced mainly in the liver by the urea cycle, serves as the disposal form of ammonia, which is toxic, particularly to the brain and central nervous system. Normally, little ammonia (or NH_4^+) is present in the blood. The concentration ranges between

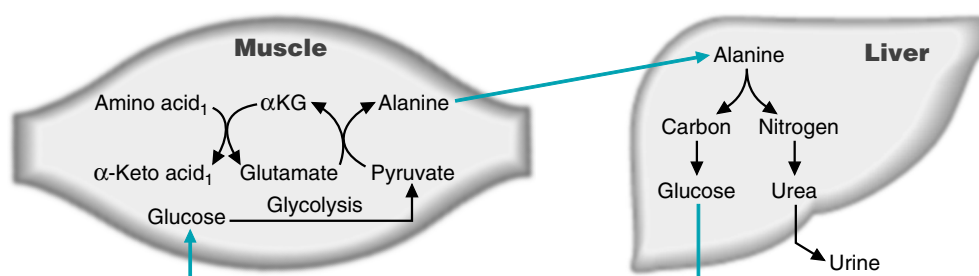


Fig. 38.10. The glucose/alanine cycle. Within the muscle, amino acid degradation leads to the transfer of nitrogens to α -ketoglutarate and pyruvate. The alanine formed travels to the liver, where the carbons of alanine are used for gluconeogenesis and the alanine nitrogen is used for urea biosynthesis. This could occur during exercise, when the muscle uses blood-borne glucose (see Chapter 47).

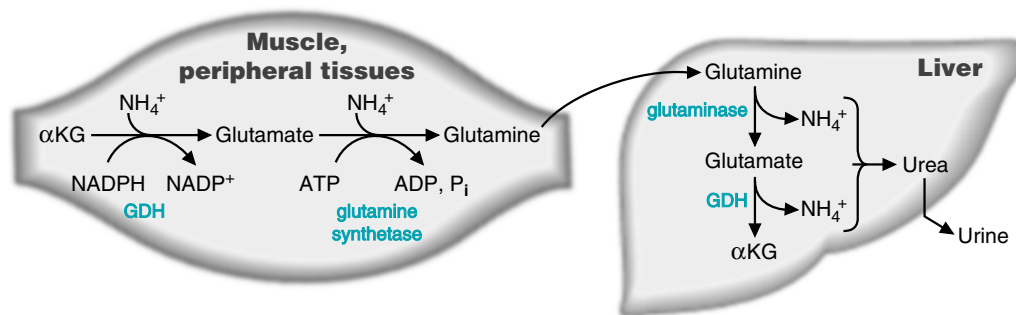


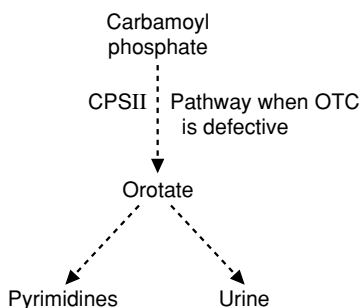
Fig. 38.11. Synthesis of glutamine in peripheral tissues and its transport to the liver. Within the liver, glutaminase converts glutamine to glutamate. Note how α -ketoglutarate can accept two molecules of ammonia to form glutamine. GDH = glutamate dehydrogenase.



Percy Veere's symptoms and laboratory abnormalities did not slowly subside over the next 6 weeks as they usually do in uncomplicated viral hepatitis A infections. Instead, his serum total bilirubin, ALT, AST, and alkaline phosphatase levels increased further. His vomiting became intractable, and his friend noted jerking motions of his arms (asterixis), facial grimacing, restlessness, slowed mentation, and slight disorientation. He was admitted to the hospital with a diagnosis of hepatic failure with incipient hepatic encephalopathy (brain dysfunction caused by accumulation of various toxins in the blood), a rare complication of acute type A viral hepatitis alone. The possibility of a superimposed acute hepatic toxicity caused by the use of acetaminophen was considered.



When ornithine transcarbamoylase (OTC) is deficient, the carbamoyl phosphate that normally would enter the urea cycle accumulates and floods the pathway for pyrimidine biosynthesis. Under these conditions, excess orotic acid (orotate), an intermediate in pyrimidine biosynthesis, is excreted in the urine. It produces no ill effects but is indicative of a problem in the urea cycle.



30 and 60 μM . Ammonia is rapidly removed from the blood and converted to urea by the liver. Nitrogen travels in the blood mainly in amino acids, particularly alanine and glutamine.

A. Reactions of the Urea Cycle

Nitrogen enters the urea cycle as NH_4^+ and aspartate (Fig. 38.12). NH_4^+ forms carbamoyl phosphate, which reacts with ornithine to form citrulline. Ornithine is the compound that both initiates and is regenerated by the cycle (similar to oxaloacetate in the TCA cycle). Aspartate reacts with citrulline, eventually donating its nitrogen for urea formation. Arginine is formed in two successive steps. Cleavage of arginine by arginase releases urea and regenerates ornithine.

1. SYNTHESIS OF CARBAMOYL PHOSPHATE

In the first step of the urea cycle, NH_4^+ , bicarbonate, and ATP react to form carbamoyl phosphate (see Fig. 38.12). The cleavage of 2 ATPs is required to form the high-energy phosphate bond of carbamoyl phosphate. Carbamoyl phosphate synthetase I (CPSI), the enzyme that catalyzes this first step of the urea cycle, is found mainly in mitochondria of the liver and intestine. The Roman numeral suggests that another carbamoyl phosphate synthetase exists, and indeed, CPSII, located in the cytosol, produces carbamoyl phosphate for pyrimidine biosynthesis, using nitrogen from glutamine (see Chapter 41).

2. PRODUCTION OF ARGININE BY THE UREA CYCLE

Carbamoyl phosphate reacts with ornithine to form citrulline (see Fig. 38.12). The high-energy phosphate bond of carbamoyl phosphate provides the energy required for this reaction, which occurs in mitochondria and is catalyzed by ornithine transcarbamoylase. The product citrulline is transported across the mitochondrial membranes in exchange for cytoplasmic ornithine and enters the cytosol. The carrier for this transport reaction catalyzes an electroneutral exchange of the two compounds.

In the cytosol, citrulline reacts with aspartate, the second source of nitrogen for urea synthesis, to produce argininosuccinate (see Fig. 38.12). This reaction, catalyzed by argininosuccinate synthetase, is driven by the hydrolysis of ATP to adenosine monophosphate (AMP) and pyrophosphate. Aspartate is produced by transamination of oxaloacetate.



The urea cycle was proposed in 1932 by Hans Krebs and a medical student, Kurt Henseleit, based on their laboratory observations. It was originally called the Krebs-Henseleit cycle. Subsequently, Krebs used this concept of metabolic cycling to explain a second process that also bears his name, the Krebs (or TCA) cycle.

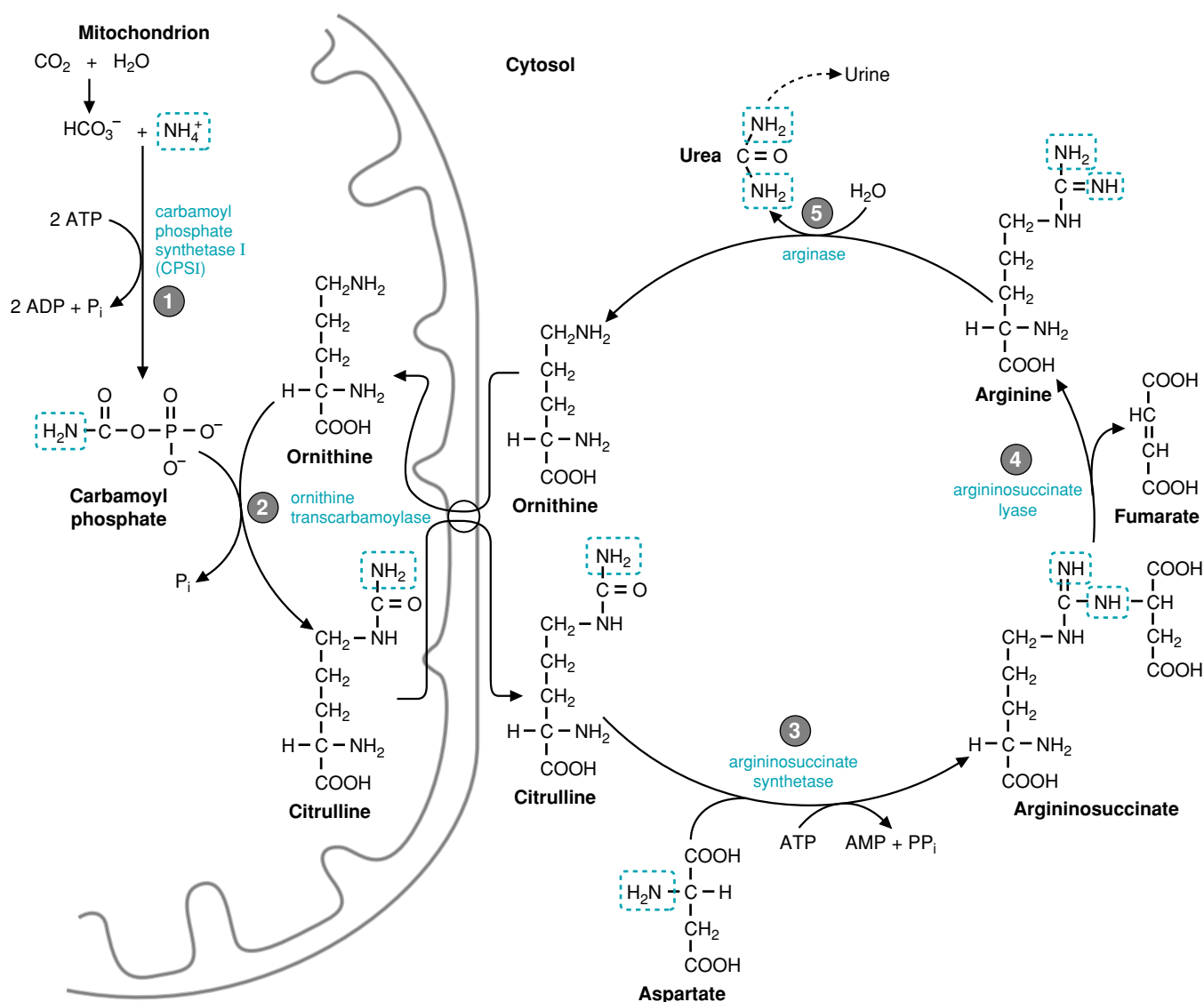


Fig. 38.12. Urea cycle. The steps of the cycle are numbered 1 to 5.

Argininosuccinate is cleaved by argininosuccinate lyase to form fumarate and arginine (see Fig. 38.12). Fumarate is produced from the carbons of argininosuccinate provided by aspartate. Fumarate is converted to malate (using cytoplasmic fumarase), which is used either for the synthesis of glucose by the gluconeogenic pathway or for the regeneration of oxaloacetate by cytoplasmic reactions similar to those observed in the TCA cycle (Fig. 38.13). The oxaloacetate that is formed is transaminated to generate the aspartate that carries nitrogen into the urea cycle. Thus, the carbons of fumarate can be recycled to aspartate.

3. CLEAVAGE OF ARGININE TO PRODUCE UREA

Arginine, which contains nitrogens derived from NH_4^+ and aspartate, is cleaved by arginase, producing urea and regenerating ornithine (see Fig. 38.12). Urea is produced from the guanidinium group on the side chain of arginine. The portion of arginine originally derived from ornithine is reconverted to ornithine.

The reactions by which citrulline is converted to arginine and arginine is cleaved to produce urea occur in the cytosol. Ornithine, the other product of the arginase

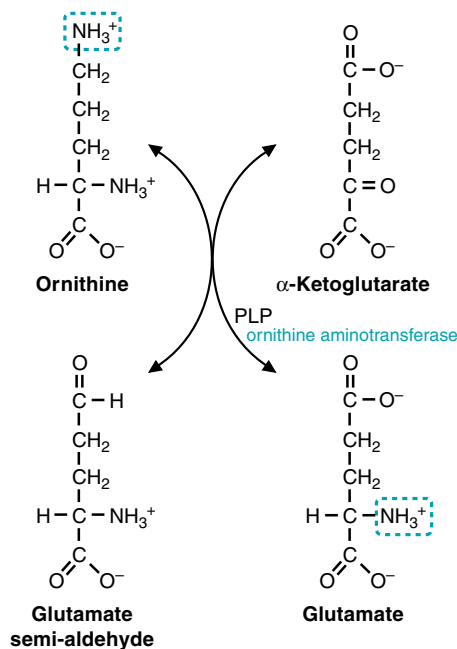


Fig. 38.14. The ornithine aminotransferase reaction. This is a reversible reaction dependent on pyridoxal phosphate, which normally favors ornithine degradation.

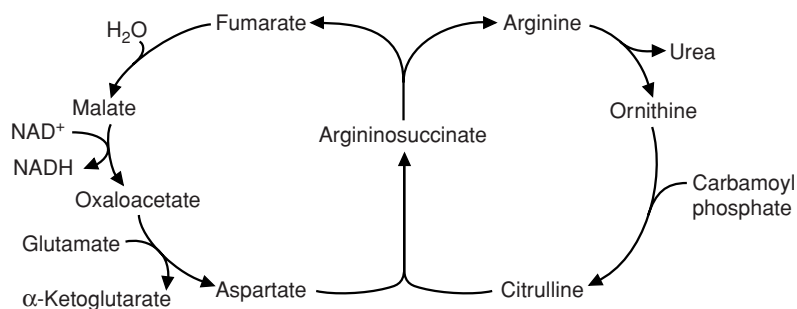


Fig. 38.13. The Krebs bicycle, indicating the common steps between the TCA and urea cycles. All reactions shown occur in the cytoplasm except for the synthesis of citrulline, which occurs within the mitochondria.

reaction, is transported into the mitochondrion in exchange for citrulline, where it can react with carbamoyl phosphate, initiating another round of the cycle.

B. Origin of Ornithine

Ornithine is an amino acid. However, it is not incorporated into proteins during the process of protein synthesis because no genetic codon exists for this amino acid. Although ornithine is normally regenerated by the urea cycle (one of the products of the arginase reaction), ornithine also can be synthesized *de novo* if needed. The reaction is an unusual transamination reaction catalyzed by ornithine aminotransferase under specific conditions in the intestine (Fig. 38.14). The usual direction of this reaction is the formation of glutamate semialdehyde, which is the first step of the degradation pathway for ornithine.

C. Regulation of the Urea Cycle

The human liver has a vast capacity to convert amino acid nitrogen to urea, thereby preventing toxic effects from ammonia, which would otherwise accumulate. In general, the urea cycle is regulated by substrate availability; the higher the rate of ammonia production, the higher the rate of urea formation. Regulation by substrate availability is a general characteristic of disposal pathways, such as the urea cycle, which remove toxic compounds from the body. This is a type of “feed-forward” regulation, in contrast to the “feedback” regulation characteristic of pathways that produce functional endproducts.

Two other types of regulation control the urea cycle: allosteric activation of CPSI by *N*-acetylglutamate (NAG) and induction/repression of the synthesis of urea cycle enzymes. NAG is formed specifically to activate CPSI; it has no other known function in mammals. The synthesis of NAG from acetyl CoA and glutamate is stimulated by arginine (Fig. 38.15). Thus, as arginine levels increase within the liver, two important reactions are stimulated. The first is the synthesis of NAG, which will increase the rate at which carbamoyl phosphate is produced. The second is to produce more ornithine (via the arginase reaction), such that the cycle can operate more rapidly.

The induction of urea cycle enzymes occurs in response to conditions that require increased protein metabolism, such as a high-protein diet or prolonged fasting. In both of these physiologic states, as amino acid carbon is converted to glucose, amino acid nitrogen is converted to urea. The induction of the synthesis of urea cycle enzymes under these conditions occurs even though the uninduced enzyme levels are far in excess of the capacity required. The ability of a high-protein diet to increase urea cycle enzyme levels is another type of “feed-forward” regulation.



The precise pathogenesis of the central nervous system (CNS) signs and symptoms that accompany liver failure (hepatic encephalopathy) in patients such as **Percy Veere** is not completely understood. These changes are, however, attributable in part to toxic materials that are derived from the metabolism of nitrogenous substrates by bacteria in the gut that circulate to the liver in the portal vein. These materials “bypass” their normal metabolism by the liver cells, however, because the acute inflammatory process of viral hepatitis severely limits the ability of liver cells to degrade these compounds to harmless metabolites. As a result, these toxins are “shunted” into the hepatic veins unaltered and eventually reach the brain through the systemic circulation (“portal-systemic encephalopathy”).

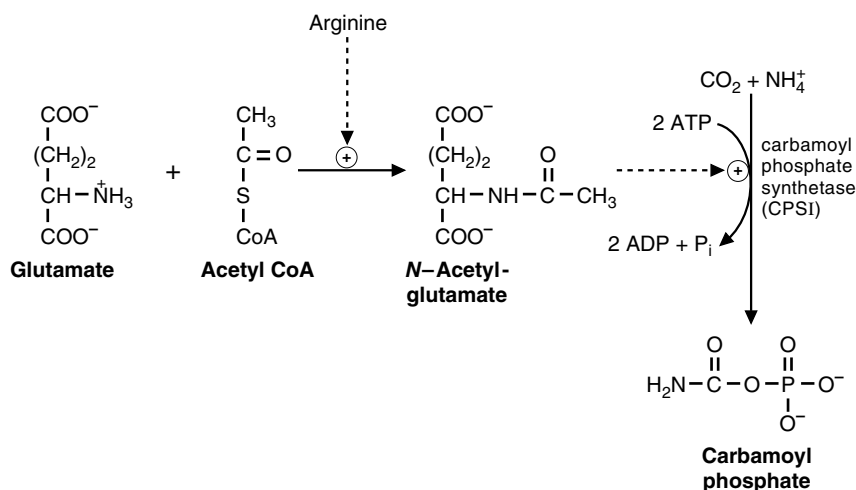


Fig. 38.15. Activation of carbamoyl phosphate synthetase I (CPSI). Arginine stimulates the synthesis of *N*-acetylglutamate, which activates CPSI.

D. Function of the Urea Cycle during Fasting

During fasting, the liver maintains blood glucose levels. Amino acids from muscle protein are a major carbon source for the production of glucose by the pathway of gluconeogenesis. As amino acid carbons are converted to glucose, the nitrogens are converted to urea. Thus, the urinary excretion of urea is high during fasting (Fig. 38.16). As fasting progresses, however, the brain begins to use ketone bodies, sparing blood glucose. Less muscle protein is cleaved to provide amino acids for gluconeogenesis, and decreased production of glucose from amino acids is accompanied by decreased production of urea (see Chapter 31).

The major amino acid substrate for gluconeogenesis is alanine, which is synthesized in peripheral tissues to act as a nitrogen carrier (see Fig. 38.10). Glucagon release, which is expected during fasting, stimulates alanine transport

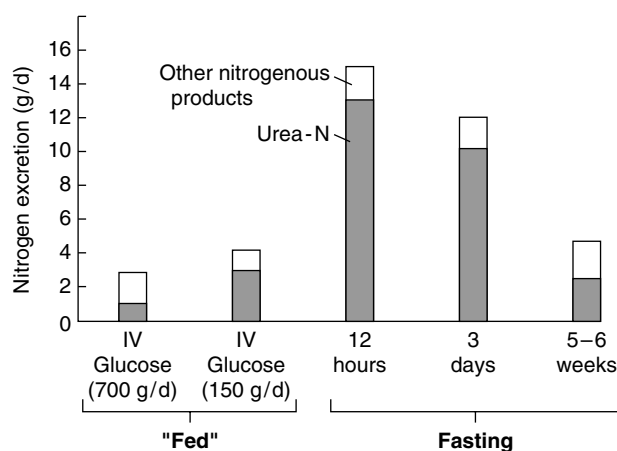


Fig. 38.16. Nitrogen excretion during fasting. Human subjects were initially given intravenous (IV) glucose as indicated, then fasted. Total nitrogen excretion was measured as well as the nitrogen in urea (dark area). Based on Ruderman NB, et al. Gluconeogenesis and its disorders in man. In: Hanson RW, Mehlerman MA, eds. Gluconeogenesis: Its Regulation in Mammalian Species. New York: John Wiley, 1976:518.



NH_4^+ is one of the toxins that results from the degradation of urea or proteins by intestinal bacteria and is not metabolized by the infected liver. The subsequent elevation of ammonia concentrations in the fluid bathing the brain causes depletion of tricarboxylic acid (TCA) cycle intermediates and ATP in the central nervous system. α -Ketoglutarate, a TCA cycle intermediate, combines with ammonia to form glutamate in a reaction catalyzed by glutamate dehydrogenase. Glutamate subsequently reacts with ammonia to form glutamine.

The absolute level of ammonia and its metabolites, such as glutamine, in the blood or cerebrospinal fluid in patients with hepatic encephalopathy correlates only roughly with the presence or severity of the neurologic signs and symptoms. γ -Aminobutyric acid (GABA), an important inhibitory neurotransmitter in the brain, is also produced in the gut lumen and is shunted into the systemic circulation in increased amounts in patients with hepatic failure. In addition, other compounds (such as aromatic amino acids, false neurotransmitters, and certain short-chain fatty acids) bypass liver metabolism and accumulate in the systemic circulation, adversely affecting central nervous system function. Their relative importance in the pathogenesis of hepatic encephalopathy remains to be determined.



In addition to producing urea, the reactions of the urea cycle also serve as the pathway for the biosynthesis of arginine. Therefore, this amino acid is not required in the diet of the adult; however, it is required in the diet for growth.



Urea is not cleaved by human enzymes. However, bacteria, including those in the human digestive tract, can cleave urea to ammonia and CO_2 , using the enzyme urease. Urease is not produced by humans.

To some extent, humans excrete urea into the gut and saliva. Intestinal bacteria convert urea to ammonia. This ammonia, as well as ammonia produced by other bacterial reactions in the gut, is absorbed into the hepatic portal vein. It is normally extracted by the liver and converted to urea. Approximately one fourth of the total urea released by the liver each day is recycled by bacteria.

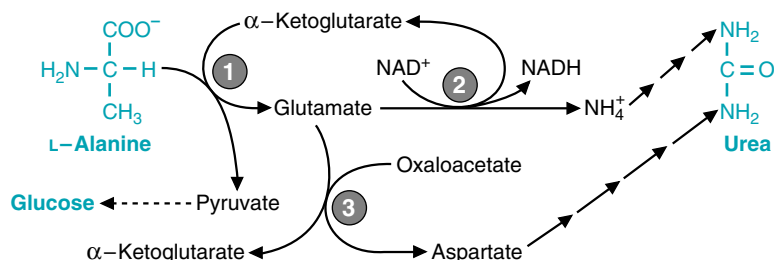


Fig. 38.17. Conversion of alanine to glucose and urea. 1. Alanine, the key gluconeogenic amino acid, is transaminated to form pyruvate, which is converted to glucose. The nitrogen, now in glutamate, can be released as NH₄⁺ (2) or transferred to oxaloacetate to form aspartate (3). NH₄⁺ and aspartate enter the urea cycle, which produces urea. In summary, the carbons of alanine form glucose and the nitrogens form urea. Two molecules of alanine are required to produce one molecule of glucose and one molecule of urea.

into the liver by activating the transcription of transport systems for alanine. Two molecules of alanine are required to generate one molecule of glucose. The nitrogen from the two molecules of alanine is converted to one molecule of urea (Fig. 38.17).

CLINICAL COMMENTS



The two most serious complications of acute viral hepatitis found in patients such as **Percy Veere** are massive hepatic necrosis leading to fulminant liver failure and the eventual development of chronic hepatitis. Both complications are rare in acute viral hepatitis type A, however, suggesting that acetaminophen toxicity may have contributed to Percy's otherwise unexpectedly severe hepatocellular dysfunction and early hepatic encephalopathy.

Fortunately, bed rest, rehydration, parenteral nutrition, and therapy directed at decreasing the production of toxins that result from bacterial degradation of nitrogenous substrates in the gut lumen (e.g., administration of lactulose, which reduces gut ammonia levels by a variety of mechanisms, the use of enemas and antibiotics to decrease the intestinal flora, a low-protein diet) prevented **Percy Veere** from progressing to the later stages of hepatic encephalopathy. As with most patients who survive an episode of fulminant hepatic failure, recovery to his previous state of health occurred over the next 3 months. Percy's liver function studies returned to normal, and a follow-up liver biopsy showed no histologic abnormalities.

BIOCHEMICAL COMMENTS



Disorders of the urea cycle are dangerous because of the accumulation of ammonia in the circulation. Ammonia is toxic to the nervous system, and its concentration in the body must be carefully controlled. Under normal conditions, free ammonia is rapidly fixed into either α -ketoglutarate (by glutamate dehydrogenase, to form glutamate) or glutamate (by glutamine synthase, to form glutamine). The glutamine can be used by many tissues, including the liver; the glutamate donates nitrogens to pyruvate to form alanine, which travels to the liver. Within the liver, as the nitrogens are removed from their carriers, carbamoyl phosphate synthetase I fixes the ammonia into carbamoyl phosphate to initiate the urea

cycle. However, when a urea cycle enzyme is defective, the cycle is interrupted, which leads to an accumulation of urea cycle intermediates before the block. Because of the block in the urea cycle, glutamine levels increase in the circulation, and because α -ketoglutarate is no longer being regenerated by removal of nitrogen from glutamine, the α -ketoglutarate levels are too low to fix more free ammonia, leading to elevated ammonia levels in the blood. So how does one reduce ammonia and glutamine levels in such patients?

The key to treating patients with urea cycle defects is to diagnose the disease early and then aggressively treat with compounds that can aid in nitrogen removal

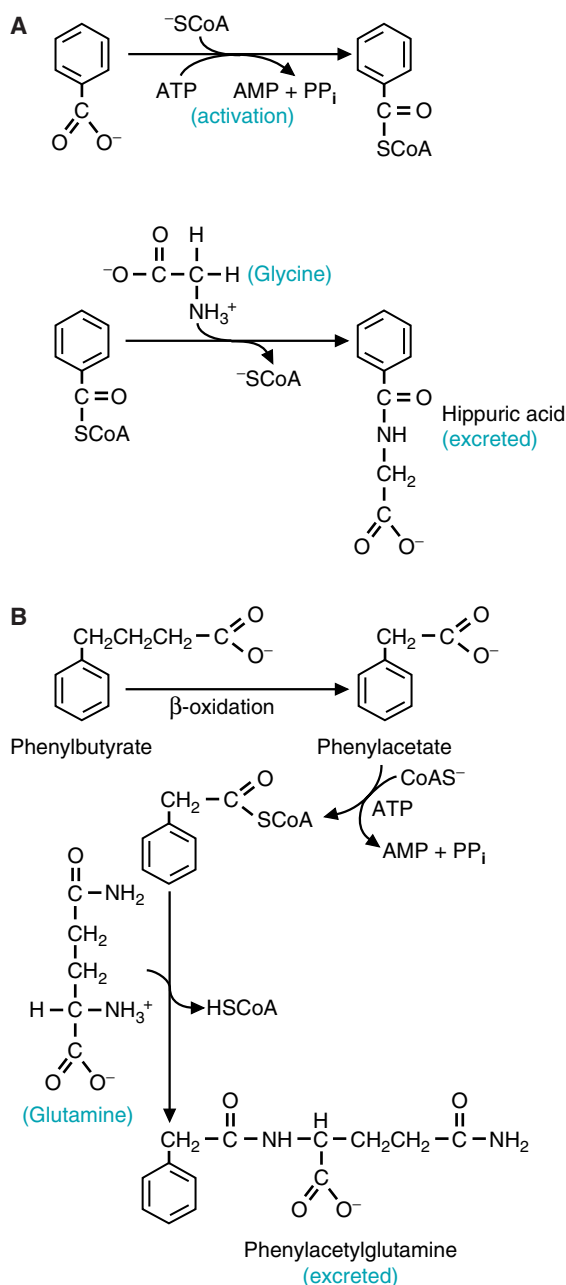


Fig. 38.18. The metabolism of benzoic acid (A) and phenylbutyrate (B), two agents used to reduce nitrogen levels in patients with urea cycle defects.

from the patient. Low-protein diets are essential to reduce the potential for excessive amino acid degradation. If the enzyme defect in the urea cycle comes after the synthesis of argininosuccinate, massive arginine supplementation has proved beneficial. Once argininosuccinate has been synthesized, the two nitrogen molecules destined for excretion have been incorporated into the substrate; the problem is that ornithine cannot be regenerated. If ornithine could be replenished to allow the cycle to continue, argininosuccinate could be used as the carrier for nitrogen excretion from the body. Thus, ingesting large levels of arginine leads to ornithine production by the arginase reaction, and nitrogen excretion via argininosuccinate in the urine can be enhanced.

Arginine therapy will not work for enzyme defects that exist in steps before the synthesis of argininosuccinate. For these disorders, drugs are used that form conjugates with amino acids. The conjugated amino acids are excreted, and the body then has to use its nitrogen to resynthesize the excreted amino acid. The two compounds most frequently used are benzoic acid and phenylbutyrate (the active component of phenylbutyrate is phenylacetate, its oxidation product. Phenylacetate has a bad odor, which makes it difficult to take orally). As indicated in Figure 38.18A, benzoic acid, after activation, reacts with glycine to form hippuric acid, which is excreted. As glycine is synthesized from serine, the body now uses nitrogens to synthesize serine, so more glycine can be produced. Phenylacetate (see Fig. 38.18B) forms a conjugate with glutamine, which is excreted. This conjugate removes two nitrogens per molecule and requires the body to resynthesize glutamine from glucose, thereby using another two nitrogen molecules.

Urea cycle defects are excellent candidates for treatment by gene therapy. This is because the defect only has to be repaired in one cell type (in this case, the hepatocyte), which makes it easier to target the vector carrying the replacement gene. Preliminary gene therapy experiments had been carried out on individuals with ornithine transcarbamoylase deficiency (the most common inherited defect in the urea cycle), but the experiments came to a halt when one of the patients died of a severe immunologic reaction to the vector used to deliver the gene. This incident has placed gene replacement therapy in the United States “on hold” for the foreseeable future.

Suggested Readings

- Brusilow S, Horwich A. Urea cycle enzymes. In: Scriver CR, Beaudet AL, Valle D, Sly WS, et al., eds. *The Metabolic and Molecular Bases of Inherited Disease*, vol II, 8th Ed. New York: McGraw-Hill, 2001:1909–1963.
- Marshall E. Gene therapy death prompts review of adenovirus vector. *Science* 1999;286:2244–2245.
- Morris SM Jr. Regulation of enzymes of the urea cycle and arginine metabolism. *Annu Rev Nutr* 2002;22:87–105.



REVIEW QUESTIONS—CHAPTER 38

1. Deficiency diseases have been described that involve each of the five enzymes of the urea cycle. Clinical manifestations may appear in the neonatal period. Infants with defects in the first four enzymes usually appear normal at birth, but after 24 hours progressively develop lethargy, hypothermia, and apnea. They have high blood ammonia levels, and the brain becomes swollen. One possible explanation for the swelling is the osmotic effect of the accumulation of glutamine in the brain produced by the reactions of ammonia with α -ketoglutarate and glutamate. Arginase deficiency is not as severe as deficiencies of the other urea cycle enzymes.

Given the following information about five newborn infants (identified as I to V) who appeared normal at birth but developed hyperammonemia after 24 hours, determine which urea cycle enzyme might be defective in each case (for each infant, choose from the same five answers, lettered A through E). All infants had low levels of blood urea nitrogen (BUN). (Normal citrulline levels are 10-20 μM .)

Infant	Urine Orotate	Blood Citrulline	Blood Arginine	Blood Ammonia
I	Low	Low	Low	High
II	-*	High ($> 1,000 \mu\text{M}$)	Low	High
III	-	-	High	Moderately high
IV	High	Low	Low	High
V	-	High ($200 \mu\text{M}$)	Low	High

* = Value not determined; low = below normal; high = above normal.

- (A) Carbamoylphosphate synthetase I
- (B) Ornithine transcarbamoylase
- (C) Argininosuccinate synthetase
- (D) Argininosuccinate lyase
- (E) Arginase

2. The nitrogens in urea are directly derived from which of the following compounds?

- (A) Ornithine and carbamoyl phosphate
- (B) Ornithine and aspartate
- (C) Ornithine and glutamate
- (D) Carbamoyl phosphate and aspartate
- (E) Carbamoyl phosphate and glutamine
- (F) Aspartate and glutamine

3. Which one of the following enzymes can fix ammonia into an organic molecule?

- (A) Alanine-pyruvate aminotransferase
- (B) Glutaminase
- (C) Glutamate dehydrogenase
- (D) Arginase
- (E) Argininosuccinate synthetase

4. Pyridoxal phosphate, which is required for transaminations, is also required for which of the following pathways?

- (A) Glycolysis
- (B) Gluconeogenesis
- (C) Glycogenolysis
- (D) TCA cycle
- (E) Fatty acid oxidation

5. The major regulated step of the urea cycle is which of the following?

- (A) Carbamoyl phosphate synthetase I
- (B) Ornithine transcarbamoylase
- (C) Argininosuccinate synthetase
- (D) Argininosuccinate lyase
- (E) Arginase

39 Synthesis and Degradation of Amino Acids



As a general rule, genetic codons exist for 20 amino acids. Only these 20 common amino acids are incorporated into proteins during the process of protein synthesis. Modifications to these amino acids occur after they are incorporated into proteins (such as the synthesis of hydroxyproline in collagen). The major exception to this rule is selenocysteine, which is an essential component of enzymes involved in scavenging free radicals (such as glutathione peroxidase-1; see Chapter 24). Selenocysteine has a selenium atom in the place of the oxygen atom in serine and is synthesized enzymatically in a reaction that requires adenosine triphosphate (ATP), selenium, and serine attached to a tRNA specific for selenocysteine. This reaction uses two high-energy bonds. The codon recognized by the tRNA-selenocysteine is a stop codon in the mRNA (UGA). The secondary structure of the mRNA allows the ribosomes and tRNA to understand which UGA is a stop codon and which requires the insertion of selenocysteine.

Because each of the 20 common amino acids has a unique structure, their metabolic pathways differ. Despite this, some generalities do apply to both the synthesis and degradation of all amino acids. These are summarized in the following sections. Because a number of the amino acid pathways are clinically relevant, we present most of the diverse pathways occurring in humans. However, we will be as succinct as possible.

Important coenzymes: *Pyridoxal phosphate* (derived from vitamin B6) is the quintessential coenzyme of amino acid metabolism. In degradation, it is involved in the removal of amino groups, principally through **transamination reactions** and in donation of amino groups for various amino acid biosynthetic pathways. It is also required for certain reactions involving the carbon skeleton of amino acids.

Tetrahydrofolate (FH₄) is a coenzyme used to transfer one-carbon groups at various oxidation states. FH₄ is used both in amino acid degradation (e.g., serine and histidine) and biosynthesis (e.g., glycine). **Tetrahydrobiopterin (BH₄)** is a cofactor required for ring hydroxylation reactions (e.g., phenylalanine to tyrosine).

Synthesis of the amino acids: Eleven of the twenty common amino acids can be synthesized in the body (Fig. 39.1). The other nine are considered “**essential**” and must be obtained from the diet. Almost all of the amino acids that can be synthesized by humans are amino acids used for the synthesis of additional nitrogen-containing compounds. Examples include glycine, which is used for porphyrin and purine synthesis; glutamate, which is required for neurotransmitter and purine synthesis; and aspartate, which is required for both purine and pyrimidine biosynthesis.

Nine of the eleven “**nonessential**” amino acids can be produced from glucose plus, of course, a source of nitrogen, such as another amino acid or ammonia. The other two nonessential amino acids, tyrosine and cysteine, require an essential amino acid for their synthesis (phenylalanine for tyrosine, and methionine for cysteine). The carbons for cysteine synthesis come from glucose; the methionine only donates the sulfur.

The carbon skeletons of the 10 nonessential amino acids derived from glucose are produced from intermediates of **glycolysis** and the **tricarboxylic acid (TCA) cycle** (see Fig 39.1). Four amino acids (**serine, glycine, cysteine, and alanine**) are produced from glucose through components of the **glycolytic pathway**. TCA cycle intermediates (which can be produced from glucose) provide carbon for synthesis of the six remaining nonessential amino acids. α -Ketoglutarate is the precursor for the synthesis of glutamate, glutamine, proline, and arginine. Oxaloacetate provides carbon for the synthesis of aspartate and asparagine.

The regulation of individual amino acid biosynthesis can be quite complex, but the overriding feature is that the **pathways are feedback regulated** such that as the **concentration of free amino acid increases**, a key biosynthetic **enzyme is allosterically or transcriptionally inhibited**. Amino acid levels, however, are

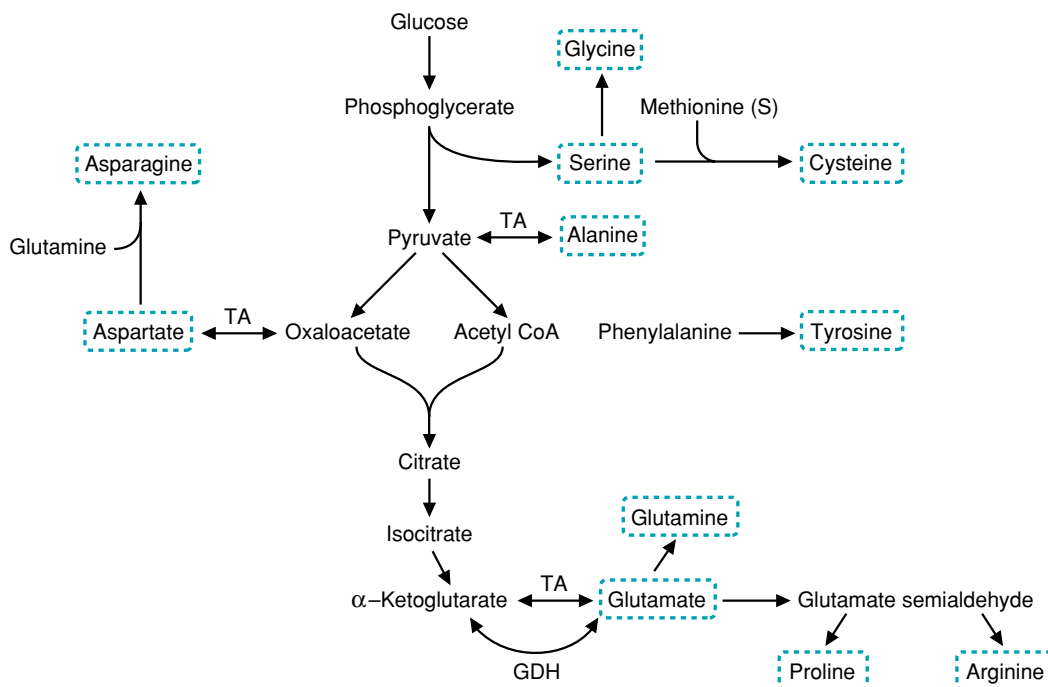


Fig. 39.1. Overview of the synthesis of the nonessential amino acids. The carbons of 10 amino acids may be produced from glucose through intermediates of glycolysis or the TCA cycle. The 11th nonessential amino acid, tyrosine, is synthesized by hydroxylation of the essential amino acid phenylalanine. Only the sulfur of cysteine comes from the essential amino acid methionine; its carbons and nitrogen come from serine. Transamination (TA) reactions involve pyridoxal phosphate (PLP) and another amino acid/ α -keto acid pair.

always maintained at a level such that the aminoacyl-tRNA synthetases can remain active, and protein synthesis can continue.

Degradation of amino acids: The *degradation pathways* for amino acids are, in general, **distinct from biosynthetic pathways**. This allows for separate regulation of the anabolic and catabolic pathways. Because protein is a fuel, almost every amino acid will have a degradative pathway that can generate NADH, which is used as an electron source for oxidative phosphorylation. However, the energy-generating pathway may involve direct oxidation, oxidation in the TCA cycle, conversion to glucose and then oxidation, or conversion to **ketone bodies**, which are then oxidized.

The fate of the carbons of the amino acids depends on the physiologic state of the individual and the tissue where the degradation occurs. For example, in the liver during fasting, the carbon skeletons of the amino acids produce glucose, ketone bodies, and CO_2 . In the fed state, the liver can convert intermediates of amino acid metabolism to glycogen and triacylglycerols. Thus, the **fate of the carbons of the amino acids parallels that of glucose and fatty acids**. The liver is the only tissue that has all of the pathways of amino acid synthesis and degradation.

As amino acids are degraded, their **carbons** are converted to (a) CO_2 , (b) compounds that produce **glucose** in the liver (pyruvate and the TCA cycle intermediates α -ketoglutarate, succinyl CoA, fumarate, and oxaloacetate), and (c) **ketone bodies** or their precursors (acetoacetate and acetyl CoA) (Fig. 39.2). For simplicity, amino acids are considered to be **glucogenic** if their carbon skeletons can be converted to a precursor of glucose and **ketogenic** if their carbon skeletons can be directly converted to acetyl CoA or acetoacetate. Some amino acids contain carbons that produce a glucose precursor and other carbons that produce acetyl CoA or acetoacetate. These amino acids are both glucogenic and ketogenic.

The amino acids synthesized from intermediates of glycolysis (**serine, alanine, and cysteine**) plus certain other amino acids (**threonine, glycine, and tryptophan**)

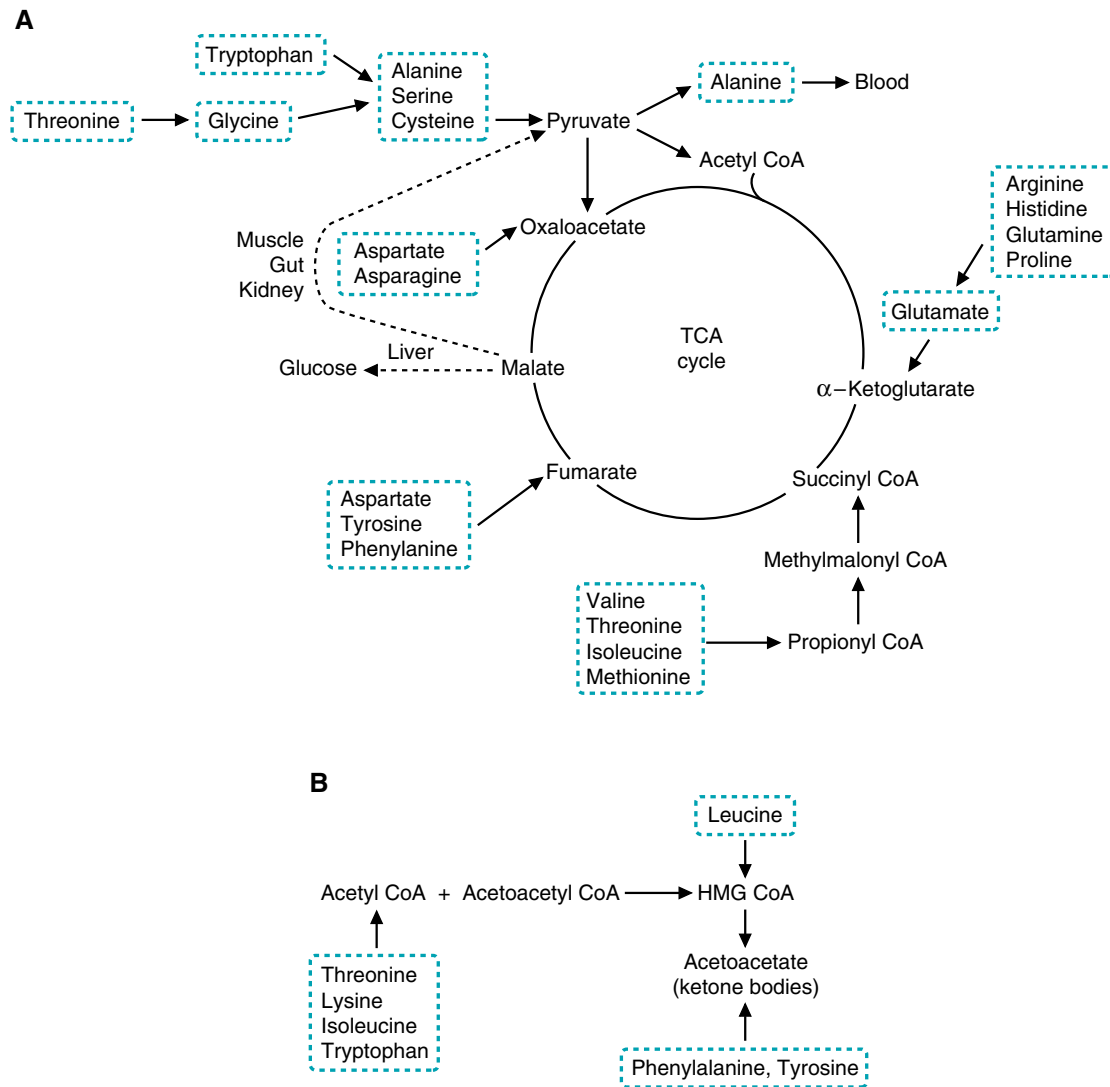


Fig. 39.2. Degradation of amino acids. A. Amino acids that produce pyruvate or intermediates of the TCA cycle. These amino acids are considered glucogenic because they can produce glucose in the liver. The fumarate group of amino acids produces cytoplasmic fumarate. Potential mechanisms whereby the cytoplasmic fumarate can be oxidized are presented in section III.C.1. B. Amino acids that produce acetyl CoA or ketone bodies. These amino acids are considered ketogenic.

produce **pyruvate** when they are degraded. The amino acids synthesized from TCA cycle intermediates (**aspartate, asparagines, glutamate, glutamine, proline, and arginine**) are reconverted to these intermediates during degradation. **Histidine** is converted to **glutamate** and then to the TCA cycle intermediate **α -ketoglutarate**. **Methionine, threonine, valine, and isoleucine** form **succinyl CoA**, and **phenylalanine** (after conversion to **tyrosine**) forms **fumarate**. Because pyruvate and the TCA cycle intermediates can produce glucose in the liver, these amino acids are glucogenic.

Some amino acids with carbons that produce glucose also contain other carbons that produce ketone bodies. **Tryptophan, isoleucine, and threonine** produce **acetyl CoA**, and **phenylalanine and tyrosine** produce **acetoacetate**. These amino acids are both glucogenic and ketogenic.

Two of the essential amino acids (**lysine and leucine**) are strictly **ketogenic**. They do not produce glucose, only acetoacetate and acetyl-CoA.



THE WAITING ROOM



Piquet Yuria, a 4-month-old female infant, emigrated from the Soviet Union with her French mother and Russian father 1 month ago. She was normal at birth but in the last several weeks was less than normally attentive to her surroundings. Her psychomotor maturation seemed delayed, and a tremor of her extremities had recently appeared. When her mother found her having gross twitching movements in her crib, she brought the infant to the hospital emergency room. A pediatrician examined Piquet and immediately noted a musty odor to the baby's wet diaper. A drop of her blood was obtained from a heel prick and used to perform a Guthrie bacterial inhibition assay using a special type of filter paper. This screening procedure was positive for the presence of an excess of phenylalanine in Piquet's blood.



Homer Sistine, a 14-year-old boy, had a sudden grand mal seizure (with jerking movements of the torso and head) in his eighth grade classroom. The school physician noted mild weakness of the muscles of the left side of Homer's face and of his left arm and leg. Homer was hospitalized with a tentative diagnosis of a cerebrovascular accident involving the right cerebral hemisphere, which presumably triggered the seizure.

Homer's past medical history was normal, except for slight mental retardation requiring placement in a special education group. He also had a downward partial dislocation of the lenses of both eyes for which he had had a surgical procedure (a peripheral iridectomy).

Homer's left-sided neurologic deficits cleared spontaneously within 3 days, but a computerized axial tomogram (CAT) showed changes consistent with a small infarction (damaged area caused by a temporary or permanent loss of adequate arterial blood flow) in the right cerebral hemisphere. A neurologist noted that Homer had a slight waddling gait, which his mother said began several years earlier and was progressing with time. Further studies confirmed the presence of decreased mineralization (decreased calcification) of the skeleton (called osteopenia if mild and osteoporosis if more severe) and high methionine and homocysteine but low cystine levels in the blood.

All of this information, plus the increased length of the long bones of Homer's extremities and a slight curvature of his spine (scoliosis), caused his physician to suspect that Homer might have an inborn error of metabolism.

I. THE ROLE OF COFACTORS IN AMINO ACID METABOLISM

Amino acid metabolism requires the participation of three important cofactors. Pyridoxal phosphate is the quintessential coenzyme of amino acid metabolism (see Chapter 38). All amino acid reactions requiring pyridoxal phosphate occur with the amino group of the amino acid covalently bound to the aldehyde carbon of the coenzyme (Fig. 39.3). The pyridoxal phosphate then pulls electrons away from the bonds around the α -carbon. The result is transamination, deamination, decarboxylation, β -elimination, racemization, and γ -elimination, depending on which enzyme and amino acid are involved.

The coenzyme FH_4 is required in certain amino acid pathways to either accept or donate a one-carbon group. The carbon can be in various states of oxidation. Chapter 40 describes the reactions of FH_4 in much more detail.

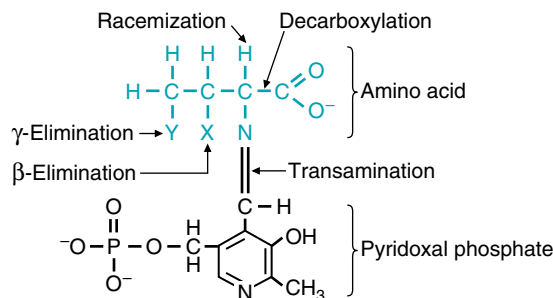


Fig. 39.3. Pyridoxal phosphate covalently attached to an amino acid substrate. The arrows indicate which bonds are broken for the various types of reactions in which pyridoxal phosphate is involved. The X and Y represent leaving groups that may be present on the amino acid (such as the hydroxyl group on serine or threonine).

The coenzyme BH_4 is required for ring hydroxylations. The reactions involve molecular oxygen, and one atom of oxygen is incorporated into the product. The second is found in water (see Chapter 24). BH_4 is important for the synthesis of tyrosine and neurotransmitters (see Chapter 48).

II. AMINO ACIDS DERIVED FROM INTERMEDIATES OF GLYCOLYSIS

Four amino acids are synthesized from intermediates of glycolysis: serine, glycine, cysteine, and alanine. Serine, which produces glycine and cysteine, is synthesized from 3-phosphoglycerate, and alanine is formed by transamination of pyruvate, the product of glycolysis (Fig. 39.4). When these amino acids are degraded, their carbon atoms are converted to pyruvate or to intermediates of the glycolytic/gluconeogenic pathway and, therefore, can produce glucose or be oxidized to CO_2 .

A. Serine

In the biosynthesis of serine from glucose, 3-phosphoglycerate is first oxidized to a 2-keto compound (3-phosphohydroxypyruvate), which is then transaminated to form phosphoserine (Fig. 39.5). Phosphoserine phosphatase removes the phosphate, forming serine. The major sites of serine synthesis are the liver and kidney.

Serine can be used by many tissues and is generally degraded by transamination to hydroxypyruvate followed by reduction and phosphorylation to form 2-phosphoglycerate, an intermediate of glycolysis that forms PEP and, subsequently, pyruvate. Serine also can undergo β -elimination of its hydroxyl group, catalyzed by serine dehydratase, to form pyruvate.

Regulatory mechanisms maintain serine levels in the body. When serine levels fall, serine synthesis is increased by induction of 3-phosphoglycerate dehydrogenase and by release of the feedback inhibition of phosphoserine phosphatase (caused by higher levels of serine). When serine levels rise, synthesis of serine decreases because synthesis of the dehydrogenase is repressed and the phosphatase is inhibited (see Fig. 39.5).

B. Glycine

Glycine can be synthesized from serine and, to a minor extent, threonine. The major route from serine is by a reversible reaction that involves FH_4 and pyridoxal phosphate (Fig. 39.6). Tetrahydrofolate is a coenzyme that transfers one-carbon groups at different levels of oxidation. It is derived from the vitamin folate and is discussed in more detail in Chapter 40. The minor pathway for glycine production involves threonine degradation (this is an aldolase-like reaction because threonine contains a hydroxyl group located two carbons from the carbonyl group).

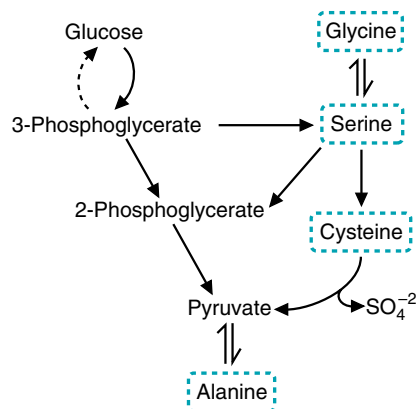


Fig. 39.4. Amino acids derived from intermediates of glycolysis. These amino acids can be synthesized from glucose. Their carbons can be reconverted to glucose in the liver.

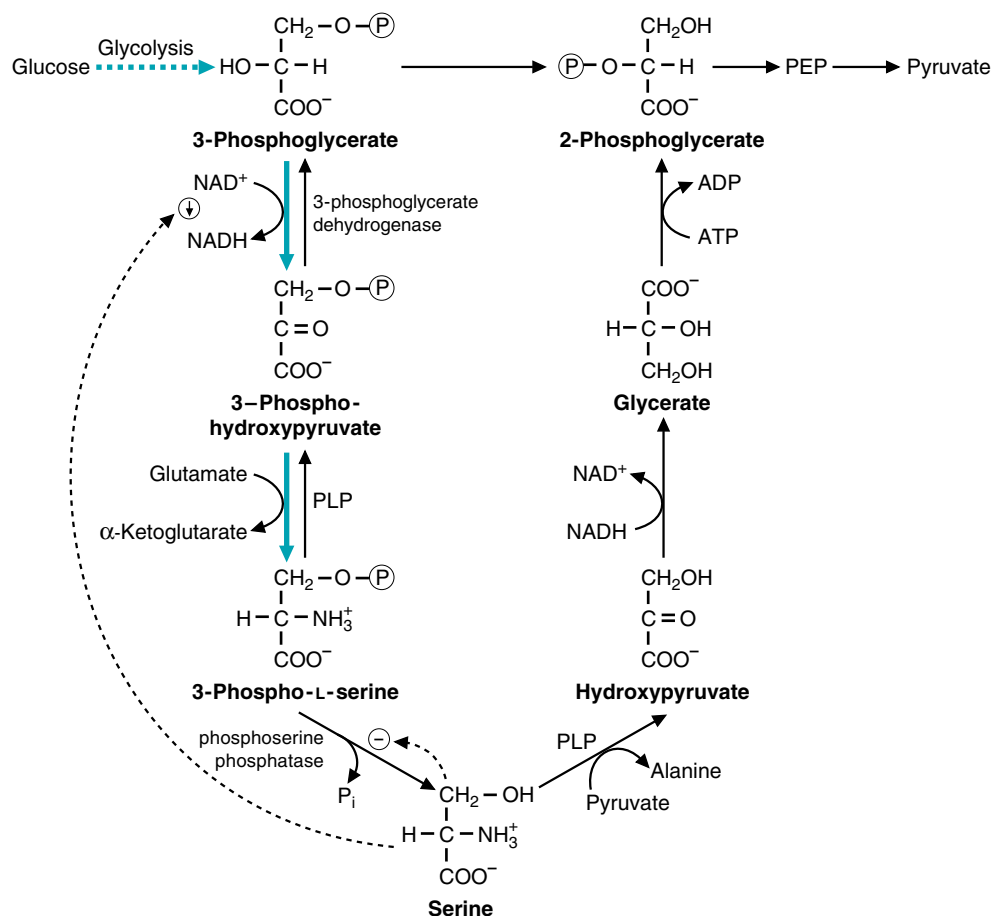


Fig. 39.5. The major pathway for serine synthesis from glucose is on the left, and for serine degradation on the right. Serine levels are maintained because serine causes repression (circled $\perp$) of 3-phosphoglycerate dehydrogenase synthesis. Serine also inhibits (circled $-$) phosphoserine phosphatase.

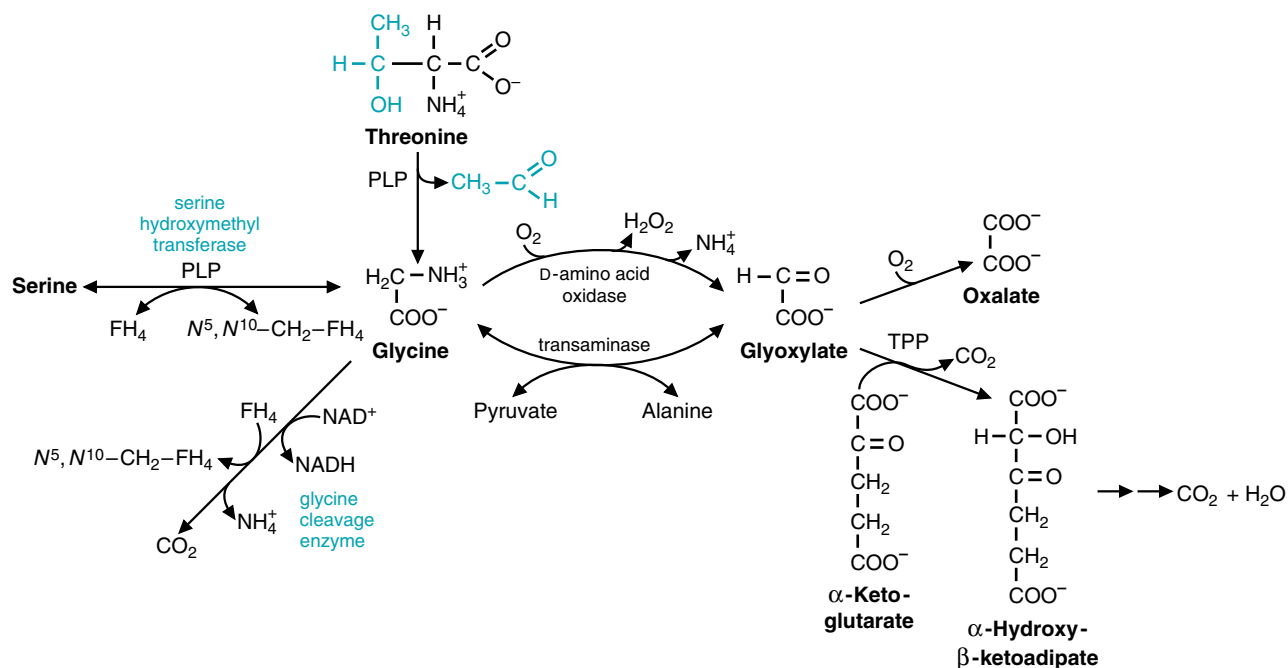


Fig. 39.6. Metabolism of glycine. Glycine can be synthesized from serine (major route) or threonine. Glycine forms serine or CO_2 and NH_4^+ by reactions that require tetrahydrofolate (FH_4). Glycine also forms glyoxylate, which is converted to oxalate or to CO_2 and H_2O .



Oxalate, produced from glycine or obtained from the diet, forms precipitates with calcium. Kidney stones (renal calculi) are often composed of calcium oxalate. A lack of the transaminase that can convert glyoxylate to glycine (see Fig. 39.6) leads to the disease primary oxaluria type I (PH 1). This disease has a consequence of renal failure attributable to excessive accumulation of oxalate in the kidney.



Cystathionuria, the presence of cystathionine in the urine, is relatively common in premature infants. As they mature, cystathionase levels rise, and the levels of cystathionine in the urine decrease.

In adults, a genetic deficiency of cystathionase causes cystathionuria. Individuals with a genetically normal cystathionase can also develop cystathionuria from a dietary deficiency of pyridoxine (vitamin B6), because cystathionase requires the cofactor pyridoxal phosphate. No characteristic clinical abnormalities have been observed in individuals with cystathionase deficiency, and it is probably a benign disorder.



Cystinuria and cystinosis are disorders involving two different transport proteins for cystine, the disulfide formed from two molecules of cysteine. Cystinuria is caused by a defect in the transport protein that carries cystine, lysine, arginine, and ornithine into intestinal epithelial cells and that permits resorption of these amino acids by renal tubular cells. Cystine, which is not very soluble in the urine, forms renal calculi (stones). **Cal Kulis**, a patient with cystinuria, developed cystine stones (see Chapter 37).

Cystinosis is a rare disorder caused by a defective carrier that normally transports cystine across the lysosomal membrane from lysosomal vesicles to the cytosol. Cystine accumulates in the lysosomes in many tissues and forms crystals, impairing their function. Children with this disorder develop renal failure by 6–12 years of age.



Alanine aminotransferase (ALT) and aspartate amino transferase (AST) are released from the liver when liver cells are injured. Measurement of these two transaminases in the serum (ALT and AST) is one of the standard laboratory tests for liver damage caused by a variety of conditions.

The conversion of glycine to glyoxylate by the enzyme D-amino acid oxidase is a degradative pathway of glycine that is medically important. Once glyoxylate is formed, it can be oxidized to oxalate, which is sparingly soluble and tends to precipitate in kidney tubules, leading to kidney stone formation. Approximately 40% of oxalate formation in the liver comes from glycine metabolism. Dietary oxalate accumulation has been estimated to be a low contributor to excreted oxalate in the urine because of poor absorption of oxalate in the intestine.

Although glyoxylate can be transaminated back to glycine, this is not really considered a biosynthetic route for “new” glycine, because the primary route for glyoxylate formation is from glycine oxidation.

Generation of energy from glycine occurs through a dehydrogenase (glycine cleavage enzyme) that oxidizes glycine to CO_2 , ammonia, and a carbon that is donated to FH_4 .

C. Cysteine

The carbons and nitrogen for cysteine synthesis are provided by serine, and the sulfur is provided by methionine (Fig. 39.7). Serine reacts with homocysteine (which is produced from methionine) to form cystathionine. This reaction is catalyzed by cystathionine β -synthase. Cleavage of cystathionine by cystathionase produces cysteine and α -ketobutyrate, which forms succinyl CoA via propionyl CoA. Both cystathionine β -synthase (β -elimination) and cystathionase (γ -elimination) require PLP.

Cysteine inhibits cystathionine β -synthase and, therefore, regulates its own production to adjust for the dietary supply of cysteine. Because cysteine derives its sulfur from the essential amino acid methionine, cysteine becomes essential if the supply of methionine is inadequate for cysteine synthesis. Conversely, an adequate dietary source of cysteine “spares” methionine; that is, it decreases the amount that must be degraded to produce cysteine.

When cysteine is degraded, the nitrogen is converted to urea, the carbons to pyruvate, and the sulfur to sulfate, which has two potential fates (see Fig. 39.7; see also Chapter 43). Sulfate generation, in an aqueous media, is essentially generating sulfuric acid, and both the acid and sulfate need to be disposed of in the urine. Sulfate is also used in most cells to generate an activated form of sulfate known as PAPS (3'-phosphoadenosine 5'-phosphosulfate), which is used as a sulfate donor in modifying carbohydrates or amino acids in various structures (glycosaminoglycans) and proteins in the body.

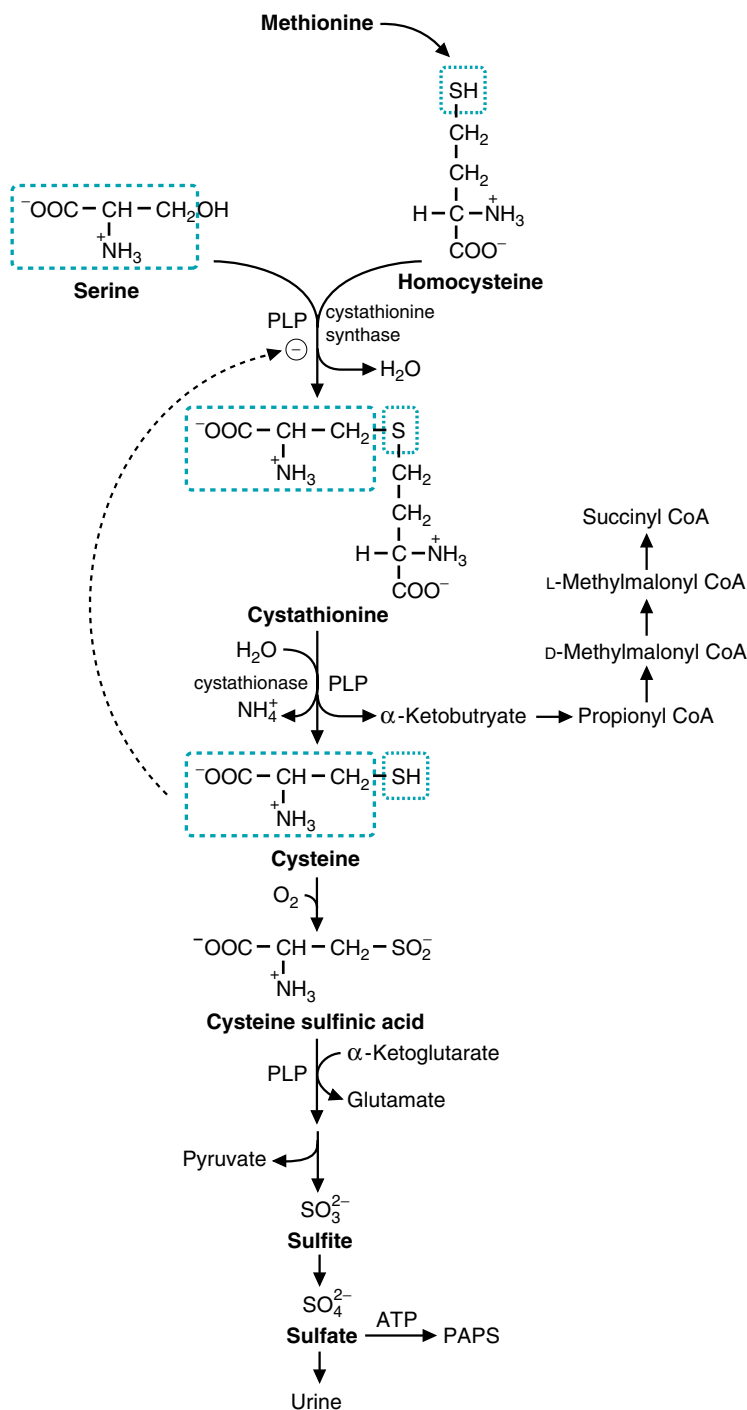
The conversion of methionine to homocysteine and homocysteine to cysteine is the major degradative route for these two amino acids. Because this is the only degradative route for homocysteine, vitamin B6 deficiency or congenital cystathionine β -synthase deficiency can result in homocystinemia, which is associated with cardiovascular disease.

D. Alanine

Alanine is produced from pyruvate by a transamination reaction catalyzed by alanine aminotransferase (ALT) and may be converted back to pyruvate by a reversal of the same reaction (see Fig. 39.4). Alanine is the major gluconeogenic amino acid because it is produced in many tissues for the transport of nitrogen to the liver.

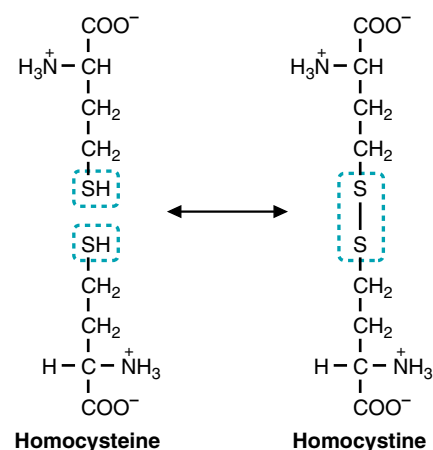
III. AMINO ACIDS RELATED TO TCA CYCLE INTERMEDIATES

Two groups of amino acids are synthesized from TCA cycle intermediates; one group from α -ketoglutarate and one from oxaloacetate (see Fig. 39.2). During degradation, four groups of amino acids are converted to the TCA cycle intermediates α -ketoglutarate, oxaloacetate, succinyl CoA, and fumarate (see Fig. 39.3A).



Homocysteine is oxidized to a disulfide, homocystine. To indicate that both forms are being considered, the term *homocyst(e)ine* is used.

Homocysteine



Because a colorimetric screening test for urinary homocystine was positive, the doctor ordered several biochemical studies on **Homer Sistine's** serum, which included tests for methionine, homocyst(e)ine (both free and protein-bound), cystine, vitamin B12, and folate. The level of homocystine in a 24-hour urine collection was also measured.

The results were as follows: the serum methionine level was 980 μM (reference range, <30); serum homocyst(e)ine (both free and protein bound) was markedly elevated; cystine was not detected in the serum; the serum B12 and folate levels were normal. A 24-hour urine homocystine level was elevated.

Based on these measurements, **Homer Sistine's** doctor concluded that he had homocystinuria caused by an enzyme deficiency. What was the rationale for this conclusion?

Fig. 39.7. Synthesis and degradation of cysteine. Cysteine is synthesized from the carbons and nitrogen of serine and the sulfur of homocysteine (which is derived from methionine). During the degradation of cysteine, the sulfur is converted to sulfate and either excreted in the urine or converted to PAPS (universal sulfate donor; 3'-phosphoadenosine 5'-phosphosulfate), and the carbons are converted to pyruvate.



If the blood levels of methionine and homocysteine are very elevated and cystine is low, cystathionine β -synthase could be defective, but a cystathionase deficiency is also a possibility. With a deficiency of either of these enzymes, cysteine could not be synthesized, and levels of homocysteine would rise. Homocysteine would be converted to methionine by reactions that require B12 and tetrahydrofolate (see Chapter 40). In addition, it would be oxidized to homocystine, which would appear in the urine. The levels of cysteine (measured as its oxidation product cystine) would be low. A measurement of serum cystathionine levels would help to distinguish between a cystathionase or cystathionine β -synthase deficiency.

A. Amino Acids Related through α -Ketoglutarate/ Glutamate

1. GLUTAMATE

The five carbons of glutamate are derived from α -ketoglutarate either by transamination or by the glutamate dehydrogenase reaction (see Chapter 38). Because α -ketoglutarate can be synthesized from glucose, all of the carbons of glutamate can be obtained from glucose (see Fig. 39.2). When glutamate is degraded, it is likewise converted back to α -ketoglutarate either by transamination or by glutamate dehydrogenase. In the liver, α -ketoglutarate leads to the formation of malate, which produces glucose via gluconeogenesis. Thus, glutamate can be derived from glucose and reconverted to glucose (Fig. 39.8).

Glutamate is used for the synthesis of a number of other amino acids (glutamine, proline, ornithine, and arginine) (see Fig. 39.8) and for providing the glutamyl moiety of glutathione (γ -glutamyl-cysteinyl-glycine; see Biochemical Comments of Chapter 37). Glutathione is an important antioxidant, as has been described previously (see Chapter 24).

2. GLUTAMINE

Glutamine is produced from glutamate by glutamine synthetase, which adds NH_4^+ to the carboxyl group of the side chain, forming an amide (Fig. 39.9). This is one of only three human enzymes that can fix free ammonia into an organic molecule; the other two are glutamate dehydrogenase and carbamoyl-phosphate synthetase I (see Chapter 38). Glutamine is reconverted to glutamate by a different enzyme, glutaminase, which is particularly important in the kidney. The ammonia it produces enters the urine and can be used as an expendable cation to aid in the excretion of metabolic acids ($\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$).

3. PROLINE

In the synthesis of proline, glutamate is first phosphorylated and then converted to glutamate 5-semialdehyde by reduction of the side chain carboxyl group to an

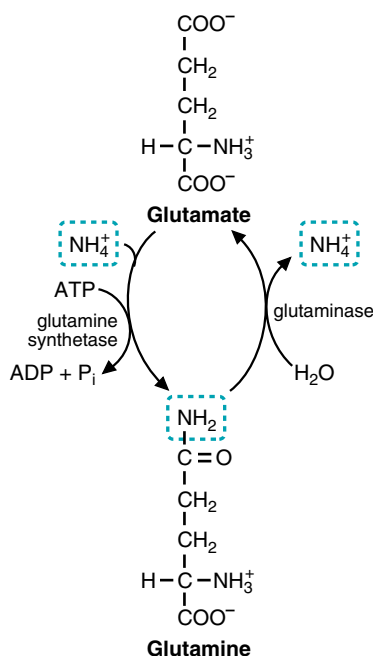


Fig. 39.9. Synthesis and degradation of glutamine. Different enzymes catalyze the addition and the removal of the amide nitrogen of glutamine.

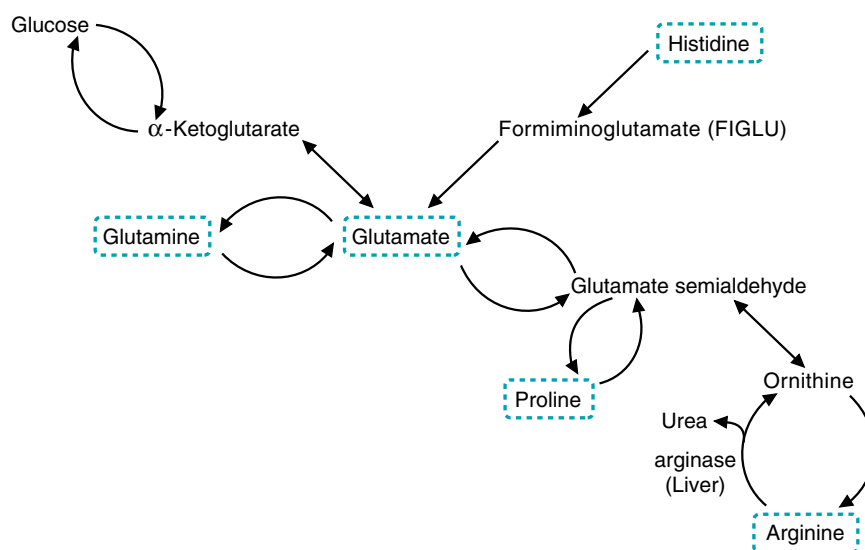


Fig. 39.8. Amino acids related through glutamate. These amino acids contain carbons that can be reconverted to glutamate, which can be converted to glucose in the liver. All of these amino acids except histidine can be synthesized from glucose.

aldehyde (Fig. 39.10). This semialdehyde spontaneously cyclizes (forming an internal Schiff base between the aldehyde and the α -amino group). Reduction of this cyclic compound yields proline. Hydroxyproline is only formed after proline has been incorporated into collagen (see Chapter 49) by the prolyl hydroxylase system, which uses molecular oxygen, iron, α -ketoglutarate, and ascorbic acid (vitamin C).

Proline is converted back to glutamate semialdehyde, which is reduced to form glutamate. The synthesis and degradation of proline use different enzymes even though the intermediates are the same. Hydroxyproline, however, has an entirely different degradative pathway (not shown). The presence of the hydroxyl group in hydroxyproline will allow an aldolase-like reaction to occur once the ring has been hydrolyzed, which is not possible with proline.

4. ARGININE

Arginine is synthesized from glutamate via glutamate semialdehyde, which is transaminated to form ornithine, an intermediate of the urea cycle (see Chapter 38 and Fig. 39.11). This activity (ornithine aminotransferase) appears to be greatest in the epithelial cells of the small intestine (see Chapter 42). The reactions of the urea cycle then produce arginine. However, the quantities of arginine generated by the urea cycle are adequate only for the adult and are insufficient to support growth. Therefore, during periods of growth, arginine becomes an essential amino acid. It is important to realize that if arginine is used for protein synthesis, the levels of ornithine will drop, thereby slowing the urea cycle. This will stimulate the formation of ornithine from glutamate.

Arginine is cleaved by arginase to form urea and ornithine. If ornithine is present in amounts in excess of those required for the urea cycle, it is transaminated to glutamate semialdehyde, which is reduced to glutamate. The conversion of an aldehyde to a primary amine is a unique form of a transamination reaction and requires pyridoxal phosphate (PLP).

5. HISTIDINE

Although histidine cannot be synthesized in humans, five of its carbons form glutamate when it is degraded. In a series of steps, histidine is converted to formiminoglutamate (FIGLU). The subsequent reactions transfer one carbon of FIGLU to the FH_4 pool (see Chapter 40) and release NH_4^+ and glutamate (Fig. 39.12).

B. Amino Acids Related to Oxaloacetate (Aspartate and Asparagine)

Aspartate is produced by transamination of oxaloacetate. This reaction is readily reversible, so aspartate can be reconverted to oxaloacetate (Fig. 39.13).

Asparagine is formed from aspartate by a reaction in which glutamine provides the nitrogen for formation of the amide group. Thus, this reaction differs from the synthesis of glutamine from glutamate, in which NH_4^+ provides the nitrogen. However, the reaction catalyzed by asparaginase, which hydrolyzes asparagine to NH_4^+ and aspartate, is analogous to the reaction catalyzed by glutaminase.

C. Amino Acids That Form Fumarate

1. ASPARTATE

Although the major route for aspartate degradation involves its conversion to oxaloacetate, carbons from aspartate can form fumarate in the urea cycle (see Chapter 38). This reaction generates cytosolic fumarate, which must be converted to malate (using cytoplasmic fumarase) for transport into the mitochondria for oxidative or anaplerotic purposes. An analogous sequence of reactions occurs in the purine nucleotide cycle. Aspartate reacts with inosine monophosphate (IMP) to

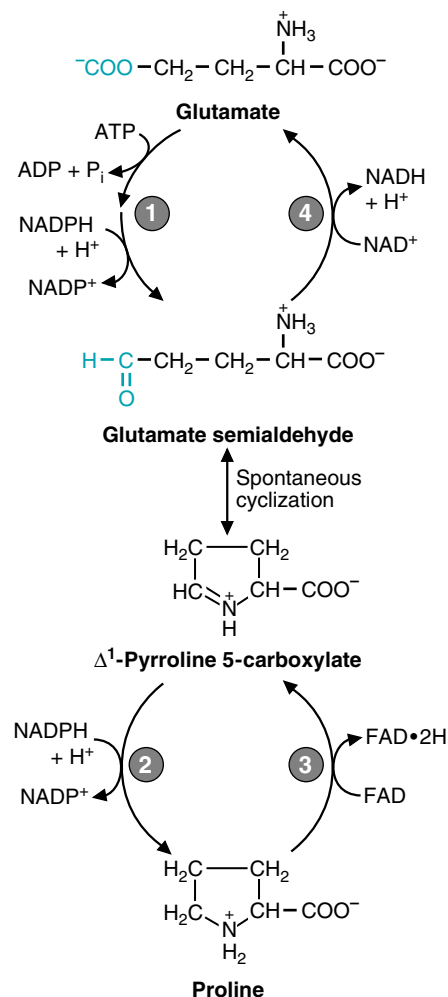


Fig. 39.10. Synthesis and degradation of proline. Reactions 1, 3, and 4 occur in mitochondria. Reaction 2 occurs in the cytosol. Synthesis and degradation involve different enzymes. The cyclization reaction (formation of a Schiff base) is nonenzymatic, i.e., spontaneous.



Certain types of tumor cells, particularly leukemic cells, require asparagine for their growth. Therefore, asparaginase has been used as an anti-tumor agent. It acts by converting asparagine to aspartate in the blood, decreasing the amount of asparagine available for tumor cell growth.

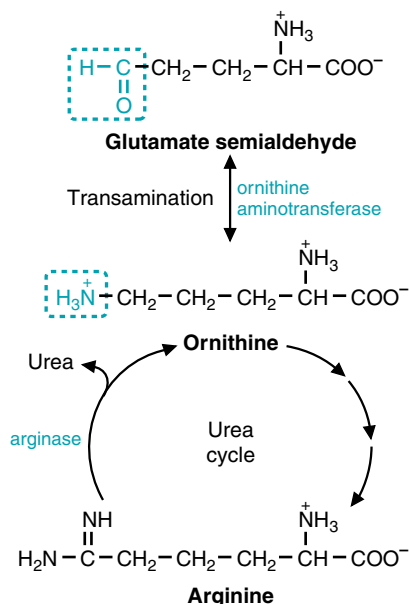


Fig. 39.11. Synthesis and degradation of arginine. The carbons of ornithine are derived from glutamate semialdehyde, which is derived from glutamate. Reactions of the urea cycle convert ornithine to arginine. Arginase converts arginine back to ornithine by releasing urea.

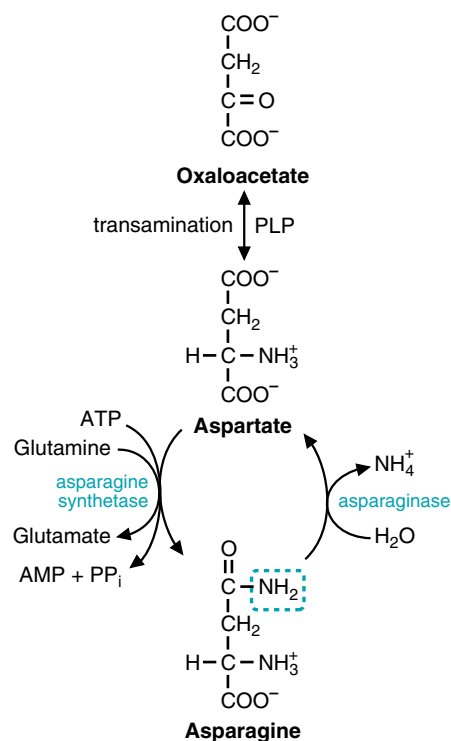


Fig. 39.13. Synthesis and degradation of aspartate and asparagine. Note that the amide nitrogen of asparagine is derived from glutamine. (The amide nitrogen of glutamine comes from NH_4^+ , see Fig. 39.9.)

form an intermediate (adenylosuccinate) which is cleaved, forming adenosine monophosphate (AMP) and fumarate (see Chapter 41).

2. PHENYLALANINE AND TYROSINE

Phenylalanine is converted to tyrosine by a hydroxylation reaction. Tyrosine, produced from phenylalanine or obtained from the diet, is oxidized, ultimately forming acetoacetate and fumarate. The oxidative steps required to reach this point are, surprisingly, not energy-generating. The conversion of fumarate to malate, followed by the action of malic enzyme, allows the carbons to be used for gluconeogenesis. The conversion of phenylalanine to tyrosine and the production of acetoacetate are considered further in section IV of this chapter.

D. Amino Acids That Form Succinyl CoA

The essential amino acids methionine, valine, isoleucine, and threonine are degraded to form propionyl-CoA. The conversion of propionyl CoA to succinyl CoA is common to their degradative pathways. Propionyl CoA is also generated from the oxidation of odd-chain fatty acids.

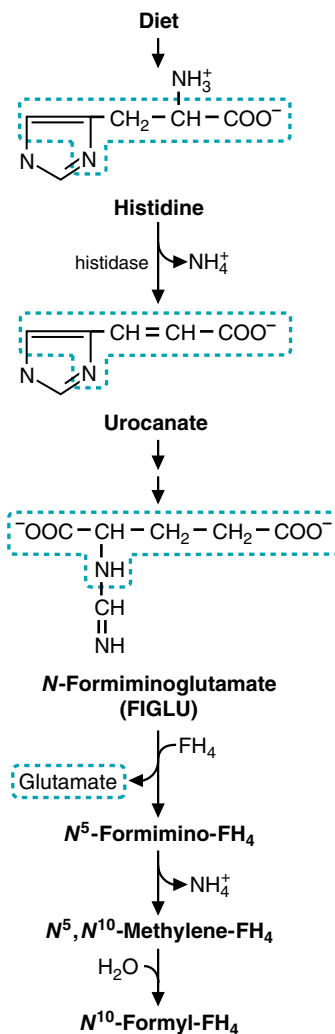


Fig. 39.12. Degradation of histidine. The highlighted portion of histidine forms glutamate. The remainder of the molecule provides one carbon for the tetrahydrofolate (FH₄) pool (see Chapter 40) and releases NH_4^+ .

Propionyl CoA is carboxylated in a reaction that requires biotin and forms D-methylmalonyl CoA. The D-methylmalonyl CoA is racemized to L-methylmalonyl CoA, which is rearranged in a vitamin B12-requiring reaction to produce succinyl CoA, a TCA cycle intermediate (see Fig. 23.11).

1. METHIONINE

Methionine is converted to *S*-adenosylmethionine (SAM), which donates its methyl group to other compounds to form *S*-adenosylhomocysteine (SAH). SAH is then converted to homocysteine (Fig. 39.14). Methionine can be regenerated from homocysteine by a reaction requiring both FH_4 and vitamin B12 (a topic that is considered in more detail in Chapter 40). Alternatively, by reactions requiring PLP, homocysteine can provide the sulfur required for the synthesis of cysteine (see Fig. 39.7). Carbons of homocysteine are then metabolized to α -ketobutyrate, which undergoes oxidative decarboxylation to propionyl-CoA. The propionyl-CoA is then converted to succinyl CoA (see Fig. 39.14).

2. THREONINE

In humans threonine is primarily degraded by a PLP-requiring dehydratase to ammonia and α -ketobutyrate, which subsequently undergoes oxidative decarboxylation to form propionyl CoA, just as in the case for methionine (see Fig. 39.14).



The conversion of α -ketobutyrate to propionyl-CoA is catalyzed by either the pyruvate or branched-chain α -keto dehydrogenase enzymes.



Homocystinuria is caused by deficiencies in the enzymes cystathionine β -synthase and cystathionase as well as by deficiencies of methyltetrahydrofolate ($\text{CH}_3\text{-FH}_4$) or of methyl-B12. The deficiencies of $\text{CH}_3\text{-FH}_4$ or of methyl-B12 are due either to an inadequate dietary intake of folate or B12 or to defective enzymes involved in joining methyl groups to tetrahydrofolate (FH_4), transferring methyl groups from FH_4 to B12, or passing them from B12 to homocysteine to form methionine (see Chapter 40).

Is **Homer Sistine's** homocystinuria caused by any of these problems?

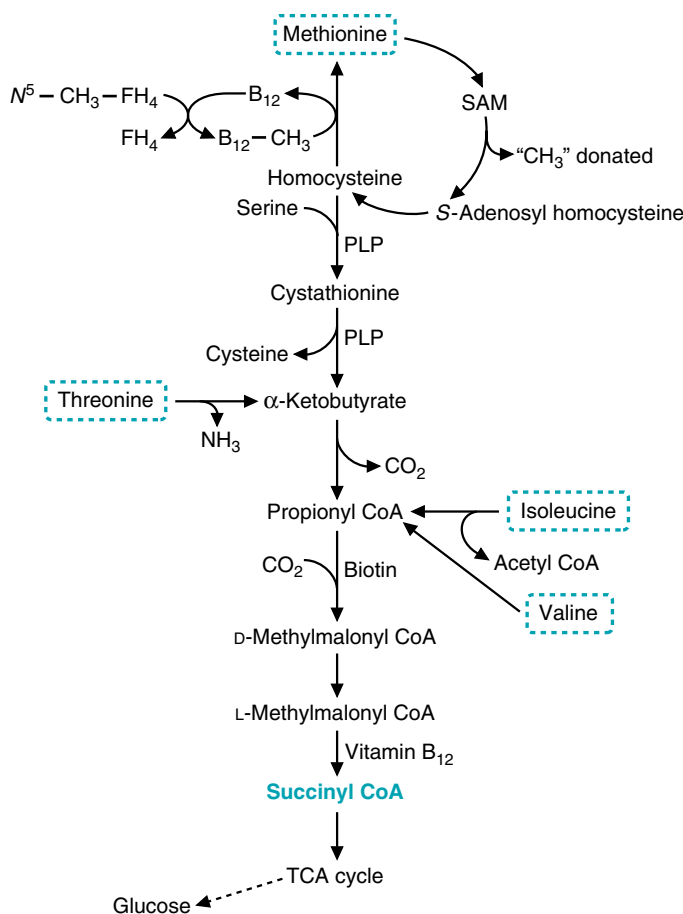


Fig. 39.14. Conversion of amino acids to succinyl CoA. The amino acids methionine, threonine, isoleucine, and valine, all of which form succinyl CoA via methylmalonyl CoA, are essential in the diet. The carbons of serine are converted to cysteine and do not form succinyl CoA by this pathway. SAM = *S*-adenosylmethionine.

A: **Homer Sistine's** methionine levels are elevated, and his B12 and folate levels are normal. Therefore, he does not have a deficiency of dietary folate or B12 or of the enzymes that transfer methyl groups from tetrahydrofolate to homocysteine to form methionine. In these cases, homocysteine levels are elevated but methionine levels are low.

A biopsy specimen from **Homer Sistine's** liver was sent to the hospital's biochemistry research laboratory for enzyme assays. Cystathionine β -synthase activity was reported to be 7% of that found in normal liver.

Thiamine deficiency will lead to an accumulation of α -keto acids in the blood because of an inability of pyruvate dehydrogenase, α -ketoglutarate dehydrogenase, and branched-chain α -keto acid dehydrogenase to catalyze their reactions (see Chapter 8). **Al Martini** had a thiamine deficiency resulting from his chronic alcoholism. His ketoacidosis resulted partly from the accumulation of these α -ketoacids in his blood and partly from the accumulation of ketone bodies used for energy production.

Q: What compounds form succinyl CoA via propionyl CoA and methylmalonyl CoA?

In maple syrup urine disease, the branched-chain α -keto acid dehydrogenase that oxidatively decarboxylates the branched-chain amino acids is defective. As a result, the branched-chain amino acids and their α -keto analogs (produced by transamination) accumulate. They appear in the urine, giving it the odor of maple syrup or burnt sugar. The accumulation of α -keto analogs leads to neurologic complications. This condition is difficult to treat by dietary restriction, because abnormalities in the metabolism of three essential amino acids contribute to the disease.

3. VALINE AND ISOLEUCINE

The branched-chain amino acids (valine, isoleucine, and leucine) are a universal fuel, and the degradation of these amino acids occurs at low levels in the mitochondria of most tissues, but the muscle carries out the highest level of branched-chain amino acid oxidation. The branched-chain amino acids make up almost 25% of the content of the average protein, so their use as fuel is quite significant. The degradative pathway for valine and isoleucine has two major functions, the first being energy generation and the second to provide precursors to replenish TCA cycle intermediates (anaplerosis). Valine and isoleucine, two of the three branched-chain amino acids, contain carbons that form succinyl CoA. The initial step in the degradation of the branched-chain amino acids is a transamination reaction. Although the enzyme that catalyzes this reaction is present in most tissues, the level of activity varies from tissue to tissue. Its activity is particularly high in muscle, however. In the second step of the degradative pathway, the α -keto analogs of these amino acids undergo oxidative decarboxylation by the α -keto acid dehydrogenase complex in a reaction similar in its mechanism and cofactor requirements to pyruvate dehydrogenase and α -ketoglutarate dehydrogenase (see Chapter 20). As with the first enzyme of the pathway, the highest level of activity for this dehydrogenase is found in muscle tissue. Subsequently, the pathways for degradation of these amino acids follow parallel routes (Fig. 39.15). The steps are analogous to those for β -oxidation of fatty acids so NADH and FAD(2H) are generated for energy production.

Valine and isoleucine are converted to succinyl CoA (see Fig. 39.14). Isoleucine also forms acetyl CoA. Leucine, the third branched-chain amino acid,

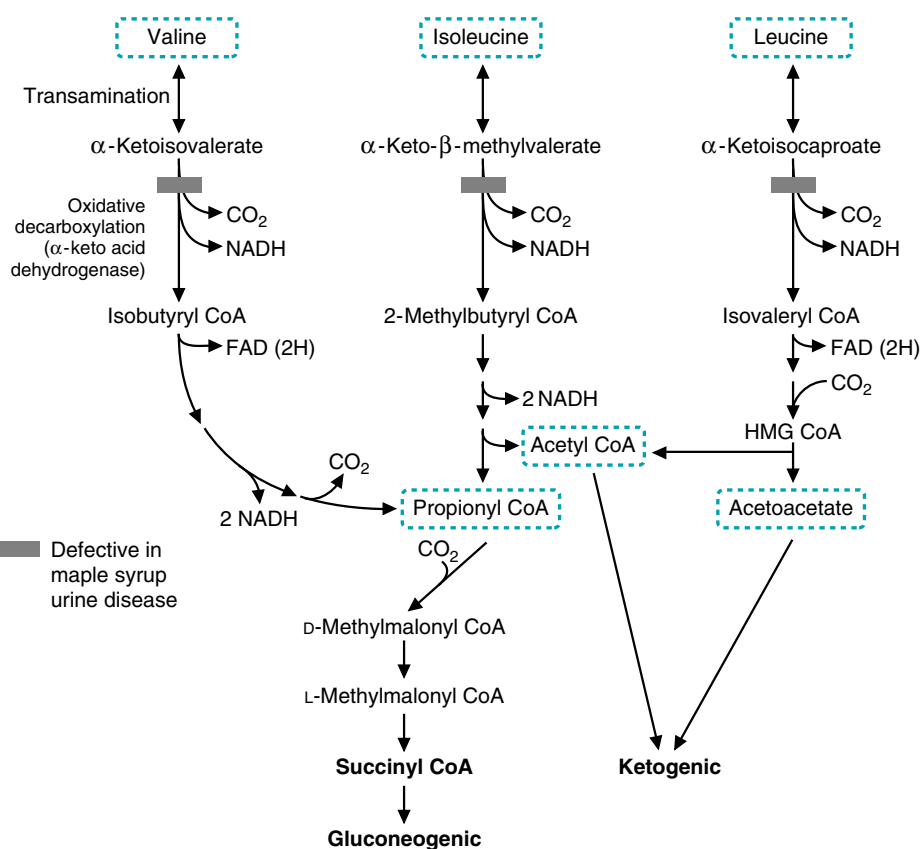


Fig. 39.15. Degradation of the branched-chain amino acids. Valine forms propionyl CoA. Isoleucine forms propionyl CoA and acetyl CoA. Leucine forms acetoacetyl CoA and acetyl CoA.

does not produce succinyl CoA. It forms acetoacetate and acetyl CoA and is strictly ketogenic.

IV. AMINO ACIDS THAT FORM ACETYL CoA AND ACETOACETATE

Seven amino acids produce acetyl CoA or acetoacetate and therefore are categorized as ketogenic. Of these, isoleucine, threonine, and the aromatic amino acids (phenylalanine, tyrosine, and tryptophan) are converted to compounds that produce both glucose and acetyl CoA or acetoacetate (Fig. 39.16). Leucine and lysine do not produce glucose; they produce acetyl CoA and acetoacetate.

A. Phenylalanine and Tyrosine

Phenylalanine is converted to tyrosine, which undergoes oxidative degradation (Fig. 39.17). The last step in the pathway produces both fumarate and the ketone body acetoacetate. Deficiencies of different enzymes in the pathway result in phenylketonuria, tyrosinemia, and alcaptonuria.

Phenylalanine is hydroxylated to form tyrosine by a mixed function oxidase, phenylalanine hydroxylase (PAH), which requires molecular oxygen and tetrahydrobiopterin (Fig. 39.18). The cofactor tetrahydrobiopterin is converted to quinoid dihydrobiopterin by this reaction. Tetrahydrobiopterin is not synthesized from a vitamin; it can be synthesized in the body from GTP. However, as is the case with other cofactors, the body contains limited amounts. Therefore, dihydrobiopterin must be reconverted to tetrahydrobiopterin for the reaction to continue to produce tyrosine.



Alcaptonuria occurs when homogentisate, an intermediate in tyrosine metabolism, cannot be further oxidized because the next enzyme in the pathway, homogentisate oxidase, is defective. Homogentisate accumulates and auto-oxidizes, forming a dark pigment, which discolors the urine and stains the diapers of affected infants. Later in life, the chronic accumulation of this pigment in cartilage may cause arthritic joint pain.



Transient tyrosinemia is frequently observed in newborn infants, especially those that are premature. For the most part, the condition appears to be benign, and dietary restriction of protein returns plasma tyrosine levels to normal. The biochemical defect is most likely a low level, attributable to immaturity, of 4-hydroxyphenylpyruvate dioxygenase. Because this enzyme requires ascorbate, ascorbate supplementation also aids in reducing circulating tyrosine levels.

Other types of tyrosinemia are related to specific enzyme defects (see Fig. 39.17). Tyrosinemia II is caused by a genetic deficiency of tyrosine aminotransferase (TAT) and may lead to lesions of the eye and skin as well as neurologic problems. Patients are treated with a low-tyrosine, low-phenylalanine diet.

Tyrosinemia I (also called tyrosinosis) is caused by a genetic deficiency of fumarylacetoacetate hydrolase. The acute form is associated with liver failure, a cabbage-like odor, and death within the first year of life.

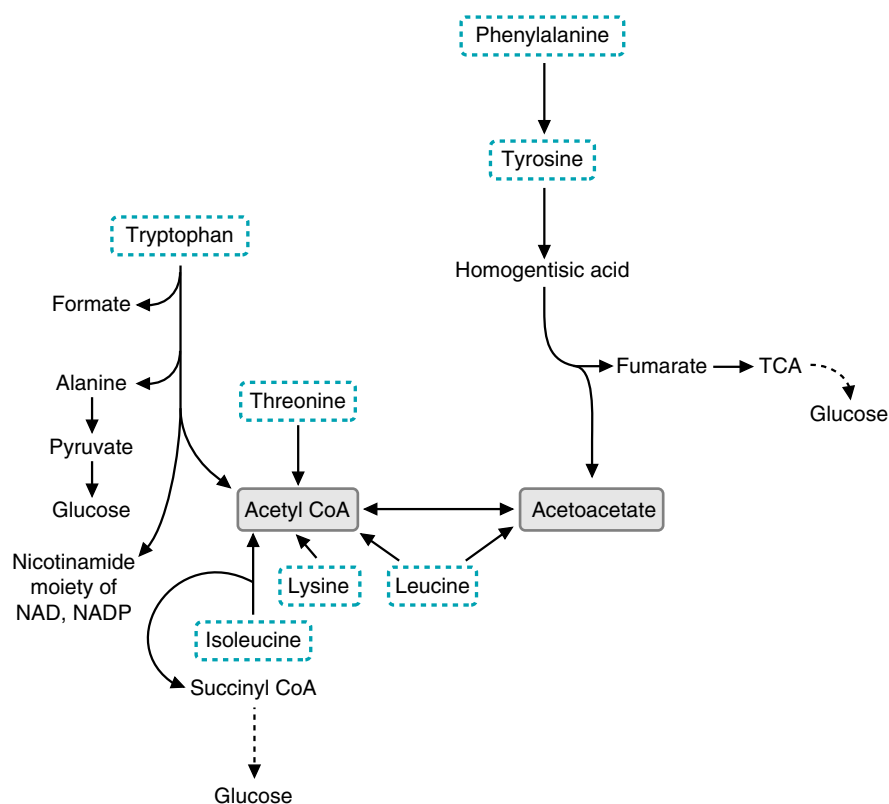


Fig. 39.16. Ketogenic amino acids. Some of these amino acids (tryptophan, phenylalanine, and tyrosine) also contain carbons that can form glucose. Leucine and lysine are strictly ketogenic; they do not form glucose.



A: In addition to methionine, threonine, isoleucine, and valine (see Fig. 39.14), the last three carbons at the ω -end of odd-chain fatty acids, form succinyl CoA by this route (see Chapter 23)



A small subset of patients with hyperphenylalaninemia show an appropriate reduction in plasma phenylalanine levels with dietary restriction of this amino acid; however, these patients still develop progressive neurologic symptoms and seizures and usually die within the first 2 years of life (“malignant” hyperphenylalaninemia). These infants exhibit normal phenylalanine hydroxylase (PAH) activity but have a deficiency in dihydropteridine reductase (DHPR), an enzyme required for the regeneration of tetrahydrobiopterin (BH_4), a cofactor of PAH (see Fig. 39.18). Less frequently, DHPR activity is normal but a defect in the biosynthesis of BH_4 exists. In either case, dietary therapy corrects the hyperphenylalaninemia. However, BH_4 is also a cofactor for two other hydroxylations required in the synthesis of neurotransmitters in the brain: the hydroxylation of tryptophan to 5-hydroxytryptophan and of tyrosine to L-dopa (see Chapter 48). It has been suggested that the resulting deficit in central nervous system neurotransmitter activity is, at least in part, responsible for the neurologic manifestations and eventual death of these patients.



If the dietary levels of niacin and tryptophan are insufficient, the condition known as pellagra results. The symptoms of pellagra are dermatitis, diarrhea, dementia, and, finally, death. In addition, abnormal metabolism of tryptophan occurs in a vitamin B6 deficiency. Kynurenine intermediates in tryptophan degradation cannot be cleaved because kynureninase requires PLP derived from vitamin B6. Consequently, these intermediates enter a minor pathway for tryptophan metabolism that produces xanthurenic acid, which is excreted in the urine.

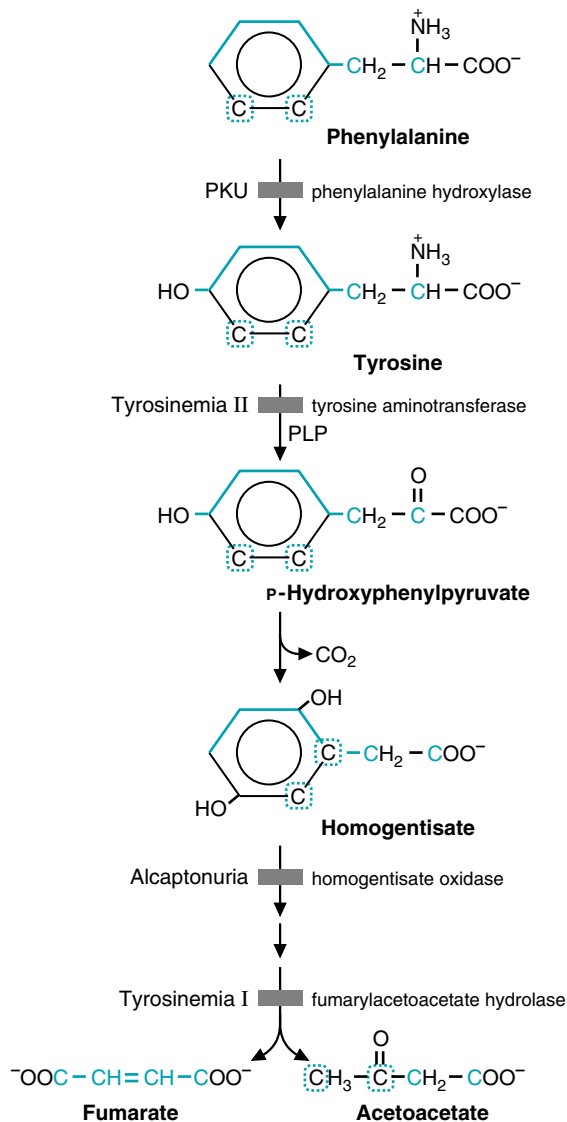


Fig. 39.17. Degradation of phenylalanine and tyrosine. The carboxyl carbon forms CO_2 , and the other carbons form fumarate or acetoacetate as indicated. Deficiencies of enzymes (gray bars) result in the indicated diseases. PKU = phenylketonuria.

B. Tryptophan

Tryptophan is oxidized to produce alanine (from the non-ring carbons), formate, and acetyl CoA. Tryptophan is, therefore, both glucogenic and ketogenic (Fig. 39.19).

NAD and NADP can be produced from the ring structure of tryptophan. Therefore, tryptophan “spares” the dietary requirement for niacin. The higher the dietary levels of tryptophan, the lower the levels of niacin required to prevent symptoms of deficiency.

C. Threonine, Isoleucine, Leucine, and Lysine

As discussed previously, the major route of threonine degradation in humans is by threonine dehydratase (see section III.D.2.). In a minor pathway, threonine degradation by threonine aldolase produces glycine and acetyl CoA in the liver.

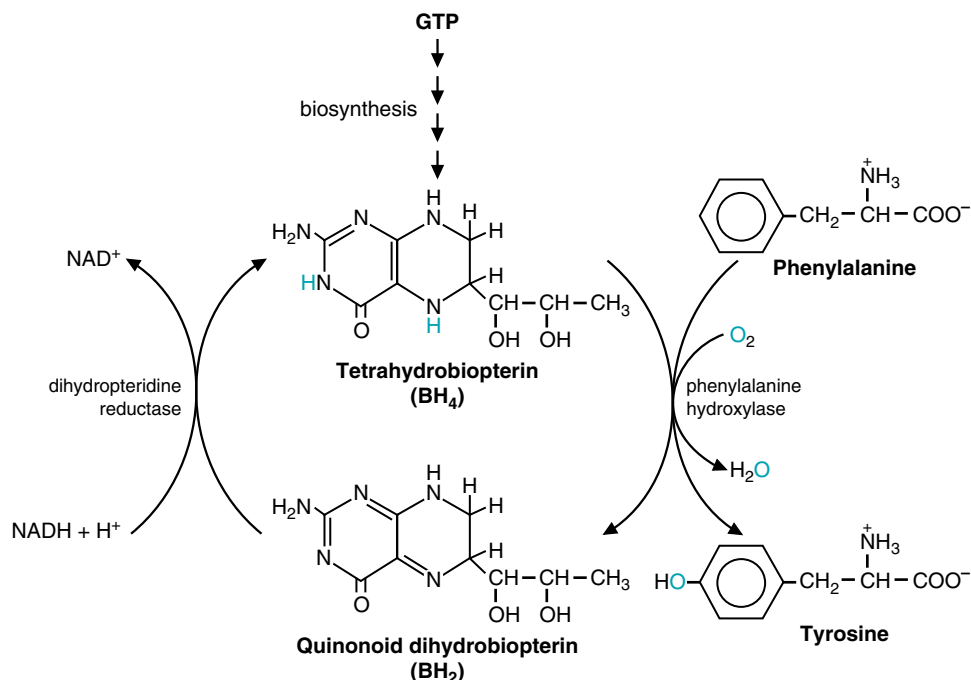


Fig. 39.18. Hydroxylation of phenylalanine. Phenylalanine hydroxylase (PAH) is a mixed-function oxidase; i.e., molecular oxygen (O₂) donates one atom to water and one to the product, tyrosine. The cofactor, tetrahydrobiopterin (BH₄), is oxidized to dihydrobiopterin (BH₂), and must be reduced back to BH₄ for the phenylalanine to continue forming tyrosine. BH₄ is synthesized in the body from GTP. PKU results from deficiencies of PAH (the classic form), dihydropteridine reductase, or enzymes in the biosynthetic pathway for BH₄.

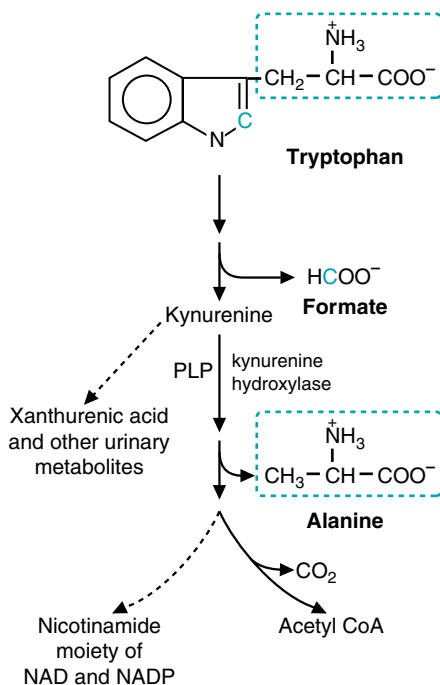
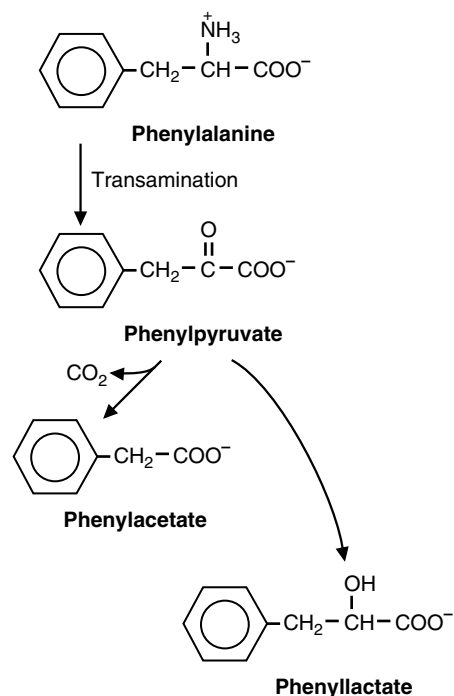


Fig. 39.19. Degradation of tryptophan. One of the ring carbons produces formate. The non-ring portion forms alanine. Kynurenine is an intermediate, which can be converted to a number of urinary excretion products (e.g., xanthurenic acid), degraded to CO₂ and acetyl CoA, or converted to the nicotinamide moiety of NAD and NADP, which also can be formed from the vitamin niacin.



On more definitive testing of **Piquet Yuria's** blood, the plasma level of phenylalanine was elevated at 18 mg/dL (reference range, <1.2). Several phenyl ketones and other products of phenylalanine metabolism, which give the urine a characteristic odor, were found in significant quantities in the baby's urine.



A liver biopsy was sent to the special chemistry research laboratory, where it was determined that the level of activity of phenylalanine hydroxylase (PAH) in Piquet's blood was less than 1% of that found in normal infants. A diagnosis of "classic" phenylketonuria (PKU) was made.

Until gene therapy allows substitution of the defective PAH gene with its normal counterpart in vivo, the mainstay of therapy in classic PKU is to maintain levels of phenylalanine in the blood between 3 and 12 mg/dL through dietary restriction of this essential amino acid.

Isoleucine produces both succinyl CoA and acetyl CoA (see section III.D.3.).

Leucine is purely ketogenic and produces hydroxymethylglutaryl CoA (HMG-CoA), which is cleaved to form acetyl CoA and the ketone body acetoacetate (see Figs. 39.15 and 39.16). Most of the tissues in which it is oxidized can use ketone bodies, with the exception of the liver. As with valine and isoleucine, leucine is a universal fuel, with its primary metabolism occurring in muscle.

Lysine cannot be directly transaminated at either of its two amino groups. Lysine is degraded by a complex pathway in which saccharopine, α -ketoadipate, and crotonyl CoA are intermediates. During the degradation pathway NADH and FADH_2 are generated for energy. Ultimately, lysine generates acetyl CoA (see Fig. 39.16) and is strictly ketogenic.

CLINICAL COMMENTS



Piquet Yuria. The overall incidence of hyperphenylalaninemia is approximately 100 per million births, with a wide geographic and ethnic variation. PKU occurs by autosomal recessive transmission of a defective PAH gene, causing accumulation of phenylalanine in the blood well above the normal concentration in young children and adults (less than 1–2 mg/dL). In the newborn, the upper limit of normal is almost twice this value. Values above 16 mg/dL are usually found in patients, such as **Piquet Yuria**, with "classic" PKU.

Patients with classic PKU usually appear normal at birth. If the disease is not recognized and treated within the first month of life, the infant gradually develops varying degrees of irreversible mental retardation (IQ scores frequently under 50), delayed psychomotor maturation, tremors, seizures, eczema, and hyperactivity. The neurologic sequelae may result in part from the competitive interaction of phenylalanine with brain amino acid transport systems and inhibition of neurotransmitter synthesis. These biochemical alterations lead to impaired myelin synthesis and delayed neuronal development, which result in the clinical picture in patients such as **Piquet Yuria**. Because of the simplicity of the test for PKU (elevated phenylalanine levels in the blood), all newborns in the United States are required to have a PKU test at birth. Early detection of the disease can lead to early treatment, and the neurologic consequences of the disease can be bypassed.

To restrict dietary levels of phenylalanine, special semisynthetic preparations such as Lofenalac or PKUaid are used in the United States. Use of these preparations reduces dietary intake of phenylalanine to 250–500 mg/day while maintaining normal intake of all other dietary nutrients. Although it is generally agreed that scrupulous adherence to this regimen is mandatory for the first decade of life, less consensus exists regarding its indefinite use. Evidence suggests, however, that without lifelong compliance with dietary restriction of phenylalanine, even adults will develop at least neurologic sequelae of PKU. A pregnant woman with PKU must be particularly careful to maintain satisfactory plasma levels of phenylalanine throughout gestation to avoid the adverse effects of hyperphenylalaninemia on the fetus.

Piquet's parents were given thorough dietary instruction, which they followed carefully. Although her pediatrician was not optimistic, it was hoped that the damage done to her nervous system before dietary therapy was minimal and that her subsequent psychomotor development would allow her to lead a relatively normal life.



Homer Sistine. The most characteristic biochemical features of the disorder affecting **Homer Sistine**, a cystathionine β -synthase deficiency, are the presence of an accumulation of both homocyst(e)ine and methionine in the blood. Because renal tubular reabsorption of methionine is highly efficient, this

amino acid may not appear in the urine. Homocystine, the disulfide of homocysteine, is less efficiently reabsorbed, and amounts in excess of 1 mmol may be excreted in the urine each day.

In the type of homocystinuria in which the patient is deficient in cystathionine β -synthase, the elevation in serum methionine levels is presumed to be the result of enhanced rates of conversion of homocysteine to methionine, caused by increased availability of homocysteine (see Fig. 39.14). In type II and type III homocystinuria, in which there is a deficiency in the synthesis of methyl cobalamin and of N⁵-methyltetrahydrofolate, respectively (both required for the methylation of homocysteine to form methionine), serum homocysteine levels are elevated but serum methionine levels are low (see Fig. 39.14).

Acute vascular events are common in these patients. Thrombi (blood clots) and emboli (clots that have broken off and traveled to a distant site in the vascular system) have been reported in almost every major artery and vein as well as in smaller vessels. These clots result in infarcts in vital organs such as the liver, the myocardium (heart muscle), the lungs, the kidneys, and many other tissues. Although increased serum levels of homocysteine have been implicated in enhanced platelet aggregation and damage to vascular endothelial cells (leading to clotting and accelerated atherosclerosis), no generally accepted mechanism for these vascular events has yet emerged.

Treatment is directed toward early reduction of the elevated levels of homocysteine and methionine in the blood. In addition to a diet low in methionine, very high oral doses of pyridoxine (vitamin B6) have significantly decreased the plasma levels of homocysteine and methionine in some patients with cystathionine β -synthase deficiency. (Genetically determined “responders” to pyridoxine treatment make up approximately 50% of type I homocystinurics.) PLP serves as a cofactor for cystathionine β -synthase; however, the molecular properties of the defective enzyme that confer the responsiveness to B6 therapy are not known.

The terms *hypermethioninemia*, *homocystinuria* (or *-emia*), and *cystathioninuria* (or *-emia*) designate biochemical abnormalities and are not specific clinical diseases. Each may be caused by more than one specific genetic defect. For example, at least seven distinct genetic alterations can cause increased excretion of homocystine in the urine. A deficiency of cystathionine β -synthase is the most common cause of homocystinuria; more than 600 such proven cases have been studied.



The pathologic findings that underlie the clinical features manifested by **Homer Sistine** are presumed (but not proved) to be the consequence of chronic elevations of homocysteine (and perhaps other compounds, e.g., methionine) in the blood and tissues. The zonular fibers that normally hold the lens of the eye in place become frayed and break, causing dislocation of the lens. The skeleton reveals a loss of bone ground substance (i.e., osteoporosis), which may explain the curvature of the spine. The elongation of the long bones beyond their normal genetically determined length leads to tall stature.

Animal experiments suggest that increased concentrations of homocysteine and methionine in the brain may trap adenosine as S-adenosylhomocysteine, diminishing adenosine levels. Because adenosine normally acts as a central nervous system depressant, its deficiency may be associated with a lowering of the seizure threshold as well as a reduction in cognitive function.

BIOCHEMICAL COMMENTS



Phenylketonuria. Many enzyme deficiency diseases have been discovered that affect the pathways of amino acid metabolism. These deficiency diseases have helped researchers to elucidate the pathways in humans, in whom experimental manipulation is, at best, unethical. These spontaneous mutations (“experiments” of nature), although devastating to patients, have resulted in an understanding of these diseases that now permit treatment of inborn errors of metabolism that were once considered to be untreatable.

Classic PKU is caused by mutations in the gene located on chromosome 12 that encodes the enzyme phenylalanine hydroxylase (PAH). This enzyme normally catalyzes the hydroxylation of phenylalanine to tyrosine, the rate-limiting step in the major pathway by which phenylalanine is catabolized.

In early experiments, sequence analysis of mutant clones indicated a single base substitution in the gene with a G to A transition at the canonical 5' donor splice site of intron 12 and expression of a truncated unstable protein product. This protein lacked the C-terminal region, a structural change that yielded less than 1% of the normal activity of PAH.

Table 39.1. Genetic Disorders of Amino Acid Metabolism

Amino Acid Degradation Pathway	Missing Enzyme	Product That Accumulates	Disease	Symptoms
Phenylalanine	Phenylalanine hydroxylase	Phenylalanine	PKU (classical)	Mental retardation
	Dihydropteridine reductase	Phenylalanine	PKU (non-classical)	Mental retardation
	Homogentisate oxidase	Homogentisic acid	Alcaptonuria	Black urine, arthritis
	Fumarylacetoacetate hydrolase	Fumarylacetoacetate	Tyrosinemia I	Liver failure, death early
Tyrosine	Tyrosine aminotransferase	Tyrosine	Tyrosinemia II	Neurologic defects
	Cystathionase	Cystathionine	Cystathioninuria	Benign
Methionine	Cystathionine β -synthase	Homocysteine	Homocysteinemia	Cardiovascular complications and neurologic problems
Glycine	Glycine transaminase	Glyoxylate	Primary oxaluria type I	Renal failure due to stone formation
Branched-chain amino acids (leucine, isoleucine, valine)	Branched-chain α -keto acid dehydrogenase	α -Keto acids of the branched chain amino acids	Maple syrup urine disease	Mental retardation

Since these initial studies, DNA analysis has shown over 100 mutations (missense, nonsense, insertions, and deletions) in the PAH gene, associated with PKU and non-PKU hyperphenylalaninemia. That PKU is a heterogeneous phenotype is supported by studies measuring PAH activity in needle biopsy samples taken from the livers of a large group of patients with varying degrees of hyperphenylalaninemia. PAH activity varied from below 1% of normal in patients with classic PKU to up to 35% of normal in those with a non-PKU form of hyperphenylalaninemia (such as a defect in BH₄ production; see Chapter 48).

The genetic diseases affecting amino acid degradation that have been discussed in this chapter are summarized in Table 39.1.

Suggested References

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REVIEW QUESTIONS—CHAPTER 39

- If an individual has a vitamin B6 deficiency, which of the following amino acids could still be synthesized and be considered nonessential?
 - Tyrosine
 - Serine
 - Alanine
 - Cysteine
 - Aspartate

2. The degradation of amino acids can be classified into families, which are named after the end product of the degradative pathway. Which of the following is such an end product?
- (A) Citrate
 - (B) Glyceraldehyde-3-phosphate
 - (C) Fructose-6-phosphate
 - (D) Malate
 - (E) Succinyl-CoA
3. A newborn infant has elevated levels of phenylalanine and phenylpyruvate in her blood. Which of the following enzymes might be deficient in this baby?
- (A) Phenylalanine dehydrogenase
 - (B) Phenylalanine oxidase
 - (C) Dihydropteridine reductase
 - (D) Tyrosine hydroxylase
 - (E) Tetrahydrofolate synthase
4. Pyridoxal phosphate is required for which of the following reaction pathways or individual reactions?
- (A) Phenylalanine $\rightarrow$ tyrosine
 - (B) Methionine $\rightarrow$ cysteine + α -ketobutyrate
 - (C) Propionyl CoA $\rightarrow$ succinyl-CoA
 - (D) Pyruvate $\rightarrow$ acetyl-CoA
 - (E) Glucose $\rightarrow$ glycogen
5. A folic acid deficiency would interfere with the synthesis of which of the following amino acids from the indicated precursors?
- (A) Aspartate from oxaloacetate and glutamate
 - (B) Glutamate from glucose and ammonia
 - (C) Glycine from glucose and alanine
 - (D) Proline from glutamate
 - (E) Serine from glucose and alanine

40 Tetrahydrofolate, Vitamin B12, And S-Adenosylmethionine

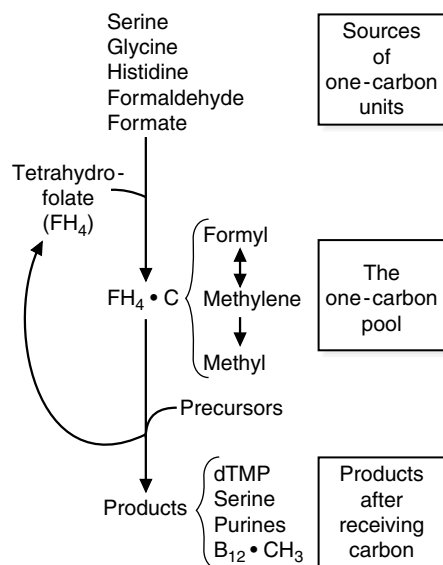


Fig. 40.1. Overview of the one-carbon pool. $\text{FH}_4\cdot\text{C}$ indicates tetrahydrofolate (FH_4) containing a one-carbon unit that is at the formyl, methylene, or methyl level of oxidation (see Fig. 40.3). The origin of the carbons is indicated, as are the final products after a one-carbon transfer.

Groups containing a single carbon atom can be transferred from one compound to another. These carbon atoms may be in a number of different oxidation states. The most oxidized form, CO_2 , is transferred by biotin. One-carbon groups at lower levels of oxidation than CO_2 are transferred by reactions involving tetrahydrofolate (FH_4), vitamin B12, and S-adenosylmethionine (SAM).

Tetrahydrofolate: Tetrahydrofolate, which is produced from the **vitamin folate**, is the primary one-carbon carrier in the body. This vitamin obtains one-carbon units from serine, glycine, histidine, formaldehyde, and formate (Fig. 40.1). While these carbons are attached to FH_4 they can be either oxidized or reduced. Because of this, folate can exist in a variety of chemical forms. Once a carbon has been reduced to the methyl level (methyl- FH_4), however, it cannot be re-oxidized. Collectively, these one-carbon groups attached to their carrier FH_4 are known as the **one-carbon pool**. The term folate is used to represent a water-soluble B-complex vitamin that functions in transferring single-carbon groups at various stages of oxidation.

The one-carbon groups carried by FH_4 are used for many biosynthetic reactions. For example, one-carbon units are transferred to the pyrimidine base of deoxyuridine monophosphate (**dUMP**) to form deoxythymidine monophosphate (**dTMP**), to the amino acid **glycine** to form **serine**, to precursors of the purine bases to produce carbons C2 and C8 of the **purine ring**, and to **vitamin B12**.

Vitamin B12: Vitamin B12 is involved in two reactions in the body. It participates in the rearrangement of the methyl group of L-methylmalonyl-CoA to form succinyl-CoA, and it transfers a methyl group, obtained from FH_4 , to homocysteine, forming methionine.

S-adenosylmethionine (SAM): SAM, produced from methionine and adenosine triphosphate (ATP), transfers the methyl group to precursors forming a number of compounds, including creatine, phosphatidylcholine, epinephrine, melatonin, methylated nucleotides, and methylated DNA.

Methionine metabolism is very dependent on both FH_4 and vitamin B12.

Homocysteine is derived from methionine metabolism and can be converted back into methionine by using both methyl- FH_4 and vitamin B12. This is the only reaction in which methyl- FH_4 can donate the methyl group. If the enzyme that catalyzes this reaction is defective, or if vitamin B12 or FH_4 levels are insufficient, homocysteine will accumulate. Elevated homocysteine levels have been linked to cardiovascular and neurologic disease. A vitamin B12 deficiency can be brought about by the lack of **intrinsic factor**, a gastric protein required for the absorption of dietary B12. A consequence of vitamin B12 deficiency is the accumulation of methyl- FH_4 and a decrease in other folate derivatives. This is known as the

methyl-trap hypothesis, in which, because of the B12 deficiency, most of the carbons in the FH_4 pool are trapped in the methyl- FH_4 form, which is the most stable. The carbons cannot be released from the folate, because the one reaction in which they participate cannot occur because of the B12 deficiency. This will therefore lead to a functional folate deficiency, even though total levels of folate are normal. A folate deficiency (whether functional or actual) will lead to **megaloblastic anemia** caused by an inability of blood cell precursors to synthesize DNA and therefore to divide. This leads to large, partially replicated cells being released into the blood to attempt to replenish the cells that have died. Folate deficiencies also have been linked to an increased incidence of **neural-tube defects**, such as spina bifida, in mothers who become pregnant while folate deficient.



THE WAITING ROOM



After resection of the cancer in his large intestine and completion of a course of postoperative chemotherapy with 5-fluorouracil (5-FU), **Colin Tuma** returned to his gastroenterologist for a routine follow-up colonoscopy. His colon was completely normal, with excellent healing at the site of the anastomosis. His physician expressed great optimism about a cure of Colin's previous malignancy but cautioned him about the need for regular colonoscopic examinations over the next few years.



Bea Twelvwlow, a 75-year-old woman, went to see her physician because of a numbness and tingling in her arms. A diet history indicated a normal and healthy diet, but Bea was not taking any supplemental vitamin pills. Laboratory results indicated a slight elevation of methylmalonic acid, and this led the physician to suspect a vitamin B12 deficiency. Direct measurement of serum B12 levels did indicate a deficiency, but the results of a Schilling test were normal.



The initial laboratory profile, determined when **Jean Ann Tonich** first presented to her physician with evidence of early alcohol-induced hepatitis, included a hematologic analysis that showed that Jean Ann was anemic. Her hemoglobin was 11.0 g/dL (reference range = 12–16 for an adult female). The erythrocyte (red blood cell) count was 3.6 million cells/mm³ (reference range = 4.0–5.2 for an adult female). The average volume of her red blood cells (mean corpuscular volume, or MCV) was 108 fL (reference range = 80–100), and the hematology laboratory reported a striking variation in the size and shape of the red blood cells in a smear of her peripheral blood (see Chapter 44). The nuclei of the circulating granulocytic leukocytes had increased nuclear segmentation (polysegmented neutrophils). Because these findings are suggestive of a macrocytic anemia (in which blood cells are larger than normal), measurements of serum folate and vitamin B12 (cobalamin) levels were ordered.

I. TETRAHYDROFOLATE (FH_4)

A. Structure and Forms of FH_4

Folates exist in many chemical forms. The coenzyme form that functions in accepting one-carbon groups is tetrahydrofolate polyglutamate (Fig. 40.2), generally just referred to as tetrahydrofolate or FH_4 . It has three major structural components, a bicyclic pteridine ring, para-aminobenzoic acid, and a polyglutamate tail consisting



The Schilling test involves the patient ingesting radioactive (Co^{60}) crystalline vitamin B12 after which a 24-hour urine sample is collected. The radioactivity in the urine sample is compared with the input radioactivity, and the difference represents the amount of B12 absorbed through the digestive tract.



Folate deficiencies frequently occur in individuals with chronic alcoholism. A number of factors are involved: inadequate dietary intake of folate; direct damage to intestinal cells and brush border enzymes, which interferes with absorption of dietary folate; a defect in the enterohepatic circulation, which reduces the absorption of folate; liver damage causing decreased hepatic production of plasma proteins; and interference with kidney resorption of folate.

Because of the possibility of a direct toxic effect of alcohol on hematopoietic tissues in **Jean Ann Tonich**, a bone marrow aspirate was performed. The aspirate contained a greater than normal number of red and white blood cell precursors, most of which were larger than normal. The large red blood cells are called megaloblasts.

These hematopoietic precursor cells when exposed to too little folate and/or vitamin B12 show slowed cell division, but cytoplasmic development occurs at a normal rate. Hence, the megaloblastic cells tend to be large, with an increased ratio of RNA to DNA. Megaloblastic erythroid progenitors are usually destroyed in the bone marrow (although some reach the circulation). Thus, marrow cellularity is often increased but production of red blood cells is decreased, a condition called "ineffective erythropoiesis."

Therefore, Jean Ann had a megaloblastic anemia, characteristic of a folate or B12 deficiency.

Jean Ann's serum folic acid level was 3.1 ng/mL (reference range = 6–15), and her serum B12 level was 154 pg/mL (reference range = 150–750). Her serum iron level was normal. It was clear, therefore, that Jean Ann's megaloblastic anemia was caused by a folate deficiency (although her B12 levels were in the low range of normal). The management of a pure folate deficiency in an alcoholic patient includes cessation of alcohol intake and a diet rich in folate.



The abbreviation fL stands for *femtoliters*. One fL is 10^{-12} milliliters (mL).



Sulfa drugs, which are used to treat certain bacterial infections, are analogs of para-aminobenzoic acid. They prevent growth and cell division in bacteria by interfering with the synthesis of folate. Because we cannot synthesize folate, sulfa drugs do not affect human cells in this way.



The current recommended dietary allowance (RDA) for folate equivalents is approximately 400 μg for adult men and women. In addition to being prevalent in green leafy vegetables, other good sources of this vitamin are liver, yeast, legumes, and some fruits. Protracted cooking of these foods, however, can destroy up to 90% of their folate content. A standard US diet provides 50 to 500 μg absorbable folate each day. Folate deficiency in pregnant women, especially during the month before conception and the month after, increases the risk of neural tube defects, such as spina bifida, in the fetus. To reduce the potential risk of neural tube defects for women capable of becoming pregnant, the recommendation is to take 400 μg of folic acid daily from a multivitamin pill. If the woman has a history of having a child with a neural tube defect, this amount is increased to 4000 $\mu\text{g}/\text{day}$ for the month before and the month after conception. Flour-containing products in the United States are now supplemented with folate to reduce the risk of neural tube defects in newborns.

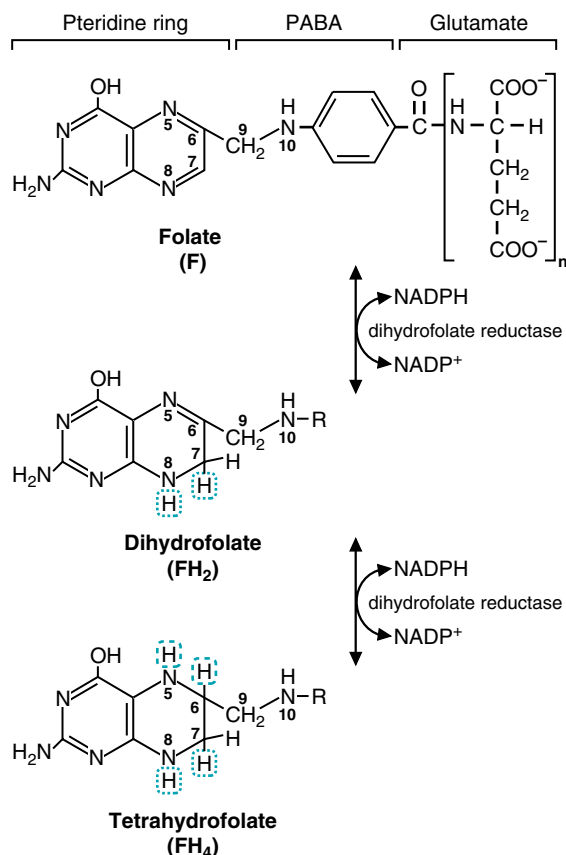


Fig. 40.2. Reduction of folate to tetrahydrofolate (FH₄). The same enzyme, dihydrofolate reductase, catalyzes both reactions. Multiple glutamate residues are added within cells ($n \sim 5$). Plants can synthesize folate, but humans cannot. Therefore, folate is a dietary requirement. R is the portion of the folate molecule shown to the right of N¹⁰. The different precursors of FH₄ are indicated in the figure. PABA = para-aminobenzoic acid.

of several glutamate residues joined in amide linkage. The one-carbon group that is accepted by the coenzyme and then transferred to another compound is bound to N⁵, to N¹⁰, or to both.

Different forms of folate may differ in the oxidation state of the one-carbon group, in the number of glutamate residues attached, or in the degree of oxidation of the pteridine ring. When the term *folate* or *folic acid* is applied to a specific chemical form, it is the most oxidized form of the pteridine ring (see Fig. 40.2). Folate is reduced to dihydrofolate and then to tetrahydrofolate by dihydrofolate reductase present in cells. Reduction is the favored direction of the reaction; therefore, most of the folate present in the body is present as the reduced coenzyme form, FH₄.

B. The Vitamin Folate

Folates are synthesized in bacteria and higher plants and ingested in green leafy vegetables, fruits, and legumes in our diet. The vitamin was named for its presence in green, leafy vegetables (foliage). Most of the dietary folate derived from natural food sources is present in the reduced coenzyme form. However, vitamin supplements and fortified foods contain principally the oxidized form of the pteridine ring.

As dietary folates pass into the proximal third of the small intestine, folate conjugases in the brush border of the lumen cleave off glutamate residues to produce the monoglutamate form of folate, which is then absorbed (see Fig. 40.2, upper